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Method Development for *O*-glycan Analysis and its Application to Parkinson's Disease

By

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18205440

A thesis submitted to University College Dublin in fulfilment of the requirements for the degree of Research Masters of Science (Major in Medicine)

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September 2020

TABLE OF CONTENTS

TABLE OF CONTENTS ii

ABSTRACT v

STATEMENT OF ORIGINAL AUTHORSHIP vi

COLLABORATIONS vii

ACKNOWLEDGEMENTS viii

PUBLICATIONS, POSTER PRESENTATIONS, AWARDS AND FUNDING ix

LIST OF FIGURES x

LIST OF TABLES xiii

ABBREVIATIONS xv

CHAPTER 1 – INTRODUCTION 1

1.1 GLYCANS AND THEIR IMPORTANCE 1
1.1.1 An Introduction to Glycans and their Classes 1
1.1.2 Glycan Biosynthesis and Roles 2
1.1.2.1 *N*-Glycans 2
1.1.2.2 *O*-Glycans 2

1.2 TECHNIQUES TO ANALYSE *O*-GLYCANS FROM GLYCOPROTEINS 5
1.2.1 *N*-Glycan Release Methods 5
1.2.2 *O*-Glycan Release Introduction 5
1.2.3 *O*-Glycan Release Methods 6
1.2.3.1 Enzymatic Release 6
1.2.3.2 Chemical Release 7
1.2.3.2.1 β-elimination 7
1.2.3.2.2 The Peeling Reaction 8
1.2.3.2.3 Reductive β-elimination 8
1.2.3.2.4 Hydrazinolysis – Non-reductive β-elimination 9
1.2.3.2.5 Ammonia Based/Mildly Alkaline Non-Reductive β-elimination 9
1.2.3.2.6 Microwave-Assisted Non-Reductive β-elimination 10
1.2.3.2.7 Oxidative Release of Natural Glycans 10
1.2.4 Glycan Labelling Techniques 10
1.2.5 Analytical Techniques to Characterize *O*-Glycans 12
1.2.5.1 Ultra-Performance Liquid Chromatography 14
1.2.5.2 Exoglycosidase Arrays 15
1.2.5.3 Mass Spectrometry 16
1.2.5.3.1 Fragmentation Approaches 17
1.2.5.4 Additional Techniques for Separation and Analysis 19
1.2.6 Analytical Methods to Characterize Intact *O*-glycoproteins/peptides 20
1.2.7 Relative vs. Absolute Quantification 23
1.2.8 Predictive Technologies 24

1.3 *O*-GLYCOMICS IN DISEASE 25
1.3.1 Glycan Expression 25
1.3.2 Relating Gene and Glycosyltransferase Expression to Glycome 26

1.4 CONCLUSION 28

CHAPTER 2 – METHOD DEVELOPMENT FOR MICROWAVE-ASSISTED NON-REDUCTVE RELEASE OF O-GLYCANS 29

2.1 INTRODUCTION 29

2.2 MATERIALS AND METHODS 30
2.2.1 Visual Schematic of the Release and Analysis Protocol 30
2.2.2 *N*-Glycan Release 30
2.2.3 *O*-glycan release (microwave-assisted non-reductive β-elimination) 31
2.2.4 HILIC-UPLC Analysis of non-reductively released (2AB labelled) glycans 31
2.3 RESULTS – OPTIMIZING NON-REDUCTIVE RELEASE METHOD 33
2.3.1 Identifying a Suitable Separation Gradient and Elution Conditions 33
2.3.2 Comparison Between Traditional Oven and Microwave-Assisted Ammonia Based Non-Reductive β-elimination Approaches 34
2.3.3 Final Optimizations Leading to the Established Protocol 35
2.3.3.1 Pre-Release Optimizations 35
2.3.3.1.1 Reduce and Alkylate 35
2.3.3.1.2 *N*-glycan Release 36
2.3.3.1.3 Sample Phase 37
2.3.3.2 Release Optimizations 38
2.3.3.2.1 Microwave Reaction Times 38
2.3.3.2.2 Microwave Reaction Temperatures 39
2.3.3.2.3 Ammonium Carbonate Concentration 40
2.3.3.3 Post-Release Optimizations 41
2.3.3.3.1 Neutralization Conditions 41
2.3.3.3.2 PGC SPE Desalting Conditions 42
2.3.4 Reproducibility studies 43
2.3.4.1 Fetuin and Porcine Gastric Mucin Standards 43
2.4 DISCUSSION 47
2.5 CONCLUSION 48
CHAPTER 3 – CHARACTERISATION AND QUANTITATION OF O-GLYCANS IN PARKINSON'S DISEASE *49*
3.1 INTRODUCTION 49
3.1.1 Introduction to Parkinson's Disease 49
3.1.2 Braak Staging 49
3.1.3 Glycosylation Changes Observed in Parkinson's Disease 50
3.1.4 Research Objectives for PD studies 51
3.2 MATERIALS AND METHODS 52
3.2.1 Initial Sample Preparation 52
3.2.2 *N*-Glycan Release 53
3.2.3 *O*-Glycan Release 54
3.2.3.1 Non-reductive *O*-glycan release 54
3.2.3.2 Reductive *O*-glycan release 54
3.2.4 *O*-Glycan Analysis 55
3.2.4.1 Liquid Chromatography 55
3.2.4.1.1 WAX-UPLC Analysis of non-reductively released (2AB labelled) glycans 55
3.2.4.1.2 HILIC-UPLC Analysis of non-reductively released (2AB labelled) glycans 55
3.2.4.2 Mass Spectrometry 56
3.2.4.2.1 Phynexus Tip Clean-up for MS 56
3.2.4.2.2 Mass Spectrometry (QToF) analysis of non-reductively released (2AB labelled) glycans 57
3.2.4.2.3 PGC-HPLC-ESI-MS (Ion-trap) analysis of reductively released glycans 58
3.2.4.3 Exoglycosidase Arrays 59
3.2.5 Statistical Analysis 60
3.2.5.1 MANOVA, Non-Parametric and Tukey Tests 61
3.2.5.2 Primary Component Analysis (PCA) 61
3.3 RESULTS – CHARACTERIZATION AND STATISTICAL ANALYSIS OF *O*-GLYCANS 62
3.3.1 Reproducibility Studies 62
3.3.2 Chromatogram Integration of 2AB-Labelled *O*-glycans 63
3.3.3 *O*-Glycan Naming Convention 64
3.3.4 WAX-UPLC Analysis 66

3.3.5 Exoglycosidase Array 67
3.3.5.1 Exoglycosidase Limitations 68
3.3.5.2 Interpreting Exoglycosidase Panel 69
3.3.6 Mass Spectrometry 70
3.3.6.1 Structural Characterization using MSMS 72
3.3.7 All *O*-Glycans Across Methods 75
3.3.8 Statistical Differences – Disease State 78
3.3.9 Statistical Differences – Gender 84
3.3.10 Statistical Differences – Age 88

3.4 DISCUSSION 91
3.4.1 *O*-glycosylation Changes in ILBD 91
3.4.1.1 Mannose Core 91
3.4.1.2 Glucuronylation 91
3.4.2 *O*-glycosylation in PD 92
3.4.2.1 Sialylation 92
3.4.2.2 Sulfation 92
3.4.3 Temporal Differences 93
3.4.4 Spatial Differences 93
3.4.5 Age and Gender Differences 93

3.5 CONCLUSION 95

CHAPTER 4 – GENERAL DISCUSSION AND CONCLUDING REMARKS 96

4.1 FUTURE DIRECTION FOR *O*-GLYCAN CHARACTERIZATION IN PD 96

4.2 GLYCAN THERAPEUTICS FOR PD 98

4.3 CONCLUSION 99

APPENDICES 100

REFERENCES 130

ABSTRACT

Glycosylation is the most common post-translational modification of proteins with glycoproteins performing roles in cell-cell interactions, protein folding and function. By observing glycosylation changes between control and diseased tissues, biomarkers may be identified and regenerative bio-implants created such as glycan-functionalized implantable hydrogels. Two main classes of glycans present on proteins are *N*-glycans (bound to asparagine residues) and *O*-glycans (bound to serine and threonine residues). Due to the availability of endoglycosidases such as PNGase F which cleaves common oligomannose, complex and hybrid *N*-glycans from glycoproteins, there is a strong understanding of *N*-glycan heterogeneity, however, no single approach exists to release *O*-glycans with the same level of proficiency due in-part to a lack of common *O*-glycan core. Enzymatic methods may be used to selectively cleave only the most basic *O*-glycan disaccharides with no further saccharide modification, however, this method is severely limited. Alternative chemical methods are used (typically under alkaline conditions where *O*-glycosylation sites are labile) to release the vast array of more complex *O*-glycan structures. The downside to this is a process termed “peeling” in which the glycan undergoes rapid stepwise monosaccharide degradation from the cleaved end of the structure.

The objectives of this Research Masters were **(i)** to critically evaluate the state-of-the-art of *O*-glycosylation, addressing strengths and weaknesses of the various forms of release and analysis, **(ii)** to develop and optimize an *O*-glycan release protocol by applying a mildly alkaline reaction solution to glycoprotein standards, under microwave radiation, in a process termed *microwave-assisted ammonia-based non-reductive β-elimination* to liberate *O*-glycans from the protein for subsequent fluorescence and mass spectrometric analysis; and **(iii)** to apply this optimized release and analysis protocol to characterize and statistically evaluate *O*-glycan changes between healthy and diseased patients with Parkinson’s Disease (PD) and Incidental Lewy Body Disease (ILBD; a progenitor disease or “pre-diagnosis” to PD).

The optimization of the *O*-glycan release protocol was successful in achieving reproducibility of release over 10 replicates of bovine fetuin standard glycoprotein, with peeling reduced to levels typically seen in commercially available released *O*-glycan standards. This instilled confidence in *O*-glycan release and analysis of PD, where 77 *O*-glycans were identified. It was discovered that sialylation upregulation was observed in nigrostriatal regions (striatum and substantia nigra) in PD. Sulfation was also observed to decrease in striatum upon PD pathogenesis, and total glucuronylation and presence of mannose core structures were found to be decreased in ILBD patients. The work set out here identified several suitable biomarkers, not yet described in the literature, for further studies into glycan therapeutics and PD research.

STATEMENT OF ORIGINAL AUTHORSHIP

I hereby certify that the submitted work is my own work, was completed while registered as a candidate for the degree of Research Masters of Science, and I have not obtained a degree elsewhere on the basis of the research presented in this submitted work.

Hayden Feher Wilkinson (September 2020)

COLLABORATIONS

Prof. Abhay Pandit: CÚRAM SFI Research Centre for Medical Devices, Ireland – Director

Professor Abhay Pandit is the director of CÚRAM who provided the funding opportunity for this degree. Acting as supervisor of PhD candidate Ana Lúcia Rebelo from National University of Ireland Galway, together they provided PD and ILBD patient samples (in glycoprotein gel form) for this work.

Prof. Niclas Karlsson: Gothenburg University, Sweden – Head of GlycoScience Research Group

Professor Niclas Karlsson acted as collaborator for in-depth mass spectrometry analytical studies for the project. This included a week trip to his laboratory for patient sample analysis. His experience in *O*-glycan studies contributed significantly to my understanding of *O*-glycosylation presented in this work.

Prof. Pauline Rudd: National Institute of Bioprocessing, Research and Training, Ireland

Professor Pauline Rudd provided oversight of this project, offering guidance and expertise in all aspects of glycobiology.

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Publications

- **Wilkinson H**, Saldova R. Current methods for the characterization of *O*-glycans – Under Review

- Colombo M, Shehni A.A, Thoma I, McGurnaghan S, Blackbourn L, **Wilkinson H**, Collier A, Patrick A, Petrie J, McKeigue P, Saldova R, Colhoun, H. Quantitative levels of serum *N*-glycans in type 1 diabetes and their association with kidney disease – Submitted

Conference Proceedings

- **Wilkinson H**, Rebelo A.L, Pandit A, Saldova R. Method development of *O*-glycan analyses for use in regenerative medicine. Glycoconj. J. 2019;36(4): 377

Poster Presentations

- 25th International Symposium on Glycoconjugates – August 25th-31st 2019 – Milan, Italy
    - Method development of *O*-glycan analyses for use in regenerative medicine

- Thomas Dillon Centenary Symposium – October 2nd 2019 – Galway, Ireland
    - Method development of *O*-glycan analyses for use in regenerative medicine

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- Dillon Symposium Threesis – Second Place – December 2nd 2019, Galway, Ireland

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LIST OF FIGURES

Figure 1. *N*- and *O*-glycans. *N*-glycans bound to asparagine residues and *O*-glycans bound to serine and threonine residues. 1
Figure 2. Production of the Thomsen-noveau (Tn) Antigen and subsequent elongation by various glycosyltransferases to generate 8 core *O*-GalNAc glycan structures. A further monosaccharide addition occurs with sialic acid giving rise to a sialyl-Tn Antigen 3
Figure 3. β-Elimination reaction of a typical mucin type *O*-glycan Core 2 including non-reduced glycan peeling reaction. 7
Figure 4. Released *O*-glycan derivatisation and end-capping strategies. 11
Figure 5. Sample exoglycosidase array showing digestion of fetuin *O*-glycans using cocktails of enzymes. 16
Figure 6. Fragmentation ions of a simple core 1 GlcNAc extended *O*-glycan. 18
Figure 7. Gene silencing of COSMC chaperone has knock-on effect of no core 1 synthesis and subseuqent core 2 synthesis and extensions. Overexpression of Tn and sTn may result. 27
Figure 8. Schematic of release protocol for both *N*- and *O*-glycans showing traditional enzymatic release of *N*-glycans and a microwave assisted non-reductive β-elimination release of *O*-glycans. 30
Figure 9. HILIC-UPLC elution conditions and separation of *O*-glycans from bovine fetuin. 33
Figure 10. HILIC-UPLC profiles of *O*-glycans from bovine fetuin glycoprotein: Heating method test. 34
Figure 11. HILIC-UPLC profiles of *O*-glycans from bovine fetuin glycoprotein: Reduce and alkylate test. 36
Figure 12. HILIC-UPLC profiles of *O*-glycans from bovine fetuin glycoprotein: PNGase F treatment test. 37
Figure 13. HILIC-UPLC profiles of *O*-glycans from bovine fetuin glycoprotein: Sample phase test. 38
Figure 14. HILIC-UPLC profiles of *O*-glycans from bovine fetuin glycoprotein: Microwave time test. 39
Figure 15. HILIC-UPLC profiles of *O*-glycans from bovine fetuin glycoprotein: Microwave temperature test. 40
Figure 16. HILIC-UPLC profiles of *O*-glycans from bovine fetuin glycoprotein: Ammonium carbonate end cap concentration test. 41
Figure 17. HILIC-UPLC profiles of *O*-glycans from bovine fetuin glycoprotein: Neutralization test. 42
Figure 18. HILIC-UPLC profiles of *O*-glycans from bovine fetuin glycoprotein: Neutralize and desalt test. 43

Figure 19. HILIC-UPLC profiles of *O*-glycans from bovine fetuin glycoprotein across 10x replicate using the optimized method. 44
Figure 20. Analysis of average peak area of fetuin (10x replicated) plotted alongside coefficient of variation (CV) for each peak. 44
Figure 21. HILIC-UPLC profiles of *O*-glycans from porcine gastric mucin across 10x replicate using the optimized method. 45
Figure 22. Analysis of average peak area of PGM (10x replicate) plotted alongside coefficient of variation (CV) for each peak. 46
Figure 23. Representative HILIC-UPLC profiles of O-glycans from pooled (A) healthy striatum, (B) PD striatum, (C) healthy substantia nigra, (D) PD substantia nigra. 62
Figure 24. Relative average glycan peak area (n=6) of healthy substantia nigra and calculated coefficient of variation (CV) for each peak. 63
Figure 25. Representative chromatograms of *O*-glycans in human striatum and substantia nigra. (A) C058A – Healthy striatum, (B) PD683A – PD striatum, (C) PD529AC6 – Healthy substantia nigra, (D) PD687BS – PD substantia nigra. 64
Figure 26. Examples of short form notation of the long form name. 65
Figure 27. WAX-UPLC Chromatograms of *O*-glycans from (A) Fetuin standard, (B) healthy striatum, (C) PD striatum, (D) healthy substantia nigra, (E) PD substantia nigra. Charge states indicated (S0, S1, S2, S3) as well as junk and free 2AB for each chromatogram. (F) Relative abundance breakdown of each charge state for each glycan pool B-E. 67
Figure 28. Exoglycosidase array showing peak shifts from subsequent digestions. 69
Figure 29. MS examples including TIC and EIC for both MS methods. 71
Figure 30. Fragmentation pattern for S3ga3[S6]**GA**. From healthy striatum. 72
Figure 31. Fragmentation pattern for the sialic-acid-sulfated *O*-glycan structure S3ga3[sS6]**GA**. From healthy striatum. 72
Figure 32. Fragmentation pattern for the Lewis-type X epitope pentasaccharide structure ga3[ga4(fa3GL6)]**GA**. From healthy striatum. 73
Figure 33. Fragmentation pattern for the sulfated Lewis X type mannose core pentasaccharide structure ga4(fa3sGL2)**m**. From healthy striatum. 73
Figure 34. Fragmentation pattern for the sulfated glucuronic acid-capped mannose core tetrasaccharide sU3ga4GL2**m**. From healthy striatum. 74
Figure 35. Fragmentation pattern for the branched mannose core septasaccharide S3ga4GL2[S3ga4GL6]**m**. From diseased striatum. 74
Figure 36. Aligning (A) Representative chromatogram (PD204P; PD substantia nigra) against (B) UND profile from PDBS digestion. 78
Figure 37. Nested tables showing % area for individual samples in each group compared against disease states in striatum. 82

Figure 38. Nested table showing % area for individual samples in each group compared against disease states in substantia nigra. 82
Figure 39. PCA plots outlining separation (difference) of healthy striatum, ILBD striatum and PD striatum. 83
Figure 40. Brain regions affected of each Braak Stage to be considered for separate analysis of glycosylation. 97

LIST OF TABLES

Table 1. Summary table of main *O*-glycan release methods. Peeling levels indicate variability even within the same methods. 6

Table 2. Summary table of main glycan separation and analysis methods and their benefits 13

Table 3. Elution conditions on HILIC-UPLC for 2AB labelled *O*-glycans. A is 50 mM ammonium formate pH 4.4, B is acetonitrile. 32

Table 4. Details of the human brain tissue of PD, ILBD and control cases used in this study. Braak stage not applicable in control patients. 53

Table 5. Elution conditions on WAX-UPLC for 2AB labelled *O*-glycans. A is 20% v/v acetonitrile, B is 0.1 M ammonium acetate pH 7.0 in 20% acetonitrile. 55

Table 6. Elution conditions on HILIC-UPLC for 2AB labelled *O*-glycans. A is 50 mM ammonium formate pH 4.4, B is acetonitrile. 56

Table 7. Elution conditions on HILIC-UPLC for 2AB labelled *O*-glycans. A is 50 mM ammonium formate pH 4.4, B is acetonitrile. 57

Table 8. Elution conditions on PGC-LC for *O*-glycan alditols. A is 10 mM ammonium bicarbonate (ABC), B is 10 mM ABC in 80% acetonitrile. 58

Table 9. Exoglycosidase information including specificty and abbreviation used throughout this work. 59

Table 10. Exoglycosidase array exoglycosidase/buffer/water volumes (in µL) for each sample replicate. 50 mM sodium acetate pH 5.5 used as buffer in all experiments except JBM digest: x – 100 mM sodium acetate. y – 2 mM Zn2+, pH 5.0. 60

Table 11. Short-hand notation for *O*-glycan naming. 65

Table 12. *O*-glycans characterized across all PD and control groups. “y” – no MS2 data. Glycans highlighted in green are major component of peak. 77

Table 13. P-values derived from comparison of each peak across different groups comparing disease states. 79

Table 14. P-values derived from comparison of glycan features across different groups comparing disease states. 80

Table 15. P-values derived from comparison of each peak across different groups comparing disease states in females. 85

Table 16. P-values derived from comparison of each peak across different groups comparing disease states in males. 86

Table 17. P-values derived from comparison of glycan features across different groups comparing disease states from females. 87

Table 18. P-values derived from comparison of glycan features across different groups comparing disease states from males. 87

Table 19. P-values derived from comparison of each peak across different groups comparing disease states separated by age (“under 85” and “85+”). 89
Table 20. P-values derived from comparison of glycan features across different groups comparing disease states separated by age (“under 85” and “85+”). 90

ABBREVIATIONS

$(NH_4)_2CO_3$	Ammonium Carbonate
2AA	2-Aminobenzoic Acid
2AB	2-Aminobenzamide
ABC	Ammonium Bicarbonate
ABS	α2-3,6,8,9 Sialidase
ACN	Acetonitrile
AD	Alzheimer's Disease
AMF	α1-3,4 Fucosidase
APTS	8-aminopyrene-1,3,6-trisulfonate
AQC	Aminoquinoline
Asn	Asparagine
BEMAD	β-elimination followed by Michael Addition
BKF	α1-2,3,4,6 Fucosidase
BTG	β1-3,4 Galactosidase
CBG	α1-3,4,6 Galactosidase
CCS	Collisional Cross Section
CE	Capillary Electrophoresis
CID	Collision Induced Dissociation
CMD	Congenital Muscular Dystrophy
COSMC	Core 1 β3-galactosyltransferase Specific Molecular Chaperone
COSY	Correlation Spectroscopy
CV	Coefficient of Variation
DFGL	Direct Fluorescent Glycan Labelling
DMS	Differential Mobility Spectrometry
DTT	Dithiothreitol
ECD	Electron Capture Dissociation
ECM	Extracellular Matrix
EIC	Extracted Ion Chromatogram
EOF	Electroosmotic Flow
ER	Endoplasmic Reticulum
ESI	Electrospray Ionization
ETD	Electron Transfer Dissociation
ExD	Electron Activated Dissociation
FAIMS	High-Field Asymmetric Waveform Ion Mobility

FLR	Fluorescence
Fuc	Fucose
Gal	Galactose
GalNAc	*N*-Acetylgalactosamine
Glc	Glucose
GlcNAc	*N*-Acetylglucosamine
GU	Glucose Units
GUH	β1-2,3,4,6-*N*-Acetylglucosaminidase
HCl	Hydrochloric Acid
HILIC	Hydrophilic Interaction Liquid Chromatography
HPLC	High Performance Liquid Chromatography
IAA	Iodoacetamide
IgG	Immunoglobulin G
ILB	Incidental Lewy Body
ILBD	Incidental Lewy Body Disease
IM	Ion Mobility
IRMPD	Infrared Multiphoton Dissociation
ISD	In-Source Decay
JBH	β1-3,4,6-*N*-Acetylglucosaminidase and β1-4-*N*-Acetylgalactosaminidase
JBM	α1-2,3,6 Mannosidase
LC	Liquid Chromatography
LiOH	Lithium Hydroxide
m/z	Mass to Charge Ratio
MALDI	Matrix Assisted Laser Desorption/Ionisation
Man	Mannose
MANOVA	Multivariate Analysis Of Variance
MIRAGE	Minimum Information Required for A Glycomics Experiment
MRM	Multiple Reaction Monitoring
MS	Mass Spectrometry
MS^1	Single dimension mass spectrometry
MS^2	Tandem mass spectrometry (fragmentation)
$NaBH_4$	Sodium Borohydride
NAN1	α2-3 Sialidase
NaOH	Sodium Hydroxide

NMR	Nuclear Magnetic Resonance
OGA	*O*-GlcNAcase
OGT	*O*-GlcNAc Transferase
PCA	Primary Component Analysis
PD	Parkinson's Disease
PDGF	Platelet-Derived Growth Factor
PGC	Porous Graphitized Carbon
PGM	Porcine Gastric Mucin
PMP	1-phenyl-3-methyl-5-pyrazolone
PNGase F	Peptide-*N*-glycosidase F
PREG	Polar Retention Effect On Graphite
PSD	Post-Source Decay
QC	Quality Control
QToF	Quadrupole Time of Flight
ROS	Reactive Oxygen Species
RP	Reverse Phase
SDS	Sodium Dodecyl Sufate
Ser	Serine
SNFG	Symbol Nomenclature For Glycans
SPE	Solid Phase Extraction
SPG	β1-4 Galactosidase
Thr	Threonine
TIC	Total Ion Chromatogram
Tn	Thomsen-noveau
TOCSY	Total Correlation Spectroscopy
TWIMS	Travelling Wave Ion Mobility Mass Spectrometry
UDP	Uridine Diphosphate
UPLC	Ultra-Performance Liquid Chromatography
UV	Ultra-Violet
WAX	Weak Anion Exchange

CHAPTER 1 – INTRODUCTION

1.1 GLYCANS AND THEIR IMPORTANCE

1.1.1 An Introduction to Glycans and their Classes

Carbohydrates are of incredible importance in eukaryotes performing a variety of functions associated with cellular metabolism needed for protein synthesis, movement and transport[1]. Carbohydrates may exist in a monomeric form (as is the case with glucose), or bound to other saccharides via glycosidic linkages (e.g sucrose, a disaccharide of glucose linked to fructose) and may be glycosylated to other macromolecules. Glycosylation is the most common post translational modification (PTM) in eukaryotes linking carbohydrates to proteins, nucleic acids, cell surfaces, secreted soluble molecules and other macromolecules[2].

The arrangement of saccharide moieties covalently linked to specific regions of these macromolecules gives rise to classes of carbohydrates called “glycans”. Nitrogen- or *N*-linked and oxygen- or *O*-linked glycans (abbreviated to *N*- and *O*-glycans) are the most commonly expressed and studied glycan modifications of proteins with each performing diverse roles in protein activity and recognition.[2, 3] These modifications are synthesised in the Golgi apparatus and endoplasmic reticulum as proteins travel along the secretory pathway and ultimately aid in biological processes involved in cellular interactions, protein folding, stability, signal transduction, as well as many other functions. [2, 4] (Glycan abbreviations and representation are shown in figure 1).

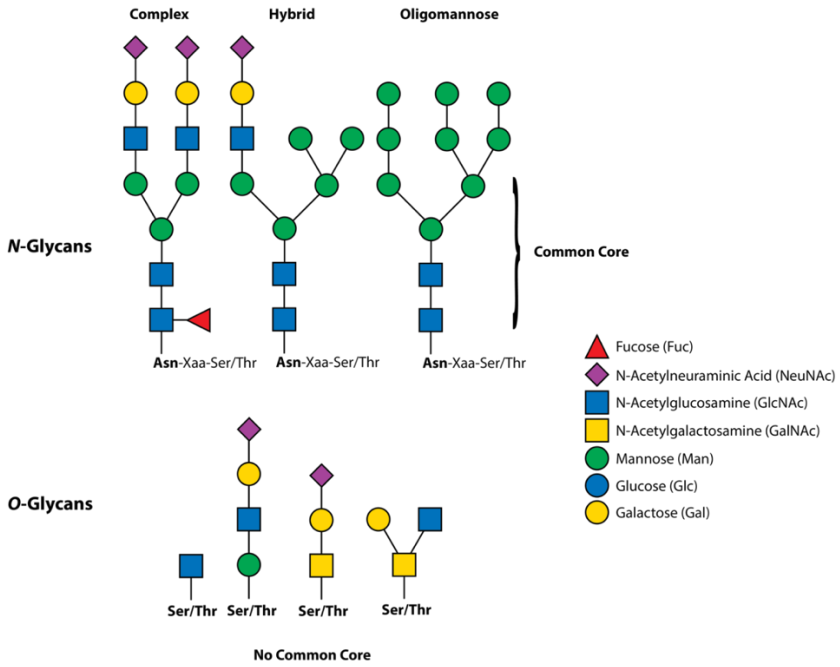


Figure 1. *N- and O-glycans. N-glycans bound to asparagine residues and O-glycans bound to serine and threonine residues. Symbol Nomenclature for Glycans (SNFG) used*[5, 6]*.*

1.1.2 Glycan Biosynthesis and Roles

1.1.2.1 *N*-Glycans

N-glycans are initially synthesised onto a lipid anchor in the endoplasmic reticulum (ER) to generate a 14 saccharide residue (Glc3Man9GlcNAc2) which is co-translationally linked to nitrogen on the side chain of some asparagine residues of the protein through the terminal GlcNAc residue of the saccharide structure. *N*-glycosylation sites are limited to binding to asparagine residues at Asn-Xaa-Ser/Thr sequons along the amino acid sequence (where Xaa represents any amino acid except proline)[7]. This structure is cleaved back and rebuilt in the ER and then further extended in the Golgi appartus through the activity of exoglycosidase (removing monosaccharides) and glycosyltransferases (adding monosaccharides) until the final glycan is created[8].

1.1.2.2 *O*-Glycans

Comparing to *N*-glycans, *O*-glycans are synthesised and integrated with proteins more ambiguously. While the sites for *O*-glycosylation are instead serine and threonine (and occasionally hydroxylysine), there is an absence of a common glycan core (initial core monosaccharide directly linked to serine and threonine) as well as a lack of a single consensus sequence for glycan attachment. In contrast to the *N*-glycosylation pathway (which oversees the attachment of a terminal GlcNAc by a single glycosyltransferase[9] initiating the development of the trimannosyl-chitibiose common core of *N*-glycans), *O*-glycosylation has the potential to be initiated by either mannose[10], galactose[11], fucose[12], glucose[12], xylose[13] or *N*-acetylglucosamine (GalNAc)[14] followed by further monosaccharide additions. These glycosylation events are mainly initiated in the Golgi apparatus after protein folding (in the ER). The most common *O*-glycosylation pathway for eukaryotic proteins, however, is through GalNAc-*O*-glycosylation occurring in the Golgi apparatus, after protein folding[15]. This process, commonly referred to as mucin-type *O*-glycosylation, is initiated by a family of ~20 GalNAc transferases which transpose GalNAc (from UDP-GalNAc) onto coil, turn and linker regions of the folded protein, generating the GalNAcα1-Ser/Thr "Tn Antigen"[16]. Glycosyltransferase activity may then generate 8 "core" glycan structures (figure 2), with cores 1-4 commonly extending further in a stepwise manner through further glycosyltransferase activity[17, 18], giving rise to heterogeniety, including topology, from linear to extensively branched glycan structures[19], which can contain blood group and lewis epitopes on their chains[20].

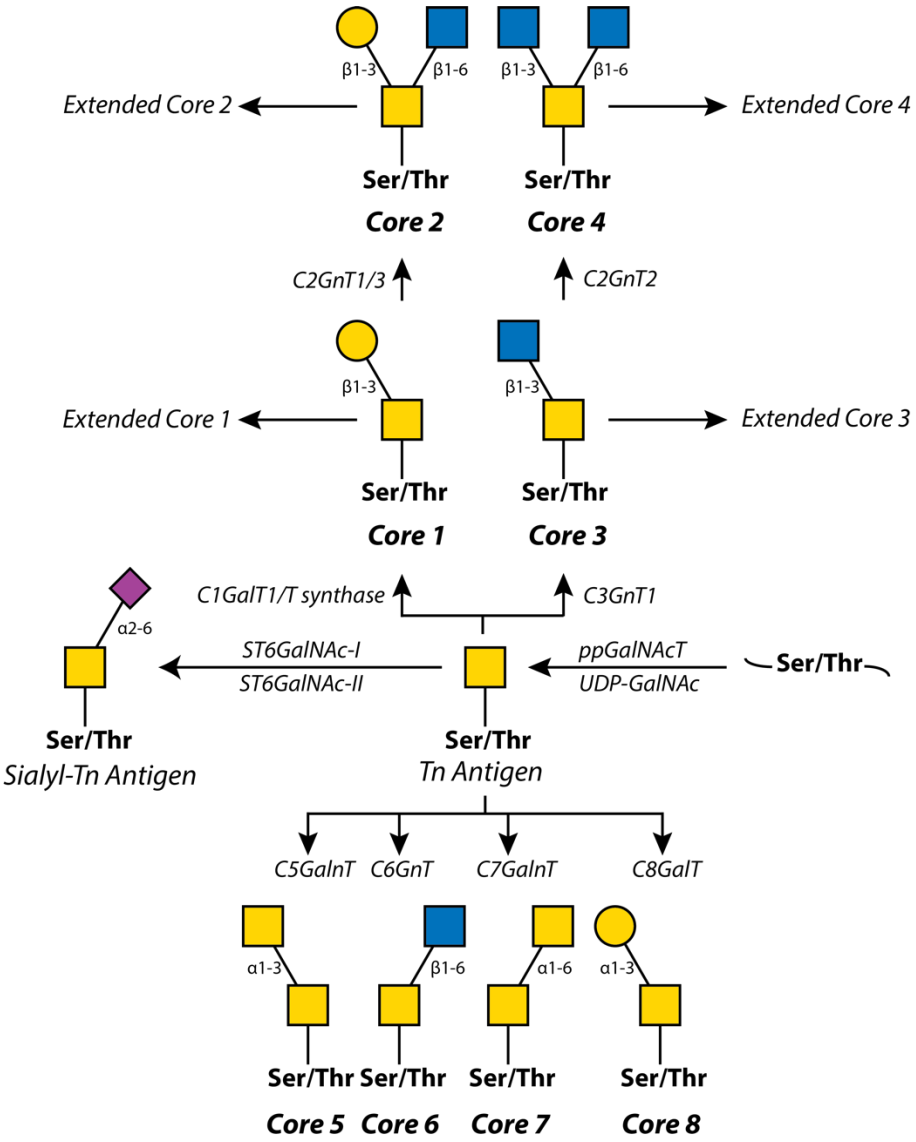


Figure 2. *Production of the Thomsen-noveau (Tn) Antigen and subsequent elongation by various glycosyltransferases to generate 8 core O-GalNAc glycan structures. A further monosaccharide addition occurs with sialic acid giving rise to a sialyl-Tn Antigen*[21]*.*

Of interest is the role that *O*-GlcNAcylated cytoplasmic and nuclear proteins play in tumorigensis, aging and neurodegenerative diseases[22-25]. *O*-GlcNAc is so far unique as it is the only *O*-glycan to be built and reversibly broken down by only two highly conserved enzymes *O*-GlcNAc transferase (OGT) and *O*-GlcNAcase (OGA), respectively. It is expressed as a single monsaccharide glycan (which may be phosphorylated) and is highly dynamic (created and broken down in minutes) operating similarly to phosporylation (which may also compete with glycans in binding phosphates to the peptide) in influencing protein activity, transcription and translation[26]. *O*-GlcNAcylation is especially important in external stimuli stress response. *O*-GlcNAc upregulation is influenced by the glucose metabolic pathway, which is differentially

regulated due to the increase of reactive oxygen species (ROS) caused by oxidative stress. This cascade of events and increased *O*-GlcNAcylation may aid in cell survival[27, 28].

Analysis of these glycans is challenging using traditional analytical approaches as *O*-linked glycans have a propensity to decorate themselves with sulfates (such as the glycosaminoglycan chondroitin sulfate[9]) and phosphates (such as phosphorylated *O*-Man structures in α-dystroglycan[29]). A result of this is that databases and reference material routinely accompany analysis, increasing obfuscation of results leading to false discoveries. To overcome this pitfall, guidelines such as the Minimum Information Required for A Glycomics Experiment (MIRAGE)[30] are followed to ensure reliable and reproducible glycomics is generated and shared among the glycomics community. Guidelines on experimental procedures are broken down into subcategories of sample preparation, mass spectrometric analysis, glycan microarray analysis, enzyme array digestions and hydrophilic interaction (HILIC) and reverse phase (RP) liquid chromatography analysis[31-34].

1.2 TECHNIQUES TO ANALYSE *O*-GLYCANS FROM GLYCOPROTEINS

1.2.1 *N*-Glycan Release Methods

A common approach for analysing glycans involves cleavage of the saccharide at or near the protein glycosylation site using endoglycosidases such as peptide-*N*-glycosidase F (PNGase F) and Endoglycosidase H.[18] These enzyme will cleave to different extents complex, hybrid, and oligomannose *N*-glycans allowing for subsequent derivatisation with fluorescent labels for fluorescence (FLR) detection using techniques such as FLR-Ultra Performance Liquid Chromatography (FLR-UPLC), Capillary Electrophoresis (CE) and Mass Spectrometry (MS); or not labelling and reducing the glycan's cleaved end instead with a reductant such as sodium borohydride ($NaBH_4$), generating an alditol, for detection using MS.

1.2.2 *O*-Glycan Release Introduction

The above enzymatic technique has allowed for detailed analyses of *N*-linked glycans and their diverse heterogeneity, however, there remains a restricted knowledge and understanding of their *O*-linked siblings. This is primarily due to the current lack of an a generic enzyme which can cleave the diverse range of *O*-glycans from protein, requiring the need for high pH solvents to break the *O*-glycosidic linkage as well as limitations to mainly reductive amination labelling techniques (if fluorescently labelling).

As a result, a number of approaches are utilized to profile the *O*-glycome including a glyco-peptide/protein level analysis of attached *O*-glycans, enzymatic and chemical *O*-glycan release methods which separate the glycan from the protein prior to analysis[35].

With respect to glycan release protocols for both *N*- and *O*-linked glycans, methods to automate high throughput processes have been proposed[36-39].

Release Method	Release Time	Analytical Methods	Sample quantity	Strengths	Weaknesses	% Peeling (literature)
Enzymatic	16 hours	FLR-UPLC, MS, CE, NMR and other fluorescent techniques	µg	Not released in basic conditions	Limited to non-extended Core 1 and 3 structures	Negligible (no alkaline degradation)
Reductive β-elimination	16 hours	MS (may be coupled with LC for separation), NMR	µg	High yield	Unable to be fluorescently derivatized	0-10%[40-42]
Non reductive β-elimination (hydrazinolysis)	6 hours	FLR-UPLC, MS, CE, NMR and other fluorescent techniques	µg	High yield	Toxic reagents and high level of expertise required	17%[43], <2-19%[44], 35-42%[45], 6-40%[46]
Non reductive β-elimination (mild alkaline)	16-40 hours (minutes using microwave)	FLR-UPLC, MS, CE, NMR and other fluorescent techniques	µg-mg	Reduced sugars may be fluorescently labelled	Variable level of peeling and lower yield	0%[47], 0-30%[48], 9-27%[49], ~60%[50]
Oxidative Release of Natural Glycans	24 hours	FLR-UPLC, MS, CE, NMR and other fluorescent techniques	g-kg	Large scale workups.	Acidic aglycons produced – difficult to reference against existing databases. Large quantities required.	None – no "reducing end" on aglycon generated

Table 1. *Summary table of main O-glycan release methods. Peeling levels indicate variability even within the same methods.*

1.2.3 *O*-Glycan Release Methods

1.2.3.1 Enzymatic Release

As mentioned, no currently known endoglycosidase can cleave all *O*-glycans from glycoproteins. Endo-α-*N*-Acetylgalactosaminidase (simplified to '*O*-glycosidase' or '*O*-glycanase') has limited activity, cleaving the unsubstituted disaccharide core 1 and 3 mucin-type *O*-glycans containing an *N*-Acetylgalactosamine bound via a β1-3 linkage to galactose (core 1) or bound via a β1-3 linkage to *N*-Acetylglucosamine (core 3) only.[51] *O*-glycans expressed in cellular systems are often heavily substituted with chains of monosaccharides, meaning these endoglycosidases have very limited use. However, by first digesting longer chains with monosaccharide-specific exoglycosidases one at a time to clip off terminal carbohydrates, such as those with sialylation and galactosylation, the core structures can be exposed for endoglycosidase release. While this will successfully cleave *O*-glycans and

maintain protein structure, structural elucidation of complex *O*-glycans (not just simple core 1 and core 3 structures scattered with one or two sialic acid residues) is not so straightforward[35].

1.2.3.2 Chemical Release

1.2.3.2.1 β-elimination

Currently, alternate chemical methods are favoured for intact glycan release. Under alkaline conditions, the *O*-glycosidic linkage between the reducing monosaccharide (GalNAc core in mucin-type glycans) and serine/threonine residues is labile and readily hydrolysed (more than the *N*-glycosidic linkage), promoting cleavage of the glycan. Figure 3 outlines the suspected reaction mechanism. The α-carbon of the protein adjacent to the saccharide bound carbon (known as the β-carbon) contains an acidic proton which, when subjected to nucleophilic attack under basic conditions, starts a (β-) elimination reaction producing a free glycan and unsaturated peptide/protein (not shown)[18]. This β-elimination reaction successfully liberates the *O*-glycan but produces knock-on effects that must to be overcome.

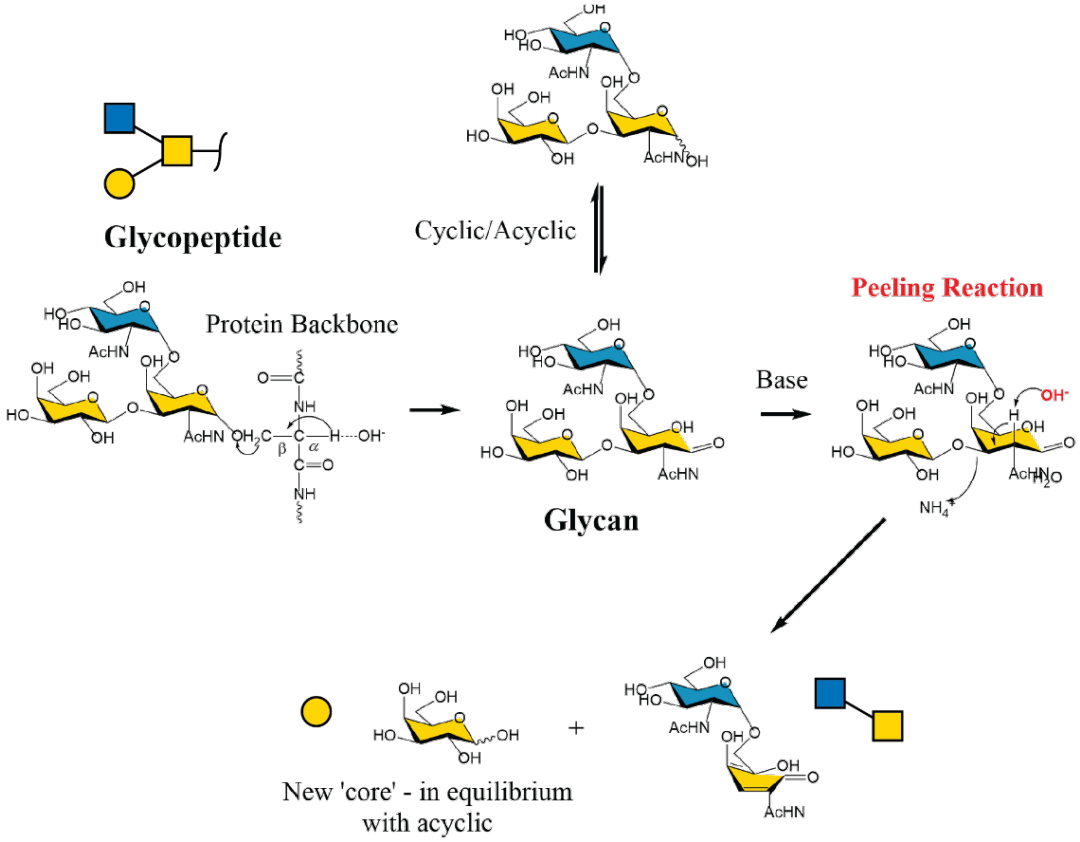


Figure 3. *β-Elimination reaction of a typical mucin type O-glycan Core 2 (SNFG structure and colours shown for comparison) including non-reduced glycan peeling reaction.*

1.2.3.2.2 The Peeling Reaction

Peeling is a process in which the released non-reduced glycan is subjected to degradation from the reducing core monosaccharide (for this example – GalNAc)[52] separating it from the proceeding sugars in the chain. This reaction typically proceeds under the same alkaline conditions required for chemical *O*-glycan release, with nucleophilic attack of the released glycan's core monosaccharide starting a condensation reaction of the carbon-2 (C-2) hydrogen of the open-ring form of the core, generating an alkene between C-2 and C-3, destabilising the β1-3 glycosidic linkage, thus separating GalNAc core from the proceeding monosaccharide[48] (figure 3). It has been observed that the β1-3 linkage extending from the core is significantly more succeptible to peeling than a β1-4 linkage (seen in *N*-glycan cores) and will continue to degrade the glycan along the β1-3 linkage until an alkali-resistant linkage is reached[53] or alkalinity is removed.

The downside of such a reaction is the generation of a glycan profile that is disingenuous to the true profile. Attempts to reduce this unwanted reaction are discussed below.

1.2.3.2.3 Reductive β-elimination

Reductive β-elimination is the most popular approach to liberate *O*-glycans from glycoproteins due to the relative simplicity and safety of its approach, as well as the comparably superior yield of glycans obtained. Glycans are released under alkaline conditions (typically in 1M NaOH) with an incubation time generally from 16-40 hours. To prevent "peeling", the reducing C1 of the glycan core is reduced into a base-stable alditol through the addition of a reducing agent such as sodium borohydride[54] in excess to the reaction mixture[55]. The reaction is quenched upon acid addition of acetic acid or HCl. Upon observation of high sodium salt content remaining at the end of this process which may interfere with downstream analyses, this method has been modified to include desalting steps as well as methods of incorporating an aqueous ammonia hydrolysing solution with a borane-ammonia reducing complex[56].

As this reducing reaction is irreversible, glycans are unsuitable for further derivatisation such as reductive amination (figure 4). As glycans contain no intrinsic chromophore themselves, reductive β-elimination prevents fluorescence detection using UPLC, High performance liquid chromatography (HPLC) and other fluorescence detection techniques, allowing only limited analyses using MS (often after permethylation), liquid chromatography-MS (LC-MS), nuclear magnetic resonance (NMR) spectroscopy and other intact glycopeptide and protein analyses[55, 57]. Methods to preserve the de-glycosylated peptide/protein from degrading using sodium sulfite to convert threonine and serine residues to cysteic acids and α-amino-β-sulfonylbutyric acids have permitted chromatographic separation and analysis to determine *O*-glycosites, however, to the detriment of degrading the released glycans[58]. Sodium borodeuteride has also been used in lieu of sodium borohydride to act as a reductant/tagging agent of serine and

threonine during the release, as well as ammonium hydroxide, to capture both glycan and de-glycosylated protein for analysis[59]. The development of β-elimination followed by Michael addition (BEMAD) using nucleophiles like dithiothreitol or biotin pentylamine have also allowed glycosylation site determination using tandem mass spectrometry[60].

Techniques of alditol "capping" have been performed more recently using pyrazolone derivatives such as 1-phenyl-3-methyl-5-pyrazolone (PMP) which are highly sensitive to UV radiation, allowing for LC-electrospray-ionisation (ESI)-MS analyses[61, 62].
Another downside to reductive β-elimination is that the resultant unsaturated peptides and proteins become unstable and are unsuitable for accurate elucidation of glycosylation site information[63].

Methods of *O*-glycan release which do allow for subsequent derivatisation have been developed to overcome some limitations of reductive β-elimination, termed non-reductive β-elimination.

1.2.3.2.4 Hydrazinolysis – Non-reductive β-elimination

Hydrazinolysis using hydrazine as the primary reaction reagent is a popular method for non-reducitve *O*-glycan release as it is considered to provide the highest yield of glycans compared to competing non-reductive release methods[64]. The presence of anhydrous hydrazine initiates a β-elimination reaction with a secondary reaction occurring at the glycan reducing end generating hydrazine derivatives. Acetic anhydride is used to re-*N*-acetylate amino groups of the glycan that are lost during the release, with subsequent reductive amination techniques for fluorescence detection following thereafter[65]. This method has been shown to produce minimal peeling products even at the relatively long reaction times required (6 hours at 60°C), however, may be unsuitable for *O*-glycan release in large-scale applications due to the toxic and explosive nature of anhydrous hydrazine (experimentation must be performed under anhydrous conditions at small scale – approximately 0.2-1mL reactions). Studies into safer hydrazine alternatives such as hydrazine monohydrate have been shown to release glycans with comparable efficiency, which, concomitantly limit gaseous hydrazine monohydrate production, minimising risk of explosion[66].

1.2.3.2.5 Ammonia Based/Mildly Alkaline Non-Reductive β-elimination

Using small alkylamines such as ethyl-, methyl- and dimethyl-amine to provide a more mildly alkaline reaction environment compared to NaOH, *O*-linked glycoproteins release their glycans in a non-reduced form (commonly referred to as ammonia-based β-elimination)[47, 67, 68]. As released glycans are, again, subject to peeling reactions, ammonium carbonate and ammonium bicarbonate are incorporated into the reaction vessel to convert the reducing

glycan into a glycosylamine intermediate species which may then be further derivatised by reductive amination or carbamate labelling strategies.

Similar studies using LiOH as a mild alkaline catalyst have been used in tandem with reductive amination in a similar manner[69].

Recently, Kameyama et. al (2019) have shown that the use of hydroxylamine combined with the superbase 1,8-diazabicyclo[5.4.0]undec-7-ene have produced comparably high *O*-glycan yields to hydrazinolysis techniques with minimal peeling[43].

1.2.3.2.6 Microwave-Assisted Non-Reductive β-elimination

The downside of both reductive and non-reductive β-elimination techniques are the long reaction times (up to 40 hours) resulting in unfavourable peeling products and time limitations to throughput. To overcome this, novel methods using microwave radiation have been established, reducing reaction time from hours to minutes, providing comparable yields of released glycans[59, 63, 68].

Deriving from this, a one pot approach has been discussed which incorporates a reductive label such as PMP into the release reaction vessel, removing the need for additional labelling steps[35, 70]. Combining all time-saving methodologies, what was once a 4-5 day protocol of glycan isolation and labelling from glycoproteins could now be as short as hours with greater reproducibility (less time = less variability from peeling).

1.2.3.2.7 Oxidative Release of Natural Glycans

A competing method for the release of *O*-glycans, *N*-glycans and Glycosphingolipids named oxidative release of natural glycans has been reported by Song et al.[71]. This method releases glycans from kilogram quantities of samples using sodium hypochlorite. It suitably liberates *N*-glycans at their reducing end from asparagine allowing subsequent tagging using tranditional non-reducing glycan labelling methods. In contrast, this method cleaves *O*-glycans from proteins less specifically due to the less labile *O*-glycosidic linkage (compared to the *N*-glycosidic linkage) instead targeting a bound portion of the serine/threonine-glycan moitey, generating an acidic aglycon. This aglycon was then labelled using a fluorescent tag ‘monoFmoc ethylenediamine’. The obvious benefit of this method is the collection of multiple glycan types with no peeling generated from β-elimination mechanisms.

1.2.4 Glycan Labelling Techniques

As previously mentioned, glycan derivatisation using a fluorescent label (for ultraviolet (UV) or fluorescent detection) is a popular choice to aid in glycan characterisation as glycans contain no natural chromophore or fluorophore suitable for detection in LC and MS systems, however, this is limited typically to non-reduced glycans.

Multiple fluorsecent labels are available for derivatisaition depending on the glycans that are procured and their yield (low yield glycan releases may require a more sensitive label to produce usable fluorescence data).

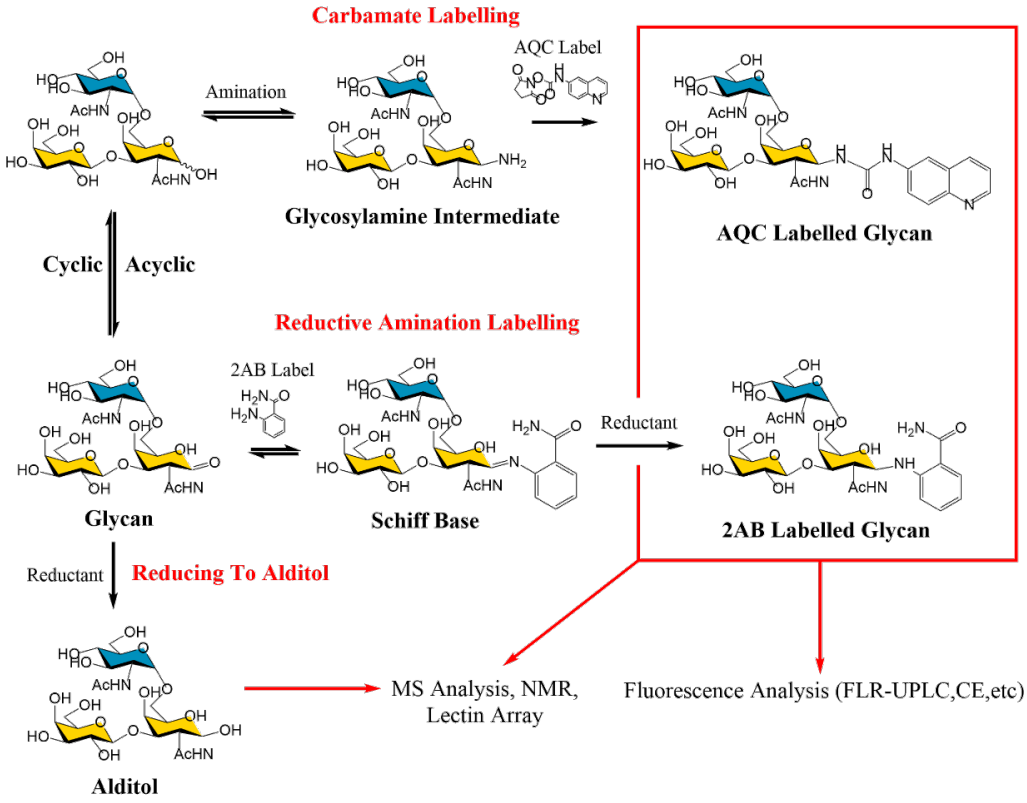


Figure 4. *Released O-glycan derivatisation and end-capping strategies. Released glycans have a reducing end sugar that is in equilibrium between cyclic and acyclic form. Acyclic glycans may be reduced into alditols (seen in reductive β-elimination) or may undergo reductive amination (with 2AB label in this instance). Cyclic glycans may be aminated to produce a glycosylamine which can undergo carbamate chemistry-based labelling (with AQC in this instance). Fluorescently labelled glycans may be analysed using many fluorescence and mass spectroscopic techniques while alditols are limited to mass spectrometry, NMR and lectin array.*

The most common approach to labelling glycans is through reductive amination where the aldehyde group of the reducing glycans are reacted in a condensation reaction with an aromatic amine in excess to drive the formation of a Schiff base[72]. For free *O*-glycans, the reducing sugar exists in equilibrium between hemiacetal form (cyclic) and open-ring *O*-glycoside form (acyclic) and free *N*-glycans exist in equilibrium between glycosylamine form, acyclic form and cyclic form. To drive aldehyde formation required for a condensation reaction with the aromatic amine, glycosylamines are reduced in acidic conditions[69], often with dilute formic acid. Aromatic amines commonly used are 2-amoinobenzamide (2AB)[73], 2-

aminobenzoic acid (2AA)[73], procainamide[74, 75], and 8-aminopyrene-1,3,6-trisulfonate (APTS)[76, 77]. In order to generate a base-stable labelled glycan, a reducing agent is incorporated such as sodium cyanoborohydride[72]. This has a downside however due to the creation of toxic hydrogen cyanide as a byproduct[78]. More recently methods of reducing the Schiff base have included replacing sodium cyanoborhydride with non-toxic 2-picoline-borane[79] or sodium triacetoxyborohydride[80].

The transient glycosylamine feature of *N*-glycans which derive from the *N*-linked deglycosylation (by PNGase F) of GlcNAc from asparagine residues also allows carbamate based labelling reactions to occur[81]. Carbamate-based labels react rapidly compared to reductive amination (minutes instead of hours) and have become a popular method of derivatising glycans[82]. Such labels include 6-aminoquinolyl-N-hydroxysuccinimidyl carbamate (AQC), InstantAB and Waters' RapiFluor-MS[83]. Furuki et al. have shown how *O*-glycans can be derivatised using RapiFluor-MS[84] (a technique typically limited to *N*-glycans) by following a glycoblotting *O*-glycan release technique utilising ammonium carbamate to generate a glycosylamine intermediate suitable for carbamate based tagging[85].

Recently, Wu et al. created a direct fluorescent glycan labelling (DFGL) method utilising copper click chemistry to synthesise fluorescent sialic acids which were then translated onto desialylated glycoproteins (fetuin and asialofetuin standards) by the appropriate sialyltransferases[86] prior to analysis. This was applicable to both *N*- and *O*-linked glycans.

Exploring *O*-GlcNAcylation more specifically, an interesting approach to the *O*-GlcNAc release workflow was proposed (and remains the gold standard) by Torres et al. (1984)[22] where [^{3}H]galactose was transpose by bovine milk galactosyltransferase onto *O*-GlcNAc glycoproteins as an impermeant probe to produce [^{3}H]galactose-β1-4-*N*-acetylglucosamine disaccharides prior to reductive β-elimination, providing a disaccharide detectable through autoradiography and more readily discernible through HPLC and MS compared to the single GlcNAc monosaccharide[87]. Other tagging methods utilising copper-click reactions with alkyne- and azide-tagged *O*-GlcNAc and an affinity probe have been proposed, as well as β-elimination followed by Michael addition (BEMAD) to determine glycosylation site [88, 89].

1.2.5 Analytical Techniques to Characterize *O*-Glycans

When considering analytical methods to identify and quantitate glycans or glycoproteins, it is important to first understand what the main goal of the project is and what complimentary methods will give sufficient results. Normally several methods are required in tandem (multiattribute glycan profiling)[90]. Through the combination of multiple separation and detection techniques, challenges in characterisation and quantitation of a vast heterogeneity of glycans may be overcome. Described in the following sections are some of the main forms of separation, characterisation and quantification of glycan structures. Integral to characterization

is the accompaniment of glycan design tool like GlycoWorkbench; a software containing not only a GlycanBuilder to help identify potential structures, but also a mass search function pulling glycan information from a range of databases[91].

Technology	Application	Modes	Benefits	Limitations
Liquid Chromatography (HPLC, UPLC)	Separation (unlabelled glycan/glycopeptides) and fluorescence detection (labelled glycans)	HILIC	High selectivity Steeper gradients	Glycosylamine formation from reducing sugar interactions with amino stationary phases
		RP	Reproducible, Linear relationship of retention/glycan size	Poor isomer separation
		PGC	Superior isomer separation	Poor reproducibility
Exoglycosidase Arrays	Probing glycan structure and linkage specificity (labelled glycans)	-	Detailed structural analysis and linkage determination	Incomplete digests make analysis difficult
Mass Spectrometry	Accurate mass determination of glycans/glycopeptides. Glycosylation site determination in glycopeptides and proteins. MS^1 unable to identify isomers and requires fragmentation which may inhibit quantitation.	ESI	'Soft' technique keeps glycans intact	Multiple charge states
		MALDI	Uniform charge for less convoluted spectra	Unsuitable for native and reductively aminated glycans. Unwanted fragmentation of labile sialic acid residues.
		Ion Mobility (IM)	Superior isomer separation (not identification) based on glycan 3D structure	Hard to discern larger branched glycans
MS Ion Fragmentation	Fragmentation of precursor glycan/glycopeptide ions to probe linkage specificity and site attachment	CID	Low energy – glycosidic link cleavage High energy – ring cleavage	Larger glycan fragmentation inefficiencies
		IRMPD	Greater efficiency of ion fragmentation	X-type ions typically not observed[92]
		ExD	Higher yield and more informative protein backbone fragmentation	Difficult to implement on quadrupole ion traps (ECD). complex scan functions required (ETD)[93]
		PSD	Fragmentation without extra precursor ion screening	Unnecessary fragmentation
		ISD	Protein fragmentation (larger amino acid sequences ~70aa)	Suited to protein fragmentation, not glycan fragmentation
CE	Glycan separation and laser-induced fluorescence detection	-	Short separation, low sample volumes required	Poor reproducibility
NMR	Glycan structural analysis	COSY TOCSY	Understand glycan receptor interactions	Challenging for complex mixtures
Lectin Affinity	Quantification of glycans and glycopeptides	Over 300 known lectins (specific epitope affinity)	Ability to absolutely quantify glycans	Poor affinity and specificity in some cases. Repetitive enrichment steps may be required.

Table 2. *Summary table of main glycan separation and analysis methods and their benefits*

1.2.5.1 Ultra-Performance Liquid Chromatography

Hydrophilic interaction liquid chromatography-ultra performance liquid chromatography (HILIC-UPLC) is a favoured method of separation for very polar compounds such as glycans which interact strongly with amino, amide and diol stationary phase groups within the LC-column permitting a high selectivity of glycans based on size and hydrophilicity, linearly separating based on their glucose units (GU) – with good reproducibility[94]. Reverse phase liquid chromatography (RP-LC) which relies primarily on hydrophobic interaction of fluorescent glycan labels and porous graphitized carbon liquid chromatography (PGC-LC) which exploits the polar retention effect on graphite (PREG)[95] are also common approaches with the latter being superior for separating glycan isomers that may not be separated using HILIC and RP stationary phases[96].

Interestingly, each technology plays to their strengths in different ways. While being the favourite, HILIC provides only partial separation of isomeric glycan species which may be particularly challenging when working with complex glycan mixtures. It does, however, offer higher selectivity of the three methods, permitting a steeper elution gradient to be applied (increasing peak capacity). Care should be taken however with amino bonded silica in HILIC stationary phases, as reducing sugars may undergo Schiff base formation with stationary amino groups, generating unwanted glycosylamines, deteriorating the column in the process. Stationary phase modification with phosphate and ammonium formate buffers ($pH \leq 6$) have sought to remedy this[97].

Typical reverse phase chemistries such as C18 have poorer isomeric glycan separation due to their limitation of hydrophobic interactions primarily with only the glycan label (or if the glycans have been permethylated – increasing hydrophobicity), requiring shallower gradients (reducing peak capacity). An upside to this, however, is the grouping of similar glycans. Coupled with the linear relationship between glycan size/composition and retention time, this technology performs well in quality control (QC) environments where only simpler glycosylation patterns (for example a single glycan with 0, 1 or 2 extra hexoses/dhexoses) need to be separated with good reproducibility[98]. On the other hand, PGC technology operates as somewhat of an outlier compared to these two methods, exhibiting characteristics of both HILIC and RP. While considered a reverse phase technology, PGC relies on retention of glycans via hydrophilic interactions with the glycan via induced dipoles of the graphitic carbon, as well as through hydrophobic interactions[99]. A 3D representation of hydrophobic and hydrophilic “patches” on glycans (shown by Karlsson et al. 2017) gives a visual approximation of how glycans are locked up in tight globular structures, with isomers each exhibiting differing surface level hydrophilicity, capable of interacting differently with the PGC stationary phase[9]. Ionic interactions, particularly with sialylated and sulfated glycans have also be shown to increase retention in PGC chromatography[100]. Where it excels in superior isomer separation

however, it has been suggested to lack in reproducibility compared with other HILIC and RP methods[98, 101]. Also, where HILIC and RP technologies fit multiple linear regression models when directly comparing retention time to monosaccharide composition, PGC doesn't fit such models, suggesting the 3D structure of glycans plays a considerable role in influencing retention time[98].

As mentioned previously, the ability to tag non-reduced glycan structures with fluorescent labels has allowed observation and optimisation of glycan separation purely within an offline LC system which is commonplace in a QC environment.

1.2.5.2 Exoglycosidase Arrays

By combining UPLC separation with fluorescence detection, chromatograms showing separated glycan peaks can be generated (through the use of programs such as Empower and Masslynx), hinting at glycan structures. These structures can be probed further by incorporating external standards such as fluroscently labelled dextran ladders into sample sets, quantitating glycan GU and thus, glycan size; which can subsequently be referenced against previously characterised glycan data. Drilling down with this knowledge, structural identification of labelled glycans can be achieved through the use of exoglycosidase arrays[102-104] which cleave the glycan with linkage and monosaccharide specificity allowing for accurate structural determination. By sequentially adding exoglycosidases in a stepwise manner to sample replicates (starting with an undigested sample replicate, adding one exoglycosidase to the next replicate, two exoglycosidases to the next and so on) until all suitable exoglycosidases have been incorporated, a panel of replicates with varying degrees of cleaved glycan structures is produced (figure 5). Piecing together chromatograms of this panel by observing peak shifts of cleaved glycans, the glycan profile may be built up like a puzzle from most digested replicate to least until the profile is confidently characterised. While this is traditionally done manually, there have been attempts to automate the process, to varying degrees of success[105-108].

Exoglycosidase arrays benefit from confirming linkage information that MS may have trouble in determining without good reference databases, however, MS confirmation serves as a method to validate these results and may determine post translational modifications otherwise missed.

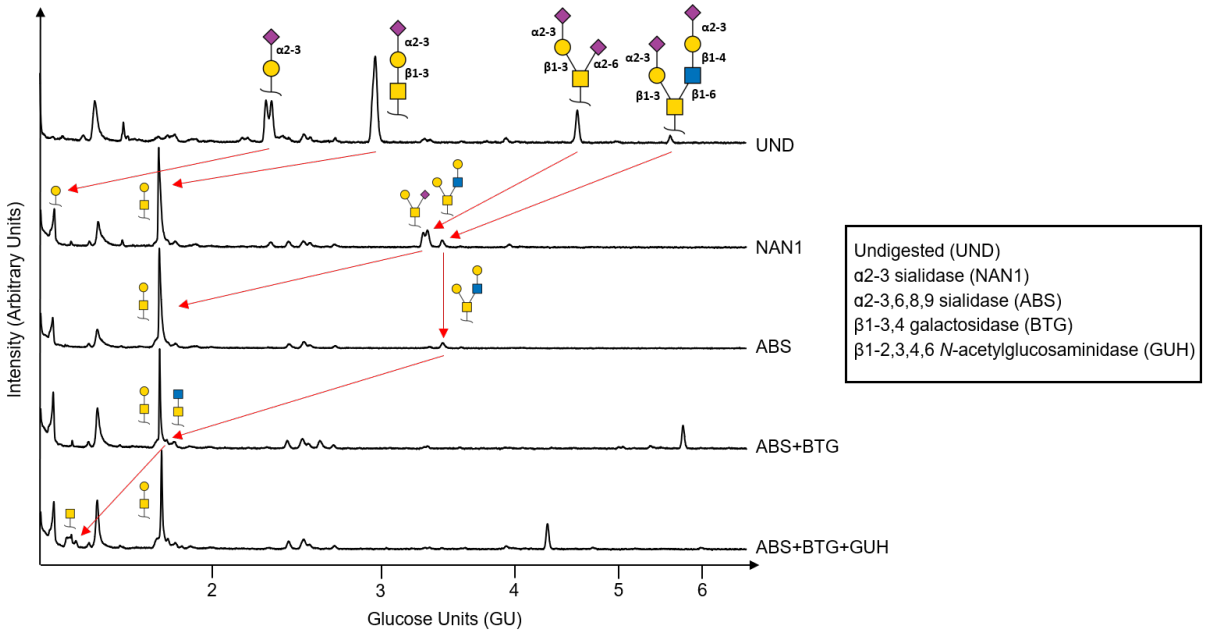


Figure 5. *Sample exoglycosidase array showing digestion of fetuin O-glycans using cocktails of enzymes The core 1 disaccharide resists galactosidase digestion (discussed in 3.3.5.1).*

1.2.5.3 Mass Spectrometry

While only non-reduced labelled glycans are suitable for LC fluoresence analysis, both non-reduced and unlabelled reduced glycans are suitable for MS detection, however, the ionisation efficiency of the glycan depends greatly on how it is derivatised.

Reduced glycan alditols from reductive β-elimination ionise well in negative ion mode[109]. On the other hand 2AB labelled reductively aminated glycans have a poor ionisation efficiency[110] so larger sample sizes are required. In order to overcome this, fluorescent labels such as procainamide and Rapifluor-MS (Waters™)contain a basic tertiary amine which has a high proton affinity, hence ionise well in positive mode ESI-MS[75, 111].

Matrix assisted laser desorption/ionisation (MALDI) MS and electrospray ionisation (ESI) MS are two commonly applied MS techniques utilised to obtain glycan mass (m/z)[92]. The relatively strong ionisation towards labile glycosidic linkages of glycans in MALDI, particularly for acidic glycan components such as sialic acid residues and sulfated glycans, causes increased internal vibrational excitation and unfavourable dissociation, however, this leads to the production of diagnostic ions which can be used to help characterise generic linkage positions of the structure[112]. The limitation to this is that terminal sialic acid loss isn't particularly useful when analysing complex glycan structures, so often glycans are permethylated to stabilise their structure[113], and in-depth characterization is instead achieved through tandem mass spectrometry approaches. Alternately, cooling nitrogen streams have also been applied in source to stabilize these structures[114]. By increasing ionization efficiency of neutral glycans through derivatization with negatively charged labels, as has been shown through the use of glycan-reactive matrices like 3-aminoquinoline/p-coumaric acid (3-AQ/CA)[115], MALDI MS is

also useful in neutral glycan analysis. Recently, Zhang et al. (2019) used Girard's reagent T on-target derivatization (GTOD) to form stable hydrazones of reducing neutral and sialylated glycans, measured in positive mode MALDI, improving ionization efficiency of neutral structures and limiting desialylation[116]. Another strength of MALDI technology is the ability to perform MALDI imaging mass spectrometry (MALDI-IMS) to spatially characterise glycans from formalin-fixed paraffin-embedded (FFPE) tissue block cross sections[117-119], however, this tends to favor *N*-glycan analysis where PNGase F may be sprayed onto tissue to release the glycans prior to the application of a matrix such as cyano-4-hydroxycinnamic acid (CHCA) for subsequent analysis.

ESI-MS on the other hand, uses a 'softer' ionisation to keep native, acidic and reductively aminated glycans intact[92], however, this creates ions with varying charge states that may convolute the mass spectrum and may need greater separation pre-ionisation[120]. By incorporating nano ESI techniques into the MS workflow, native glycans show greater sensitivity due to less suppression by hydrophobic molecules compared to traditional ESI[121].
By observing mass to charge ratios using these methodologies, generic glycan composition may be determined and coupled with exoglycosidase digestions, glycan can be successfully characterised, however in the absence of exoglycosidase techniques, linkage specificity and isomeric differentiation requires additional orthogonal approaches.

1.2.5.3.1 Fragmentation Approaches

The importance of identifying diagnostic fragmentation ions has initiated a new generation in determining glycan structure and linkage composition (for instance the positive mode GlcNAc oxonium ion from *O*-GlcNAcylation – mass to charge ratio (m/z) of 204.1 – an otherwise difficult to determine PTM, as well as Lewis type epitope fragmentation)[122, 123].

Tandem mass spectrometry techniques are incoroprated to capture fragmentation information of glycan structures, uncovering generic linkage information and other modifications. Collision induced dissociation (CID) is a common method of precursor ion fragmentation particularly for smaller glycans (such as small *O*-glycans) in which selected ions (isolated from ion traps or from ion beams in time-of flight systems) are subjected to an inert collision gas vibrationally exciting precursor glycan ions inducing bond-breaking[124-126]. This occurs typically at labile *O*-glycosidic linkages (giving rise to B, C, Y and Z fragmentation ions[127] which indicate glycan composition) and also at cross (sugar) ring linkages (giving rise to X and A fragmentation ions[127] which help indicate linkage specificity)[120] (figure 6). This technique may be further broken down into low energy and high energy CID with low energy favouring glycosidic linkage breaks and high energy favouring both glycosidic linkage breaks and cross ring breaks[120, 128, 129].

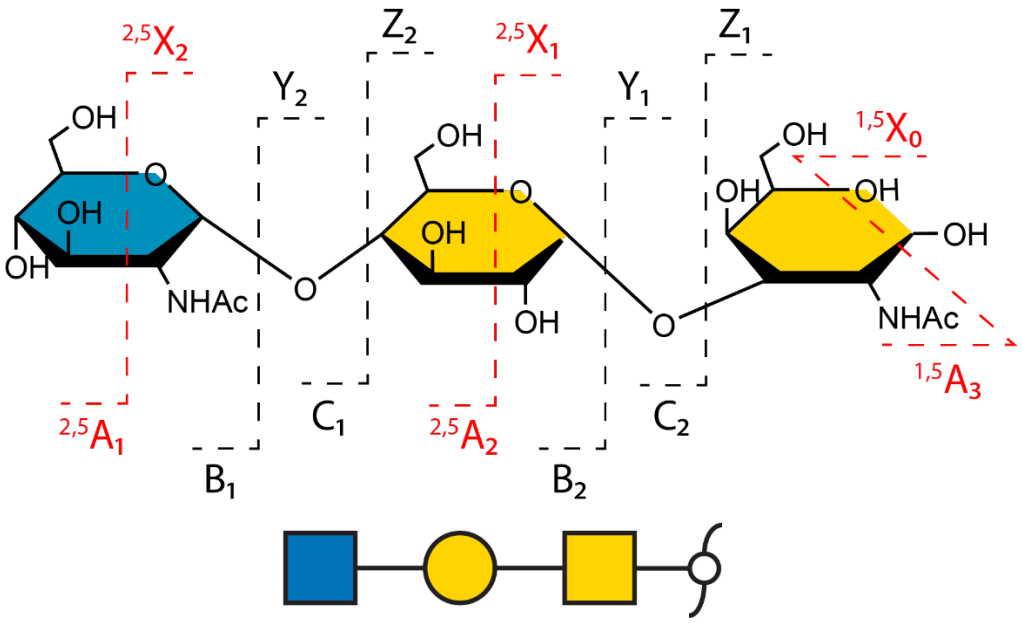


Figure 6. *Fragmentation ions of a simple core 1 GlcNAc extended O-glycan. B, C, Z and Y ions indicate glycosidic cleavage (black dashed line). A and X ions indicate cross ring cleavage (red dashed line). Superscript indicates cleavage position of carbon atoms in the ring. Subscript indicates number of residues the resulting ion contains. SNFG included for reference.*

Due to CID inefficiencies in fragmenting larger structures (such as *N*-glycans and large mucin-type *O*-glycans)[130], techniques such as infrared multiphoton dissociation (IRMPD) are incorporated that do not require a collision gas and have greater efficiency of ion fragmentation[131].

More modern techniques of ion fragmentation have been incorporated into tandem MS systems including the class of electron activated dissociation (ExD) techniques such as electron capture dissociation (ECD) and electron transfer dissociation (ETD) among others[132] which seek to produce more informative fragment ions at high yield[133].

Ion mobility mass spectrometry (IM-MS) is a method with great potential for separating isomeric glycans post-ionisation as they travel through an inert gas phase, separating based on three dimensional shape and collisional cross section (CCS), a function of drift time of ions as they pass through the gas phase[134-136]. This is often performed using travelling wave ion mobility mass spectrometry (TWIMS) techniques. Limitations, however, to separation of larger biantennary glycan structures using TWIMS have been suggested[137, 138] which has given rise to development of techniques such as differential mobility spectrometry (DMS) or high-field asymmetric waveform ion mobility (FAIMS) which allows chemical modification of the travelling ion cell, enhancing separation[137, 139].

The requirement for reference spectra for precursor and fragment ion interpretation has driven the development of a multitude of glycan and glycoproteomic databases. UniCarbDB[140], GlycoEpitope[141], Glycobase and Glycostore[142] are examples of databases that categorise thousands of glycan structures (*O*- and *N*-Glycans, GSLs, glycosaminoglycans) based on mass (precursor ion and fragmentation ions/patterns), retention time, and glucose units across a range of derivatization methods. Such databases individually store analytical and metadata which are collectively fed into integrative projects such as GlyTouCan[143], GlyConnect[144] and GlyGen[145] that act as global repositories where glycans may be uniquely identified (with the goal to achieve the same degree of absolute uniformity seen in well established proteomic and genomic nomenclatures). Taking GlyConnect as an example, glycan structures may be probed further by their association to taxonomy, tissue type and prevalence in disease, providing more useful information than simple glycan characterization.

Stemming from this assembly of databases, umbrella repositories like GlyCosmos[146] have been created to help bridge the gap between glycomics data and other omics disciplines and life sciences.

1.2.5.4 Additional Techniques for Separation and Analysis

Although LC, LC-MS and MS are the most commonly used technologies for the analysis of glycans, there are a multitude of other technqiues often used in tandem with these. Capillary electrophoresis (CE) is a well established method for the separation and analysis of oligosaccharides. In CE, glycans travel in an electrolyte through an electric field (influenced by electroosmotic flow – EOF, and electrophoretic mobility) and are separated based on charge-to-size ratio[147]. With the exception of sulfated and sialylated glycans which posess intrinsic charge and separate well with CE, neutral glycans have inhibited mobility. To overcome this, glycans are often labelled with APTS (via reductive amination), providing an additional 3 negative charges as well as a fluorophore allowing superior separation and subsequent fluorescence detection[148, 149]. The benefits of using CE analysis are centred around the ability to apply a high electric field, resulting in shorter separation times whilst maintaining a high level of separation. Small injection volumes (femtolitre range) compared to traditional LC (microlitre range) coupled with this shorter run-time also drive down the cost of operation[147]. The downside to this technology are the requirements for reference standards and additional exoglycosidase digestion workflows to determine monsaccharide and linkage specificity. Furthermore, CE exhibits poor reproducibility due to inconsistent flow rates[150]. Nevertheless, like the aforementioned methods, including tandem approaches to quantification, such as CE-MS, have sought to aid this[151-154].

Nuclear magnetic resonance spectroscopy has also been used for structural determination *N*- and *O*-glycans[155, 156], Due to the high levels of protons and carbons on the glycans, ^{1}H and ^{13}C NMR spectroscopy (1D NMR) are used and alongside 2D coupling correlation methods like correlation spectroscopy (COSY) and total correlation spectroscopy (TOCSY) to identify isolated glycans and glycans in simple mixtures[18, 157]. NMR is also a useful tool in understanding glycan-receptor interactions and recognition events[158]. This method, however, requires mg quantities of glycoprotein, making it not viable when sample amount is limited[159].

Specific glycan binding proteins called lectins are a class of proteins which have been incorporated into microarrays to allow for analysis of specific glycan epitopes of the global glycan profiles of free glycan and glycoprotein mixtures[160]. Fluorescently tagged lectins are often covalently attached to a polymer stationary phase and a glycan mobile phase is passed across. While there are upwards of 300 lectins that have been uncovered in nature, they are often separated into monosaccharide affinity binding groups (mannose, galactose and GalNAc, glucose and GlcNAc, fucose and sialic acid)[161]. While not suitable for sensitive structural determination of all glycan species within a sample, lectin mircroarrays have been used for initial identification of glycosylation differences between biological samples and act as a preliminary screening for specific glycans and antigenic glycan epitopes – providing valuable information in disease biomarker identification[162], as well for identifying substrates for bacteria to bind[163].

1.2.6 Analytical Methods to Characterize Intact *O*-glycoproteins/peptides

The desire to detail site specific glycosylation patterns on glycoproteins is increasing due to the need to understand the role glycans play within biologcal systems[164]. This has driven the interest in “bottom up” glycomic approaches that eliminate the need to remove (and poteintially degrade) a free glycan from the glycoprotein. This has been particularly prudent for the presence of *O*-GlcNAc and site determination that may have evaded conventional glycan release techniques. Additionally, the dynamic nature of glycan buildup and breakdown of *O*-GlcNAc for instance requires a detailed understanding of glycan attachment site.

Bottom up or “shotgun glycoproteomics” utilises a protease (such as trypsin or pronase) to cleave the glycoprotein into a specific or non-specific peptide length.

The diversity of glycopeptide ions and fragment ions that are generally aquired from complex tissue samples and mixtures by MS requires technically challenging manual assignments, so bioinformatic automation is becoming more sought after, drawing from databases such as UniProt[165] and PeptideAtlas[166].

Following glycopeptide enrichment (using techniques like lectin weak affinity chromatography – common for *O*-GlcNAc peptide analysis) [167-169], hydrazine solid phase extraction

techniques[170, 171] , anitbody immunopurification[122], as well as normal and reverse phase chromatographic separation (eg. HILIC and C18)[172] for complex sample deconvolution, glycopeptide structures are subjected to fragmentation. Low energy CID from ESI ion-trap and quadrupole time of flight (QToF) systems of glycopeptides has been suggested to produce primarily B and Y glycosidic-cleaved ions of glycans, with higher energy CID targeting B and Y peptide backbone fragments instead. MALDI methods have, via higher energy CID or post-source decay (PSD) MALDI-MS (which incoroprates a laser-induced decomposition of molecular ions as they pass the drift region)[173], generated a combination of fragmentation ions from both the glycan and peptide[174, 175].

ECD has also been shown using low energy electrons (<0.2 eV) to preferentially cleave the protein backbone of glycopeptides, helping outlining site specific binding of both *N*-glycans[176, 177] and *O*-glycans[178-180]. Comparably, ETD has been suggested as a technique for similar fragmentation of the peptide backbone[172]. ETD has the additional benefit of maintaing *O*-GlcNAc linkages to the protein compared to fragmentation techniques such as CID. In this instance, due to the fact that *O*-GlcNAc is a labile PTM group similar to phosphorylation, it is particularly susceptible to glycosidic cleavage by CID, making site determination difficult[88].

"Top down" glycomics or glycoproteomics utlises a similar approach to bottom up glycomics for glycan identification, instead targeting in-tact glycoproteins, allowing for an accurate identfication of glycoprotein sequence and glycosylation site[181]. Due to the conserved nature of *N*-glycoslyation at Asn residues along a consensus senquence[182], *N*-glycan site determination remains relatively straightforward compared to their *O*-glycan counterparts, which don't target any common codon and bind to serine and threonine more unpredictably[173]. This leads to a high level of heterogeniety of the sample, convoluting the spectra and making analysis difficult. First and second dimension protein fractionation and separation by anion exchange and RP-LC, for example, have worked to mitigate this[183, 184]. Of these enrichment methods, LWAC (of native or desialylated glycoproteins[185]) is most commonly used to enrich glycan-presenting residues on proteins, and ETD is applied for similar reasons to the bottom up approach, to obtain peptide-based fragmentation allowing glycosite determination. More comprehensive approaches have been employed, combining ETD protocols alongside CID using, LWAC, anion exchange and HILIC chromatographic separation to more completely cover GalNAc linked glycoproteins and their fragmentation patterns[186, 187]. Yu et al. in 2017 proposed an electron-transfer/higher-energy collision dissociation (EThcD) fragmentation approach, combining data from ETD (favouring peptide fragmentation) and HCD (favouring glycan fragmentation) into one contiguous, and more informative, spectrum[188].

The development of "SimpleCell" technology on engineered cell lines (*O*-glycan truncation to Tn and STn antigens using zinc-finger nuclease), allowed more uniform enrichment of, now non-heterogeneous, *O*-glycans using LWAC to more readily discern glycosylation sites[189].

Regarding non-engineered tissues, chemical approaches to deglycosylate O-glycan chains using trifluoromethanesulfonic acid (TFMS) have been used to produce the same Tn antigen homogeneity for simpler glycosite determination[190]. As mentioned breifly in the *O*-GlcNAc labelling section, BEMAD has also been employed, utilising 2-mercaptoethanol as a nucleophile for de-GalNAcylated proteins from the β-elimination process, with subsequent MS/MS detection to characterise glycosites[67].

The ability to target *O*-glycosylation sites via PSD MALDI-MS has historically been used as it cleaves peptide bonds more readily than glycosidic bonds (as seen with CID fragmentation), allowing for site specific linkage determination. An upside to PSD is the ability to capture data from fragmented analytes that are generated by the decay naturally in the ionisation from MS1 rather than through a selection process of precursor ions which then must be subjected to further fragmentation[191].

Advances in in-source decay (ISD) MALDI-MS technology has allowed an increased level of glycoprotein analysis. Here molecular ions fragment *within* the ion source to amino acids up to 70 amino acid residues in length. From ISD, peptide ladder fragments may be selected for PSD for further fragmentation. This is, however, limited to pure protein samples[173] and provides a lack of sensitivity.

ESI-MS on the other hand, generates multiply charged ions of the glycoprotein, facilitating more effiecient dissociation[183]. Furthermore, ESI-MS allows separation of precusor ions of a specific charge state (obtained from a quadrupole, for example) for fragmentation, compared to MALDI, where the fragmentation occurs in the ionisation source[192].

Other methods of fragmentation including the aforementioned ExD electron based fragmentation have similarly been used to more efficiently fragment the protein backbone compared to traditional CID technologies, enhancing the localisation of post translational modification sites. It has been suggested that, regardless of fragmentation (CID, PSD and ExD), top-down analyses preserve glycosidic linkages due to preferential fragmentation of protein backbone linkages, arising from the higher-order structures of gaseous protein ions (~8kDa) driving this type of fragmentation[183, 193, 194].
The convolution of mass spectra using these approaches, which is often much more complex than bottom-up spectra has driven the need for proteomic databases (like the ProSight suite[195, 196] and PRIDE[197]) that is sensitive to post translational modifications like glycosylation.

1.2.7 Relative vs. Absolute Quantification

There is debate in the glycoscience community between relative and absolute quantification of glycosylation. While one may argue that absolute quantification will provide a superior picture of the total glycan heterogeniety in a sample (particularly important for free oligosaccharides such as non anchored glycans like human milk oligosaccharides), relative quantification remains the most time effective method and is suitable enough to determine the regulation of glycosylation and glycosylation machinery, helping with biomarker identification for disease[198]. The big push for absolute quantification (which will provide glycosylation efficiency of glycoproteins for example) stems from the proteomics field where absolute quantification of protein in a sample is required. However, disease research in a clinical setting is heavily focused on observing the relative changes of specific glycans (such as antigenic-epitope expressed glycans) between biological samples.

Regardless of the experimental necessity for type of quantification, analytical techniques and sample preparation are able to be tweaked to help discriminate between absolute and relative abundances. For fluorescence based techniques, released glycans are fluorescently labelled at a ratio of 1:1 (glycan:label), providing a direct linear relationship to molar concentration when referenced against an internal "spiked" standard (such as Man3) with known concentration or an external standard calibration curve[199-202]. Indeed this may be suitable for the absolute quantification of *N*-glycans which are wholly (or close to) liberated from the glycoprotein by PNGase F, however, *O*-glycan yields are often much lower suggesting there is still glycan remaining on the protein following release, ultimately making absolute quantification more of a challenge. Other difficulties arise through glycan loss during sample preparation (i.e. through loss in post-labelling cleanup steps). Lectin based fluorescence methodologies on the other hand provide a more accurate analysis for both free and bound glycan/glycoproteins of absolute quantification, but lack in detailed glycan characterisation and may be influenced by the fact that one lectin may interact with multiple glycans with similar epitopes. MS based absolute quantification operates in somewhat of a similar way to fluorescence based techniques in that, due to the fact that glycans ionise slightly differently, derivatisation (i.e permethylation) helps normalise glycan ionisation of the global pool for a better absolute determination[203]. Incorporation of internal standards also helps overcome these limitations[204], however, there exists no complete set to account for every type of glycan species. Multiple reaction monitoring (MRM) has also been investigated to produce absolute quantification of free human milk oligosaccharides[205]. Taking a relative quantification approach to MS, extracted ion chromatograms (EICs) of specific glycans taken of MS1 raw data may be integrarated and peak areas relatively compared to eachother to help determine glycan expression.

1.2.8 Predictive Technologies

While there is a large heterogeniety amongst glycans that may be synthesised from the 90+ glycosyltransferases in humans (for instance), there are much less that are observed experimentally. One reason for this reduction in heterogeneity is the presence of biosythetic glycan pathways with specific glycosyltransferase genes responsible for glycan-building organised at pathway specific steps, orchestrating the creation of glycans in a somewhat ordered manner (albeit still diverse)[206]. In 2005 Kawano et al. generated a reaction pattern library of bond formation patterns of glycosyltransferase reactions and correlated them against reaction patterns observed in the glycan database "KEGG GLYCAN". With 81% accuracy in prediction against the database they turned their focus on human carcinoma cell glycans, successfully predciting silalic acid and sialyl lewis type epitopes[207]. Other theoretical glycosylation reaction patterns and glycosylation predictions for *O*-glycans have since been investigated to some success with McDonald et al. (2016) identifyinig 25 distinct enzymes resposible for mucin-type *O*-glycan extension[208, 209].

With respect to *O*-glycosylation site prediction, there have been numerous technologies to determine *O*-linked glycosylation sites of proteins, such as GlycoPP[210], GlycoMine[211], GPP[212] and SPRINT-Gly[213] based on protein sequence data from databases like *O*-GlycBase[214]. Support vector machines (SMVs) or random forest (RF) are employed to predict whether a site is *O*-glycosylated. Work by Nishikawa et al. (2010) trained two SMVs to target, separately, clustered regions of *O*-glycosylation and isolated regions and identified that intrinsically disorded structures (IDs) on extracellular proteins were a preferred *O*-glycosylation site[215].

In some workflows, for instance SPRINT-Gly, deep learning neural networks have also been employed. As has been discussed in the aforementioned work, however, was that while hundreds of glycosylation sites were predicted from a few hundred human proteins, sensitivity and precision of prediction were low, relative to *N*-glycosylation site prediction, due to limitations in overall dataset size available for *O*-glycans from the databases the computation drew from[213]. This is somewhat expected as, in addition to dataset size, *N*-glycosylation is sequon specific whereas *O*-glycosylation along amino acid chains is more ambiguous, providing more of a computational challenge (although possible sequence indicators have been suggested in addition to *O*-glycosylation ocurring only at accessible sites on folded proteins as well as in specific regions clustered on mucins and in mucin-like domains of proteins, recognised differentially by different members of the GalNAc-transferase family)[216-219].

1.3 *O*-GLYCOMICS IN DISEASE

1.3.1 Glycan Expression

Emphasis is placed on the glycosylation profiles of proteins, regardless of the challenges in glycan sample preparation and analysis, as changes in these profiles brought on by changes in the highly competitive processes of glycosyltransferases are associated with various cancers, inflammatory diseases and similar conditions.[4, 220] These changes can be interpreted as potential disease indicators or 'biomarkers', and subsequent biotherapeutics may be developed like regenerative implants that aim to re-establish a healthy glyco-environment through modulating glycan expression. For instance, by identifying glycan structures and modulating glycosyltransferase expression in cell culture systems, therapeutic antibodies may be fine tuned to elicit specific biological functions (biantennary *N*-glycans on Immunoglobulin G (IgG) structures) which helps facilitate homogeniety in antibody production[221].

Glycan interactions typically occur through direct and indirect means. Direct interactions occur directly between the glycan and a substrate such as glycan binding proteins (GBPs) which help promote cell signalling, adhesion and other interactions. Indirect interactions are those typically observed in glycoprotein folding and stability[222]. Hence, it can be understood that abberant changes in glycan expression, may negatively affect the protein, cell and tissue.

Lewis, sulfo and sialyl-Lewis fucosylated extrinsic *O*-glycosylation features, as well as Tn and sTn antigens, have been understood to play important roles in leukocyte trafficking during inflammation[223] and are associated with cancer. Abberant Lewis type *O*-glycosylation has also been seen in pediatric cancers like neuroblastoma[224].

Regarding GalNAc core differentiation, Core 1 and 2 GalNAc core (mucin-type) structures are the most commonly expressed mucin-type glycan structures expressed in humans, with core 2 commonly forming the template for polyLacNAc extension (-Gal-GlcNAc-) on either left or right chain of the glycan which can accomodate lewis and blood group epitopes. In addition to abberant lewis epitope glycosylation that may subsequently occur, upregulation of core 2 structures have been associated with HIV-1 infected cells and in Wiskott-Aldrich syndrome[225-227].

With respect to neuroinflammation, *O*-GlcNAcylation has been extensively studied for its impact in neurodegeneration[228] and is discussed in further detail, alongside other neurological glycosylation changes (such as abberant *O*-mannose glycosylation of α-dystroglycan causing α-dystroglycanopathies)[229], in chapter 3.

Identifying correct and accurate glycan biomarkers relies heavily on achieving reproducibility of results from the sample workup process (e.g. non-reductive β-elimination), which is expanded on in chapter 2.

1.3.2 Relating Gene and Glycosyltransferase Expression to Glycome

Glycosyltransferase expression has helped to identify glycans present and the effect up/down-regulation of glycosyltransferases has on disease. For instance, studies have suggested that in breast carcinomas, upregulation of ST3GAL1 (sialyltransferase to galactose) has been linked to tumor grade and under-glycosylated MUC1 glycoprotein[230, 231]. Another sialyltransferase, ST6GalNAc-I, which plays an important role in the synthesis of the Sialyl-Tn antigen (commonly expressed on the surface of tumor cells), has been shown to be overexpressed in ovarian, gastric and late stage pancreatic cancer[232-235].

Modifying gene expression, for instance through epigentic silencing of the COSMC chaperone molecule imporatant in core 1 (β1-3 galactosidase) extension has been linked to inhibition of core 1 and 2 synthesis leading to an upregulation of Tn and sialyl-Tn antigens (figure 7)[236]. Inhibiting *O*-glycan elongation is embryonically lethal, with glycan truncation (typically these overregulated T, sialyl-Tn (sTn) and T antigens) in humans leading to developmental and immune dysfunction[225]. Cancers have generally been associated with the truncation of *O*-glycans[237, 238]. Specifically, the truncated Tn antigen is expressed in over 90% of breast, pancreatic and lung carcinomas, and in over 50% of colon and cervical carcinomas. In contrast it is expressed in less than 20% of normal tissue. This truncation is caused by misbehaving glycosyltransferases, most notably C1GalT1 or “T-synthase” responsible for Tn antigen elongation to the core 1 (T antigen) disaccharide for subsequent elongation, and C2GnT1-3 responsible for core 2 creation, and subsequent elongation[225, 239].

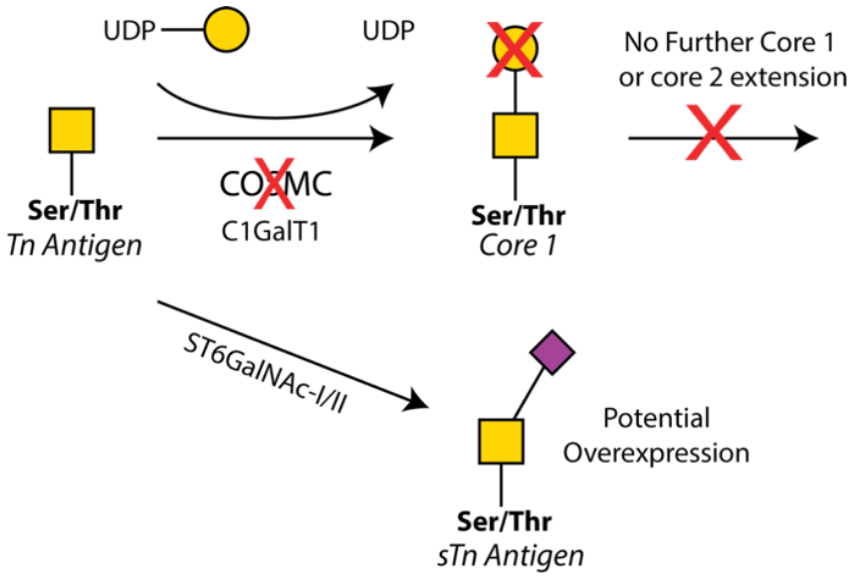


Figure 7. *Gene silencing of COSMC chaperone has knock-on effect of no core 1 synthesis and subseuqent core 2 synthesis and extensions. Overexpression of Tn and sTn may result.*

1.4 CONCLUSION

The aim of this chapter was to describe *O*-glycosylation by covering biological synthesis, analytical methodologies used to characterise *O*-glycans, *O*-glycan biomarker discovery, and how *O*-glycans relate to disease, providing context for the following chapters. Chapter 2 will focus on optimizing a microwave-assisted, non-reductive β-elimination *O*-glycan release. Utilizing this optimized method, chapter 3 will describe the characterization and statistical analysis of complex *O*-glycan samples from striatum and substantia nigra regions of Incidental Lewy Body (ILB) and Parkinson's Disease (PD) patients, drawing relevant conclusions.

CHAPTER 2 – METHOD DEVELOPMENT FOR MICROWAVE-ASSISTED NON-REDUCTVE RELEASE OF *O*-GLYCANS

2.1 INTRODUCTION

As discussed in chapter 1, peeling reactions occurring unintentionally during the β-elimination reaction process are a significant problem in characterization and analysis of released non-reduced *O*-glycans. While reducing these oligosaccharides during the release reaction with excess $NaBH_4$ driving formation of alditols limits this peeling, analytical techniques are restricted to mainly mass spectrometry-based approaches, posing a number of problems. The greatest limitation to alditol formation is the lack of fluorescence of natively non-fluorescent glycans making purely fluorescence-UPLC (FLR-UPLC) analysis and comparison across individual samples difficult.

The focus placed on non-reductive β-elimination methods, allowing subsequent fluorescent labelling, have similarly needed to overcome this hurdle of peeling. Hydrazinolysis minimizes peeling while yielding a high concentration of released glycan, however is relatively hazardous and may not be applicable to labs lacking expertise. Oven-based β-elimination methods using an ammonia-based reaction solution (ammonium hydroxide or 20-30% ammonia in solution) at 60-70°C for 16-40 hours also provide a relatively high yield of released glycans, but increase peeling. “End-cap” reagents that inhibit peeling through temporary reducing-end glycosylamine formation (such as ammonium carbonate, ammonium bicarbonate and ammonium carbamate) have sought to limit this, however, peeling still remains an issue.

The method developed in our lab for this work aims to provide usable glycomics data with minimal peeling by utilizing a microwave-assisted non-reductive β-elimination approach. By using mic rowave radiation, reaction rate is increased (reducing reaction time), meaning a reaction that is typically performed overnight can be performed in as little as 12 minutes. The benefit of this is a reduction in time permissible for suspected peeling reactions to occur. This hypothesis is based on previous preliminary work conducted in our lab by Dr. Mark Hilliard.

2.2 MATERIALS AND METHODS

H_2O (water) in the following experments is from a Milli-Q® (Millipore) system with resistivity of 18.2 MΩ.cm (at 25°C).

Acetonitrile (ACN) in the following experments is from Honeywell Fluka (#34888).

For release method optimisation as well as reproducibility studies, fetuin glycoprotein (#F3385, Sigma Aldrich) and porcine gastric mucin (#M1778, Sigma Aldrich) were used.

2.2.1 Visual Schematic of the Release and Analysis Protocol

While multiple methods for the release and analysis of glycans were performed (outlined in detail below), the focus on optimisation of a microwave assisted non-reductive β-elimination method to release *O*-glycans formed the first component of this work and is highlighted in figure 8.

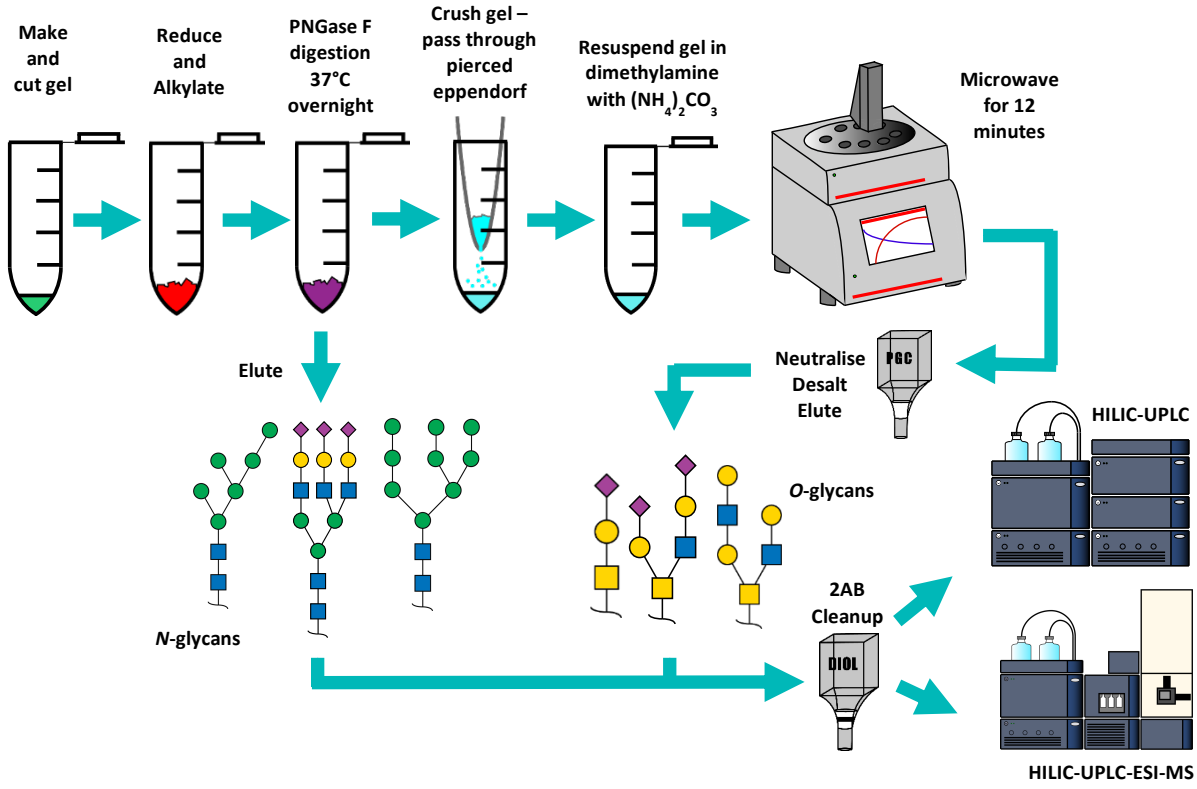


Figure 8. *Schematic of release protocol for both N- and O-glycans showing traditional enzymatic release of N-glycans and a microwave assisted non-reductive β-elimination release of O-glycans.*

2.2.2 *N*-Glycan Release

Tissue/glycoprotein samples were broken down and set in bis-acrylamide protogel (#EC8901LTR, Fisher Scientific)/10% sodium dodecyl sulfate (SDS; #436413 Sigma Aldrich)

mix and cut into ~1mm^2 pieces. Protein containing gels were reduced and alkylayed using 50 mM dithiothreitol (DTT; #D0632, Sigma Aldrich) and 10 mM iodoacetamide (IAA; #I1149, Sigma Aldrich) respectively, in 2mL tubes, dried down in a speedvac (Savant, Fisher Scientific), and resuspended in 100 µL of 1 U/400 µL PNGase F (#P0709, New England Biolabs) in 20 mM $NaHCO_3$ (#09830, Sigma Aldrich). After 16hr digestion at 37°C, *N*-glycans were eluted from gels and the eluent dried. Oligosaccharides were fluorescently labelled through reductive amination by first reducing in 20 µL 1% formic acid (#F0507, Sigma Aldrich) for 40 minutes, drying down, and resuspending in 5 µL 2-aminobenzamide (#A89804, 2AB; Sigma Aldrich) solution for 2 hours at 65°C. Excess label was removed using a 10 mg bed weight Hypersep Diol Solid-Phase Extraction (SPE) cartridge (#60300-630, Fisher Scientific). First, SPE cartridges were washed with 3 x 1 mL 95% ACN, 3 x 1 mL water and then 3 x 1 mL 95% ACN. Samples were diluted in 900 µL ACN, loaded on to the cartridge and washed with 3 x 1 mL 95% ACN. Labelled glycans were eluted with 2 x 0.9 mL water. Glycan samples were dried down prior to analysis.

2.2.3 *O*-glycan release (microwave-assisted non-reductive β-elimination)

Remaining wetted gels from *N*-glycan release were crushed by passing them through a small hole in the bottom of a 0.2 mL tube into a 1.5 mL tube (under 14,000 rpm centrifugal force), dried down, resuspended in 250 uL 40% dimethlyamine in water (#426458, Sigma Aldrich) containing 0.1 g/mL ammonium carbonate (#AC401130250, Fisher Scientific) and transferred to G4 vials (Anton Paar). Tubes were subjected to 12 minute microwave radiation at 70°C in a Monowave 450 microwave reactor at 600W (Anton Paar). Reaction solution inlcuding gel was transferred to 2 mL tubes and neutralised with 1M HCl (#H1758 Sigma Aldrich) using a micro pH electrode (#HI-1093B, Hanna Instruments). Samples were desalted using 10 mg bed weight Hypersep Hypercarb SPE (#60302-606, Fisher Scientific). First, SPE cartridges were washed with 3 x 1 mL 100% ACN, 3 x 1 mL 50% ACN and then 3 x 1 mL water. Samples were loaded on to the cartridge and washed with 3 x 1 mL water, and glycans were eluted with 2 x 0.9 mL 25% ACN/75% 50 mM ammonium formate (prepared with 50 mM formic acid; #F0507 Sigma Aldrich, adjusted to pH 4.4 with ammonium hydroxide solution; #338818 Sigma Aldrich) and dried. Glycans were labelled with 2AB and excess label was removed using a 10 mg bed weight Hypersep Diol SPE (#60300-630, Fisher Scientific) applying the same process used in the *N*-glycan release desribed above in the section 2.2.2.

2.2.4 HILIC-UPLC Analysis of non-reductively released (2AB labelled) glycans

Non-reductive β-eliminated 2AB labelled *O*-glycans were resusupended in 20 µL 88% ACN and 19 µL was injected into an Aquity H-Class hydrophilic-interaction liquid chromatography-ultra-performance liquid chromatography (HILIC-UPLC) system (Waters Corporation) coupled with a fluorescence detector (Waters Corporation). The oligosaccharides were separated on

an ACQUITY UPLC Glycan BEH Amide, 130Å, 1.7 µm column (#186004742, Waters Corporation) at a flow rate of 561 µL/min and a column temperature of 40 °C.
The oligosaccharides were eluted using buffer A (50 mM ammonium formate, prepared with 50 mM formic acid; #F0507 Sigma Aldrich, adjusted to pH 4.4 with ammonium hydroxide solution; #338818 Sigma Aldrich) and buffer B (acetonitrile) over a 30 minute run using a gradient shown in the table below.

Time (min)	Flow (µL/min)	% A	% B
Intial	561.0	12.0	88.0
1.47	561.0	12.0	88.0
25.00	561.0	47.6	52.4
25.60	300.0	70.0	30.0
26.80	300.0	70.0	30.0
28.00	300.0	12.0	88.0
30.00	561.0	12.0	88.0

Table 3. *Elution conditions on HILIC-UPLC for 2AB labelled O-glycans. A is 50 mM ammonium formate pH 4.4, B is acetonitrile.*

Fluorescence was measured at 420 nm, with excitation at 330 nm. External calibration was perfomed using 2AB labelled glucose oligomers (#31388, Sigma Aldrich), creating a dextran ladder with retention times of all identified glycan peaks expressed in glucose units (GUs).

2.3 RESULTS – OPTIMIZING NON-REDUCTIVE RELEASE METHOD

2.3.1 Identifying a Suitable Separation Gradient and Elution Conditions

To best identify *O*-glycans from pre- to post-release optimization stages, a suitable separation gradient for *O*-glycans needed to be established. Historically the focus on 2AB labelled *N*-glycans within our lab meant *N*-glycan separation gradients for HILIC-UPLC had already been established to favor these larger glycan structures. As *O*-glycans are typically smaller, a similar gradient of elution/separation was unsuitable and so an elution gradient was developed. Figure 9 compares the *N*-glycan elution gradient and the developed *O*-glycan method on bovine fetuin *O*-glycan separation.

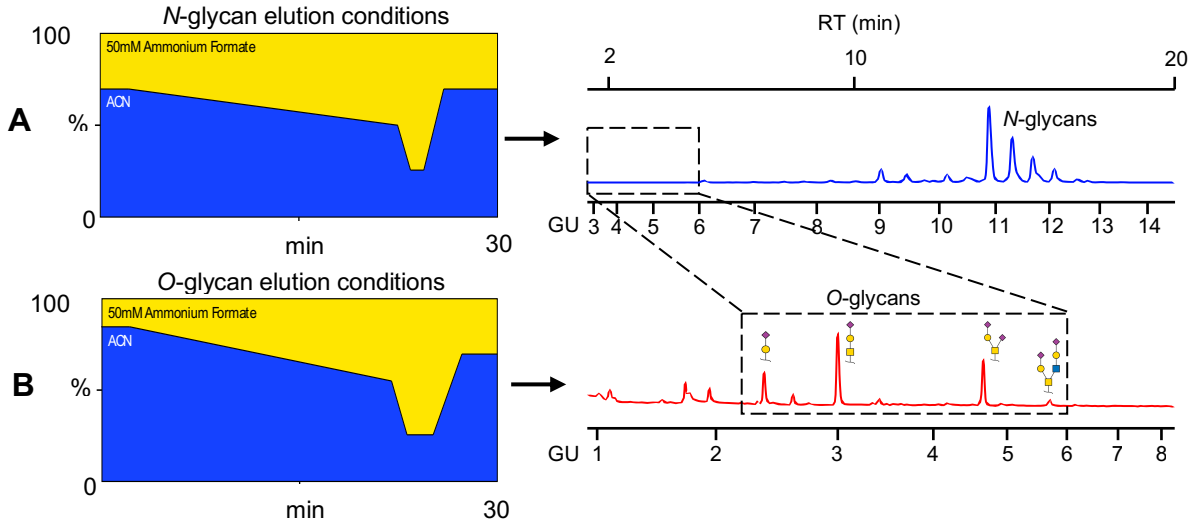


Figure 9. *HILIC-UPLC elution conditions and separation of O-glycans from bovine fetuin. (A) N-glycan method. (B) O-glycan method. Dotted boxes indicate the same GU region, spread over different timepoints of each of the two methods.*

By beginning the UPLC run with a more organic mobile phase (88% ACN in the *O*-glycan method compared to 70% in the *N*-glycan method), elution of the slightly less hydrophilic *O*-glycans (compared to *N*-glycans) proceeds more slowly, better separating those species at lower GUs. With respect to the *O*-glycan method, employing a gradient from 88% ACN to 53% ACN over 25 minutes preferentially separates *O*-glycan species over *N*-glycan species which all elute close together later in the run (after 20 minutes – not shown).

While the *O*-glycan profile of bovine fetuin is relatively simple, this increase in baseline separation is particularly important for more complex *O*-glycan samples and samples that contain a wider array of glycans with various cores that have similar glycan composition.

2.3.2 Comparison Between Traditional Oven and Microwave-Assisted Ammonia Based Non-Reductive β-elimination Approaches

A traditional oven-based approach to β-elimination was compared with a microwave assisted approach in order to validate the benefit of using the microwave reactor for future glycan release experiments.

In this instance, a duplicate study was performed, each for a 16-hour oven-based release (at 70°C), a 40-hour oven-based release and a 12-minute microwave assisted method (figure 10). The protocol for sample preparation is outlined in the materials and methods section 2.4.1 with the only difference being method of heat/reaction source for glycan release (oven vs. microwave).

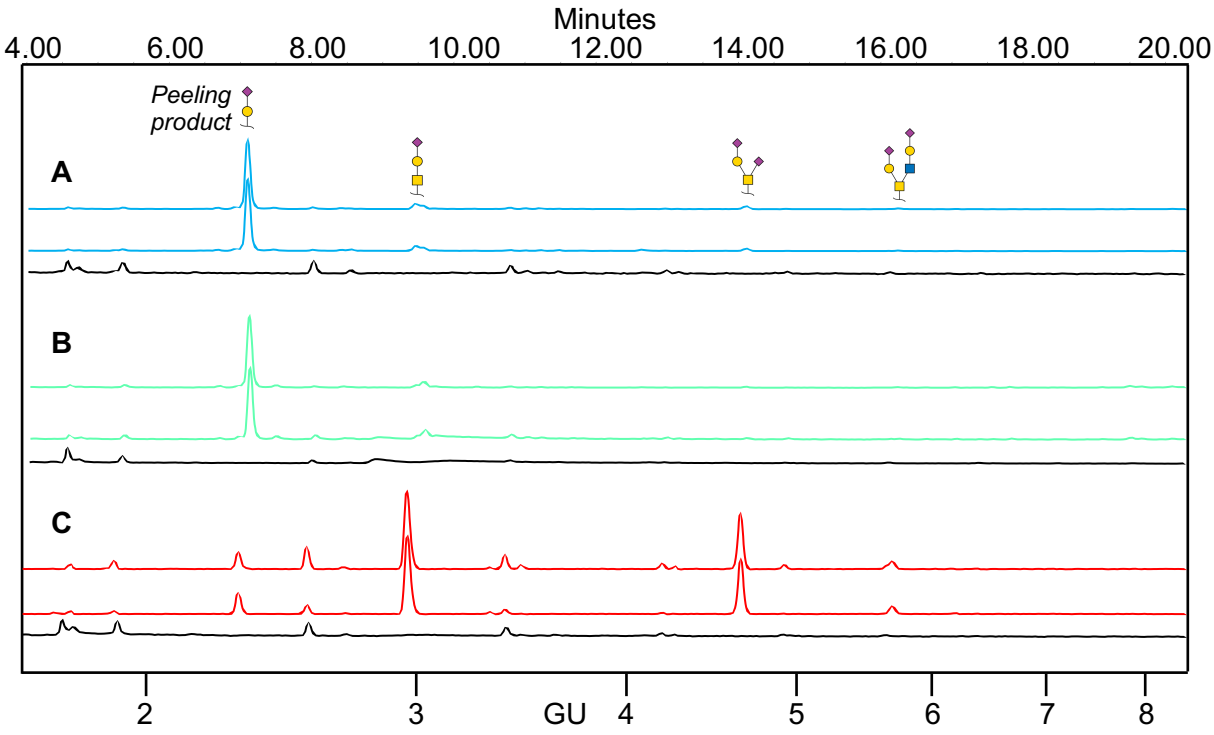


Figure 10. *HILIC-UPLC profiles of O-glycans from bovine fetuin glycoprotein. (A) 16 hour non-reductive β-elimination at 70°C using a heating oven, (B) 40 hour non-reductive β-elimination at 70°C using a heating oven, and (C) 12 minute microwave-assisted non-reductive β-elimination at 70°C using the Monowave 450 system. Duplicate studies are shown together in matching colour with their respective negative controls underneath (in black). Major O-glycans highlighted.*

From these profiles, it was confirmed that an oven-based release method produced a profile with high proportion of peeling, indicated by the large peak (labelled “peeling product”) at GU 2.4 in **A** and **B**, corresponding to 2AB labelled NeuNAcα2-3Gal which had degraded from the

larger glycan structures. As the duplicate profiles in **C** (microwave-assisted method) indicated less peeling peak area and higher relative area of intact glycans, this heating/release source was taken through further optimizations.

2.3.3 Final Optimizations Leading to the Established Protocol

A number of parameters were tested to maximize yield whilst minimizing peeling. As peeling seemed to directly correlate to reaction time, a balance of reaction time vs. degree of peeling needed to be met.

For all optimizations the profiles obtained from the protocol outlined in section 2.2.3 will act as the control to which all modification to the experiment are compared against.

This section will be broken down into subsections. The first will address pre-release optimizations, the second will address release optimizations and the third will address post-release optimizations.

Optimization experiments were performed in duplicate on 1 mg fetuin to which 50% of released glycans were injected onto the UPLC. Duplicates are grouped using the same color with respective blanks directly underneath (in black).

2.3.3.1 Pre-Release Optimizations

2.3.3.1.1 Reduce and Alkylate

Reduction and alkylation are typically used for most proteomic and glycoproteomic workflows[240] in order to denature and unfold the protein to reduce hindrance for enzymatic digestion of glycans. Disulfide bridges between internal cysteine residues on the protein are first reduced, typically with dithiothreitol (DTT) and then free SH groups are alkylated, typically with iodoacetamide (IAA)[241]. While this benefits enzymatic digestion of internal *N*-glycans, its role in chemical de-*O*-glycosylation is less defined. Figure 11 identifies the profiles of *O*-glycans released with and without reducing and alkylating, outlining an increase in relative area of the peeling product by not reducing and alkylating.

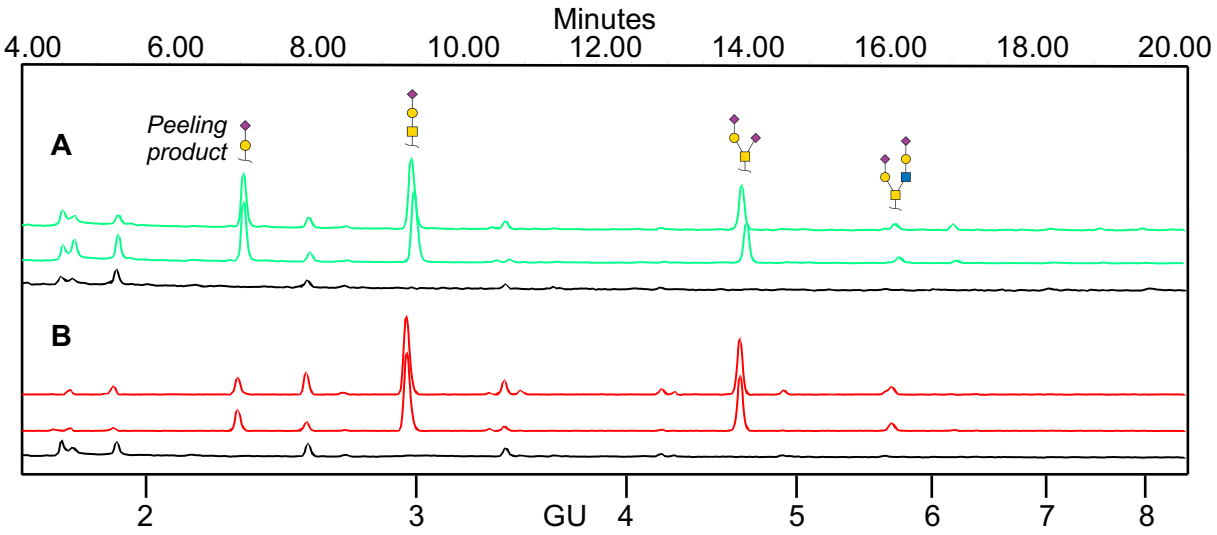


Figure 11. *HILIC-UPLC profiles of O-glycans from bovine fetuin glycoprotein. (A) O-glycan profile from non-reduced and alkylated fetuin glycoprotein, (B) control. Duplicate studies are shown together in matching colour with their respective negative controls underneath (in black). Major O-glycans highlighted.*

As fetuin contains 12 cysteine residues (all involved in disulfide bridging)[242] it is expected that reducing and alkylating may have some impact on yield. It can be speculated that denaturation of the protein provides access to a larger population of *O*-glycans and that by not denaturing, chemical release may only have access to those glycans which have a higher propensity to peel.

2.3.3.1.2 *N*-glycan Release

Pretreatment of the glycoprotein with PNGase F to remove *N*-glycans was also assessed. For Chapter 3 of this project, samples had been provided in gel form post-*N*-glycan release, however, future projects may present samples with *N*-glycans still attached. Figure 12 shows a duplicate set of results outlining the effect releasing/not releasing *N*-glycans first has on *O*-glycan release.

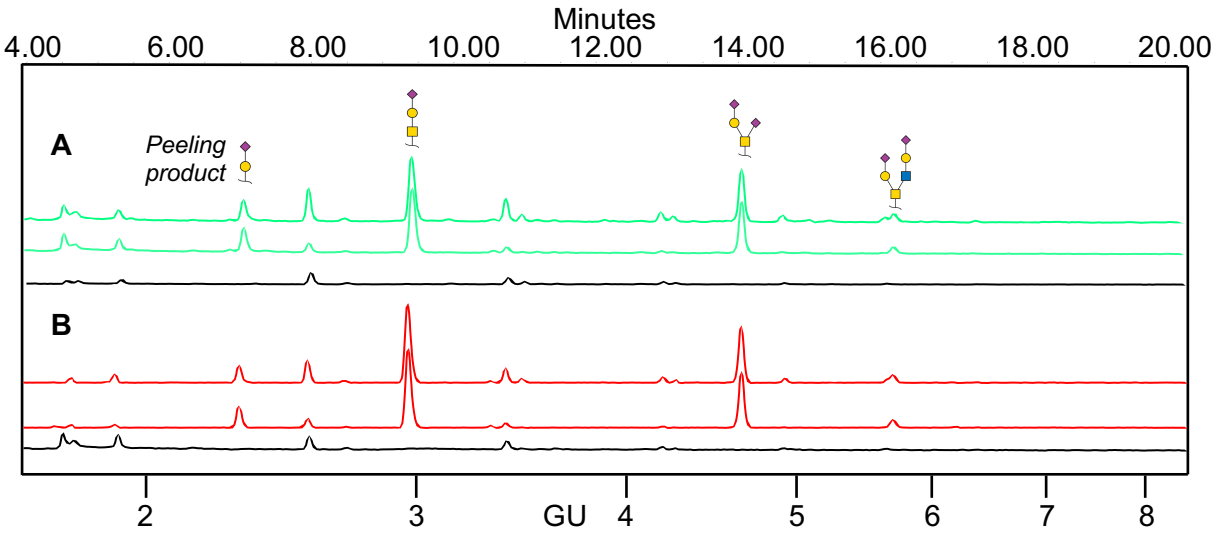


Figure 12. *HILIC-UPLC profiles of O-glycans from bovine fetuin glycoprotein. (A) No treatment with PNGase F before O-glycan release, (B) control. Duplicate studies are shown together in matching colour with their respective negative controls underneath (in black). Major O-glycans highlighted.*

The results show a similar profile with respect to *O*-glycans released, however, slightly higher levels of surrounding junk are seen in the non-PNGase F digested sample, indicating possible degradation of larger glycans or other contaminating products.

Maximizing the benefit of a glycomic workflow, more information is provided on total protein glycosylation by first releasing and eluting out *N*-glycans. Furthermore, as *N*-glycans are partially released during the β-elimination reaction, the benefit of removing them aids in deconvoluting UPLC chromatograms and mass spectra for characterization.

2.3.3.1.3 Sample Phase

A final modification of the pre-release protocol involved identifying the ideal phase of the sample, between gel (protein set in bis-acrylamide) and liquid (protein in solution) forms. Two processes of gel preparation were compared against an in-solution process. The in-solution process differs from the gel-based process in that the protein is not reduced or alkylated and the *N*-glycans have not been released first. Instead, the glycoprotein is placed directly into the G4 reaction tube for the microwave. The remainder of the protocol was followed identically. For the gel preparations, one duplicate set was cut into 1 mm^2 pieces as described in the *N*-glycan release method (section 2.2.2) without further crushing and the other duplicate set was crushed further by passing through a pierced tube under centrifuge (the control). The reasoning behind the experiment was that due to the rapid nature of adding alkaline β-elimination solution to the gel for a 12-minute digestion, absorption and swelling of the gel (and hence interaction with the protein it contained) needed to be quick. By comparing these *O*-

glycan chromatograms against the chromatograms of a traditional solution-based approach (figure 13), ideal conditions could be identified.

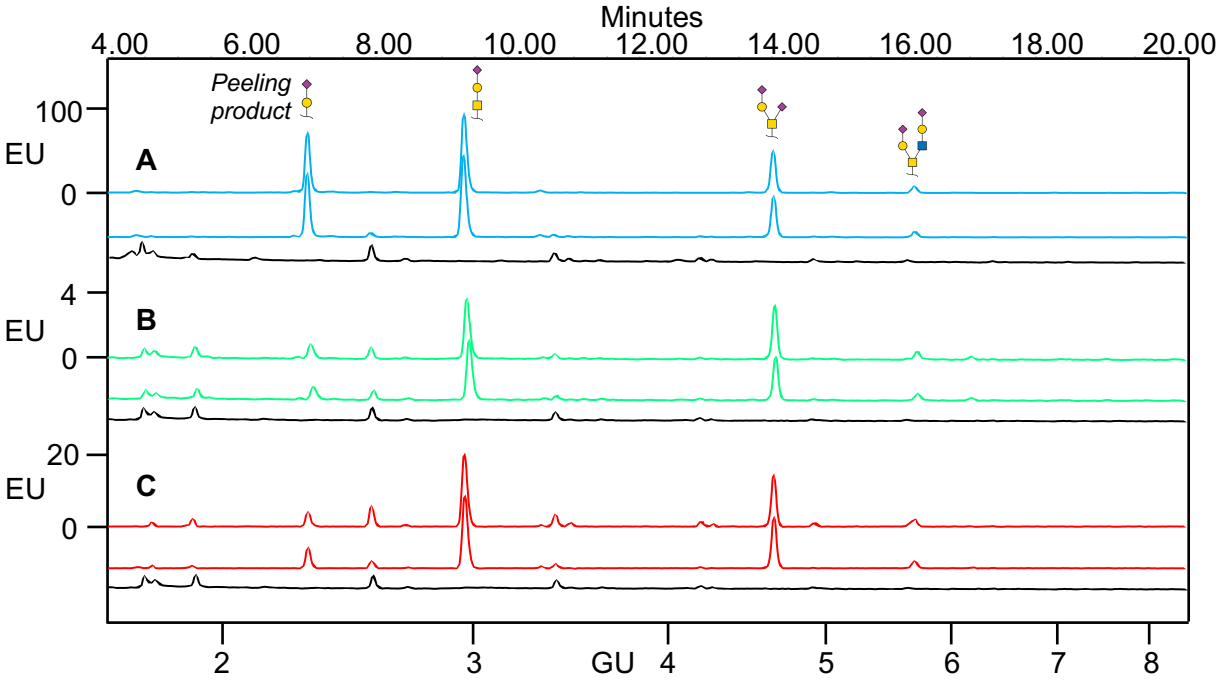


Figure 13. *HILIC-UPLC profiles of O-glycans from bovine fetuin glycoprotein. (A) In-solution release, (B) in-gel release (large pieces), and (C) control (crushed gel). EU (fluorescence intensity) values given on the upper chromatogram of each duplicate study A,B and C. Duplicate studies are shown together in matching colour with their respective negative controls underneath (in black). Major O-glycans highlighted.*

Fluorescence intensity is highest for *O*-glycans obtained from an in-solution release (indicating better yield), however, relative area of the peeling product is also highest. In-gel release from larger gel pieces provided an acceptable profile with minimal peeling, however, fluorescence was lowest. In-gel (crushed) release was observed to provide a suitable in-between for these two methods.

2.3.3.2 Release Optimizations

2.3.3.2.1 Microwave Reaction Times

Microwave reaction times were compared to identify the ideal balance between reaction speed, yield, peeling, reproducibility and overall workflow when dealing with large sample sets. Figure 14 identifies the coefficient of variation (CV) for each peak in each reaction time duplicate chromatogram.

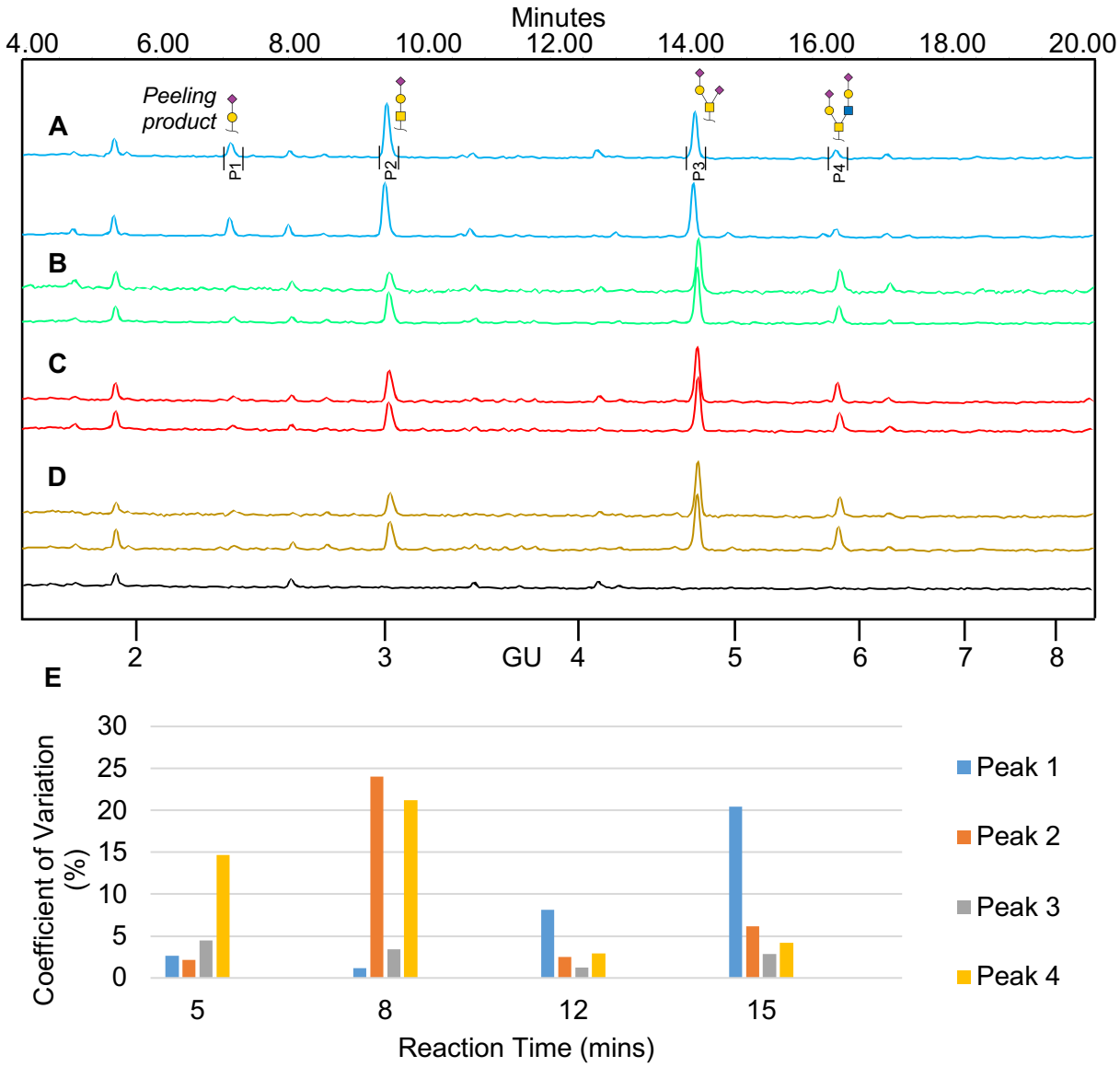


Figure 14. *HILIC-UPLC profiles of O-glycans from bovine fetuin glycoprotein after microwave digestion for (A) 5 minutes (sample integration shown; P=Peak), (B) 8 minutes (C) 12 minutes (control) and (D) 15 minutes. (E) Coefficient of variation of reaction time duplicates to identify most reproducible sample set. Duplicate studies are shown together in matching colour with the negative control underneath (in black). Major O-glycans highlighted.*

A 12-minute reaction was identified as the ideal reaction time and also showed the least peeling (A: 12.45%, B: 5.20%, **C: 4.46%**, D: 5.16%) and lowest overall CV.

2.3.3.2.2 Microwave Reaction Temperatures

Reaction temperatures of 60°C, 70°C and 80°C were also tested to determine ideal conditions. The results from figure 15 demonstrate that chromatograms are similar between all 3 temperatures. According to the literature, temperatures around 60-70°C are common[53, 68]. As

70°C reaction temperature had been previously used in our lab for release, this temperature was maintained through optimization and final sample analysis.

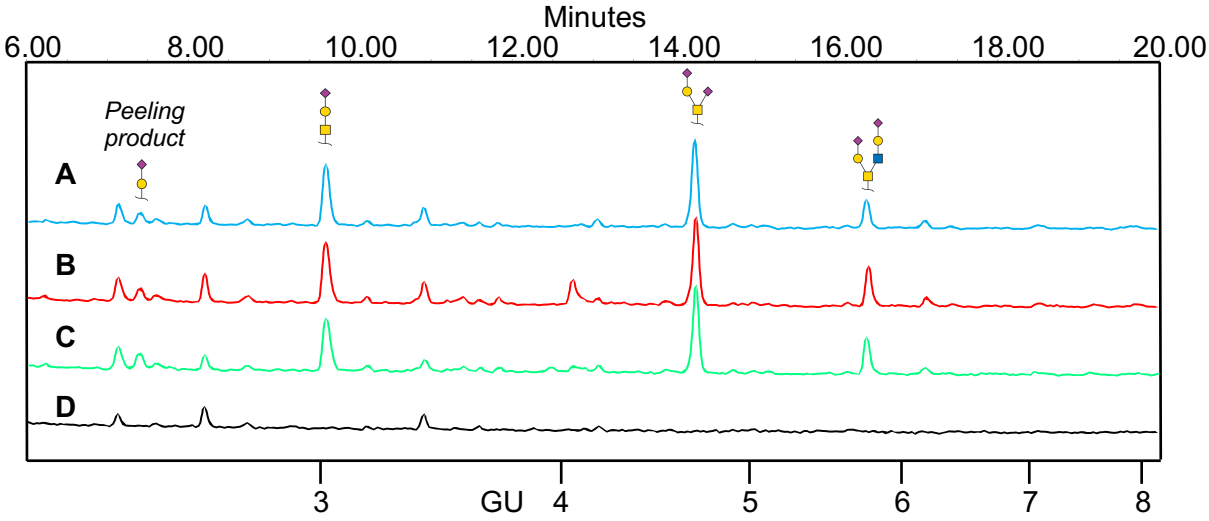


Figure 15. *HILIC-UPLC profiles of O-glycans from bovine fetuin glycoprotein after 12-minute microwave digestion at (A) 60°C, (B) 70°C (control) (C) 80°C. (D) Blank. Major O-glycans highlighted.*

2.3.3.2.3 Ammonium Carbonate Concentration

As ammonium carbonate is suspected to act as the 'end-cap' agent to prevent alkali-catalyzed peeling from the reducing terminus of the glycan (by converting the glycan into a more stable glycosylamine intermediate species), comparisons of ammonium carbonate concentration in the reaction vessel was performed. Figure 16 identifies five different concentrations of ammonium carbonate that were tested.

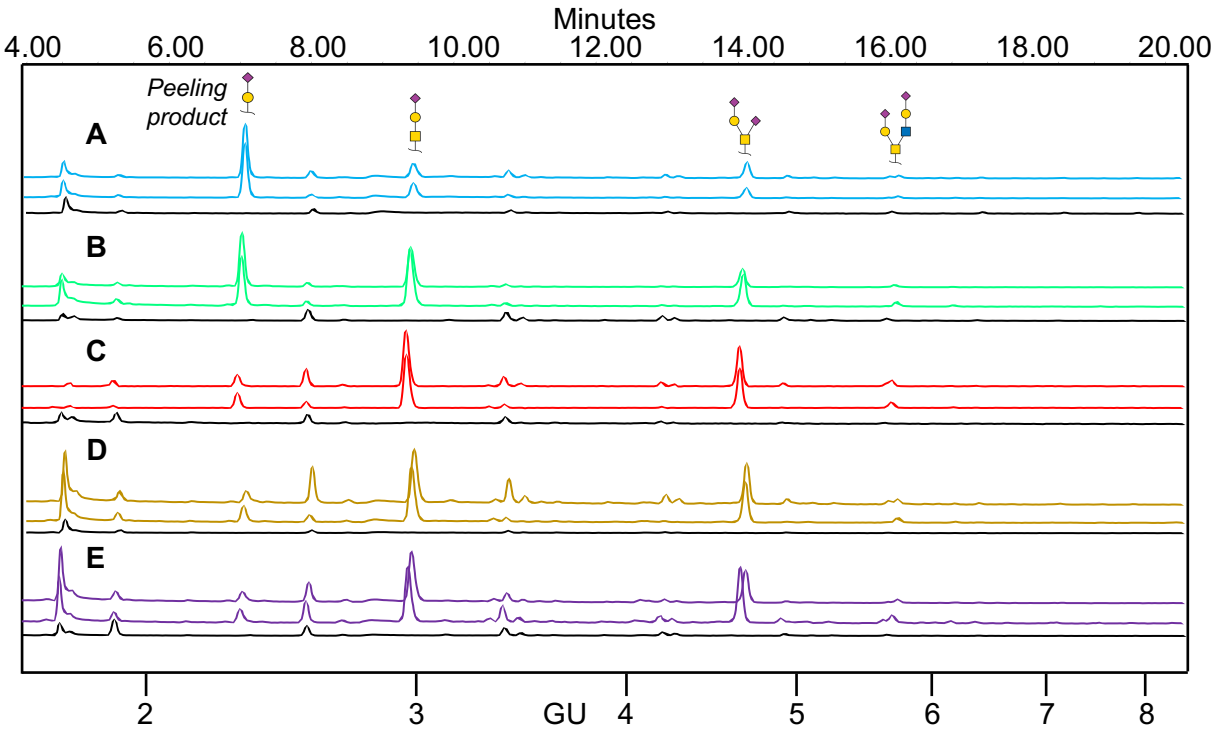


Figure 16. *HILIC-UPLC profiles of O-glycans from bovine fetuin glycoprotein released using (A) 0 g/mL $(NH_4)_2CO_3$, (B) 0.05 g/mL $(NH_4)_2CO_3$, (C) 0.1 g/mL $(NH_4)_2CO_3$ (control), (D) 0.15 g/mL $(NH_4)_2CO_3$ and (E) 0.2 g/mL $(NH_4)_2CO_3$. Duplicate studies are shown together in matching colour with their respective negative controls underneath (in black). Major O-glycans highlighted.*

The results of this experiment show that increasing concentration from 0.1 g/mL $(NH_4)_2CO_3$ onwards minimized peeling of the three larger glycans, however, by increasing the concentration of ammonium carbonate past this point, the peak at GU 1.8 started increasing in area, indicating the increase in concentration of a potential Hex-HexNAc structure (most likely the core 1 T antigen). This may be due to degradation or de-sialylation of the larger glycans or potential increase of the release of this structure under increased ammonium carbonate conditions. Based on this, 0.1 g/mL $(NH_4)_2CO_3$ was the selected concentration taken forward in optimization.

2.3.3.3 Post-Release Optimizations

2.3.3.3.1 Neutralization Conditions

As released non-reduced glycans are subject to peeling in the alkaline conditions of the reaction mixture, the benefit of neutralizing the reaction immediately after the 12-minute digestion is obvious. Figure 17 shows the comparison of profiles produced from neutralizing

(or more specifically, pH adjusting) the reaction solution to pH 6.5, 7.0 and 7.5 prior to desalting, 2AB labelling, and analysis.

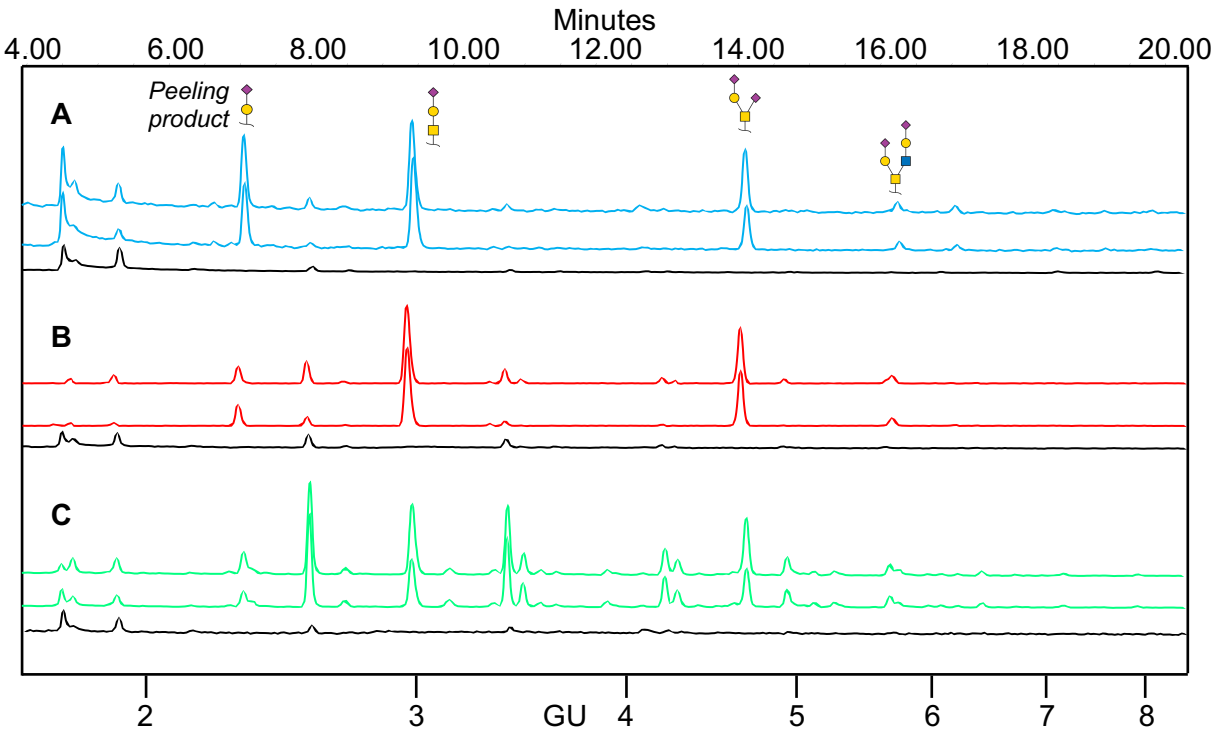


Figure 17. *HILIC-UPLC profiles of O-glycans from bovine fetuin glycoprotein where reaction solution was neutralized to (A) pH 6.5, (B) pH 7.0 (control), and (C) pH 7.5. Duplicate studies are shown together in matching colour with their respective negative controls underneath (in black). Major O-glycans highlighted.*

While both adjustments to pH 7.0 and 7.5 produce a profile with minimal peeling compared with pH 6.5, the pH 7.5 profile contains a lot of additional degradation products or other artifacts and so wasn't selected for further optimizations. As expected, neutralization to pH 7.0 provided the most reproducible results.

2.3.3.3.2 PGC SPE Desalting Conditions

Following neutralization, the sample solution required desalting to remove neutralization salts and other unwanted products. This was found to be a particularly important step, as without neutralization and/or without desalting, peeling products dominated the profiles (figure 18).

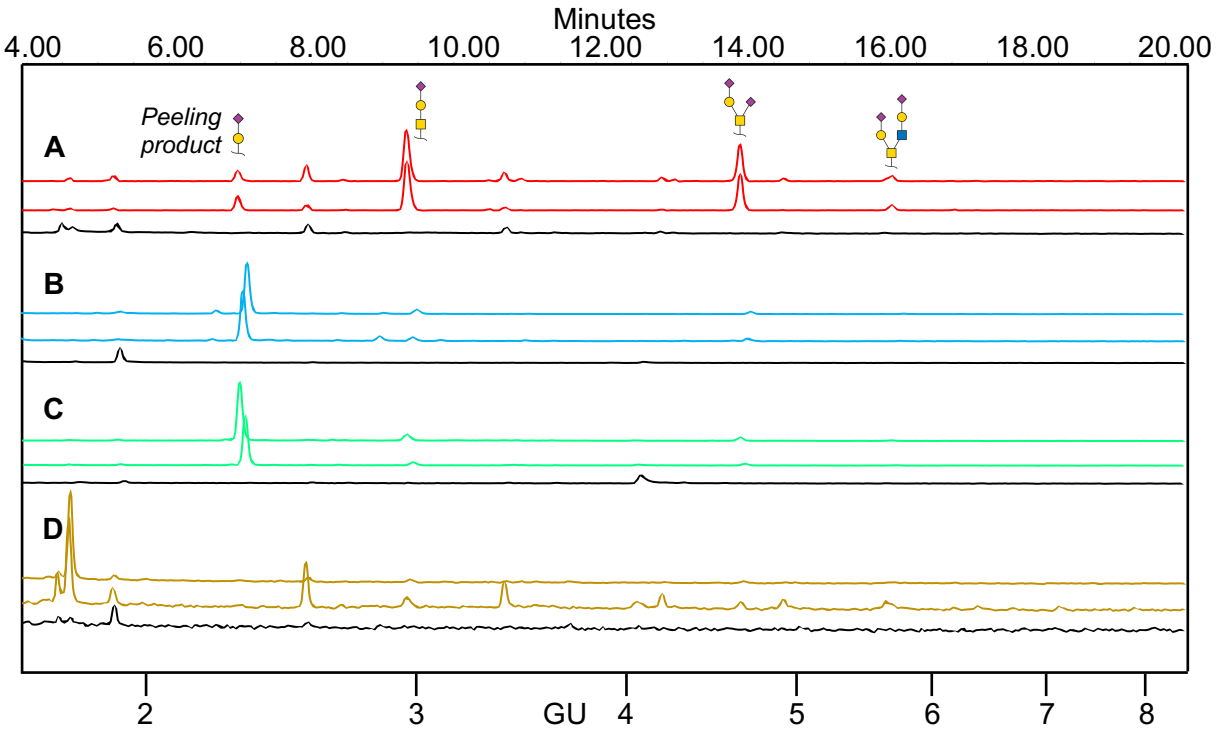


Figure 18. *HILIC-UPLC profiles of O-glycans from bovine fetuin glycoprotein where reaction solution was (A) neutralized and desalted (control), (B) not neutralized or desalted, (C) neutralized but not desalted, and (D) not neutralized but desalted. Duplicate studies are shown together in matching colour with their respective negative controls underneath (in black). Major O-glycans highlighted.*

This indicated that desalting was critical, but only after correct neutralization procedures from the previous experiment were followed.

2.3.4 Reproducibility studies

2.3.4.1 Fetuin and Porcine Gastric Mucin Standards

With all optimizations in place, 10x standard replicate runs were compared to test for reproducibility of the optimized method. This was performed primarily on fetuin and then also on porcine gastric mucin (PGM). While fetuin has a relatively simple *O*-glycan profile, PGM is a lot more complex with a high degree of *O*-glycan heterogeneity[46, 243, 244], and might better represent other unknown complex samples.

Figures 19 and 20 (respectively) show replicate chromatograms as well as the coefficient of variation and average relative area for each glycan peak in a 10x replicate study on fetuin.

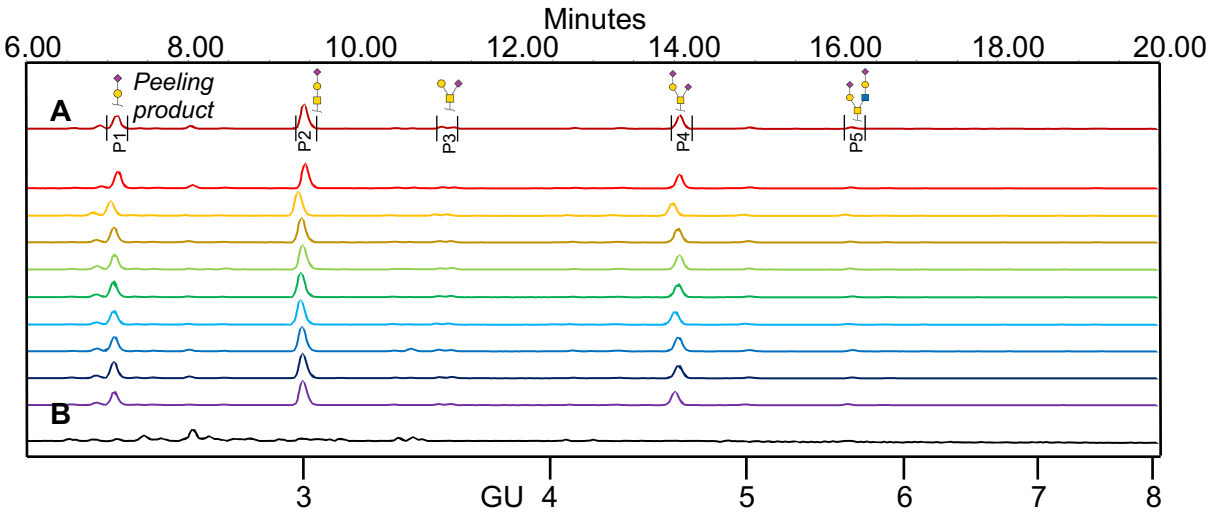


Figure 19. *HILIC-UPLC profiles of O-glycans from bovine fetuin glycoprotein across (A) 10x replicate using the optimized method (sample integration shown; P=Peak). (B) Blank. Major O-glycans highlighted.*

The inclusivity of the mono-sialylated structure at GU 3.5 was incorporated into the calculations for reproducibility due to its consistent presence in this particular experiment, whereas previous optimization work focused on only the 4 main *O*-glycans of fetuin that were, at the time, more easily resolved from the chromatogram baseline.

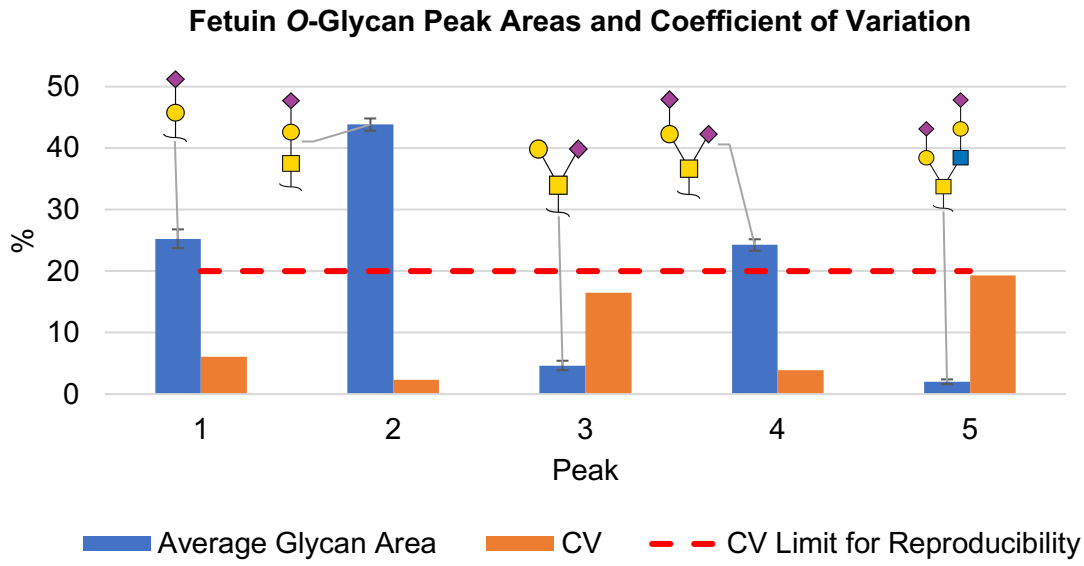


Figure 20. *Analysis of average peak area of fetuin (10x replicated) plotted alongside coefficient of variation (CV) for each peak. An upper threshold for a CV of 20% was applied for the acceptability for reproducibility. (% Area and % CV plotted on same axis).*

Figures 21 and 22 (respectively) show replicate chromatograms as well as the coefficient of variation and average relative area for each glycan peak in a 10x replicate study on PGM.

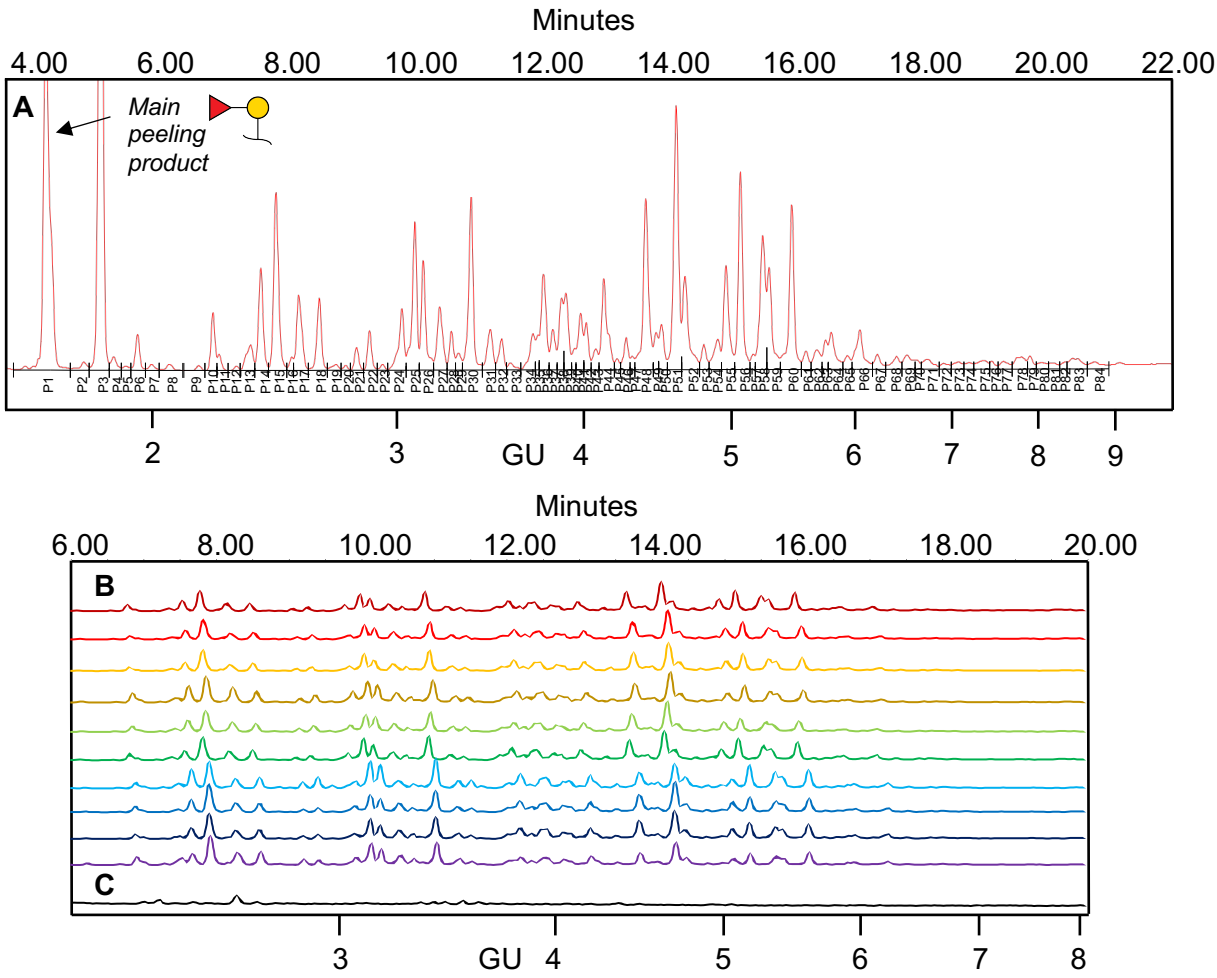


Figure 21. *HILIC-UPLC profiles of O-glycans from porcine gastric mucin. (A) sample integration of PGM profile (extended to show higher and lower GU; P=Peak), (B) 10x replicate using the optimized method. (C) Blank.*

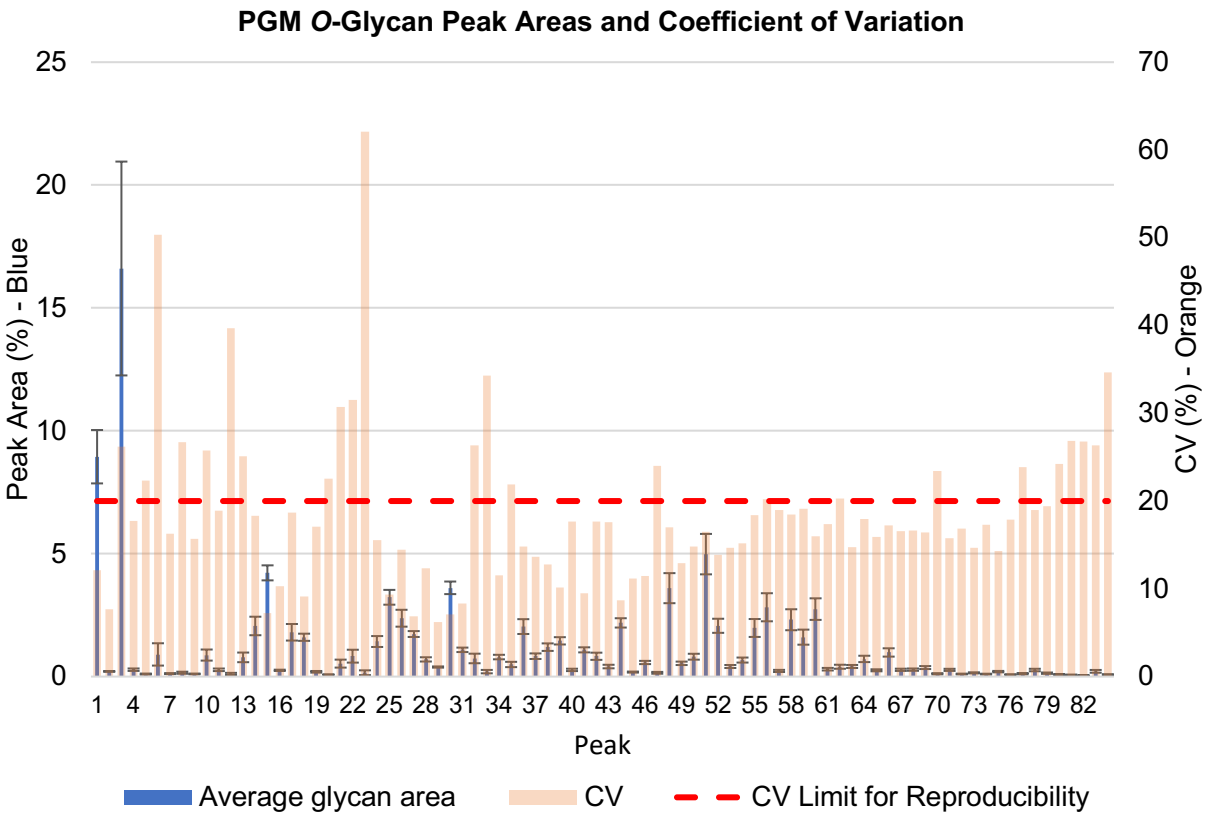


Figure 22. *Analysis of average peak area of PGM (10x replicate) plotted alongside coefficient of variation (CV) for each peak. An upper threshold for a CV of 20% was applied for the acceptability for reproducibility.*

The results from these reproducibility studies indicate most major peaks in both fetuin and PGM had a CV below 20%. The discrepancies in PGM CV calculations most likely stem from peeling products (particularly fucoseα1-2galactose; identified in figure 21 as the main peeling product) arising from the greater heterogeneity and higher relative abundance of peeling-susceptible glycans (typically expressed in mucin samples). This emphasizes the difficulty in minimizing peeling using an ammonia-based non-reductive β-elimination method, which remains a significant problem within the literature[53, 65].

2.4 DISCUSSION

While results were comparable to similar ammonia-based non-reductive β-elimination techniques seen in the literature[47, 62, 64], peeling from these methods remained a pressing issue. Through experimentation and optimization, peeling was reduced to levels expected from a typical hydrazine based non-reductive release (approx. 15-20%), which fell more in line with non-reductively released *O*-glycans commercially available (Ludger™). Moving forward to clinical samples in chapter 3 (PD, ILBD, control), the main release and analysis methods were performed using this optimized ammonia-based non reductive release with 2AB labelling, however, reductive elimination was also performed for ion-trap mass spectrometry experiments, which was advantageous as peeling here was negligible (no fluorescence detection however).

Due to the high number of variables that were tested in this work, there are many substitutions that could be made to limit peeling from non-reductive β-elimination even further (whether it be identifying a more suitable alkaline reaction solution, or faster acting labelling reagent for example). Another method to explore in the future is the creation of a flow system that removes *O*-glycans from the alkaline reaction solution immediately after release to limit peeling, as has been observed in the literature[245]. Furthermore, low yields remained an issue, particularly during exoglycosidase digestion studies where larger sample sizes were required. In future work, release efficiency studies will need to be performed and the method further optimized.
As mentioned in chapter 1, glycoproteomic approaches to glycan analysis by leaving the glycan attached to the peptide glycosylation site have proven valuable in characterizing not only the glycans but also their relative position as they appear *in vivo*. Similarly, as glycoproteomic databases are built out, predictive technologies are also being employed to help make obsolete the non-reductive and reductive release systems.

Nonetheless, as discussed in chapter 3, peeling data is semi-usable if the components on the peeling products contain specific features of interest (fucosylation, sialylation, etc.), as these contribute nonetheless to total relative area of all the peaks with that specific feature.

2.5 CONCLUSION

Based on these results, optimization of the protocol through testing and monitoring each metric of the release workflow led to an *O*-glycan release with minimal variation between sample replicates. Driving down peeling has also provided insight into non-degraded glycan profiles that can better be used to statistically evaluate profile differences between samples.

The following chapter will discuss characterization and statistical evaluation of individual clinical samples between healthy, ILBD and PD states to identify potential biomarkers and/or significant glycan differences associated with these diseases.

CHAPTER 3 – CHARACTERISATION AND QUANTITATION OF *O*-GLYCANS IN PARKINSON'S DISEASE

3.1 INTRODUCTION

3.1.1 Introduction to Parkinson's Disease (PD)

Parkinson's Disease is a neurodegenerative disorder affecting millions globally, symptomized by imipaired balance, rigidity and shaking with symptoms over time expanding to dementia, depression, and further immobilization[246].

PD is linked to the degeneration of dopaminergic neurons in the *substantia nigra pars compacta* due in large part to inflammation, oxidative stress and mitochondrial dysfunction[247] as well as dysfunction of the ubiquitin-proteasome pathway[248]. This results in the loss of the striatal neurotransmitter dopamine, modulating behaviour and motivation and causing symptoms of bradykinesia, poor rigidity and inhibited muscle contraction through altered signalling down the spinal cord[249, 250]. This influence on the nigrostriatal dopaminergic pathway, connecting the substantia nigra to the striatum, has driven the desire to identify potential biomarkers by mapping the glycomes of these regions.

Pathogenesis of the disease may be affected by a number of environmental and genetic factors[251] including from air pollution[252, 253] and as a result of pharmaceutical product use (statins and non steroidal anti-inflammatory drugs for instance)[254, 255], as well as through altered expressions of genes such as SNCA (encoding for the α-synuclein protein) and LRRK2 (a kinase encoding enzyme)[256].

Of particular interest is the role that the presynaptic neuronal protein α-synuclein plays in PD neuropoathy. During PD degeneration, Lewy neurites (spindle-like structures) and globular Lewy bodies form through toxic α-synuclein aggregation causing neuroinflammation, degeneration (having a deleterious effect on ajacent cells), lesions and ultimately cause cell death. Inclusive in this pathogenesis is the idea that movement and misfolding of α-synuclein from it's homeostatic position within the brain may also contribute to the disease[257, 258].

3.1.2 Braak Staging

Braak et al. proposed in 2003 a staging concept for PD diagnosis based on synuclein pathology and abnornal synuclein manifestation as it traverses the brain and has since become widely accepted[259]. Six stages separate the degree of Parkinsonism with stages 1 and 2 limited to lesions in the dorsal motor nucleus of the medulla oblongata (stages 1 and 2) and locus coeruleus (stage 2). This is often declared a pre-clinical or early stage of PD and is diagnosed often as Incidental Lewy Body Disease (ILDB)[260], however, competing evidence is suggesting

ILDB may be instead associated with aging moreso than PD as lewy bodies are seen in ~10% of clinically normal people[260, 261]. Stages 3 and 4 are when a clincal diagnosis of PD is presumed as toxic α-synuclein pathology reaches the substantia nigra and amygdala (stages 3 and 4) and the mesocortex (stage 4). Stages 5 and 6 diagnoses are given when the prefrontal (stages 5 and 6) and primary neocortices (stage 6) are affected[262].

3.1.3 Glycosylation Changes Observed in Parkinson's Disease

Similar to glycosylation changes seen in other neurodegenerative diseases such as Alzheimer's and Huntington's disease, the role abberrant glycosylation plays in cellular overload of dysfunctional proteins, as well as protein aggregation leading to cell death[247] has driven the need to identify diagnostic glycan biomarkers to replace traditional time-consuming diagnosis of the disease[263].

N-glycosylation of serum in PD has identified general patterns of lower sialylation and higher fucosylation in tri-antennary glycans in PD affected patients[263] and differing abundances of a few individual structures in PD IgG *N*-glycans (including lower relative abundance of an oligomannose structure and increase in a core fucosylated monogalactosylated structure)[264].

O-GlcNAcylation has been studied extensively on α-synuclein proteins associated with PD, outlining that increases in *O*-GlcNacylation of these proteins reuslt in a decrease of toxic α-synuclein aggregation[228, 265, 266].

With respect to non-*O*-GlcNAc *O*-linked glycosylation within the brain, up to 30% of *O*-glycans contain a core mannose residue (derived primarily from glycosylation of α-dystroglycan – a membrane-associated protein) which extend in a relatively conserved manner, often by addition of *N*-acetylglucosamine, galactose, fucose, sialic acid and occasionally the uncommon 3-*O*-sulfated glucuronic acid residue, generating a potential 50-60 structures[267, 268].

The glycosylation of α-dystroglycan is known to play a vital role in correct muscle function, with hypoglycoslyation causing congenital muscular dystrophy as well as being associated with cancer metastasis[269].

Concerning mucin-type *O*-glycans, core 1 structures have been observed to be differntially regulated in PD model mice, in midbrain, stratum and hippocampus regions with an upregulation/accumulation of core 1 glycans on microtubule-associated protein 6 (MAP6) in the disease state[250]. This upregulation has been hypothesised to damage neurons within the domainergic pathway, however confirmation of this as well as a more comprehensive analysis on relative abundances of the entire released *O*-glycomes of these regions has yet to be performed.

Additional carbohydrate classes called gangliosides are also differentially regulated during pathogenesis of PD. GM1 ganglioside deficiencies in nigral dopaminergic neurons of PD patients have been suggested to promote pathogenesis, and as a result, may act as a target for therapeutics (through administration of LIGA-20 – a GM1 analog)[270, 271].

3.1.4 Research Objectives for PD studies

This review forms the basis of a comparative study on *O*-glycosylation in PD, from characterisation of *O*-glycans in the striatum and substantia nigra regions between control and PD patients (collected post-mortem). The method used to release the *O*-glycans is a microwave assisted non-reductive β-elimination approach on tissue/glycoprotein-containing gels. Samples were provided in gel form following *N*-glycan release.

In the proceeding sections of this thesis, experimental procedure, analysis of glycan profiles and results will be derived and relevant significant differences discussed.

3.2 MATERIALS AND METHODS

H_2O (water) in the following experments is from a Milli-Q® (Millipore) system with resistivity of 18.2 MΩ.cm (at 25°C).

Acetonitrile (ACN) in the following experments is from Honeywell Fluka (#34888).

3.2.1 Initial Sample Preparation

For positive controls fetuin glycoprotein (#F3385, Sigma Aldrich) was used.

For reproducibility studies, as well as final studies on human tissue, frozen autopsied striatum and substantia nigra from patients with Parkinson's Disease or Incidental Lewy Bodies Disease (ILBD), healthy matched controls and associated clinical and neuropathological data were provided by the Parkinson's UK Brain Bank, funded by Parkinson's UK, a charity registered in England and Wales (258197) and in Scotland (SC037554). The age, gender and Braak stages of the different samples are described in table 4 below.

The frozen brain tissue was homogenised in RIPA buffer (#R0278, Sigma Aldrich) and Complete Protease Inhibitor Cocktail (#5056489001, Roche) through mechanical disruption using Qiagen TissueLyser LT (Qiagen), at frequency of 40Hz, 4°C, for an 8 minute duration. The homogenates were centrifuged for 20 minutes at 13,000 rpm, 4°C, and the supernatants were collected and kept at -80°C until further analysis. This work on tissue samples was performed by Ana Lúcia Rebelo (PhD candidate at the National University of Ireland, Galway) and samples were provided (for this arm of the project) de-*N*-glycosylated and in gel form.

Brain Bank identifier number	Gender	Age at death (years)	Braak stage	Group
PD204	F	86	3	**PD**
PD572	M	72	3	
PD576	F	80	3	
PD579	M	76	4	
PD590	F	82	3	
PD591	M	77	4	
PD596	M	85	4	
PD612	M	78	4	
PD683	F	83	4	
PD687	M	86	3	
PD709	F	88	3	
PD712	F	87	4	
PDC007	F	104	ILBD	**Incidental Lewy Bodies Disease (ILBD)**
PDC082	F	77	ILBD	
PDC108	F	82	ILBD	
PDC034	M	90	N/A	**Healthy**
PDC040	F	61	N/A	
PDC053	F	89	N/A	
PDC085	F	82	N/A	
PDC088	F	96	N/A	
PDC105	M	95	N/A	
PDC107	F	87	N/A	
PDC111	F	88	N/A	
PD291A	F	87	N/A	
C045	M	77	N/A	
C054	M	66	N/A	
C064	F	63	N/A	
C074	F	84	N/A	
C075	M	88	N/A	
C083	F	87	N/A	
C084	F	84	N/A	
C085	F	81	N/A	
PD529	F	70	N/A	

Table 4. *Details of the human brain tissue of PD, ILBD and control cases used in this study. Braak stage not applicable in control patients.*

3.2.2 *N*-Glycan Release

Despite receiving the protein de-*N*-glycosylated, the *N*-glycan release method has been included so treatment of the sample is understood. Tissue samples were broken down and set in bis-acrylamide protogel (#EC8901LTR, Fisher Scientific)/10% sodium dodecyl sulfate (SDS; #436413 Sigma Aldrich) mix and cut into ~1mm^2 pieces. Protein containing gels were reduced and alkylayed using 50 mM dithiothreitol (DTT; #D0632, Sigma Aldrich) and 10 mM iodoacetamide (IAA; #I1149, Sigma Aldrich) respectively, in 2mL tubes, dried down in a

speedvac (Savant, Fisher Scientific), and resuspended in 100 µL of 1 U/400 µL PNGase F (#P0709, New England Biolabs) in 20 mM $NaHCO_3$ (#09830, Sigma Aldrich). After 16hr digestion at 37°C, *N*-glycans were eluted from gels and the eluent dried. Oligosaccharides were fluorescently labelled through reductive amination by first reducing in 20 µL 1% formic acid (#F0507, Sigma Aldrich) for 40 minutes, drying down, and resuspending in 5 µL 2-aminobenzamide (#A89804, 2AB; Sigma Aldrich) solution for 2 hours at 65°C. Excess label was removed using a 10 mg bed weight Hypersep Diol SPE (#60300-630, Fisher Scientific). First, SPE cartridges were washed with 3 x 1 mL 95% ACN, 3 x 1 mL water and then 3 x 1 mL 95% ACN. Samples were diluted in 900 µL ACN, loaded on to the cartridge and washed with 3 x 1 mL 95% ACN. Labelled glycans were eluted with 2 x 0.9 mL water. Glycan samples were dried down prior to analysis.

3.2.3 *O*-Glycan Release

3.2.3.1 Non-reductive *O*-glycan release

Remaining wetted gels from *N*-glycan release were crushed by passing them through a small hole in the bottom of a 0.2 mL tube into a 1.5 mL tube (under 14,000 rpm centrifugal force), dried down, resuspended in 250 uL 40% dimethlyamine in water (#426458, Sigma Aldrich) containing 0.1 g/mL ammonium carbonate (#AC401130250, Fisher Scientific) and transferred to G4 vials (Anton Paar). Tubes were subjected to 12 minute microwave radiation at 70°C in a Monowave 450 microwave reactor at 600W (Anton Paar). Reaction solution inlcuding gel was transferred to 2 mL tubes and neutralised with 1M HCl (#H1758 Sigma Aldrich) using a micro pH electrode (#HI-1093B, Hanna Instruments). Samples were desalted using 10 mg bed weight Hypersep Hypercarb SPE (#60302-606, Fisher Scientific). First, SPE cartridges were washed with 3 x 1 mL 100% ACN, 3 x 1 mL 50% ACN and then 3 x 1 mL water. Samples were loaded on to the cartridge and washed with 3 x 1 mL water, and glycans were eluted with 2 x 0.9 mL 25% ACN/75% 50 mM ammonium formate (prepared with 50 mM formic acid; #F0507 Sigma Aldrich, adjusted to pH 4.4 with ammonium hydroxide solution; #338818 Sigma Aldrich) and dried. Glycans were labelled with 2AB and excess label was removed using a 10 mg bed weight Hypersep Diol SPE (#60300-630, Fisher Scientific) applying the same process used in the *N*-glycan release.

3.2.3.2 Reductive *O*-glycan release

O-glycans were released using β-elimination and desalted according to Schulz et al. 2002.[272] Briefly, Remaining gels from *N*-glycan release were dried down, resuspended in 50 mM NaOH (#338818, Sigma Aldrich) and 0.5 M $NaBH_4$ (#71320, Sigma Aldrich) enough to cover gels, and incubated for 16 hours at 50°C. Samples were neutralised with glacial acetic acid (#49199, Honeywell Fluka) and desalted with 25 µL AG50WX8 cation-exchange resin (#1421441,

Sigma Aldrich) on a C18 ZipTip (#Z720070, Millipore). Borate salts were eliminated through repeated evaporations using 1% acetic acid in methanol (#34860, Sigma Aldrich) in a speedvac (Savant, Fisher Scientific).

3.2.4 *O*-Glycan Analysis

3.2.4.1 Liquid Chromatography

3.2.4.1.1 WAX-UPLC Analysis of non-reductively released (2AB labelled) glycans

Non-reductive β-eliminated 2AB labelled *O*-glycans were resusupended in 50 µL water and 20 µL was injected into a Aquity H-Class Ultra Performance Liquid Chromatography system (Waters Corporation) coupled with a fluorescence detector (Waters Corporation). The oligosaccharides were separated on a DEAE anion exchange 75 x 7.5 mm i.d, 10 µm particle size column (#186002180, Waters Corporation) at a flow rate of 750 µL/min and a column at ambient temperature.

The oligosaccharides were eluted using buffer A (20% v/v acetonitrile) and buffer B (0.1 M ammonium acetate pH 7.0; #A2706, Sigma Aldrich, in 20% acetonitrile) over a 30 minute run using a gradient shown in the table below.

Time (min)	Flow (µL/min)	% A	% B
Intial	750.0	100.0	0.0
5.00	750.0	100.0	0.0
20.00	750.0	0.0	100.0
22.50	750.0	0.0	100.0
23.00	750.0	100.0	0.0
30.00	750.0	100.0	0.0

Table 5. *Elution conditions on WAX-UPLC for 2AB labelled O-glycans. A is 20% v/v acetonitrile, B is 0.1 M ammonium acetate pH 7.0 in 20% acetonitrile.*

Fluorescence was measured at 420 nm, with excitation at 330 nm. External referencing was perfomed by comparing fluorescence trace against 2AB labelled fetuin *O*-glycans (#CLIBO-FETUIN-01, Ludger).

3.2.4.1.2 HILIC-UPLC Analysis of non-reductively released (2AB labelled) glycans

Non-reductive β-eliminated 2AB labelled *O*-glycans were resusupended in 20 µL 88% ACN and 19 µL was injected into an Aquity H-Class hydrophilic-interaction liquid chromatography-

ultra-performance liquid chromatography (HILIC-UPLC) system (Waters Corporation) coupled with a fluorescence detector (Waters Corporation). The oligosaccharides were separated on an ACQUITY UPLC Glycan BEH Amide, 130Å, 1.7 µm column (#186004742, Waters Corporation) at a flow rate of 561 µL/min and a column temperature of 40 °C.
The oligosaccharides were eluted using buffer A (50 mM ammonium formate, prepared with 50 mM formic acid; #F0507 Sigma Aldrich, adjusted to pH 4.4 with ammonium hydroxide solution; #338818 Sigma Aldrich) and buffer B (acetonitrile) over a 30 minute run using a gradient shown in the table below.

Time (min)	Flow (µL/min)	% A	% B
Intial	561.0	12.0	88.0
1.47	561.0	12.0	88.0
25.00	561.0	47.6	52.4
25.60	300.0	70.0	30.0
26.80	300.0	70.0	30.0
28.00	300.0	12.0	88.0
30.00	561.0	12.0	88.0

Table 6. *Elution conditions on HILIC-UPLC for 2AB labelled O-glycans. A is 50 mM ammonium formate pH 4.4, B is acetonitrile.*

Fluorescence was measured at 420 nm, with excitation at 330 nm. External calibration was perfomed using 2AB labelled glucose oligomers (#31388, Sigma Aldrich), creating a dextran ladder with retention times of all identified glycan peaks expressed in glucose units (GUs).

3.2.4.2 Mass Spectrometry

3.2.4.2.1 Phynexus Tip Clean-up for MS

2AB labelled oligosaccharides required additional desalting prior to MS to avoid excess contamination/degradation of the UPLC column and ion source. 10 µL normal phase Phytip® Columns (#PTR-91-10-09, Phynexus Inc.) were conditioned with 3 x 1 mL 95% ACN, 3 x 1 mL 20% ACN and 3 x 1 ml 95% ACN again (to waste). Glycans were loaded onto the columns through 10x in/out cycles after the column had been pre-condtioned with 3 x 1 mL 95% ACN, 3 x 1 mL 20% ACN and 3 x 1 ml 95% ACN again (discarded to waste). Phytips were washed through 10x uptake and discard cycles with 0.85 mL 95% ACN and glycans were eluted from the tips with 1 mL 20% ACN. Eluent was dried down for further analysis.

3.2.4.2.2 Mass Spectrometry (QToF) analysis of non-reductively released (2AB labelled) glycans

Non-reductive β-eliminated 2AB labelled *O*-glycans were resusupended in 10 µL 88% ACN and 9 µL was injected into an Aquity H-Class hydrophilic-interaction liquid chromatography-ultra-performance liquid chromatography (HILIC-UPLC) system (Waters Corporation) with a BEH Glycan column (1.0 x 150 mm, 1.7 µm particle size; #186004850, Waters Corporation) and an AQUITY fluorescence detector coupled inline with a Waters Xevo G2 QTof system (Waters Corporation).

The flow rate was 150 µL/min and column temperature was maintained at 60 °C. The oligosaccharides were eluted using buffer A (50 mM ammonium formate, prepared with 50 mM formic acid; #F0507 Sigma Aldrich, adjusted to pH 4.4 with ammonium hydroxide solution; #338818 Sigma Aldrich) and buffer B (acetonitrile) over a 30 minute run using a gradient shown in the table below.

Time (min)	Flow (µL/min)	% A	% B
Intial	150.0	12.0	88.0
1.00	150.0	12.0	88.0
25.00	150.0	47.0	53.0
25.50	150.0	70.0	30.0
25.55	100.0	70.0	30.0
26.50	100.0	70.0	30.0
27.00	100.0	12.0	88.0
30.0	150.0	12.0	88.0

Table 7. *Elution conditions on HILIC-UPLC for 2AB labelled O-glycans. A is 50 mM ammonium formate pH 4.4, B is acetonitrile.*

Fluorescence was measured at 420 nm, with excitation at 330 nm. For MS acquisition data, the instrument was operated in negative ion sensitivity mode with a capillary voltage of 1.80 kV. The ion source block and nitrogen desolvation gas temperatures were set at 120 °C and 400°C, respectively. The desolvation gas was set to a flow rate of 600 L/h. The cone voltage was maintained at 50V. Fullscan data for glycans were acquired over m/z range of 450 to 2500. Data collection and processing were controlled by MassLynx 4.1 software (Waters Corporation).

3.2.4.2.3 PGC-HPLC-ESI-MS (Ion-trap) analysis of reductively released glycans

Reductive β-eliminated *O*-glycan alditols were resuspended in 20 µL water and 2 µL was injected onto a liquid chromatography-electrospray ionization tandem mass spectrometry (LC-ESI/MS) system with a 10 cm × 250 µm column packed in-house with 5 µm porous graphite particles (#35005-104630, Thermo-Hypersil). The flow rate was 5 µL/min and the column was at ambient temperature. The oligosaccharides were eluted using buffer A (10 mM ammonium bicarbonate (ABC); #09830, Sigma Aldrich) and buffer B, (10 mM ABC in 80% acetonitrile) using a gradient shown in the table below.

Time (min)	Flow (µL/min)	% A	% B
Initial	5.0	100.0	0.0
46.00	5.0	55.0	45.0
54.00	5.0	0.0	100.0
78.00	5.0	100.0	0.0

Table 8. *Elution conditions on PGC-LC for O-glycan alditols. A is 10 mM ammonium bicarbonate (ABC), B is 10 mM ABC in 80% acetonitrile.*

The samples were analyzed in negative ion mode on a LTQ linear ion trap mass spectrometer (Thermo Electron), with an IonMax standard ESI source equipped with a stainless steel needle kept at –3.5 kV. Compressed air was used as nebulizer gas. The heated capillary was kept at 270°C, and the capillary voltage was –50 kV. Full scan (*m/z* 380-1800, two microscan, maximum 100 ms, target value of 30,000) was performed, followed by data-dependent MS^2 scans (two microscans, maximum 100 ms, target value of 10,000) with normalized collision energy of 30%, isolation window of 2.5 units, activation q=0.25 and activation time 30 ms). The threshold for MS^2 was set to 300 counts. Data acquisition was conducted with Xcalibur 2.0.7 software (Thermo Scientific).

3.2.4.3 Exoglycosidase Arrays

The table below shows information of the exoglycosidases used for experimentation.

Abbr.	Enzyme source	Specificity	Conc.	Supplier	Prod. No.
ABS	Cloned from *Arthrobacter ureafaciens* and expressed in *E. coli*	α2-3,6,8,9 Sialidase	20000 units/mL	New England Biolabs	#P0722
NAN1	Recombinant from *Streptococcus pneumoniae*, expressed in *E. coli*	α2-3 Sialidase	25 unit/mL	Prozyme	#GK80021
SPG	Cloned from *Streptococcus pneumoniae* and expressed in *E. coli*	β1-4 Galactosidase	8000 units/mL	New England Biolabs	#P0745
BTG	Cloned from bovine testis and expressed in *Pichia pastoris*	β1-3,4 Galactosidase	8000 units/mL	New England Biolabs	#P0746
AMF	Cloned from the sweet almond tree (*Prunus dulcis*) and expressed in *Pichia pastoris*	α1-3,4 Fucosidase	4000 units/mL	New England Biolabs	#P0769
BKF	Cloned from *bovine kidney* and expressed in *E. coli*	α1-2,3,4,6 Fucosidase	8000 units/mL	New England Biolabs	#P0748
GUH	Cloned from *Streptococcus pneumoniae* and expressed in *E. coli*	β1-2,3,4,6-N-Acetylglucosaminidase	4000 units/mL	New England Biolabs	#P0744
JBH	Cloned from *Streptomyces plicatus* and overexpressed in *E. coli*	β1-3,4,6-N-Acetylglucosaminidase β1-4-N-Acetylgalactosaminidase	4000 units/mL	New England Biolabs	#P0721
CBG	Cloned from green coffee bean and expressed in *E. coli*	α1-3,4,6 Galactosidase	8000 units/mL	New England Biolabs	#P0747
JBM	Cloned from *Canavalia ensiformis* (Jack Bean) and expressed in *Pichia pastoris*	α1-2,3,6 Mannosidase	2000 units/mL	New England Biolabs	#P0768

Table 9. *Exoglycosidase information including specificty and abbreviation used throughout this work.*

2AB labelled glycan pools from each sample group were divided evenly into 11 replicates and dried. Each replicate received a different cocktail of 10 µL total volume exoglycosidase/buffer/water to generate the array. The buffer used in all cases is 50 mM sodium acetate buffer, pH 5.5 (#B1727, New England Biolabs) with the exception of JBM digest (replicate 11.1/2). Initially glycans in this replicate must first undergo digestion with previous exoglycosidases (11.1), with a secondary digestion required using two different buffers; ^{x}50 mM sodium acetate buffer, pH 4.5 (#B1703, New England Biolabs) , and ^{y}2 mM Zn^{2+}, pH 5.0 (#B0768, New England Biolabs) (11.2).

The volumes for each digestion vessel performed in 0.2 mL tubes are shown (in µL) in the table below.

Replicate	Water	Buffer	ABS	NAN1	SPG	BTG	AMF	BKF	GUH	JBH	CBG	JBM
1. (UND)	9	1	0	0	0	0	0	0	0	0	0	0
2. (NAN1)	7	1	0	2	0	0	0	0	0	0	0	0
3. (ABS)	8	1	1	0	0	0	0	0	0	0	0	0
4. (ABS+SPG)	7	1	1	0	1	0	0	0	0	0	0	0
5. (ABS+BTG)	7	1	1	0	0	1	0	0	0	0	0	0
6. (ABS+BTG+ AMF)	6	1	1	0	0	1	1	0	0	0	0	0
7. (ABS+BTG+ AMF+BKF	5	1	1	0	0	1	1	1	0	0	0	0
8. (ABS+BTG+ AMF+BKF+ GUH)	4	1	1	0	0	1	1	1	1	0	0	0
9. (ABS+BTG+ AMF+BKF+ GUH+JBH)	2	1	1	0	0	1	1	1	1	2	0	0
10. (ABS+BTG+ AMF+BKF+ GUH+JBH+ CBG)	1	1	1	0	0	1	1	1	1	2	1	0
11.1. (ABS+BTG+ AMF+BKF+ GUH+JBH+ CBG)	1	1	1	0	0	1	1	1	1	2	1	0
11.2. (ABS+BTG+ AMF+BKF+ GUH+JBH+ CBG+JBM)	6	$1^x + 1^y$	0	0	0	0	0	0	0	0	0	2

Table 10. *Exoglycosidase array exoglycosidase/buffer/water volumes (in µL) for each sample replicate. 50 mM sodium acetate pH 5.5 used as buffer in all experiments except JBM digest: x – 100 mM sodium acetate. y – 2 mM Zn^2+, pH 5.0.*

Reaction tubes were incubated for 16 hours at 37°C. Clean-up was performed using Nanosep® Centrifugal Devices with Omega™ Membrane 10K (#516-8491, Pall). Initially, exoglycosidases were inactivated by incubating each digestion for 15 minutes at 65°C. 500 µL water was loaded onto the centrifugal filter device and centrifuged at 13,000 rpm for 10 minutes. Filtered eluent was discarded and the wash steps repeated. Digestions were then loaded onto the filter and centrifuged at 13,000 rpm for 2 minutes. The digestion tubes were washed with 20 µL water and this was added to filter and centrifuged again at 13,000 rpm for 2 minutes. A final 100 µL water was added to the filter and centrifuged at 13,000 rpm for 5 minutes. The collected eluent was dried down for further analysis using HILIC-UPLC as described in the method in section 2.5.1.2.

3.2.5 Statistical Analysis

Individual clinical sample chromatograms were integrated into 51 peaks (figure 12) using Empower software (v3; Waters Corporation) to capture relative % peak area. Peaks with similar features (e.g. sialylation, sulfation) had their % areas combined for feature analysis.

3.2.5.1 MANOVA, Non-Parametric and Tukey Tests

As the sum of the % area values were constrained to a constant of 100%, the relative area values were logit transformed using Excel (Microsoft) using the formula:

$$log\left(\frac{Peak\ Area}{(100 - Peak\ Area)}\right)$$

Statistical analysis was performed on these logit values using SPSS statistical software for Windows (version 24.0; IBM) and output was captured into Excel for manipulation.

First, peak data was checked for normal distribution using a Kolmogorov-Smirnov non-parametric test.

Multivariate analysis of variance (MANOVA) with post-hoc analysis using Tukey's Honest Significant difference test was then performed to identify significant peak differences across groups ($p < 0.05$ was considered statistically significant).

3.2.5.2 Primary Component Analysis (PCA)

PCA was performed on % peak area and % feature area of each sample/group using the Perseus software platform[273] to identify relatendess of patient groups and samples.

3.3 RESULTS – CHARACTERIZATION AND STATISTICAL ANALYSIS OF *O*-GLYCANS

3.3.1 Reproducibility Studies

Now that experiments on fetuin and PGM had seen suitable levels of reproducibility, reproducibility studies for PD/control clinical samples could be performed.

Initially, individual clinical samples were pooled into 4 respective groups - healthy striatum, PD striatum (containing Braak stages 1-4), healthy substantia nigra, and PD substantia nigra (containing Braak stages 1-4). Each healthy group acted as controls for their specific brain regions. Figure 23 shows representative chromatograms taken from approximately 1 mg protein of each group.

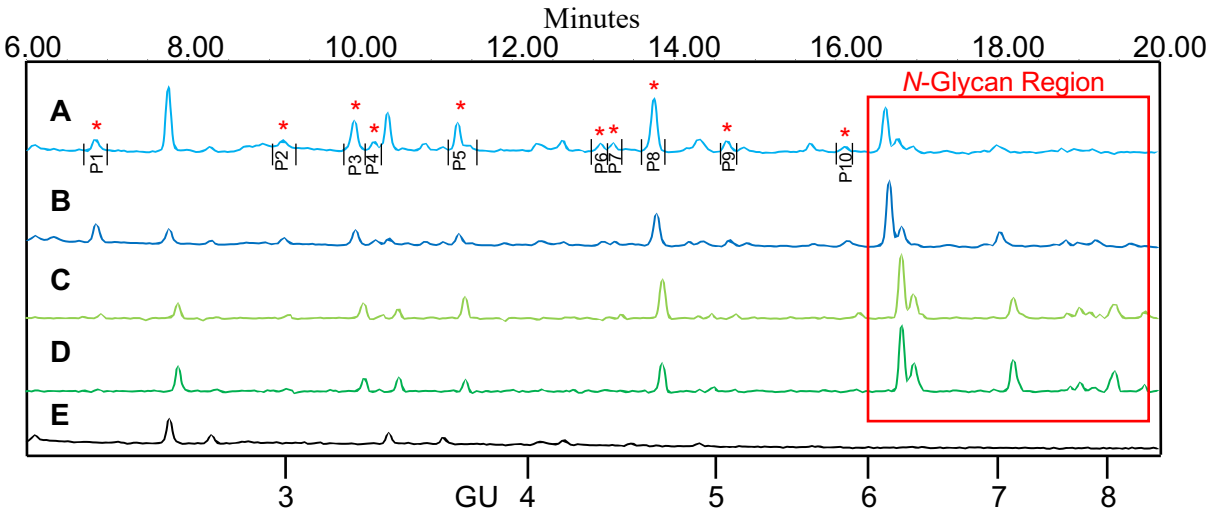


Figure 23. *Representative HILIC-UPLC profiles of O-glycans from pooled (A) healthy striatum (sample integration shown; P=Peak), (B) PD striatum, (C) healthy substantia nigra, (D) PD substantia nigra, and (E) blank. Red asterisks identify peaks taken for reproducibility analysis. The region highlighted by the red box was expected to contain only N-glycans.*

As each group produced a similar profiles of potential *O*-glycans, the healthy substantia nigra pool was used as the representative sample for reproducibility studies. As shown in figure 23, the 10 glycan features highlighted with an asterisk were used for CV calculations (results in figure 24). This selection was based on these peaks appearing in all samples and not in the blank. 6 replicates had their *O*-glycans released, labelled and run on HILIC-UPLC.

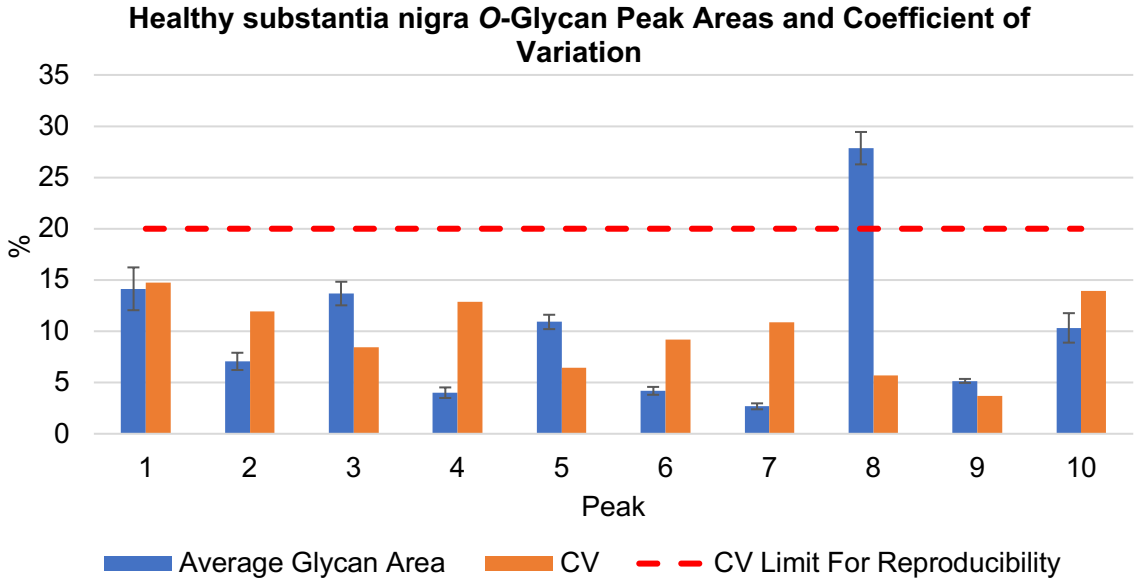


Figure 24. *Relative average glycan peak area (n=6) of healthy substantia nigra and calculated coefficient of variation (CV) for each peak. (% Area and % CV plotted on same axis).*

From these results, it was determined that reproducibility was achieved (and is lower than 15% for all peaks). Based on figure 23, there is a region excluded from CV calculations as it was expected that this region contained predominantly *N*-glycans (presumed from GU values compared with GlycoStore). These may have derived from the β-elimination reaction, or more likely, through incomplete elution of released *N*-glycans from the gel (section 3.2.2).

3.3.2 Chromatogram Integration of 2AB-Labelled *O*-glycans

The chromatograms in figure 25 identifies all 2AB-labelled *O*-glycan peaks integrated for each clinical sample (integration shown on representative sample; PD687BS) based on results from exoglycosidase arrays (2AB labelled *O*-glycans), MS analysis (QToF; 2AB labelled *O*-glycans) and MSMS analysis (ion-trap; reduced *O*-glycan alditols). GUs were assigned from GU 1 to 13 (1-6.5 shown) by quantitating against an external 2AB-labelled dextran ladder.

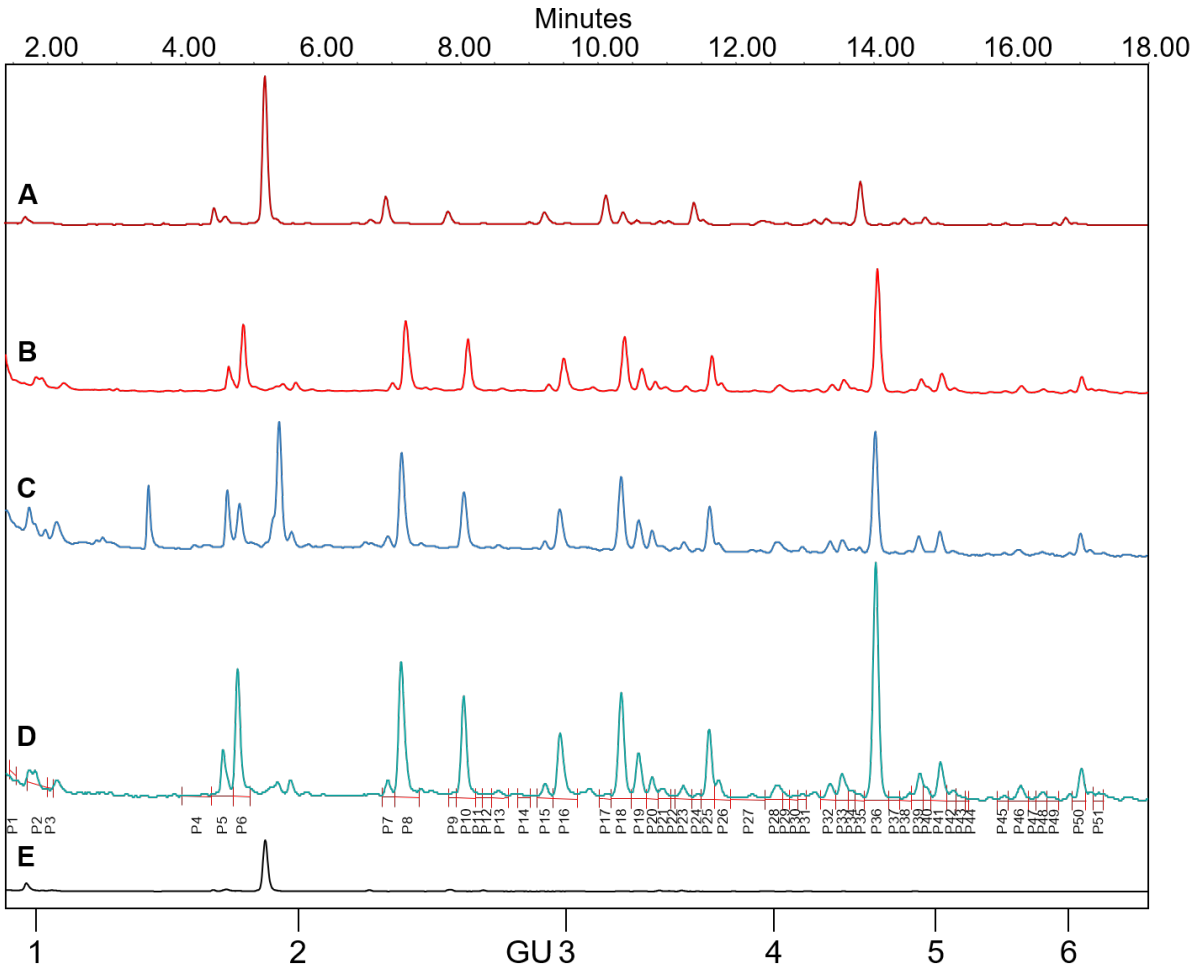


Figure 25. *Representative chromatograms of O-glycans in human striatum and substantia nigra. (A) C058A – Healthy striatum, (B) PD683A – PD striatum, (C) PD529AC6 – Healthy substantia nigra, (D) PD687BS – PD substantia nigra (with integration example), (E) Blank. P=peak.*

88 individual glycan structures were identified with 77 derived from brain tissue and 11 from other sources discussed in more detail in section 3.3.7, occupying 51 peaks. The blank in figure 25 shows a large peak at GU 1.96 which was present at high abundance in healthy striatum. This was suspected to come from a contamination of the 2AB label solution. Other smaller contaminants were present either from the blank or from degradation products of the release reaction (discussed in 3.3.5.2), explaining the absence of continuous integration between P1 and P51.

3.3.3 *O*-Glycan Naming Convention

Due to their extensive heterogeneity, no consensus naming convention exists for *O*-glycans like their oligomannose, hybrid and complex *N*-glycan siblings. A short-hand notation system was therefore designed for this project to illustrate *O*-glycan structures more easily. Table 11 shows the legend used for this naming convention with examples provided in figure 26.

Component	**Short-Hand Notation**
Galactose (Gal)	ga
Glucose (Glc)	gl
N-acetylgalactosamine (GalNAc)	GA
N-acetylglucosamine (GlcNAc)	GL
Mannose (Man)	m
Fucose (Fuc)	f
Sulfate; SO3 (written directly before the monosaccharide)	s
Sialic acid (α2-3 linked)	S3
Sialic acid (α2-6 linked)	S6
Glucuronic acid (GlcA)	U
Bold lettering	Core monosaccharide of the glycan
Inside square brackets	Right chain from core
Outside square brackets	Left chain from core
No square brackets	Linear glycan with no branching
Inside round brackets	Rightmost monosaccharide is in the chain, leftmost monosaccharide branches from it
α1-2 linked	a2
α1-3 linked	a3
α1-4 linked	a4
β1-2 linked	2
β1-3 linked	3
β1-4 linked	4
β1-6 linked	6

Table 11. *Short-hand notation for O-glycan naming.*

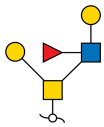
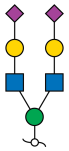
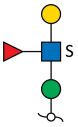
Short	**SNFG Structure**	**Long**
ga3[ga4(fa3GL6)]**GA**		GlcNAc linked to right chain – fucose branches from it Beta linkage Alpha linkage **Galβ1-3[Galβ1-4(Fucα1-3GlcNAcβ1-6)]GalNAc** Left chain Right chain Core
S3ga4GL2[S6ga4GL6]**m**		Alpha 2-3 linked sialic Alpha 2-6 linked sialic **NeuNAcα2-3Galβ1-4GlcNAcβ1-2[NeuNAcα2-6Galβ1-4GlcNAcβ1-6]Man** Left chain Right chain Core
ga4(fa3sGL2)**m**		Sulfated structure **Galβ1-4(Fucα1-3-SO3-GlcNAcβ1-2)Man** Fucose not in linear chain branches from GlcNAc

Figure 26. *Examples of short form notation of the long form name. SNFG structures shown.*

3.3.4 WAX-UPLC Analysis

WAX-UPLC analysis was performed on each of the four glycan pools using the method described in 3.3.4. Using WAX-UPLC, charge states (Sx; x=charge) of the total glycan population for each pool were determined.

Chromatograms for each glycan pool including their % composition of each charge state are given in figure 27. The results indicated that the total glycan population in striatum and substantia nigra ranged across S0, S1, S2 and S3 charge states. Given that mucin-type *O*-glycans are commonly expressed as linear or branched with one left and one right chain, estimations to the degree of sialylation of glycans could be given, as sialic acids typically terminate chains (one sialic acid contributes one negative charge). Sulfation also contributes a negative charge so was assumed to be present, especially for S3 species, as these glycans would already have both chains occupied by sialic acid yet still contain one unaccounted for negative charge. Interestingly, according to this result, neutral glycans dominate the chromatogram, however they were found to be at a much lower relative abundance in UPLC and MS analysis. This may in part be due to S0 species eluting with other junk contaminants within the sample. Based on WAX-UPLC, across the pools 65-82% of glycans were found to be neutral, 13-24% of glycans contained one charge, 5-11% contained two charges and 0-1% contained 3 charges.

With this preliminary knowledge, work on detailed structural characterisation using exoglycosidase arrays and MS could proceed.

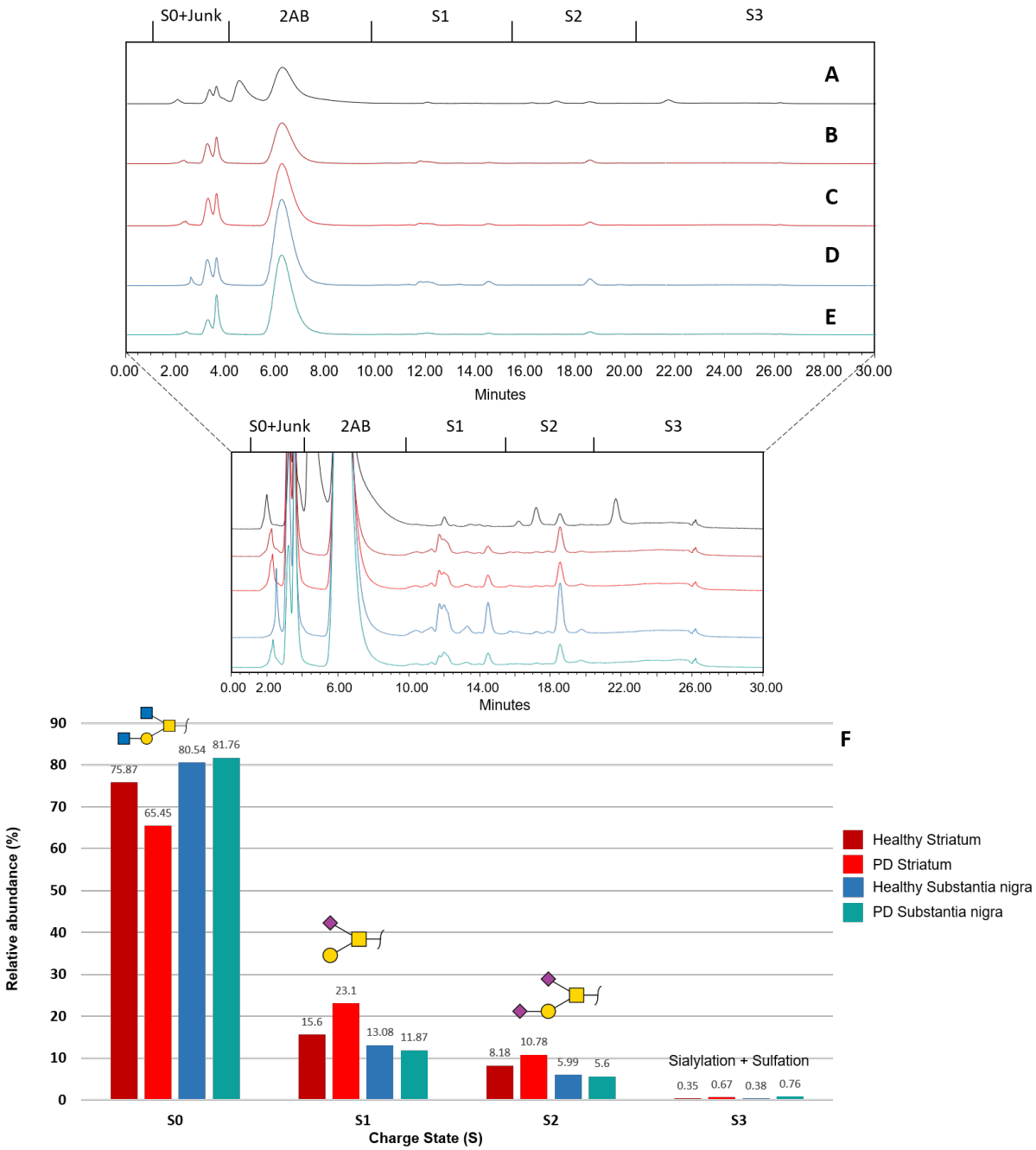


Figure 27. *WAX-UPLC Chromatograms of O-glycans from (A) Fetuin standard, (B) healthy striatum, (C) PD striatum, (D) healthy substantia nigra, (E) PD substantia nigra. Charge states indicated (S0, S1, S2, S3) as well as junk and free 2AB for each chromatogram. (F) Relative abundance breakdown of each charge state for each glycan pool B-E. Example SNFG glycan structures shown.*

3.3.5 Exoglycosidase Array

An exoglycosidase digestion panel was performed on each glycan pool (appendix I-VIII) to confirm glycan structure and saccharide linkage information. As discussed in chapter 1, starting from an undigested sample, replicates are subjected to an exoglycosidase digestion panel which will cleave specific monosaccharides depending on the exoglycosidase used, in

a stepwise manner. As glycan size is expressed in terms of its GU, removing a monosaccharide will result in a glycan size change, and hence its GU value will shift. By evaluating the series of characteristic shifts associated with a glycan peak after subsequent digestions as well as the change in integrated area under the glycan peak, its structure may be elucidated.

3.3.5.1 Exoglycosidase Limitations

During data interpretation, there was growing evidence to suggest some enzymes were poorly digesting some glycan structures or simply not digesting them at all. It was determined that BTG (β1-3,4 galactosidase) had very poor activity on cleaving β1-3 linked galactose if it was linked directly to a core GalNAc (the mucin-type core 1 disaccharide). This did not seem to affect other disaccharides that contained different linkages or were composed of different monosaccharides, such as GlcNAcβ1-6GalNAc or GlcNAcβ1-2man. Furthermore, there was severe inhibition of all galactosidases on β-linked galactose if the galactose was linked directly to a GlcNAc that was also linked to a fucose. Galactose in this instance would only digest once the internal fucose was removed using AMF (α1-3,4 fucosidase) or BKF (α1-2,3,4,6 fucosidase). This appeared not to be the case for ga3[ga4(fa3GL6)]**GA** however, which de-galactosylated without initial fucosidase treatment. Also present in some samples was a lactose (Galβ1-4Glc) contaminating peak which can be seen in some chromatograms at GU 1.96. In this case, SPG which targets specifically β1-4 linked galactose had little to no enzymatic activity, however BTG with its broader specificity for β1-4 and β1-3 could cleave the structure. Some of these observations have previously been reported by Royle et al., 2002[274] for *O*-glycans and Guile et al., 1998[275] for *N*-glycans. It is suggested that, for the case of the non-digestible disaccharides, the process of reductively aminating with 2AB forces the core monosaccharide into its open-ring conformation, which may inhibit recognition by specific exoglycosidases. In future studies, experiments with labelling reagents that leave the core monosaccharide in its closed-ring cyclic form (such as AQC or RapiFluor-MS) will need to be performed and compared with this method.

Figure 28 shows a representative exoglycosidase array (PD substantia nigra) that was used for glycan characterisation of each glycan pool. Two glycan structures which were identified as having these exoglycosidase inhibitions are highlighted.

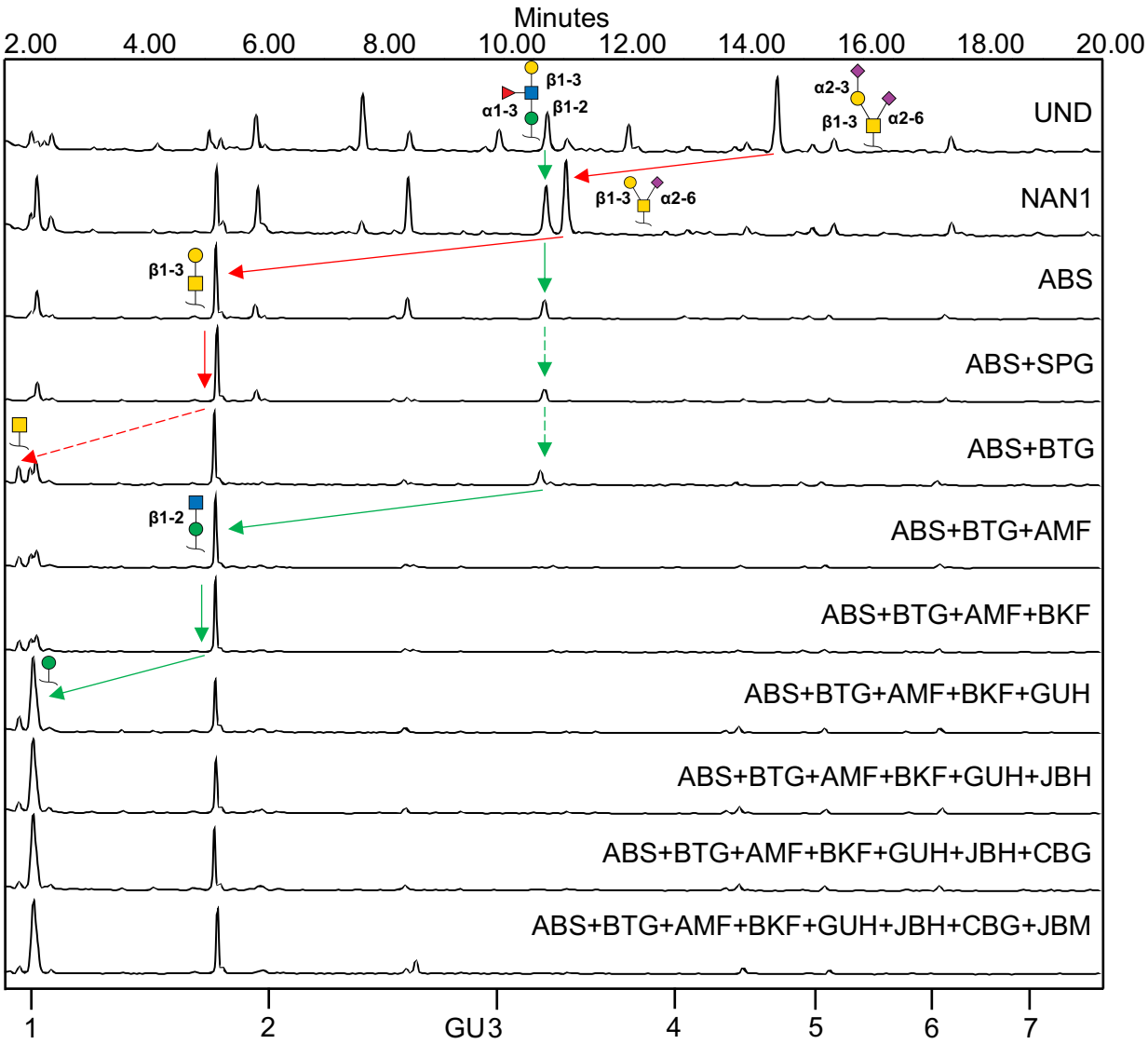


Figure 28. *Exoglycosidase array showing peak shifts from subsequent digestions. Outlined are two glycans ga4(f3GL2)**m** and S3ga3[S6]**GA** with exoglycosidase digestion (solid arrow line) and inhibited exoglycosidase activity (dashed arrow line) observed throughout the panel.*

3.3.5.2 Interpreting Exoglycosidase Panel

Traditionally for *N*-glycans, depending on the monosaccharide that is being cleaved from the glycan, there is a characteristic peak shift in response to the overall change of lipophilic surface the monosaccharides of the glycan present to the matrix of the stationary phase (e.g α2-3 linked sialic acid removal causes a GU shift of 0.7, α2-6 linked sialic acid removal causes a GU shift of 1.15 and galactose removal causes a shift of 0.8-0.9 GU). GU shifting also occurred for *O*-glycans upon digestion, however, there were slightly varied characteristic shifts. For instance, during digestion when a glycan changed from being branched (with a left and right chain extending from the core monosaccharide) to linear (seen in figure 28 during the ABS digestion of ga3[S6]**GA to** ga3**GA;** following the red arrows), there was generally a much larger GU shift observed (GU 3.3 to 1.8 in this instance). Similarly, when a hexose-hexose or *N*-

acetylhexosamine-hexose disaccharide digested to a core monosaccharide, the shift would always be approximately 0.8 GU (from GU ~1.8 to ~1.0), regardless of the monosaccharide that was being cleaved.

Another observation made was that monosaccharide peak area tended to fluctuate (separately to the change in area caused by the digestion of larger glycans), indicating the loss of some monosaccharides, potentially during the exoglycosidase clean-up stage. As characterisation using exoglycosidase arrays relies heavily on the shifting of peaks and peak areas, this made characterisation more challenging.

A final point to be made was that 20-30% of peak area across digestion panels for all groups didn't appear to shift. This was attributed to a number of factors. Firstly, it was determined through WAX-UPLC and MS that there were a number of sulfated glycans present in all samples. In the literature, exoglycosidase treatment has been observed to halt upon reaching a sulfated monosaccharide[276], unless a suitable sulfatase is used to free the glycan of its sulfate groups, allowing further digestion. This means that once a glycan has a terminal sulfated monosaccharide exposed, it will not digest or shift further. These observations were also made in this study with the exception of sulfated sialic acid residues, which appeared to cleave irrespective of sulfation. A large proportion of these non-shifting peaks, however, were attributed to non-characterizable degradation products of the protein/tissue arising from the alkaline-driven β-elimination reaction, or from other contamination sources.

3.3.6 Mass Spectrometry

Two methods to capture MS data were utilized (QToF-MS and ion-trap-MSMS), primarily to confirm structures which were identified using exoglycosidase digestions, but also to identify additional structures that weren't resolved through the digestion panel.

The benefit of analysing 2AB-labelled *O*-glycans using FLR-UPLC-MS (QToF) means that a direct comparison can be made with UPLC chromatograms from individual clinical samples (helpful when performing statistical evaluation). In this instance, samples are first separated using UPLC, generating an FLR chromatogram. The samples then pass online to the ESI-MS where a total ion chromatogram (TIC) is generated. These two chromatograms can be transposed on top of each other using Masslynx software allowing each FLR chromatogram peak to be mined using the corresponding TIC to determine glycan masses associated with that peak. Where the TIC peak signal is too weak to appear above the baseline, an EIC may by isolating specific masses. By utilizing this, glycans that are found using exoglycosidase arrays may be confirmed with MS, and, retrospectively, glycan masses found in the TIC or EIC may be used to help with exoglycosidase characterization.

PGC-MSMS (ion-trap) on glycan alditols from reductive β-elimination provided extra linkage information (as long there are reference spectra of specifically linked structures already characterised in databases; UniCarbDB used in this project), identifying structures that were

difficult to determine with exoglycosidase digestions and FLR-UPLC-MS. The use of PGC reverse phase chromatography using this method, alongside the lack of 2AB-labelling, meant a direct comparison could not be made with individual clinical sample UPLC chromatograms of 2AB labelled *O*-glycans, however, the superior isomer separation provided by using PGC and the increase in MS sensitivity of alditols in negative ion mode allowed characterization of additional structures (e.g. core 1 and core 8 isomers). Working similarly to the FLR-UPLC-MS analysis by comparing the TICs and EICs (using Xcalibur software), MSMS analysis with the ion-trap provided MS^2 fragmentation information to confirm linkage specificity to help validate results from exoglycosidase digestions. An example of both MS and MSMS applications is shown in figure 29. It can be seen that the TIC in QToF-MS appears to show only baseline. EICs were most important in this instance.

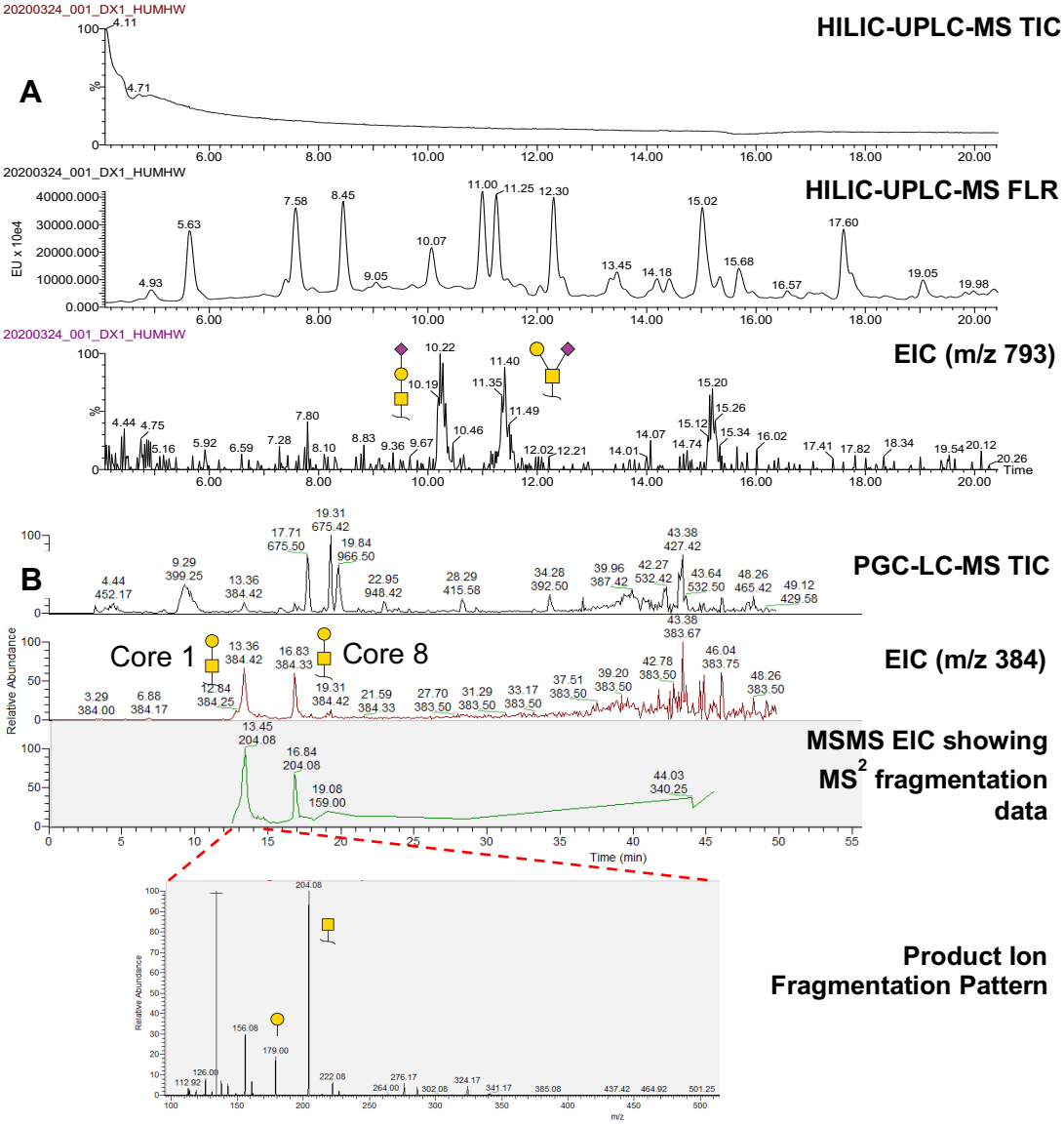


Figure 29. *MS examples including TIC and EIC for both MS methods. (A) QToF-MS showing TIC, FLR and EIC (m/z 793), and (B) ion-trap MSMS showing TIC, EIC (m/z 384) and MS^2 fragment information. Product ion fragmentation pattern shown for the core 1 disaccharide.*

3.3.6.1 Structural Characterization using MSMS

MSMS (or MS^2) provided additional orthogonal information of generic glycan composition and also linkage information of the glycans. Whereas 2-AB labelled *O*-glycans could be characterized by matching to a glycan structure repository like GlycoStore (based on GU value), MSMS fragmentation data from glycan alditols was referenced against the growing UniCarb-DB database for confirmation of structure and linkage specificity. Figures 30-35 show example MSMS fragmentation patterns for some glycans of interest from the glycan pools.

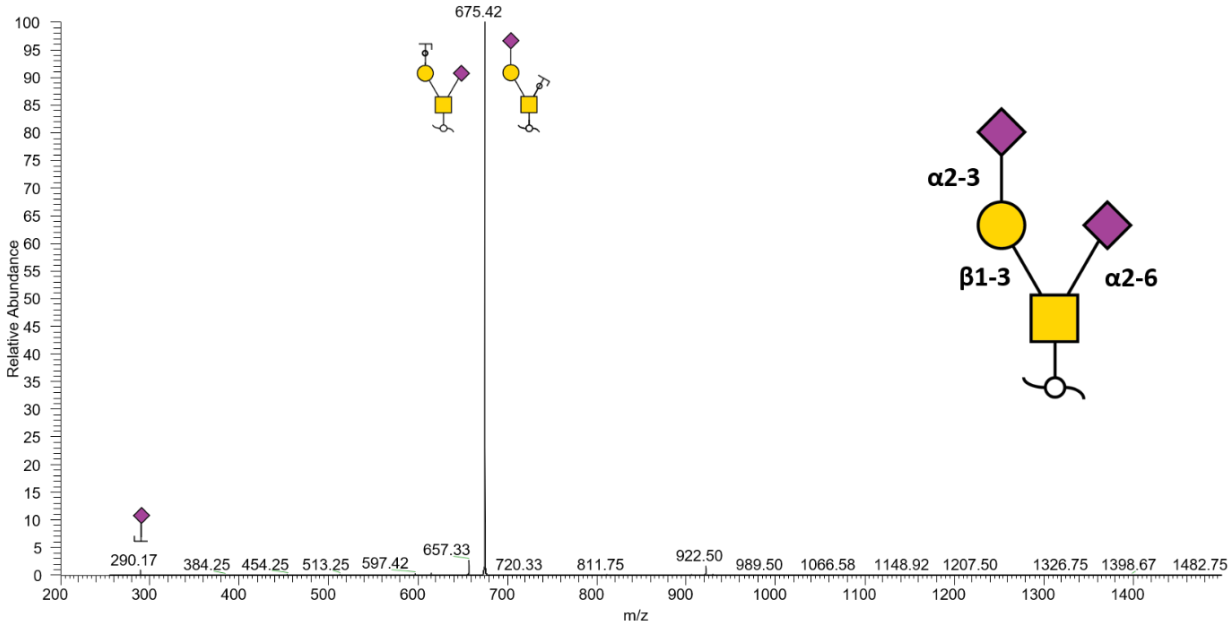


Figure 30. *Fragmentation pattern for S3ga3[S6]**GA**. From healthy striatum.*

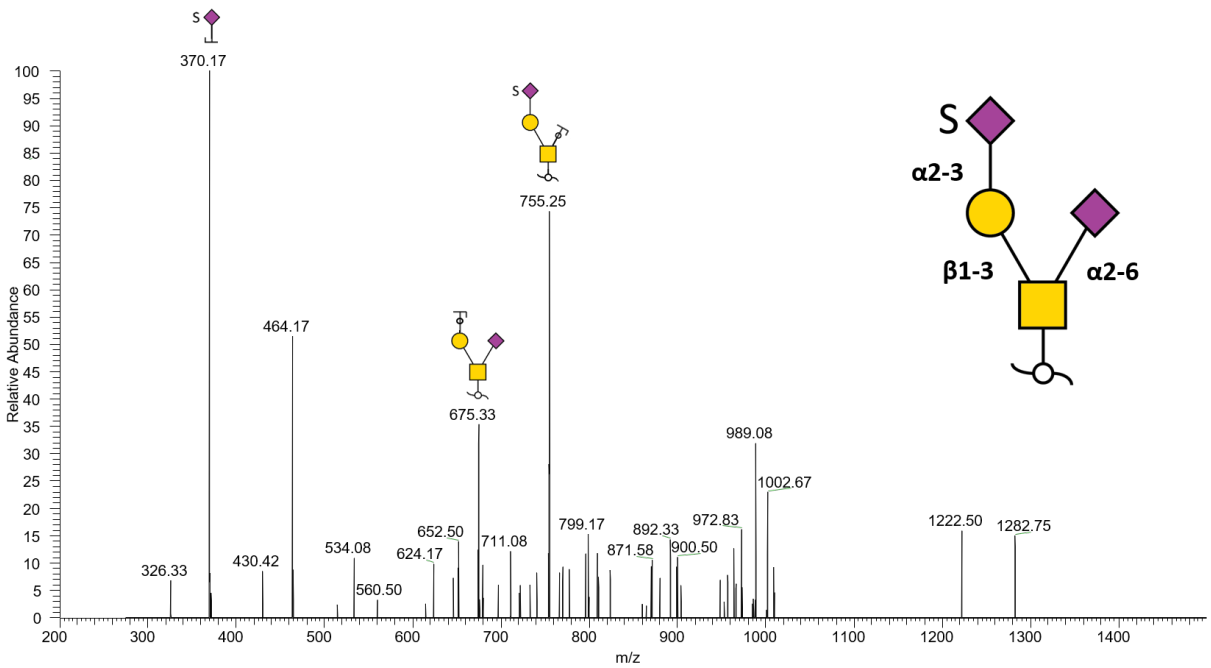


Figure 31. *Fragmentation pattern for the sialic-acid-sulfated O-glycan structure S3ga3[sS6]**GA**. From healthy striatum.*

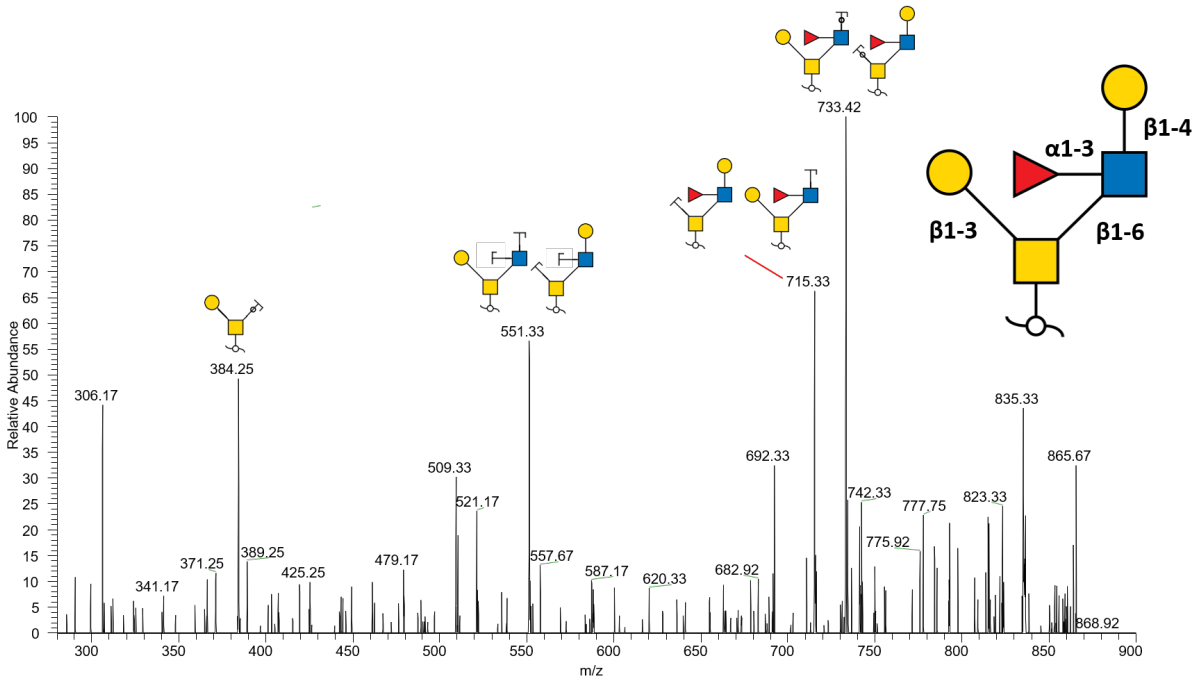


Figure 32. *Fragmentation pattern for the Lewis-type X epitope pentasaccharide structure ga3[ga4(fa3GL6)]**GA**. From healthy striatum.*

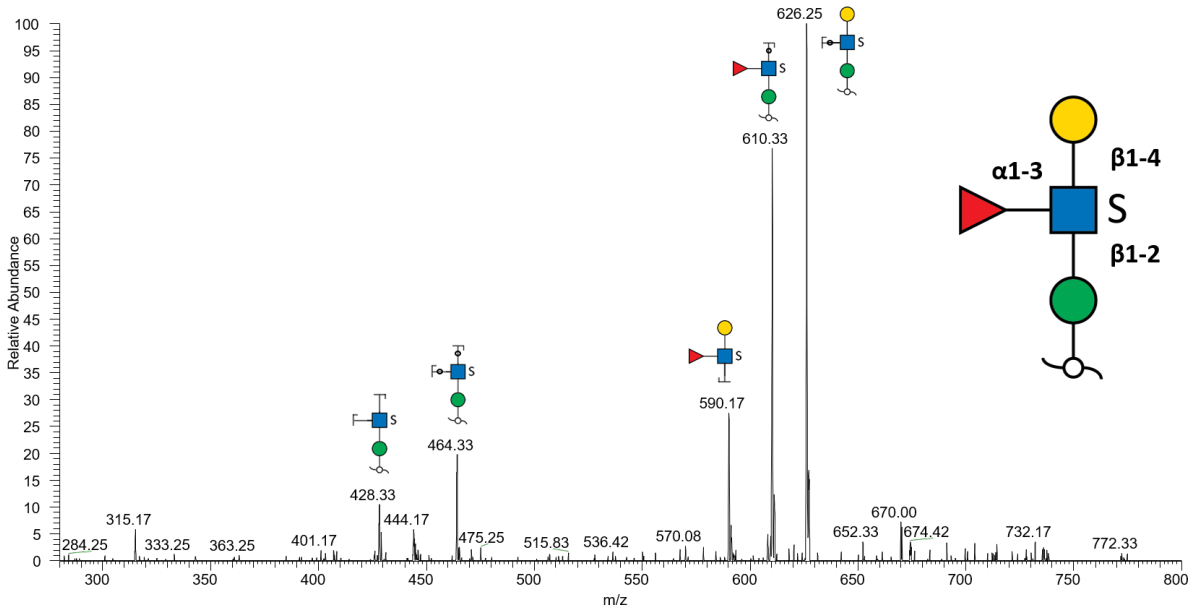


Figure 33. *Fragmentation pattern for the sulfated Lewis X type mannose core pentasaccharide structure ga4(fa3sGL2)**m**. From healthy striatum.*

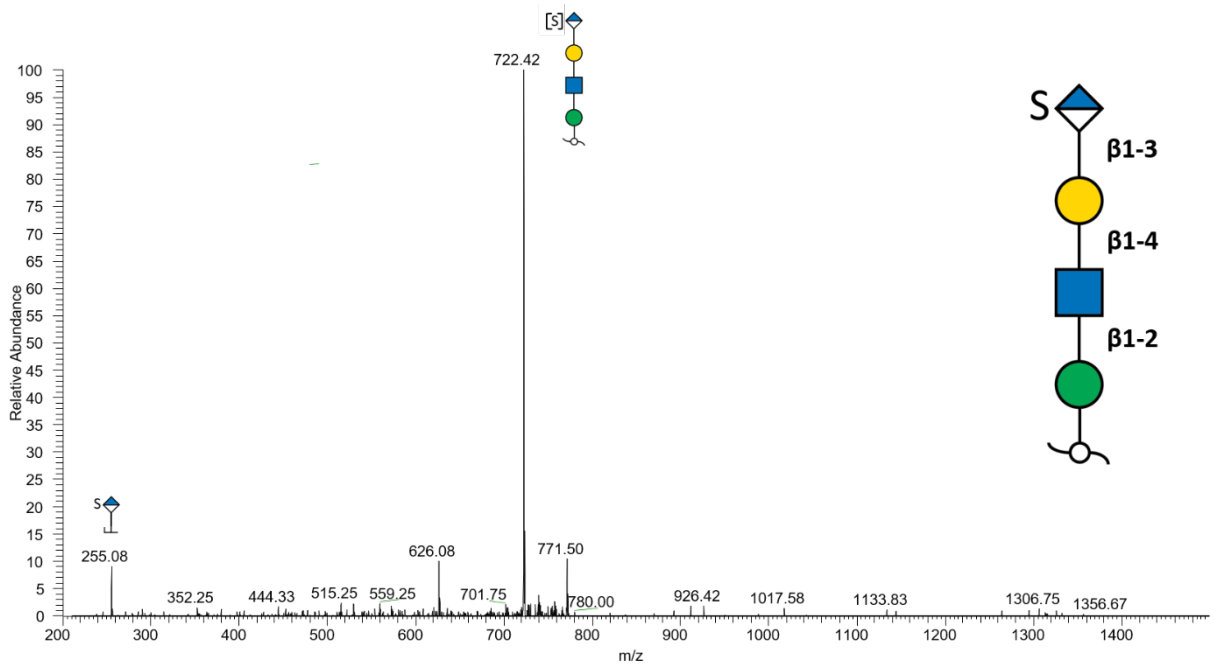


Figure 34. *Fragmentation pattern for the sulfated glucuronic acid-capped mannose core tetrasaccharide sU3ga4GL2**m**. From healthy striatum.*

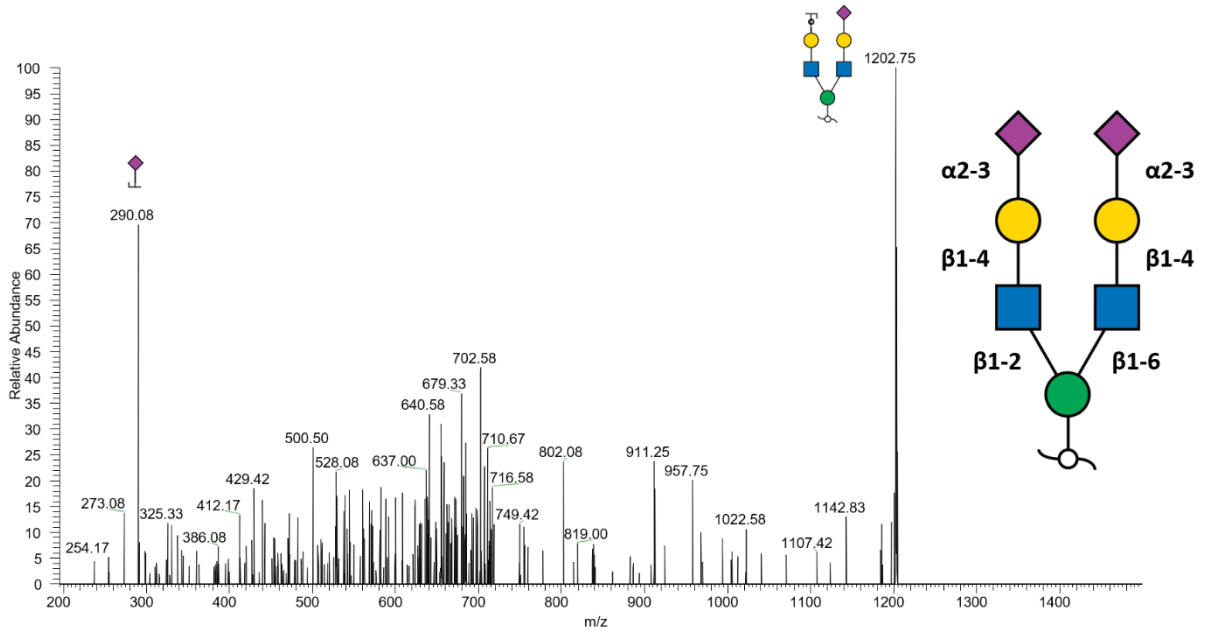


Figure 35. *Fragmentation pattern for the branched mannose core septasaccharide S3ga4GL2[S3ga4GL6]**m**. From diseased striatum.*

While some profiles outlined in these figures identify structures and their generic linkage information, figures 32, 34 and 35 show fragmentation gathered from only a single MS^2 scan. Ideally, in future experiments, acquisition of data for specific glycan masses could be optimized by selecting these masses to take into MS^2 scanning *prior* to MS analysis. For the case of

many other structures that have been characterized in this work, there was no MS^2 fragmentation spectra from the ion-trap analysis, so orthogonal analyses using exoglycosidase arrays and QToF-MS have helped to confirm structures. This is outlined in 3.3.7.
With respect to charge states of glycan ions found in MS analysis, ion-trap MS typically produced glycan precursor ions with an $[M-H^+]^{-1}$ charge state whereas QToF-MS analysis produced glycans over $[M-H^+]^{-1}$, $[M-2H^+]^{-2}$ and $[M-3H^+]^{-3}$ states.

3.3.7 All *O*-Glycans Across Methods

Combined, the analyses uncovered 88 *O*-glycan structures across a range of glycan cores. 55 glycans (62.5%) were the more common, mucin-type GalNAc core glycans. 21 (23.9%) were mannose core structures, 9 (10.2%) were glucose core structures (most likely contaminating glycans from a compromised 2AB labelling solution), 2 (2.3%) were galactose core structures (peeling products) and 1 (1.1%) was a sialic core structure (disialic acid glycan). Therefore, considering glucose core structures are contaminating products and galactose core structures are degraded products, a total of 77 intact *O*-glycans have been identified in this work to have derived from striatum and substantia nigra brain tissue in humans.

The breakdown of *O*-glycans per peak and their average GU values are given in table 12. The results from the digestion panels across all four groups required translation to the individual clinical samples to allow for statistical analysis, so, the results in table 12 were generated with the following considerations. Where the glycan was present in an undigested (UND) run in the exoglycosidase array, the GU was given as an average from the four groups. In some instances, there were glycans that were *only* detected in the undigested samples using mass spectrometry, however, these structures appeared as digestion *products* throughout the exoglycosidase panel. The GU values associated with these digestion products were used in this case and averaged accordingly. With a GU designated to each glycan structure, glycan structures were designated to each peak in the individual samples (figure 36). In two instances, contaminating products (suspected glucose core glycans) were the only structures which occupied their specific peak. These peaks were highlighted as “contam.” and are not considered for statistical evaluation. Other glucose core contaminating glycans have also been shown as these were identified regardless of their absence in brain tissue. These structures, however, are not considered for statistical evaluation. Other features labelled with “merged” were groups of two or three peaks that required merging to normalise integration across all individual sample chromatograms (in some chromatograms multiple peaks were resolved, but in others these peaks co-eluted). Graphically similar structures are presented, however, proposed linkages may differ. In addition, isomers and similar structures may present the same GU value (most likely due to the GU provided being an average over all four exoglycosidase panels). Care should be taken when observing results in table 12.

Peak	GU	SNFG	Short-form	Structure	Confirmed with MS/Exo Digestions (Appendix I-XI)
P1	0.98		GA	GalNAc	Exo
P2	1.02		gl	Glc	Exo
P3	1.04		ga	Gal	Exo
P4	1.70		s3ga3**GA**	SO3-3(6)Galβ1-3GalNAc	Exo + Ion Trap + QToF
	1.70		s6ga3**GA**	SO3-6(3)Galβ1-3GalNAc	Exo + Ion Trap + QToF
	1.70		S6s**GA**	SO3-(NeuAcα2-6)GalNAc	Exo + QToF
P5	1.76		S3**S**	NeuAcα2-3NeuAc	Exo + ^{y}Ion Trap + QToF
P6	1.78		ga3**GA**	Galβ1-3GalNAc	Exo + Ion Trap
	1.78		gaa3**GA**	Galα1-3GalNAc	Exo + Ion Trap
	1.79		GL3**GA**	GlcNAcβ1-3GalNAc	Exo + Ion Trap + QToF
contam.	1.96		ga4**gl**	Galβ1-4Glc	Exo
P7	2.42		S6**GA**	NeuAcα2-6GalNAc	Exo + Ion Trap + QToF
P8	2.42		S3**ga**	NeuAcα2-3Gal	Exo + ^{y}Ion Trap + QToF
P9	2.56		ga3GL2**m**	Galβ1-3GlcNAcβ1-2Man	Exo + Ion Trap + QToF
P10	2.62		ga4GL2**m**	Galβ1-4GlcNAcβ1-2Man	Exo + Ion Trap + QToF
	2.62		gl4ga4**gl**	Glcβ1-4Galβ1-3Glc	Exo + ^{y}Ion Trap + QToF
P11	2.64		ga3[GL6]**GA**	Galβ1-3[GlcNAcβ1-6]GalNAc	Exo + ^{y}Ion Trap
P12	2.71		GL3ga3**GA**	GlcNAcβ1-3Galβ1-3GalNAc	Exo + ^{y}Ion Trap
P13	2.75		(fa2ga3)**gl**	(Fucα1-2Galβ1-3)Glc	Exo
	2.77		GL4ga3**GA**	GlcNAcβ1-4Galβ1-3GalNAc	Exo + ^{y}Ion Trap
contam.	2.80		gl3ga4**gl**	Galβ1-3Galβ1-3Glc	Exo + ^{y}Ion Trap + QToF
P14	2.81		ga3GL3**GA**	Galβ1-3GlcNAcβ1-3GalNAc	Exo + ^{y}Ion Trap
P15	2.96		S6ga3**GA**	NeuAcα2-6Galβ1-3GalNAc	Exo + Ion Trap + QToF
P16	3.05		S3ga3**GA**	NeuAcα2-3Galβ1-3GalNAc	Exo + Ion Trap + QToF
P17	3.23		sga4(fa3GL2)**m**	SO3-Galβ1-4(Fucα1-3GlcNAcβ1-2)Man	Exo + Ion Trap + QToF
	3.27		(fa2ga3)(fa4**GA**)	(Fucα1-2Galβ1-3)(Fucα1-4GalNAc)	Exo
P18	3.28		ga4(fa3GL2)**m**	Galβ1-4(Fucα1-3GlcNAcβ1-2)Man	Exo + Ion Trap + QToF
	3.28		ga3[ga4GL6]**GA**	Galβ1-3[Galβ1-4GlcNAcβ1-6]GalNAc	Exo + ^{y}Ion Trap
P19	3.38		ga3[S6]**GA**	Galβ1-3(NeuAcα2-6)GalNAc	Exo + Ion Trap + QToF
P20	3.40		sga3[GL6]**GA**	SO3-Galβ1-3[GlcNAcβ1-6]GalNAc	Exo + QToF
P21	3.45		ga4(fa3s6GL2)**m**	Galβ1-4(SO3-Fucα1-3GlcNAcβ1-2)Man	Exo + Ion Trap + QToF
P22	3.48		ga4[GL6]ga4**gl**	Galβ1-4[GlcNAcβ1-6]Galβ1-4Glc	Exo + ^{y}Ion Trap
	3.50		sga3[S6]**GA**	SO3-Galβ1-3[NeuAcα2-6]GalNAc	Exo + Ion Trap
P23	3.52		ga3[GL6]ga4**gl**	Galβ1-3[GlcNAcβ1-6]Galβ1-4Glc	Exo + ^{y}Ion Trap
	3.52		ga4GL4ga3**GA**	Galβ1-4GlcNAcβ1-4Galβ1-3GalNAc	Exo + ^{y}Ion Trap
	3.56		ga3GL4ga3**GA**	Galβ1-3GlcNAcβ1-4Galβ1-3GalNAc	Exo + ^{y}Ion Trap
P24	3.63		(fa3GL2)[GL6]**m**	Fucα1-3-GlcNAcβ1-2[GlcNAcβ1-6]Man	Exo
P25	3.63		S6ga3GL2**m**	NeuAcα2-6Galβ1-3GlcNAcβ1-2Man	Exo + Ion Trap + QToF
	3.68		sga3[sga4GA6]**GA**	SO3-Galβ1-3[SO3-Galβ1-4GlcNAcβ1-6]GalNAc	Exo + QToF
	3.69		ga3[(fa3GA6)]**GA**	Galβ1-3[(Fucα1-3GlcNAcβ1-6)]GalNAc	Exo
	3.72		S3ga4GL2**m**	NeuAcα2-3Galβ1-4GlcNAcβ1-2Man	Exo + Ion Trap + QToF
	3.74		S6ga4GL2**m**	NeuAcα2-6Galβ1-4GlcNAcβ1-2Man	Exo + Ion Trap + QToF
P26	3.78		S3ga3GL2**m**	NeuAcα2-3Galβ1-3GlcNAcβ1-2Man	Exo + Ion Trap + QToF
P27	3.95		ga3[sS6]**GA**	Galβ1-3[SO3-NeuAcα2-6]GalNAc	Exo + Ion Trap
P28	4.03		ga3[GL4ga3GL6]**GA**	Galβ1-3[GlcNAcβ1-4Galβ1-3GlcNAcβ1-6]GalNAc	Exo + ^{y}Ion Trap
	4.05		GA4GL2[GA4GL6]**m**	GalNAcβ1-4GlcNAcβ1-2[GalNAcβ1-4GlcNAcβ1-6]Man	Exo + QToF
P29	4.08		GL4ga3[ga3GL6]**GA**	GlcNAcβ1-4Galβ1-3(Galβ1-3GlcNAcβ1-6)GalNAc	Exo + ^{y}Ion Trap
	4.09		(fa3GL2)[(fa3GL6)]**m**	(Fucα1-3GlcNAcβ1-2)((Fucα1-3GlcNAcβ1-6)]Man	Exo
P30	4.11		S3GA4ga4**gl**	NeuAcα2-3GalNAcβ1-4Galβ1-4Glc	Exo
	4.13		S6GA4ga4**gl**	NeuAcα2-6GalNAcβ1-4Galβ1-4Glc	Exo
	4.13		ga4(fa3GL2)[GL6]**m**	Galβ1-4(Fucα1-3-GlcNAcβ1-2)[GlcNAcβ1-6]Man	Exo + ^{y}Ion Trap + QToF
	4.13		GL4ga3[ga4GL6]**GA**	GlcNAcβ1-4Galβ1-3[Galβ1-4GlcNAcβ1-6]GalNAc	Exo + ^{y}Ion Trap

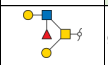
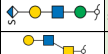
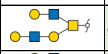
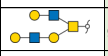
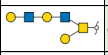
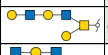
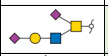
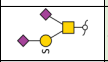
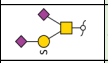
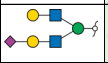
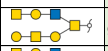
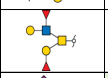
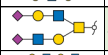
P31	**4.15**		ga3[GL4ga4GL6]**GA**	Galβ1-3[GlcNAcβ1-4Galβ1-4GlcNAcβ1-6]GalNAc	Exo + ^{y}Ion Trap
P32	**4.36**		ga3[ga3[(fa4GL6)]**GA**	Galβ1-3[Galβ1-4(Fucα1-4GlcNAcβ1-6)]GalNAc	Exo + QToF
	4.40		ga3[S3ga4GL6]**GA**	Galβ1-3[NeuAcα2-3Galβ1-4GlcNAcβ1-6]GalNAc	Exo + Ion Trap + QToF
	4.40		ga3[ga4(fa3GL6)]**GA**	Galβ1-3[Galβ1-4(Fucα1-3GlcNAcβ1-6)]GalNAc	Exo + Ion Trap + QToF
P33	**4.42**		S3ga4(fa3GL2)**m**	NeuAcα2-3Galβ1-4(Fucα1-3GlcNAcβ1-2)Man	Exo + QToF
	4.46		ga3[ga3GA4GL6]**GA**	Galβ1-3[Galβ1-3GalNAcβ1-4GlcNAcβ1-6]GalNAc	Exo + ^{y}Ion Trap
	4.48		sU3ga4GL2**m**	SO3-GlcAβ1-3Galβ1-4GlcNAcβ1-2Man	Exo + Ion Trap + QToF
P34	**4.54**		sU3ga3GL2**m**	SO3-GlcAβ1-3Galβ1-3GlcNAcβ1-2Man	Exo + Ion Trap + QToF
P35	**4.57**		S3ga3[ga4GL6]**GA**	NeuAcα2-3Galβ1-3[Galβ1-4GlcNAcβ1-6]GalNAc	Exo + Ion Trap
	4.61		S6ga4(fa3s6GL2)**m**	NeuAcα2-6Galβ1-4(Fucα1-3-SO3-GlcNAcβ1-2)Man	Exo + ^{y}Ion Trap
P36	**4.70**		S3ga3[S6]**GA**	NeuAcα2-3Galβ1-3(NeuAcα2-6)GalNAc	Exo + Ion Trap + QToF
	4.75		S6-(fa2ga3)[GL6]**GA**	NeuAcα2-6-(Fucα1-2Galβ1-3)[GlcNAcβ1-6]GalNAc	Exo
P37	**4.85**		ga3GL4ga3[ga3GL6]**GA**	Galβ1-3GlcNAcβ1-4Galβ1-3[Galβ1-3GlcNAcβ1-6]GalNAc	Exo + QToF
	4.86		ga4GL4ga3[ga3GL6]**GA**	Galβ1-4GlcNAcβ1-4Galβ1-3[Galβ1-3GlcNAcβ1-6]GalNAc	Exo + QToF
P38	**4.89**		ga3[ga4GL4ga3GL6]**GA**	Galβ1-3[Galβ1-4GlcNAcβ1-4Galβ1-3GlcNAcβ1-6]GalNAc	Exo + QToF
	4.92		ga3[ga3GL4ga3GL6]**GA**	Galβ1-3[Galβ1-3GlcNAcβ1-4Galβ1-3GlcNAcβ1-6]GalNAc	Exo + QToF
P39	**4.99**		ga3[(fa3GL4)ga4GL6]**GA**	Galβ1-3[(Fucα1-3GlcNAcβ1-4)Galβ1-4GlcNAcβ1-6]GalNAc	Exo
	5.01		S3ga3GL3[S6]**GA**	NeuAcα2-3Galβ1-3GlcNAcβ1-3[NeuAcα2-6]GalNAc	Exo + ^{y}Ion Trap + QToF
P40	**5.13**		S3sga3[S6]**GA** (isomer 1)	NeuAcα2-3(SO3-)Galβ1-3(NeuAcα2-6)GalNAc	Exo + Ion Trap + QToF
P41	**5.13**		S3sga3[S6]**GA** (isomer 2)	NeuAcα2-3(SO3-)Galβ1-3(NeuAcα2-6)GalNAc	Exo + Ion Trap + QToF
P42	**5.23**		S3ga4GL2[ga4GL6]**m**	NeuAcα2-3Galβ1-4GlcNAcβ1-2[Galβ1-4GlcNAcβ1-6]Man	Exo + ^{y}Ion Trap
P43	**5.24**		S3g3[S6ga4GL6]**GA**	NeuAcα2-3Galβ1-3[NeuAcα2-6Galβ1-4GlcNAcβ1-6]GalNAc	Exo + Ion Trap
P44	**5.30**		S3ga3[S3ga4GL6]**GA**	NeuAcα2-3Galβ1-3[NeuAcα2-3Galβ1-4GlcNAcβ1-6]GalNAc	Exo + Ion Trap
P45	**5.57**		ga3GA3ga3[GA3ga4GL6]**GA**	Galβ1-3GalNAcβ1-3Galβ1-3[GalNAcβ1-3Galβ1-4GlcNAcβ1-6]GalNAc	Exo
	5.61		ga4GA3ga3[GA3ga4GL6]**GA**	Galβ1-4GalNAcβ1-3Galβ1-3[GalNAcβ1-3Galβ1-4GlcNAcβ1-6]GalNAc	Exo
P46	**5.65**		S6-ga3[ga4(fa3GL6)]**GA**	NeuAcα2-6-Galβ1-3[Galβ1-4(Fucα1-3GlcNAcβ1-6)]GalNAc	Exo + ^{y}Ion Trap + QToF
	5.68		(fa2ga3)[ga4(fa3GL6)]**GA**	Fucα1-2Galβ1-3[Galβ1-4(Fucα1-3GlcNAcβ1-6)]GalNAc	Exo + QToF
	5.79		sS3ga3[S6]**GA**	[SO3] + NeuAcα2-3Galβ1-3(NeuAcα2-6)GalNAc	Exo + Ion Trap + QToF
P47	**5.83**		S3-ga3GL4ga3[ga4GL6]**GA**	NeuAcα2-3-Galβ1-3GlcNAcβ1-4Galβ1-3[Galβ1-4GlcNAcβ1-6]GalNAc	Exo
	5.87		S3GL4ga3[S3ga4GL6]**GA**	NeuAcα2-3GlcNAcβ1-4Galβ1-3[NeuAcα2-3Galβ1-4GlcNAcβ1-6]GalNAc	Exo + ^{y}Ion Trap
P48	**5.89**		S3-ga3[ga3GL4ga4GL6]**GA**	NeuAcα2-3-Galβ1-3[Galβ1-3GlcNAcβ1-4Galβ1-4GlcNAcβ1-6]GalNAc	Exo
	5.92		S6GL4ga3[S3ga3GL6]**GA**	NeuAcα2-6GlcNAcβ1-4Galβ1-3[NeuAcα2-3Galβ1-4GlcNAcβ1-6]GalNAc	Exo + ^{y}Ion Trap
P49	**5.94**		S3sga3[S3sga4GL6]**GA**	NeuAcα2-3-SO3-Galβ1-3[NeuAcα2-3-SO3-Galβ1-4GlcNAcβ1-6]GalNAc	Exo
P50	**6.20**		S3GA4GL2[S3GA4GL6]**m**	NeuAcα2-3GalNAcβ1-4GlcNAcβ1-2[NeuAcα2-3GalNAcβ1-4GlcNAcβ1-6]Man	Exo + QToF
P51	**6.41**		S3ga4GL2[S3ga4GL6]**m**	NeuAcα2-3Galβ1-4GlcNAcβ1-2[NeuAcα2-3Galβ1-4GlcNAcβ1-6]Man	Exo + Ion Trap
	6.47		S3ga4GL2[S6ga4GL6]**m**	NeuAcα2-3Galβ1-4GlcNAcβ1-2[NeuAcα2-6Galβ1-4GlcNAcβ1-6]Man	Exo + Ion Trap

Table 12. *O-glycans characterized across all PD and control groups. "y" – no MS^2 data. Glycans highlighted in green are major component of peak. Glycans highlighted in red are suspected glucose core contaminant glycans not considered to be derived from brain tissue.*

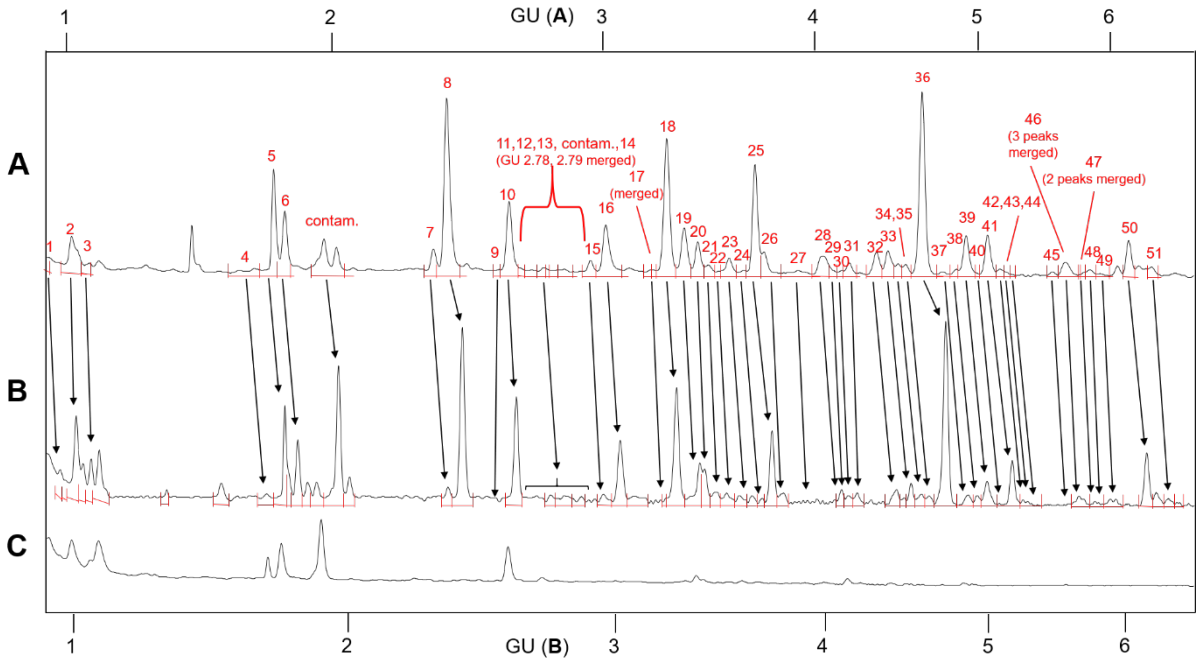


Figure 36. *Aligning (A) Representative chromatogram (PD204P; PD substantia nigra) against (B) UND profile from PDBS digestion. (C) Blank. Peak numbers and comments shown above each peak.*

3.3.8 Statistical Differences – Disease State

6 groups were compared to identify statistical significance using the method described in 3.2.5.1. These groups are: healthy striatum, ILBD (Braak stage 1-2) striatum, PD (Braak stage 3-4) striatum, healthy substantia nigra, ILBD substantia nigra and PD substantia nigra. By separating based on the degree of PD over differing levels of Braak staging, more accurate significant differences could be seen, due to the fact that ILBD-associated complications haven't reached substantia nigra or striatum brain regions yet which might negatively influence the results of PD (Braak stage 3-4) if pooled together.

Kolmogorov-Smirnov non-parametric test was run to which 63% of peaks were found to be normally distributed.Peak differences were compared across all relevant groups and plotted as p-values in table 13. Areas of significance ($p < 0.05$) have been highlighted in either red or green. Red indicates a decrease in relative peak area of the leading group (e.g the healthy group in healthy vs. early PD) and green indicates an increase.

Statistical significance in glycan features were also measured across peaks whose glycans expressed that particular feature (table 14). The features which were analysed for significance across the groups were: sialylation, fucosylation, sulfation, mannose core differences, glucuronylation and peeling.

Peak	Healthy vs Diseased Striatum		ILBD vs PD Striatum	Healthy vs Diseased Nigra		ILBD vs PD Substantia Nigra	Healthy Striatum vs Nigra	PD Striatum vs Nigra	
	ILBD	PD		ILBD	PD			ILBD	PD
1	0.009	0.003	0.872	1.000	0.985	0.989	0.000	0.999	0.388
2	0.038	0.231	0.001	0.715	1.000	0.792	1.000	0.014	0.357
3	1.000	0.105	0.524	0.462	0.987	0.735	0.997	0.524	0.685
4	0.065	0.035	0.000	0.975	0.063	0.939	0.843	0.227	0.795
5	0.661	0.774	0.982	0.998	0.932	0.925	0.023	0.258	0.000
6	0.613	0.759	0.196	0.928	0.183	0.998	0.008	0.998	0.294
7	0.376	0.068	0.009	0.967	0.076	0.960	0.800	0.101	0.928
8	0.708	0.000	0.001	0.198	0.217	0.936	0.997	0.036	0.379
9	0.173	0.999	0.124	0.788	0.886	0.989	0.064	0.577	0.502
10	0.999	0.064	0.767	0.983	0.082	0.939	0.223	0.980	0.319
11	0.090	0.008	0.000	0.246	0.116	0.987	0.732	0.026	0.257
12	0.128	0.224	0.004	0.993	0.459	0.996	0.702	0.483	0.557
13	0.365	0.014	0.003	0.989	0.662	1.000	0.506	0.805	0.019
14	0.240	0.135	0.007	0.996	0.902	1.000	0.607	0.714	0.081
15	0.941	1.000	0.977	0.702	0.969	0.934	0.051	0.581	0.051
16	0.891	0.777	0.461	0.989	0.217	0.979	0.154	1.000	0.830
17	1.000	0.988	1.000	0.992	0.997	0.960	0.990	1.000	0.650
18	0.067	1.000	0.092	0.997	0.988	0.966	0.050	0.991	0.466
19	0.660	1.000	0.573	1.000	0.558	0.842	0.779	0.998	1.000
20	0.796	0.213	1.000	1.000	0.994	0.997	0.995	0.890	0.035
21	0.857	0.540	1.000	0.985	0.899	1.000	0.029	0.931	0.337
22	0.985	0.000	0.000	0.971	0.004	0.690	0.018	0.141	0.992
23	0.992	0.001	0.039	0.347	0.778	0.840	1.000	0.405	0.060
24	0.982	0.007	0.067	1.000	1.000	1.000	0.782	1.000	0.001
25	0.400	0.986	0.228	0.992	1.000	0.993	0.065	0.978	0.045
26	0.669	1.000	0.754	1.000	0.881	0.997	0.271	1.000	0.100
27	0.999	0.177	0.475	1.000	0.999	0.999	0.766	0.999	0.045
28	0.230	1.000	0.325	0.934	0.371	0.329	0.517	1.000	0.993
29	0.905	0.000	0.000	1.000	0.009	0.320	0.006	0.195	0.992
30	0.275	1.000	0.399	0.996	0.240	0.487	0.198	1.000	0.999
31	0.845	0.107	0.090	0.976	0.952	0.825	0.789	1.000	0.081
32	0.867	0.001	0.719	0.959	0.869	1.000	0.088	0.942	0.986
33	0.081	1.000	0.064	0.980	0.999	0.996	0.007	1.000	0.007
34	0.345	0.435	0.037	0.703	0.001	0.880	0.803	0.290	0.867
35	0.998	0.707	0.798	0.993	0.259	0.979	0.712	0.982	0.000
36	0.222	0.000	0.000	0.444	0.166	0.999	0.816	0.104	0.007
37	0.999	1.000	1.000	0.952	0.871	1.000	0.231	0.766	0.102
38	0.067	0.190	0.749	1.000	1.000	1.000	0.222	0.855	1.000
39	1.000	0.626	0.837	0.358	0.010	1.000	0.998	0.401	0.285
40	0.954	0.000	0.000	0.884	0.000	0.260	0.414	0.213	0.921
41	0.198	0.000	0.000	0.615	0.001	0.907	0.733	0.167	0.689
42	0.440	0.423	0.053	0.780	1.000	0.830	0.000	0.997	0.000
43	0.762	0.896	0.394	0.559	0.566	0.986	0.593	0.585	0.959
44	0.996	0.863	1.000	0.401	0.492	0.960	0.122	0.971	0.995
45	0.833	0.837	0.425	0.896	0.637	1.000	0.171	0.978	0.517
46	0.996	0.996	1.000	0.716	0.964	0.945	0.998	0.949	0.987
47	0.914	0.190	0.998	0.896	0.994	0.766	0.668	0.930	0.998
48	0.936	0.935	0.688	0.994	1.000	0.989	1.000	1.000	0.957
49	0.998	0.205	0.928	0.920	1.000	0.952	0.763	0.995	0.848
50	0.996	0.982	0.943	0.850	0.809	0.442	0.998	0.983	1.000
51	0.986	0.987	0.908	0.431	0.996	0.658	0.681	0.125	0.879

Table 13. *P-values derived from comparison of each peak across different groups comparing disease states. Red indicates a decrease in relative abundance in the leading group (e.g the healthy group in healthy vs. PD) and green indicates an increase – only statistically significant areas highlighted.*

Feature	Healthy vs Diseased Striatum		ILBD vs PD Striatum	Healthy vs Diseased Nigra		ILBD vs PD Substantia Nigra	Healthy Striatum vs Nigra	PD Striatum vs Nigra	
	ILBD	PD		ILBD	PD			ILBD	PD
Sialylation	0.110	0.000	0.000	0.151	0.038	0.989	0.344	0.038	0.027
Fucosylation	0.083	0.998	0.171	0.999	0.958	0.960	0.135	0.970	0.935
Sulfation	0.999	0.001	0.056	1.000	0.967	0.993	0.957	0.991	0.134
Mannose Core	0.017	0.965	0.075	1.000	1.000	1.000	0.287	0.672	0.892
Glucuronylation	0.039	0.958	0.012	1.000	0.744	0.975	0.011	0.942	0.149
Peeling	0.554	0.001	0.002	0.379	0.398	0.970	1.000	0.060	0.381

Table 14. *P-values derived from comparison of glycan features across different groups comparing disease states. Red indicates a decrease in relative abundance in the leading group (e.g the healthy group in healthy vs. PD) and green indicates an increase – only statistically significant areas highlighted. Main glycan of each peak used for analysis.*

These results indicate a number of significant changes. Presence of mannose core glycans (p = 0.017) and glucuronylation (p = 0.039) are significantly decreased in ILBD striatum compared with healthy controls. Notedly, of the 21 mannose core structures which have been identified, no individual glycan was significantly different between groups, however, the area differences in the overall cohort were enough to make this result significant. This was the case for total glucuronylation also (2 glycans).

At the later stages of PD pathology (Braak stages 3-4), striatum expressed an increase in overall sialylation (p = 0.000) and a decrease in total sulfation (p = 0.001) compared with healthy controls. To note here is the significant increase in overall peeling (p = 0.001). While not truly representative of any physiological glycosylation *in vivo*, there is some value gained from this, somewhat, unwanted result. The majority of this peeling feature (> 90%) is derived from the glycan product S3**ga** (peak 8) mentioned throughout this work. Despite peeling being an unwanted side-reaction, the sialylation of this structure contributes to the calculation of overall sialylation, as it originates from larger mucin-type sialylated *O*-glycan structures. Including this peeling product, S3ga3[S6]**GA** (peak 36) and the sulfated S3sga3[S6]**GA** (isomer 2) (peak 41; could not determine sulfation position on the galactose, hence the description “isomer 1” and “isomer 2” for the two species identified) were all significantly increased in PD (all with p = 0.000), while S6s**GA**/s3ga3**GA**/s6ga3**GA** (peak 4; p = 0.035) sga3[S6]**GA** (peak 22; p = 0.000), ga3[S3ga4GL6]**GA** (peak 32; p = 0.001) and the sulfated S3sga3[S6]**GA** (isomer 1) (peak 40; p = 0.000) were decreased. These up- and down-regulated sulfated structures are the same structures that contribute to significant sulfation changes.

Comparing ILBD to PD striatum, there were similar changes observed to healthy vs. PD striatum (sialylation and peeling increased in PD), with the addition of increased glucuronylation in PD ($p = 0.012$) coming from the significant increase of sU3ga3GL2**m** (peak 34; $p = 0.037$). While sialylation remained significantly increased in PD ($p = 0.000$) there was a notable decrease in abundance of the sialyl-Tn antigen S6**GA** (peak 7; $p = 0.009$).

Interestingly, little significant difference was observed in any *O*-glycosylation features between healthy (control), ILBD or PD substantia nigra, except for an increase in overall sialylation ($p = 0.038$) in PD substantia nigra compared with the healthy control group. This was attributed to an increase in S3ga3GL3[S6]**GA** (peak 39; $p = 0.010$) and S3sga3[S6]**GA** (isomer 2) (peak 41; $p = 0.000$) in the PD group and a decrease in sga3[S6]**GA** (peak 22; $p = 0.004$) and S3sga3[S6]**GA** (isomer 1) (peak 40, $p = 0.001$). There was, however, no significant difference in any feature when comparing ILBD and PD substantia nigra.

Comparing healthy controls across the different spatial regions, there were significantly higher relative abundances of glucuronylation ($p = 0.011$) in healthy striatum compared with substantia nigra. With respect to significance of individual glucuronic acid containing structures, sU3ga4GL2**m** (peak 33; $p = 0.007$) was at a higher abundance in healthy striatum.

When observing temporally the glycosylation of both regions after the onset of ILBD, there were higher levels of sialylation in substantia nigra compared with striatum indicated by an increase in S3**ga** (peak 8; $p = 0.036$), however, upon the onset of later Braak stage 3-4 PD in both groups, sialylation ($p = 0.15$) was significantly higher in striatum. Significant individual glycan differences driving higher levels of sialyation for PD striatum came from increases S3ga4GL2**m** (peak 25; $p = 0.045$), S3ga3[S6]**GA** (peak 36; $p = 0.007$) and S3ga4GL2[ga4GL6]**m** (peak 42; $p = 0.000$), however, decreases in S3**S** (peak 5; $p = 0.000$), ga3[sS6]**GA** (peak 27; $p = 0.045$) and S3ga3[ga4GL6]**GA** (peak 35; $p = 0.000$) were also seen.

Graphical representations of significant feature differences across disease states are shown in figure 37 for striatum and figure 38 for substantia nigra outlining the degree of significance. All groups were compared with each other and only groups with significant changes are highlighted (All comparison plots in appendix XII-XIII).

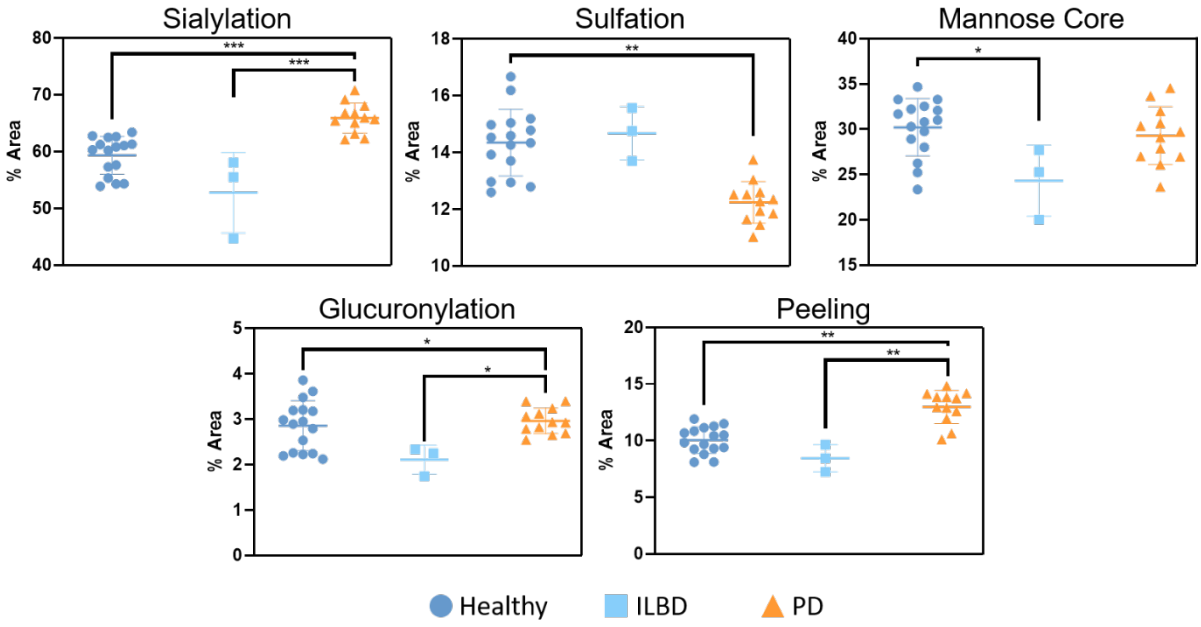


Figure 37. *Nested tables showing % area for individual samples in each group compared against disease states in striatum. Significance across groups is written as follows: * (p < 0.05), ** (p < 0.01), *** (p < 0.001).*

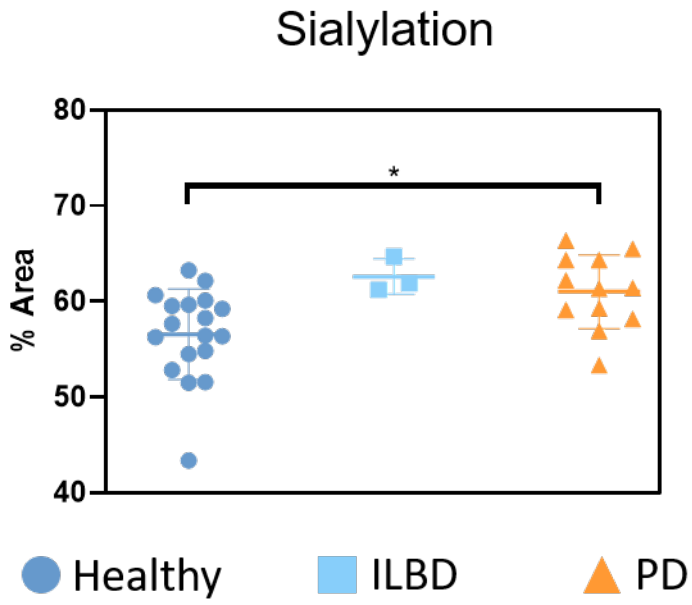


Figure 38. *Nested table showing % area for individual samples in each group compared against disease states in substantia nigra. * (p < 0.05).*

Primary Component Analysis (PCA) was performed to identify how groups with certain features/components were related. % peak area of each sample in a group were compared against other groups (e.g. healthy striatum vs PD striatum), as well as % feature area (sialylation, fucosylation, etc). The greater the separation on the PCA plot between samples of each group, the less they were related, or, the greater their difference. It was observed across

most PCA plots that groups were all closely related and there was little separation (appendix XIV-XXIII). The main exception to this was the comparison of healthy striatum, ILBD striatum and PD striatum. Figure 39 illustrates the separation of each of these groups on the PCA plot (for both peak and feature analysis) and identifies the components that drove this difference.

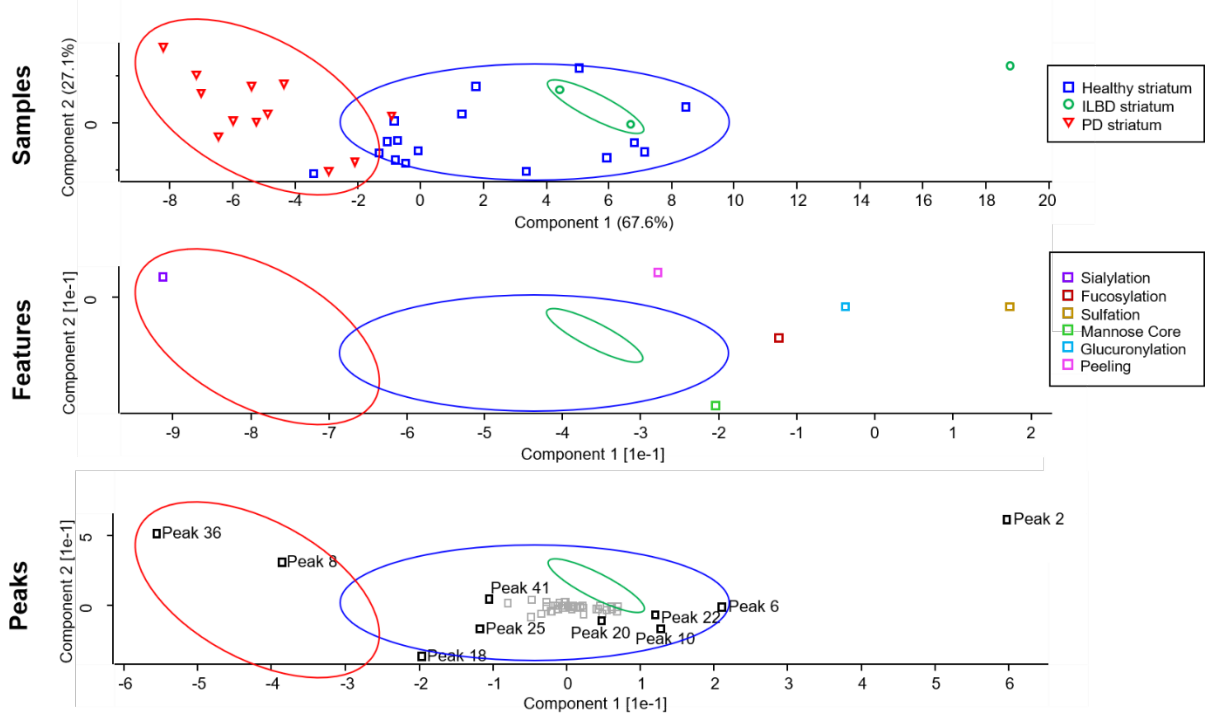


Figure 39. *PCA plots outlining separation (difference) of healthy striatum, ILBD striatum and PD striatum. Grouped areas were superimposed over feature and peak PCA plots to identify components driving the difference.*

As can be seen from these results, sialylation is the main component separating healthy and PD striatum (derived from peaks 36; S3ga3[S6]**GA** and peak 8; S3**ga**). The overlap, or lack of separation, of healthy and ILBD striatum samples indicates that ILBD and healthy control share a lot of similarities (peak areas as well as features), however, this could be attributed to the small sample set for ILBD (n = 3). Peaks 6 (ga3**GA**, gaa3**GA**, GL3**GA**), 10 (ga4GL2**m**), 18 (ga4(fa3GL2)**m**), 20 (sga3[GL6]**GA**), 22 (sga3[S6]**GA**), 25 (S3ga4GL2**m**) and 41 (S3sga3[S6]**GA** (isomer 2)), can be seen to be attributed to differences in healthy striatum. These results match closely the results obtained from MANOVA analysis with respect to sialylated glycan signatures in PD striatum, however, PCA analysis was able to emphasise further a more prominent difference in S3ga3[S6]**GA** (peak 36) and S3**ga** (peak 8) glycan peaks between healthy and PD compared with MANOVA analysis which also considered S3sga3[S6]**GA** (isomer 2) (peak 41) to be equally significant (all 3 peaks were equally significant; p = 0.000).

3.3.9 Statistical Differences – Gender

Gender analysis was performed using the same MANOVA and tukey analysis method, however, 10 groups were compared. The initial 6 groups were divided into males and females with the exception of ILBD groups which contained only female patients.

When comparing the effect of gender on glycosylation (both in healthy and diseased states), there were no significant statistical differences in any glycan features measured. Interestingly, there were only 2 significant *glycan peak* differences observed across all comparisons. There was an increase in S6ga3**GA** (peak 15; $p = 0.013$) in PD males (stage 3-4) in the striatum when compared with females of the same group. There was also an increase in S3ga3GL3[S6]**GA** (peak 39; $p = 0.045$) in healthy males in the substantia nigra when compared with females of the same group.

Comparisons were also made to determine how gender was associated with glycosylation changes upon the onset of ILBD or PD. Glycosylation changes of each peak for females and males are shown in tables 15 and 16, respectively. These are similarly given as p-values where significance is highlighted. All 6 features from the analysis were also compared for all groups. Tables 17 and 18 outline significant changes of glycan features in females and males, respectively, upon the onset of disease.

Peak	Healthy vs Diseased Striatum		ILBD vs PD Striatum	Healthy vs Diseased Nigra		ILBD vs PD Substantia Nigra	Healthy Striatum vs Nigra	Diseased Striatum vs Nigra	
	ILBD	PD		ILBD	PD			ILBD	PD
1	0.035	0.003	1.000	1.000	0.999	1.000	0.001	1.000	1.000
2	0.118	0.989	0.033	0.613	0.998	0.967	0.999	0.025	0.999
3	1.000	0.112	0.325	0.373	0.924	0.975	1.000	0.734	0.692
4	0.075	0.677	0.003	1.000	0.830	1.000	0.908	0.440	0.951
5	0.934	1.000	0.997	1.000	0.781	0.944	0.355	0.486	0.019
6	0.948	1.000	0.893	1.000	1.000	1.000	0.721	1.000	0.834
7	0.426	0.179	0.006	0.972	0.263	0.999	1.000	0.188	1.000
8	0.946	0.051	0.024	0.478	0.923	0.991	1.000	0.110	0.923
9	0.177	0.960	0.035	0.942	0.958	1.000	0.087	0.790	0.465
10	0.999	0.864	1.000	1.000	1.000	1.000	1.000	0.999	0.910
11	0.167	0.045	0.000	0.516	0.977	0.980	0.927	0.065	0.121
12	0.154	0.345	0.002	1.000	0.844	1.000	0.647	0.726	0.522
13	0.476	0.107	0.004	1.000	1.000	1.000	0.848	0.955	0.015
14	0.282	0.258	0.004	1.000	0.997	1.000	0.718	0.909	0.153
15	0.996	0.407	0.998	0.906	1.000	0.997	0.196	0.749	1.000
16	0.980	1.000	0.969	1.000	1.000	1.000	0.967	1.000	0.954
17	1.000	0.998	1.000	1.000	1.000	1.000	0.998	1.000	0.903
18	0.265	1.000	0.389	1.000	1.000	1.000	0.623	1.000	0.892
19	0.961	1.000	0.960	1.000	1.000	0.998	1.000	1.000	1.000
20	0.729	0.249	1.000	1.000	0.798	0.999	0.934	0.976	0.180
21	0.777	0.890	1.000	0.997	1.000	1.000	0.042	0.992	0.891
22	0.999	0.000	0.001	0.997	0.244	0.990	0.393	0.323	0.992
23	0.999	0.021	0.056	0.641	0.998	0.970	1.000	0.681	0.167
24	0.991	0.154	0.118	1.000	0.936	1.000	0.994	1.000	0.003
25	0.876	0.993	0.540	1.000	1.000	1.000	0.858	0.999	0.575
26	0.996	1.000	0.989	1.000	0.993	1.000	0.991	1.000	0.749
27	0.996	0.582	0.412	1.000	1.000	1.000	0.913	1.000	0.173
28	0.608	1.000	0.697	1.000	0.203	0.355	0.782	1.000	0.987
29	0.987	0.000	0.000	1.000	0.066	0.790	0.468	0.346	0.998
30	0.438	0.998	0.890	1.000	1.000	0.999	0.403	1.000	1.000
31	0.984	0.512	0.277	0.970	1.000	0.997	1.000	1.000	0.373
32	0.999	0.102	0.895	0.983	0.920	1.000	0.955	0.995	1.000
33	0.302	0.998	0.142	0.995	1.000	1.000	0.321	1.000	0.162
34	0.662	1.000	0.466	0.723	0.082	1.000	0.873	0.483	0.990
35	1.000	0.988	0.973	0.997	0.417	0.998	0.999	1.000	0.040
36	0.430	0.023	0.001	0.897	0.992	1.000	0.999	0.240	0.155
37	0.997	1.000	1.000	0.942	0.925	1.000	0.593	0.929	0.541
38	0.181	0.965	0.817	1.000	1.000	1.000	0.665	0.979	1.000
39	1.000	0.998	0.998	0.147	0.055	1.000	0.998	0.561	0.879
40	0.998	0.000	0.001	0.983	0.012	0.813	0.857	0.417	1.000
41	0.505	0.081	0.004	0.851	0.247	1.000	0.980	0.353	0.947
42	0.866	0.937	0.349	0.965	1.000	0.959	0.034	1.000	0.012
43	0.693	1.000	0.696	0.733	0.983	0.997	0.295	0.819	0.991
44	0.999	0.953	1.000	0.764	0.986	0.997	0.374	0.999	1.000
45	0.923	1.000	0.894	0.982	0.996	1.000	0.275	0.999	0.955
46	1.000	1.000	1.000	0.656	0.717	1.000	0.981	0.995	1.000
47	0.850	0.072	0.998	0.765	0.488	1.000	0.085	0.984	0.325
48	1.000	0.527	0.514	0.997	0.762	1.000	0.999	1.000	0.160
49	0.997	0.063	0.886	0.991	0.807	1.000	0.695	1.000	0.123
50	0.999	1.000	0.995	0.778	1.000	0.967	1.000	0.999	1.000
51	0.993	0.999	0.925	0.327	0.912	0.968	1.000	0.220	1.000

Table 15. *P-values derived from comparison of each peak across different groups comparing disease states in females. Red indicates a decrease in relative abundance in the leading group (e.g the healthy group in healthy vs. PD) and green indicates an increase – only statistically significant areas highlighted.*

Peak	Healthy vs Diseased Striatum		ILBD vs PD Striatum	Healthy vs Diseased Nigra		ILBD vs PD Substantia Nigra	Healthy Striatum vs Nigra	Diseased Striatum vs Nigra	
	ILBD	PD		ILBD	PD			ILBD	PD
1	-	0.850	-	-	1.000	-	0.010	-	0.299
2	-	0.647	-	-	0.917	-	0.974	-	0.447
3	-	0.970	-	-	0.975	-	0.620	-	1.000
4	-	0.155	-	-	0.355	-	1.000	-	1.000
5	-	0.953	-	-	0.999	-	0.395	-	0.079
6	-	0.987	-	-	0.003	-	0.001	-	0.782
7	-	0.755	-	-	0.999	-	0.319	-	0.820
8	-	0.108	-	-	0.840	-	1.000	-	0.953
9	-	1.000	-	-	1.000	-	0.999	-	1.000
10	-	0.439	-	-	0.002	-	0.017	-	0.739
11	-	0.602	-	-	0.394	-	1.000	-	1.000
12	-	0.945	-	-	0.999	-	1.000	-	1.000
13	-	0.468	-	-	0.478	-	0.996	-	0.991
14	-	0.885	-	-	1.000	-	1.000	-	0.982
15	-	0.891	-	-	1.000	-	0.813	-	0.005
16	-	0.843	-	-	0.002	-	0.063	-	1.000
17	-	1.000	-	-	1.000	-	1.000	-	1.000
18	-	1.000	-	-	0.976	-	0.427	-	0.990
19	-	1.000	-	-	0.375	-	0.515	-	1.000
20	-	1.000	-	-	0.716	-	0.064	-	0.650
21	-	1.000	-	-	1.000	-	0.994	-	0.896
22	-	0.001	-	-	0.454	-	0.350	-	1.000
23	-	0.347	-	-	0.996	-	1.000	-	0.967
24	-	0.171	-	-	0.482	-	0.975	-	0.668
25	-	1.000	-	-	1.000	-	0.197	-	0.522
26	-	0.976	-	-	1.000	-	0.369	-	0.593
27	-	0.550	-	-	0.986	-	1.000	-	0.837
28	-	1.000	-	-	1.000	-	1.000	-	1.000
29	-	0.001	-	-	0.996	-	0.018	-	1.000
30	-	1.000	-	-	0.387	-	0.998	-	0.987
31	-	0.900	-	-	0.500	-	0.547	-	0.862
32	-	0.055	-	-	1.000	-	0.104	-	1.000
33	-	1.000	-	-	1.000	-	0.142	-	0.421
34	-	0.747	-	-	0.416	-	1.000	-	1.000
35	-	0.993	-	-	1.000	-	0.864	-	0.119
36	-	0.029	-	-	0.216	-	0.961	-	0.438
37	-	0.999	-	-	1.000	-	0.935	-	0.746
38	-	0.740	-	-	1.000	-	0.973	-	0.984
39	-	0.912	-	-	0.999	-	0.442	-	0.779
40	-	0.000	-	-	0.017	-	0.986	-	0.982
41	-	0.091	-	-	0.073	-	0.999	-	0.999
42	-	0.995	-	-	1.000	-	0.123	-	0.010
43	-	0.670	-	-	0.995	-	0.999	-	1.000
44	-	1.000	-	-	0.927	-	0.989	-	1.000
45	-	0.851	-	-	0.992	-	1.000	-	0.980
46	-	0.997	-	-	0.966	-	0.258	-	1.000
47	-	1.000	-	-	0.030	-	0.805	-	0.785
48	-	0.996	-	-	0.525	-	0.891	-	0.878
49	-	1.000	-	-	0.979	-	1.000	-	0.976
50	-	0.997	-	-	0.125	-	0.966	-	0.998
51	-	1.000	-	-	0.910	-	0.117	-	0.988

Table 16. *P-values derived from comparison of each peak across different groups comparing disease states in males. Red indicates a decrease in relative abundance in the leading group (e.g the healthy group in healthy vs. PD) and green indicates an increase – only statistically significant areas highlighted. Cells marked with a "-" have no data due to absence of males in ILBD group.*

Feature	Healthy vs Diseased Striatum		ILBD vs PD Striatum	Healthy vs Diseased Nigra		ILBD vs PD Substantia Nigra	Healthy Striatum vs Nigra	Diseased Striatum vs Nigra	
	ILBD	PD		ILBD	PD			ILBD	PD
Sialylation	0.316	0.117	0.002	0.523	0.987	0.972	0.988	0.083	0.331
Fucosylation	0.369	1.000	0.632	1.000	1.000	1.000	0.855	0.999	0.999
Sulfation	1.000	0.022	0.101	1.000	1.000	1.000	0.933	1.000	0.417
Mannose Core	0.083	1.000	0.224	1.000	1.000	1.000	0.742	0.908	0.996
Glucuronylation	0.196	0.997	0.081	1.000	0.999	1.000	0.280	0.996	0.674
Peeling	0.898	0.137	0.036	0.785	0.994	0.996	1.000	0.161	0.945

Table 17. *P-values derived from comparison of glycan features across different groups comparing disease states from females. Red indicates a decrease in relative abundance in the leading group (e.g the healthy group in healthy vs. PD) and green indicates an increase – only statistically significant areas highlighted. Main glycan of each peak used for analysis.*

Feature	Healthy vs Diseased Striatum		ILBD vs PD Striatum	Healthy vs Diseased Nigra		ILBD vs PD Substantia Nigra	Healthy Striatum vs Nigra	Diseased Striatum vs Nigra	
	ILBD	PD		ILBD	PD			ILBD	PD
Sialylation	-	0.060	-	-	0.030	-	0.465	-	0.507
Fucosylation	-	1.000	-	-	0.987	-	0.545	-	1.000
Sulfation	-	0.305	-	-	0.675	-	1.000	-	0.943
Mannose Core	-	0.999	-	-	1.000	-	0.990	-	1.000
Glucuronylation	-	1.000	-	-	0.979	-	0.400	-	0.887
Peeling	-	0.260	-	-	0.734	-	1.000	-	0.930

Table 18. *P-values derived from comparison of glycan features across different groups comparing disease states from males. Red indicates a decrease in relative abundance in the leading group (e.g the healthy group in healthy vs. PD) and green indicates an increase – only statistically significant areas highlighted. Main glycan of each peak used for analysis. Cells marked with a "-" have no data due to absence of males in ILBD group.*

These results indicate that an increase in sulfation in PD striatum compared with healthy control is driven by females ($p = 0.022$ in females compared to $p = 0.305$ in males). In contrast, increased sialylation in PD substantia nigra compared with healthy control is driven by males ($p = 0.030$ in males compared to $p = 0.987$ in males). Other significant differences (when not considering gender) are driven by a combination of male and female glycosylation changes that, when viewed separately by gender, are not significant in themselves. To note, significant sialylation changes between ILBD and PD in striatum seem to be driven by glycosylation changes in females, however this is due to the fact that there were no males in the ILBD cohort and may not represent actual changes observed were the cohort mixed.

3.3.10 Statistical Differences – Age

By incorporating age as a covariate into the MANOVA analysis, it was identified that there was no statistical significance due to age of individual glycans, both when comparing the 6 groups without differentiating between gender ($p = 0.209$) or the 10 groups when considering gender ($p = 0.381$).

There appeared to be some overall significance when comparing glycan features across the groups when using age as a covariate ($p = 0.020$ for analysis of groups with and without separating gender). Due to the nature of SPSS not allowing a multivariate post-hoc test when using covariates and tukey tests not permissible when treating age as sub-groups (groups require more than two patients which was not possible due to limited sample size), patient groups for each brain region, control and diseased, were split in two – “under 85 years old” and “85 and over”.

MANOVA analysis was re-performed both on individual glycan peaks and on glycan features using these newly assigned age groups. With respect to individual glycan peaks, when comparing patient groups without differentiating gender, there was some significance ($p = 0.036$), however there was no significance when separating the groups by gender ($p = 0.396$). It was identified that there were no significant glycosylation differences between age groups (e.g. healthy striatum under 85 vs 85+), however, there were significance individual glycan differences observed when comparing glycosylation changes upon the onset of disease, at each age group. p-values of glycan peak differences from each group are shown in table 19 (significant peaks highlighted) for age groups “under 85” and “85+”.

With respect to glycan feature differences, there were no significant differences in glycan features ($p = 0.214$) when considering age and gender, however, by considering only age, there were significant differences ($p = 0.003$). Table 20 identifies p-values of feature differences between groups separated by age. These results indicate that the sialylation increase in PD striatum compared with healthy control is more-so correlated with patients 85 years and older. Note: ILBD has been excluded from age comparisons as there were less than 2 patients 85+ years old (not enough to perform post-hoc tukey test).

Peak	Under 85				85+			
	Healthy vs PD Striatum	Healthy vs PD Nigra	Healthy Striatum vs Nigra	PD Striatum vs Nigra	Healthy vs PD Striatum	Healthy vs PD Nigra	Healthy Striatum vs Nigra	PD Striatum vs Nigra
1	0.065	1.000	0.000	0.312	0.123	1.000	0.049	1.000
2	0.642	1.000	1.000	0.746	0.888	1.000	1.000	0.909
3	0.954	1.000	0.999	0.834	0.244	0.991	0.908	1.000
4	0.545	0.506	0.997	0.999	0.325	0.624	0.982	0.934
5	0.972	1.000	0.092	0.019	0.994	0.876	0.787	0.064
6	1.000	0.961	0.471	0.935	0.780	0.435	0.070	0.575
7	0.623	0.820	0.927	0.995	0.462	0.330	1.000	0.999
8	0.097	0.741	1.000	0.896	0.012	0.806	0.998	0.803
9	1.000	0.462	0.081	1.000	0.962	1.000	0.921	0.346
10	0.930	0.738	0.554	0.868	0.166	0.455	0.920	0.800
11	0.390	0.306	0.897	0.957	0.148	0.959	0.999	0.552
12	0.834	0.641	0.998	1.000	0.562	0.998	0.875	0.392
13	0.621	0.244	0.588	0.948	0.034	1.000	0.998	0.007
14	0.913	0.833	0.719	0.876	0.219	1.000	0.999	0.163
15	1.000	0.833	0.903	0.247	1.000	1.000	0.092	0.685
16	0.901	0.080	0.119	0.974	0.999	1.000	0.989	0.992
17	1.000	1.000	1.000	1.000	0.990	0.992	0.988	0.413
18	0.985	1.000	0.542	0.992	0.993	0.997	0.208	0.464
19	1.000	0.945	0.531	1.000	0.997	0.823	1.000	0.998
20	0.984	1.000	0.825	0.195	0.391	1.000	0.996	0.669
21	0.957	1.000	0.313	0.944	0.910	0.841	0.332	0.580
22	0.000	0.347	0.213	1.000	0.000	0.092	0.522	1.000
23	0.022	0.817	1.000	0.485	0.116	1.000	1.000	0.318
24	0.321	0.724	0.803	0.471	0.062	0.846	1.000	0.002
25	1.000	0.999	0.522	0.485	0.913	1.000	0.416	0.304
26	0.993	0.941	0.531	0.432	1.000	0.999	0.901	0.584
27	0.451	0.945	0.997	0.780	0.681	0.998	0.882	0.056
28	0.996	0.989	0.936	1.000	1.000	0.270	0.667	0.999
29	0.000	0.201	0.328	1.000	0.000	0.376	0.153	1.000
30	1.000	0.955	0.939	1.000	0.999	0.444	0.430	0.996
31	0.936	0.950	0.599	0.654	0.201	1.000	1.000	0.384
32	0.001	0.915	0.216	0.991	0.527	1.000	0.773	1.000
33	1.000	1.000	0.426	0.312	1.000	1.000	0.080	0.112
34	0.924	0.021	0.947	0.957	0.873	0.243	0.998	1.000
35	0.678	0.998	0.784	0.013	1.000	0.243	1.000	0.118
36	0.016	0.469	0.711	0.065	0.008	0.779	1.000	0.357
37	1.000	1.000	0.930	0.812	1.000	0.864	0.559	0.352
38	0.771	1.000	0.927	1.000	0.658	1.000	0.475	1.000
39	0.920	0.587	0.813	0.550	0.974	0.082	0.988	0.963
40	0.000	0.034	0.392	0.999	0.000	0.001	0.999	0.985
41	0.023	0.038	0.991	0.970	0.020	0.035	0.900	0.917
42	0.972	1.000	0.041	0.006	0.728	1.000	0.073	0.016
43	0.995	1.000	1.000	0.995	0.974	0.122	0.134	0.999
44	1.000	0.950	0.765	1.000	0.964	0.855	0.533	0.997
45	0.813	0.946	0.936	0.839	1.000	0.906	0.181	0.902
46	0.997	0.958	1.000	1.000	1.000	1.000	1.000	1.000
47	0.722	1.000	0.158	0.997	0.405	0.992	1.000	0.773
48	0.984	1.000	0.998	1.000	0.998	1.000	0.986	0.848
49	0.425	0.998	0.929	1.000	0.951	0.981	0.998	0.880
50	0.982	0.919	0.995	0.981	1.000	1.000	0.842	0.983
51	1.000	1.000	0.980	0.999	0.977	0.950	0.957	0.980

Table 19. *P-values derived from comparison of each peak across different groups comparing disease states separated by age ("under 85" and "85+"). Red indicates a decrease in relative abundance in the leading group (e.g the healthy group in healthy vs. PD) and green indicates an increase – only statistically significant areas highlighted. Cells marked with a "-" have no data due to absence of males in ILBD group.*

Feature	Under 85				85+			
	Healthy vs PD Striatum	Healthy vs PD Nigra	Healthy Striatum vs Nigra	PD Striatum vs Nigra	Healthy vs PD Striatum	Healthy vs PD Nigra	Healthy Striatum vs Nigra	PD Striatum vs Nigra
Sialylation	0.157	0.406	0.572	0.328	0.008	0.395	0.964	0.281
Fucosylation	0.909	1.000	0.813	1.000	0.999	0.837	0.276	0.954
Sulfation	0.202	1.000	1.000	0.646	0.050	0.992	0.997	0.733
Mannose Core	0.953	1.000	0.964	1.000	1.000	1.000	0.385	0.876
Glucuronylation	0.999	0.924	0.442	0.887	0.996	0.998	0.155	0.410
Peeling	0.086	0.772	1.000	0.925	0.111	0.973	1.000	0.759

Table 20. *P-values derived from comparison of glycan features across different groups comparing disease states separated by age ("under 85" and "85+"). Red indicates a decrease in relative abundance in the leading group (e.g the healthy group in healthy vs. PD).*

3.4 DISCUSSION

3.4.1 *O*-glycosylation Changes in ILBD

As a preface to the following sections, it should be understood that in depth glycan characterization and quantitation of striatum and substantia nigra in PD is lacking in the literature, particularly for *O*-glycans, with work focused heavily on *O*-GlcNAcylation of α-synuclein but not total *O*-glycosylation of brain tissue. Work has been performed on serum *N*-glycosylation[263] as well as IgG *N*-glycosylation[264] in the past, so these were used for comparisons. Where there is no suitable comparison found for the specific glycan features, similar neurodegenerative or other diseases/disorders were compared. Furthermore, the analyses that were performed provided relative quantitation on the total glycome released from total glycoproteins. In order to obtain absolute quantitation "spiking" the samples with an internal standard of known concentration would have been needed. In addition, to determine whether overall protein concentration increases relative to increased glycan concentration, or whether glycosylation simply increases or decreases upon the onset of disease, glyco-proteomics studies are required (worth investigating in future work).

3.4.1.1 Mannose Core

Although a diverse class of glycans making up approximately 28.5% of the total glycome in each group, mannose core glycans have been linked to congenital muscular dystrophies (CMDs) such as Walker Warburg syndrome, Fukuyama CMD and CMD 1C/1D, with underglycosylation of these structures on α-dystroglycan inhibiting this cell adhesion molecule from correctly binding extracellular matrix proteins[268, 277]. While its link to ILBD and PD is ambiguous in the literature, the downregulation of mannose core glycans in ILBD striatum may help outline possible causes for any dystrophic changes that may be seen in ILBD patients.

3.4.1.2 Glucuronylation

Glucuronylation has been included in feature analysis due to its uncommon nature in typical *O*-glycome analyses of mucin-type samples, as well as the fact that it plays an important role in cell-cell and cell-matrix neuronal adhesion, and performing functions associated with nervous system development[268, 278-280]. The presence of sulfated glucuronic acid at the end of the mannose-core glycans (called the Human Natural Killer-1 antigen) in this project (2 glycans: sU3ga4GL2**m** and sU3ga3GL2**m**), gives an indication to the degree of glucuronylation, or more specifically sulfoglucuronylation. The apparent downregulation of this feature in ILBD provides an interesting insight into its potential role in neurodegeneration, however, more work will need to be done to clarify its importance.

3.4.2 *O*-glycosylation in PD

3.4.2.1 Sialylation

In contrast to previous work conducted on PD *N*-glycosylation where it was found that sialylation of *N*-glycans was decreased in PD[264], sialylation of *O*-glycans appeared to increase in PD states in striatum. Translated to function, the lowering of (primarily) monosialylated *N*-glycans in IgG has been suggested to lower inflammatory activity via antibody-dependent cell cytotoxicity due to decreased capacity to ligate Fcγ-RIIIa on natural killer cells[264]. Further, the degree of sialylation is known to mediate specific receptor interactions with sialic acid binding ligands like siglecs and selectins which inhibit complement components binding to cell surfaces, inducing complement-mediated proinflammatory cascades[281]. This interesting contrast observed for *O*-glycans will need to be investigated in future work to identify the difference, however, one substantial observation made across MANOVA and PCA analyses was that the sialylation increase in striatum of PD was driven by an increase in S3ga3[S6]**GA**. This "hypersialylated" structure has been associated with various forms of cancer[282, 283] and the increase in α2-6 linked sialic acid containing structures like this one have also been correlated to an enhanced degenerative state in the brain[284]. Similar outcomes were observed in other neurodegenerative diseases like Alzheimer's Disease (AD), where sialyation events occurring in the *N*-glycome matched that in PD, however, the total *O*-glycome of AD seems yet to be elucidated also[285].

Observing changes in PD substantia nigra, sialylation was increased relative to healthy control, however, the core 3 extended structure s3ga3GL3[S6]**GA** and the "hypersialylated" structure observed in PD striatum (S3sga3[S6]**GA** (isomer 2)) were upregulated. The hypothesis of increased α2-6 linked sialic acid causing neurodegeneration is reinforced by this result.

3.4.2.2 Sulfation

Sulfated glycans help fine tune biological activity for glycans and glycoproteins and have been linked to disease and inflammation. For instance, the increased production of cystic fibrosis associated sulfated mucins have been shown to increase inflammation through the production of viscous mucus causing respiratory failure[286]. With respect to brain physiology, studies have determined that sulfotransferase deficiency is linked to abnormal myelination and axonal degeneration of the peripheral nervous system (*N*-glycosylation experiment in mice)[287]. This axonal demyelination causes permanent neurological disability seen in human myelin diseases[287]. Again, while studies on sulfation of *N*- and *O*-glycans are limited in the literature with respect to PD, the similarities here, of decrease in sulfation in PD striatum compared with healthy control is worth investigating in future work.

3.4.3 Temporal Differences

While ILBD expressed an upregulation of mannose core and glucuronylated structures, overall differences between ILBD and healthy controls according to PCA analyses were minimal, indicating the groups were similar, compared against PD (most notable in figure 39). This is somewhat expected. In the beginning of this chapter it was noted that ILBD and lewy body formation had association with aging rather than PD. Furthermore, according to Braak Staging also discussed, substantia nigra regions remain unaffected in ILBD (stage 1 and 2) and only affect this region in later stage (3 and 4) patients.

The other main observation was that, even at later stage PD, striatum appeared more affected than substantia nigra brain regions. While significant alterations in *O*-GlcNAcylation of α-synuclein in substantia nigra have been heavily analyzed and reviewed, and contributes significantly to lewy body formation and neuronal degradation in the substantia nigra[247, 265, 288], alterations in the *total O*-glycome of this region don't follow this pattern; substantia nigra *O*-glycome change is minimal. This indicates that future biomarker identification and regenerative therapeutic targets for *O*-glycans in PD patients may be of increased benefit by observing solely striatum, where greater glycosylation changes occur.

3.4.4 Spatial Differences

The results indicate that striatum and substantia nigra share much the same glycan structures, albeit at different relative abundances. Of all features, only glucuronylation was different between healthy control striatum (higher relative abundance) and substantia nigra.

In ILBD comparisons between regions, sialylation was higher in substantia nigra, however in PD comparisons, sialylation was higher in striatum.

Due to their nigrostriatal connection and proximity in the brain[289], the similarities are not unexpected, however, the cause of the differences and the sudden "flip" between lower to higher sialylation in striatum relative to healthy controls upon pathogenesis from ILBD to PD will require further investigation.

3.4.5 Age and Gender Differences

There were correlations with age and PD progression but no significant changes between different ages at the same physiological state. This differs to work performed on *N*-glycans when comparing age to which there were numerous sialylated glycans that were differentially expressed[290]. In this work, it was found that the main difference of age was that patients aged 85+ were found to be more significant contributors to sialylation in PD striatum when they were analyzed separately from their total cohort.

The main feature differences with regards to gender were that sialylation in PD substantia nigra was driven by males whereas sulfation increases in striatum were driven by females.

Furthermore, peak analysis identified only 2 peaks that were significantly differentially expressed (S6ga3**GA** increased in PD striatum males and S3ga3GL3[S6]**GA** increase in healthy substantia nigra males). This is an agreement with previous work on serum *N*-glycosylation in PD where greater significant sialylation alterations were observed in males[263].

In both gender and age analyses there were restrictions on statistical analysis that could not be performed due to the small sample size (n = 3) of ILBD (all females and only 1 over 85 years of age). Future analyses will require larger cohorts of ILBD for more meaningful analysis. Alternately, due to their similarities in glycosylation, ILBD may be pooled with healthy controls.

3.5 CONCLUSION

This chapter indicated and discussed several significant changes between healthy and disease states of Parkinson's disease as well as differences based on gender and age, indicating potential biomarkers that may be targeted for therapeutics. The changes were observed statistically by comparing overall glycosylation between males and females and glycosylation between patients aged 85+ or under 85, as well as by observing how age or gender affects glycosylation changes upon the onset of disease.

CHAPTER 4 – GENERAL DISCUSSION AND CONCLUDING REMARKS

4.1 FUTURE DIRECTION FOR *O*-GLYCAN CHARACTERIZATION IN PD

As mentioned previously, studies have historically focused on *O*-GlcNAcylation more than *O*-glycosylation, however, the results presented in this work outline the complexity of the *O*-glycome of striatum and substantia nigra in PD as well as significant glycosylation changes occurring after onset of the disease. While 77 glycan structures associated with brain tissue were identified, considerations were made. For instance, as observed in table 12 of chapter 3, characterization of some glycan structures were made using only a single characterization method (exoglycosidase digestion *or* QToF experiments *or* ion-trap experiments) and lacked confirmation from complementary methods. This may have been due in part to sample concentration limitations, limiting what could be detected on some systems, but may also be due to differences in sensitivity of detection particularly for MS. In negative ion mode, detection of negatively charged glycans (those with sulfation and sialylation) provided greater sensitivity compared with neutral glycans[84]. Furthermore, as observed in figure 29 in chapter 3, the available glycan sample was so low that a TIC of the glycans (from QToF experiment on 2AB labelled glycans) could not be resolved from the baseline (requiring mining of EICs), ultimately limiting what could be detected in MS. In future work, orthogonal methods of characterization for instance CE or NMR may be used to help validate results. Furthermore, if persisting with non-reductive *O*-glycan release methods, incorporating a more sensitive glycan label (like Procainamide[75] or RapiFlour-MS[110]) may help resolve glycan structures lost in the baseline of UPLC chromatograms and MS spectra. This, coupled with the obvious need to limit peeling through further optimization, may aid in characterization. Finally, in exoglycosidase studies, exoglycosidases used were primarily sourced from New England Biolabs. As it was observed that activity of these enzymes were limited in some instances, employing exoglycosidases from other suppliers (such as Prozyme) may help in-part or completely remove these inadequacies.

Regarding understanding glycosylation changes during PD pathogenesis and Braak Staging, it may be worthwhile to observe glycosylation changes across the different brain regions as they are subsequently impacted by the disease, separating based on individual Braak stage rather than pooling multiple stages together (figure 40), to help identify other biomarkers for the disease.

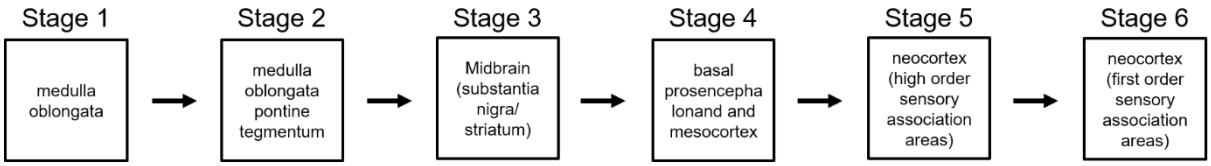


Figure 40. *Brain regions affected of each Braak Stage to be considered for separate analysis of glycosylation.*

4.2 GLYCAN THERAPEUTICS FOR PD

Glycan therapeutics cover a diverse range of glycan classes and therapeutic methodologies. From the commercialization in 1936 of the sulfated glycosaminoglycan heparin as an anticoagulant, to more recent findings of dermatan sulfate that target and inhibit activity of platelet-derived growth factor (PDGF) associated in neurological disorders[291, 292], the field for glycan therapeutics is diverse and is rapidly growing[293]. Concentrating on PD specifically, and in the context of remedying aberrant glycosylation in PD glycoproteins and associated tissue, we will focus on tissue-level modulation of glycan expression using functionalized bioimplants. Recent work published by Rebelo et al. (2020) has identified and applied a glyco-functionalized collagen-based hydrogels to primary cultures to modulate their glycoprofile[294]. This work fits in-line with modern tissue engineering efforts targeting the extracellular matrix (ECM) of brain tissue where aberrant glycosylation events may be occurring[295, 296]. By adapting previous studies of modulating or re-establishing glyco-environments using natural polysaccharides in the central nervous system[297, 298], it is expected that similar work may be possible in the future focusing instead on implantable therapeutic *glycoprotein*-based hydrogels that may act as biological cues in the ECM of neuro-degenerated brain regions to re-establish and recapitulate healthy conditions. The insight provided in this project, particularly of the *O*-glycans and features of interest that were determined in ILBD and PD, will hopefully be used in the future as functionalized components of these implantable hydrogels.

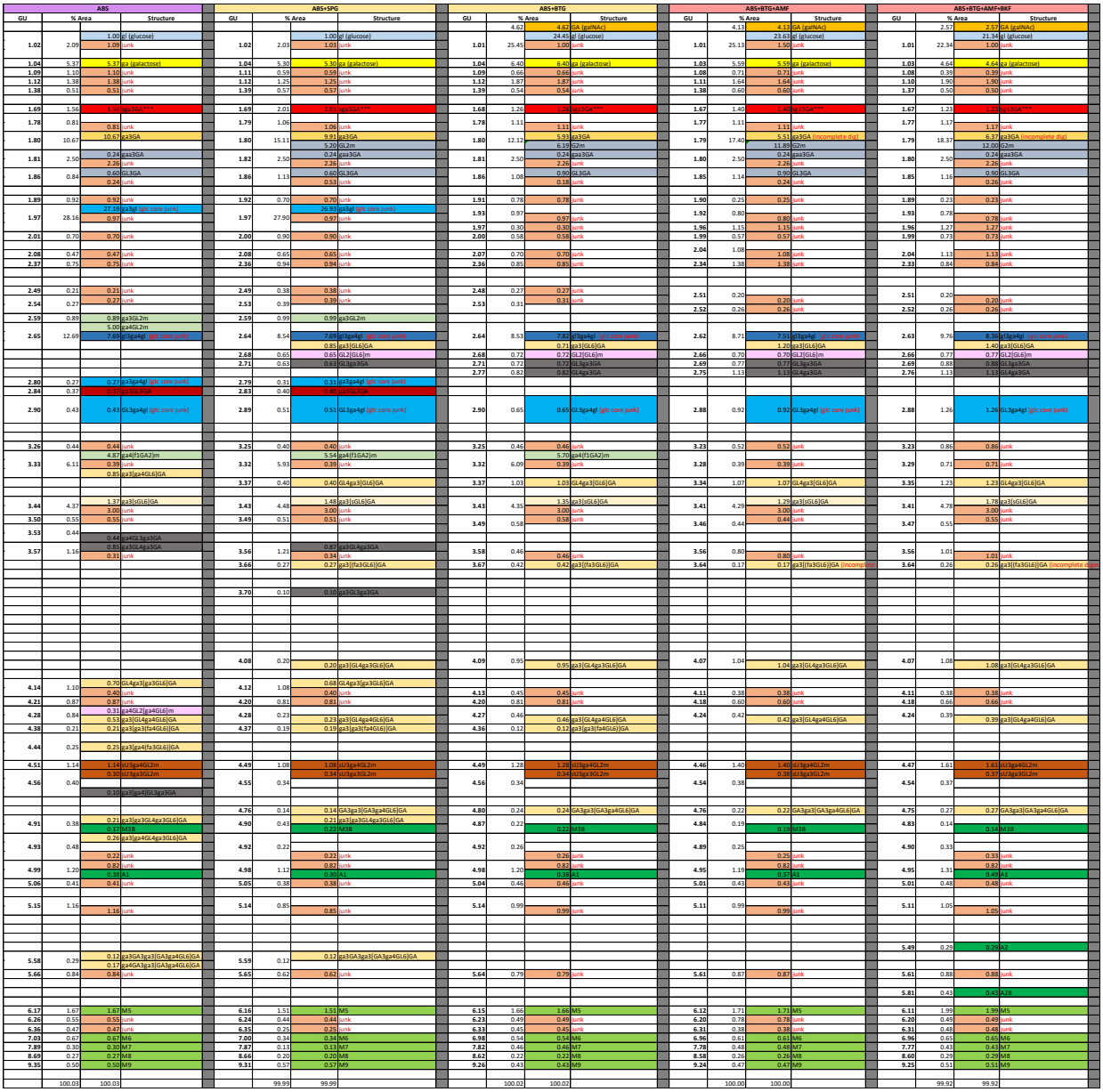
4.3 CONCLUSION

This project has shown how an optimization to an ammonia-based non-reductive β-elimination, by incorporating microwave radiation, produced *O*-glycans with minimal unwanted degradation products, allowing statistical analysis to produce meaningful results. This method was applied to study *O*-glycosylation in PD tissues, which identified *O*-glycan changes and potential biomarkers that can be targeted in the development of glyco-therapeutic treatments. Furthermore, we hope that the profiling of the *O*-glycomes from striatum and substantia nigra regions in human brain (control and PD) will help in detailing pathogenesis of PD, as well as other similar neurodegenerative diseases.

APPENDICES

Structure (long name)	Structure (image)	RT	BLANK				UND				NAN1			
			GU	% Area		Structure	GU	% Area		Structure	GU	% Area		Structure
galNAc														
glucose		1.97							1.10	gl (glucose)			1.05	gl (glucose)
			1.02	8.89	8.89	junk	**1.02**	2.29	1.19	junk	**1.02**	2.21	1.16	junk
mannose														
galactose		2.06					**1.04**	0.28	0.28	ga (galactose)	**1.04**	4.38	4.38	ga (galactose)
		2.16					**1.07**	1.15	1.15	junk	**1.09**	1.29	1.29	junk
		2.28					**1.10**	0.86	0.86	junk	**1.12**	1.45	1.45	junk
		4.01					**1.54**	1.51	1.51	junk				
Galβ1-3-SO3-GalNAc														
NeuNAc-α2-3 NeuNAc/NeuAcα2-3-SO3-GalNAc		4.94					**1.77**	2.11	1.61	S3S***	**1.78**	0.50		
			1.73	1.69	1.69	junk			0.50	junk			0.50	junk
Galβ1-3GalNAc							**1.80**	0.00			**1.80**	4.43	4.43	ga3GA
GlcNAcβ1-2Man														
Galα1-3GalNAc		5.08					**1.81**	2.94	0.68	gaa3GA	**1.82**	2.53	0.27	gaa3GA
									2.26	junk			2.26	junk
GlcNAcβ1-3GalNAc/GlcNAcβ1-6GalNAc		5.25	**1.78**	3.92	3.92	junk	**1.85**	0.82	0.60	GL3GA	**1.87**	0.73	0.50	GL3GA
									0.22	junk			0.23	junk
		5.36					**1.88**	1.14	1.14	junk				
Galβ1-3Glc		5.69	**1.93**	65.83	65.83	junk			25.69	ga3gl (glc core junk)			26.32	ga3gl (glc core junk)
							1.97	26.66	0.97	junk	**1.98**	27.29	0.97	junk
		5.84				junk	**2.00**	0.91	0.91	junk	**2.01**	1.68	1.68	junk
		6.08					**2.07**	0.87	0.87	junk	**2.08**	0.44	0.44	junk
		7.23	**2.32**	2.73	2.73	junk	**2.36**	1.70	1.70	junk	**2.37**	1.11	1.11	junk
NeuAcα2-3Gal		7.48					**2.43**	4.92	3.42	S3ga*	**2.44**	1.50		
NeuAcα2-6GalNAc									1.50	S6GA			1.50	S6GA
		7.71					**2.49**	0.65	0.65	junk	**2.49**	0.29	0.29	junk
		7.86	**2.45**	1.45	1.45	junk	**2.53**	0.47	0.47	junk	**2.54**	0.34	0.34	junk
Galβ1-3GlcNAcβ1-2Man		8.10					**2.59**	0.20	0.20	ga3GL2m	**2.60**	0.58	0.58	ga3GL2m
Galβ1-4GlcNAcβ1-2Man		8.29	**2.62**	3.7					1.40	ga4GL2m			4.64	ga4GL2m
Glcβ1-4Galβ1-3Glc					3.70	junk	**2.64**	9.09	7.69	a3ga4gl (glc core junk)	**2.65**	12.33	7.69	a3ga4gl (glc core junk)
Galβ1-3[GlcNAcβ1-6]GalNAc														
GlcNAcβ1-2[GlcNAcβ1-6]Man														
GlcNAcβ1-3Galβ1-3GalNAc														
GlcNAcβ1-4Galβ1-3GalNAc														
Galβ1-4Galβ1-3Glc		8.83	**2.75**	2.11	2.11	junk	**2.79**	0.47	0.47	ga3ga4gl (glc core junk)	**2.80**	0.34	0.34	ga3ga4gl (glc core junk)
Galβ1-3GlcNAcβ1-3GalNAc		8.98					**2.83**	0.27	0.27	ga3GL3GA	**2.84**	0.27	0.27	ga3GL3GA
GlcNAcβ1-4Galβ1-3Glc											**2.90**	0.50	0.50	GL3ga4gl (glc core junk)
NeuAcα2-6Galβ1-3GalNAc		9.51					**2.98**	0.65	0.65	S6ga3GA	**2.99**	0.55	0.55	S6ga3GA
NeuAcα2-3Galβ1-3GalNAc		9.81					**3.07**	3.02	3.02	S3ga3GA				
			3.31	0.41	0.41	junk			0.19	junk	**3.26**	0.28	0.28	junk
Galβ1-4(Fucα1-3GlcNAcβ1-2)Man		10.64					**3.32**	5.31	4.92	ga4(f1GA2)m			5.21	ga4(f1GA2)m
			3.33	0.4	0.40	junk			0.20	junk	**3.33**	6.45	0.39	junk
Galβ1-3[Galβ1-4GlcNAcβ1-6]GalNAc													0.85	ga3[ga4GL6]GA
GlcNAcβ1-4Galβ1-3[GlcNAcβ1-6]GalNAc														
Galβ1-3[NeuAcα2-6]GalNAc		10.93	**3.39**	0.62	0.62	junk	**3.41**	1.23	1.23	ga3[S6]GA	**3.42**	6.57	6.57	ga3[S6]GA
Galβ1-3[SO3-GlcNAcβ1-6]GalNAc		10.99	**3.42**	0.43	0.43	junk	**3.43**	4.75	1.75	ga3[sGL6]GA	**3.44**	4.60	1.60	ga3[sGL6]GA
			3.49	1.39	1.39	junk			3.00	junk			3.00	junk
		11.15	**3.54**	0.93	0.93	junk	**3.48**	0.59	0.59	junk	**3.50**	0.58	0.58	junk
SO3-Galβ1-3[NeuAcα2-6]GalNAc		11.27					**3.52**	0.33			**3.53**	0.50	0.25	sga3[S6]GA
Galβ1-4GlcNAcβ1-3Galβ1-3GalNAc									0.33	ga4GL3ga3GA			0.25	ga4GL3ga3GA
Galβ1-3GlcNAcβ1-4Galβ1-3GalNAc		11.39	**3.6**	1.29	1.29	junk	**3.56**	1.10	0.85	ga3GL4ga3GA	**3.58**	1.11	0.86	ga3GL4ga3GA
									0.25	junk			0.25	junk
Galβ1-3[(Fucα1-3GlcNAcβ1-6)]GalNAc														
NeuAcα2-6Galβ1-3GlcNAcβ1-2Man		11.61					**3.63**	0.53	0.53	S6ga3GL2m	**3.68**	0.32	0.32	S6ga3GL2m
NeuAcα2-3Galβ1-4GlcNAcβ1-2Man		11.93					**3.74**	3.16	2.89	S3ga4GL2m	**3.72**	0.27		
Galβ1-3GlcNAcβ1-3Galβ1-3GalNAc														
NeuAcα2-6Galβ1-4GlcNAcβ1-2Man									0.27	S6ga4GL2m			0.27	S6ga4GL2m
NeuAcα2-3Galβ1-3GlcNAcβ1-2Man		12.08					**3.80**	0.79	0.79	S3ga3GL2m				
			3.85	0.34	0.34	junk								
			4.01	0.31	0.31	junk								
SO3-Galβ1-3[NeuAcα2-6]GalNAc		12.55					**3.97**	0.30	0.30	ga3[s6]GA	**3.99**	0.62	0.47	ga3[s6]GA
Galβ1-3GlcNAcβ1-3[NeuAcα2-6]GalNAc													0.15	ga3GL4[s6]GA
			4.05	0.77	0.77	junk								
Galβ1-3[GlcNAcβ1-4Galβ1-3GlcNAcβ1-6]GalNAc														
NeuAcα2-3GalNAcβ1-4Galβ1-3Glc		12.88					**4.09**	0.50	0.50	S3GL3ga4gl (glc core junk)				
GlcNAcβ1-4Galβ1-3[Galβ1-3GlcNAcβ1-6]GalNAc		12.96					**4.12**	1.04	0.60	GL4ga3[ga3GL6]GA	**4.13**	1.05	0.60	GL4ga3[ga3GL6]GA
			4.16	0.46	0.46	junk			0.44	junk			0.45	junk
		13.13	**4.25**	0.18	0.18	junk	**4.19**	0.79	0.79	junk	**4.21**	0.97	0.97	junk
Galβ1-4GlcNAcβ1-2[Galβ1-4GlcNAcβ1-6]Man		13.32	**4.27**	0.39	0.39	junk	**4.27**	0.33			**4.28**	0.60	0.30	ga4GL2[ga4GL6]m
Galβ1-3[GlcNAcβ1-4Galβ1-4GlcNAcβ1-6]GalNAc		13.47							0.33	ga3[GL4ga4GL6]GA			0.30	ga3[GL4ga4GL6]GA
Galβ1-3[(Galβ1-3Fucα1-4GlcNAcβ1-6)]GalNAc		13.55					**4.36**	0.22	0.22	ga3[ga3(fa4GL6)]GA	**4.39**	0.27	0.27	ga3[ga3(fa4GL6)]GA
Galβ1-3[NeuAcα2-3Galβ1-4GlcNAcβ1-6]GalNAc		13.69							0.26	ga3[S3ga4GL6]GA				
Galβ1-3[(Galβ1-4Fucα1-3GlcNAcβ1-6)]GalNAc							**4.42**	0.80	0.32	ga3[ga4(fa3GL6)]GA	**4.41**	0.30	0.30	ga3[ga4(fa3GL6)]GA
NeuAcα2-3Galβ1-4(Fucα1-3GlcNAcβ1-2)Man									0.22	S3ga4(fa3GL2)m				
SO3-GlcAβ1-3Galβ1-4GlcNAcβ1-2Man		13.83					**4.48**	1.18	1.18	sU3ga4GL2m	**4.51**	1.14	1.14	sU3ga4GL2m
SO3-GlcAβ1-3Galβ1-3GlcNAcβ1-2Man		14.04							0.32	sU3ga3GL2m			0.32	sU3ga3GL2m
NeuAcα2-3Galβ1-3[Galβ1-4GlcNAcβ1-6]GalNAc							**4.56**	0.67	0.25	S3ga3[ga4GL6]GA	**4.58**	0.42		
Galβ1-3[Galβ1-3]GlcNAcβ1-3Galβ1-3GalNAc			**4.48**	0.33	0.33	junk			0.10	ga3[ga4]GL3ga3GA			0.10	ga3[ga4]GL3ga3GA
NeuAcα2-3Galβ1-3[NeuAcα2-6]GalNAc		14.37					**4.71**	6.08	6.08	S3ga3[S6]GA				
GalNAcβ1-3Galβ1-3[GalNAcβ1-3Galβ1-4GlcNAcβ1-6]GalNAc														
Galβ1-3[Galβ1-3GlcNAcβ1-4Galβ1-3GlcNAcβ1-6]GalNAc		14.71					**4.86**	0.16			**4.92**	0.38	0.25	ga3[ga3GL4ga3GL6]GA
M3B									0.14	M3B			0.11	M3B
Galβ1-4[Galβ1-3GlcNAcβ1-4Galβ1-3GlcNAcβ1-6]GalNAc		14.80							0.28	ga3[ga4GL4ga3GL6]GA			0.26	ga3[ga4GL4ga3GL6]GA
NeuAcα2-6GlcNAcβ1-4Galβ1-3[Galβ1-4GlcNAcβ1-6]GalNAc			**4.91**	0.14	0.14	junk	**4.90**	0.48			**4.94**	0.56	0.10	S6GL4ga3[ga3GL6]GA
									0.20	junk			0.20	junk
		14.92	**4.95**	0.54	0.54	junk	**4.96**	1.16	0.82	junk	**5.00**	1.24	0.82	junk
α1									0.34	A1			0.42	A1
		15.07					**5.03**	0.36	0.36	junk	**5.06**	0.47	0.47	junk
NeuAcα2-3Galβ1-3GlcNAcβ1-3[NeuAcα2-6]GalNAc							**5.05**	0.20	0.20	S3ga3GL3[S6]GA				
NeuAcα2-3(SO3-)Galβ1-3[NeuAcα2-6]GalNAc		15.25							0.29	S3sga3[S6]GA				
							5.11	1.20	0.91	junk	**5.16**	1.01	1.01	junk
NeuAcα2-3(SO3-)Galβ1-3[NeuAcα2-6]GalNAc		15.41					**5.19**	0.06	0.06	S3sga3[S6]GA				
NeuAcα2-3Galβ1-4GlcNAcβ1-2[Galβ1-4GlcNAcβ1-6]Man		15.48					**5.23**	0.30	0.30	S3-ga4GL2[ga4GL6]m				
NeuAcα2-3Galβ1-3[NeuAcα2-3Galβ1-4GlcNAcβ1-6]GalNAc		15.62					**5.30**	0.07	0.07	S3ga3[S3ga4GL6]GA				
α2														
Galβ1-3GalNAcβ1-3Galβ1-3[GalNAcβ1-3Galβ1-4GlcNAcβ1-6]GalNAc		16.15					**5.57**	0.26	0.11	ga3GA3ga3[GA3ga4GL6]GA	**5.59**	0.39	0.16	ga3GA3ga3[GA3ga4GL6]GA
Galβ1-4GalNAcβ1-3Galβ1-3[GalNAcβ1-3Galβ1-4GlcNAcβ1-6]GalNAc									0.15	ga4GA3ga3[GA3ga4GL6]GA			0.23	ga4GA3ga3[GA3ga4GL6]GA
		16.24					**5.62**	0.53	0.53	junk	**5.66**	0.82	0.82	junk
NeuAcα2-3Galβ1-3[NeuAcα2-3Galβ1-4GlcNAcβ1-6]GalNAc		16.46					**5.74**	0.13	0.13	S3ga3[sS6]GA				
NeuAcα2-3-Galβ1-3[Galβ1-3GlcNAcβ1-4Galβ1-4GlcNAcβ1-6]GalNAc		16.66					**5.84**	0.15	0.15	S3-ga3[ga3GL4ga4GL6]GA				
NeuAcα2-6GlcNAcβ1-4Galβ1-3[NeuAcα2-3Galβ1-4GlcNAcβ1-6]GalNAc		16.75					**5.89**	0.12	0.12	S6GL4ga3[S3ga3GL6]GA				
M5		17.16					**6.12**	1.08	1.08	M5	**6.18**	1.56	1.56	M5
		17.30					**6.20**	0.30	0.30	junk	**6.26**	0.55	0.55	junk
		17.48					**6.31**	0.14	0.14	junk	**6.37**	0.31	0.31	junk
M6		18.52					**6.96**	0.13	0.13	M6	**7.03**	0.59	0.59	M6
M7		19.35					**7.52**	0.10	0.10	M7	**7.87**	0.28	0.28	M7
M8		20.82					**8.60**	0.10	0.10	M8	**8.70**	0.29	0.29	M8
M9		21.60					**9.25**	0.29	0.29	M9	**9.35**	0.63	0.63	M9
				99.25	99.25			100.00	100.00			99.87	99.87	

Appendix I. *Exoglycosidase digestion panel for healthy striatum pool.*



ABS				ABS+SPG				ABS+BTG				ABS+BTG+AMF				ABS+BTG+AMF+BKF			
GU	% Area		Structure	GU	% Area		Structure	GU	% Area		Structure	GU	% Area		Structure	GU	% Area		Structure
		1.00	gl (glucose)			1.00	gl (glucose)		4.62	4.62	GA (galNAc)		4.13	4.13	GA (galNAc)		2.57	2.57	GA (galNAc)
1.02	2.09	1.09	unk	1.02	2.03	1.03	unk	1.01	25.45	24.45	gl (glucose)	1.01	25.13	23.63	gl (glucose)	1.01	22.34	21.34	gl (glucose)
										1.00	unk			1.50	unk			1.00	unk
1.04	5.37	5.37	ga (galactose)	1.04	5.30	5.30	ga (galactose)	1.04	6.40	6.40	ga (galactose)	1.03	5.59	5.59	ga (galactose)	1.03	4.64	4.64	ga (galactose)
1.09	1.10	1.10	unk	1.11	0.59	0.59	unk	1.09	0.66	0.66	unk	1.08	0.71	0.71	unk	1.08	0.39	0.39	unk
1.12	1.38	1.38	unk	1.12	1.25	1.25	unk	1.12	1.87	1.87	unk	1.11	1.64	1.64	unk	1.10	1.90	1.90	unk
1.38	0.51	0.51	unk	1.39	0.57	0.57	unk	1.39	0.54	0.54	unk	1.38	0.60	0.60	unk	1.37	0.50	0.50	unk
1.69	1.56	1.56	ga3GA***	1.69	2.01	2.01	ga3GA***	1.68	1.26	1.26	ga3GA***	1.67	1.40	1.40	ga3GA***	1.67	1.23	1.23	ga3GA***
1.78	0.81	0.81	unk	1.79	1.06	1.06	unk	1.78	1.11	1.11	unk	1.77	1.11	1.11	unk	1.77	1.17	1.17	unk
1.80	10.67	10.67	ga3GA	1.80	15.11	9.91	ga3GA	1.80	12.12	5.93	ga3GA	1.79	17.40	5.51	ga3GA (incomplete dig)	1.79	18.37	6.37	ga3GA (incomplete dig)
						5.20	G2m			6.19	G2m			11.89	G2m			12.00	G2m
1.81	2.50	0.24	ga23GA	1.82	2.50	0.24	ga23GA	1.81	2.50	0.24	ga23GA	1.80	2.50	0.24	ga23GA	1.80	2.50	0.24	ga23GA
		2.26	unk			2.26	unk			2.26	unk			2.26	unk			2.26	unk
1.86	0.84	0.60	GL3GA	1.86	1.13	0.60	GL3GA	1.86	1.08	0.90	GL3GA	1.85	1.14	0.90	GL3GA	1.85	1.16	0.90	GL3GA
		0.24	unk			0.53	unk			0.18	unk			0.24	unk			0.26	unk
1.89	0.92	0.92	unk	1.92	0.70	0.70	unk	1.91	0.78	0.78	unk	1.90	0.25	0.25	unk	1.89	0.23	0.23	unk
		27.19	ga3gl (glc core junk)			26.93	ga3gl (glc core junk)	1.93	0.97			1.92	0.80			1.93	0.78		
1.97	28.16	0.97	unk	1.97	27.90	0.97	unk			0.97	unk			0.80	unk			0.78	unk
								1.97	0.30	0.30	unk	1.96	1.15	1.15	unk	1.96	1.27	1.27	unk
2.01	0.70	0.70	unk	2.00	0.90	0.90	unk	2.00	0.58	0.58	unk	1.99	0.57	0.57	unk	1.99	0.73	0.73	unk
												2.04	1.08						
2.08	0.47	0.47	unk	2.08	0.65	0.65	unk	2.07	0.70	0.70	unk			1.08	unk	2.04	1.13	1.13	unk
2.37	0.75	0.75	unk	2.36	0.94	0.94	unk	2.36	0.85	0.85	unk	2.34	1.38	1.38	unk	2.33	0.84	0.84	unk
2.49	0.21	0.21	unk	2.49	0.38	0.38	unk	2.48	0.27	0.27	unk								
2.54	0.27	0.27	unk	2.53	0.39	0.39	unk	2.53	0.31	0.31	unk	2.51	0.20			2.51	0.26		
												2.52	0.26	0.26	unk	2.52	0.26	0.26	unk
2.59	0.89	0.89	ga3GL2m	2.59	0.99	0.99	ga3GL2m												
		5.00	ga4GL2m																
2.65	12.69	7.69	gl3ga4gl (glc core junk)	2.64	8.54	7.69	gl3ga4gl (glc core junk)	2.64	8.53	7.82	gl3ga4gl (glc core junk)	2.62	8.71	7.51	gl3ga4gl (glc core junk)	2.63	9.76	8.36	gl3ga4gl (glc core junk)
						0.85	ga3(GL6)GA			0.71	ga3(GL6)GA			1.20	ga3(GL6)GA			1.40	ga3(GL6)GA
				2.68	0.65	0.65	GL2(GL6)m	2.68	0.72	0.72	GL2(GL6)m	2.66	0.70	0.70	GL2(GL6)m	2.66	0.77	0.77	GL2(GL6)m
				2.71	0.63	0.63	GL3ga3GA	2.71	0.72	0.72	GL3ga3GA	2.69	0.77	0.77	GL3ga3GA	2.69	0.88	0.88	GL3ga3GA
								2.77	0.82	0.82	GL4ga3GA	2.75	1.13	1.13	GL4ga3GA	2.76	1.13	1.13	GL4ga3GA
2.80	0.27	0.27	GL3ga4gl (glc core junk)	2.79	0.31	0.31	ga3(ga4gl (glc core junk)												
2.84	0.37	0.37	ga3ga2 3GL2	2.83	0.40	0.40	ga23GA2												
2.90	0.43	0.43	GL3ga4gl (glc core junk)	2.89	0.51	0.51	GL3ga4gl (glc core junk)	2.90	0.65	0.65	GL3ga4gl (glc core junk)	2.88	0.92	0.92	GL3ga4gl (glc core junk)	2.88	1.26	1.26	GL3ga4gl (glc core junk)
3.26	0.44	0.44	unk	3.25	0.40	0.40	unk	3.25	0.46	0.46	unk	3.23	0.52	0.52	unk	3.23	0.86	0.86	unk
		4.87	ga4(f1GA2)m			5.54	ga4(f1GA2)m			5.70	ga4(f1GA2)m								
3.33	6.11	0.39	unk	3.32	5.93	0.39	unk	3.32	6.09	0.39	unk	3.28	0.39	0.39	unk	3.29	0.71	0.71	unk
		0.85	ga3[ga4GL6)GA																
				3.37	0.40	0.40	GL4ga3(GL6)GA	3.37	1.03	1.03	GL4ga3(GL6)GA	3.34	1.07	1.07	GL4ga3(GL6)GA	3.35	1.23	1.23	GL4ga3(GL6)GA
		1.37	ga3(sGL6)GA			1.48	ga3(sGL6)GA			1.35	ga3(sGL6)GA			1.29	ga3(sGL6)GA			1.78	ga3(sGL6)GA
3.44	4.37	3.00	unk	3.43	4.48	3.00	unk	3.43	4.35	3.00	unk	3.41	4.29	3.00	unk	3.41	4.78	3.00	unk
3.50	0.55	0.55	unk	3.49	0.51	0.51	unk			0.58	unk			0.44	unk			0.55	unk
3.53	0.44							3.49	0.58			3.46	0.44			3.47	0.53		
		0.44	ga4GL3ga3GA																
3.57	1.16	0.85	ga3GL4ga2GA			0.87	ga3GL4ga3GA												
		0.31	unk	3.56	1.21	0.34	unk	3.58	0.46	0.46	unk	3.56	0.80	0.80	unk	3.56	1.01	1.01	unk
				3.66	0.27	0.27	ga3[(fa3GL6)GA	3.67	0.42	0.42	ga3[(fa3GL6))GA	3.64	0.17	0.17	ga3[(fa3GL6))GA (incomplete dig)	3.64	0.26	0.26	ga3[(fa3GL6))GA (incomplete dig)
				3.70	0.10	0.10	ga3GL3ga3GA												
				4.08	0.20	0.20	ga3(GL4ga3GL6)GA	4.09	0.95	0.95	ga3(GL4ga3GL6)GA	4.07	1.04	1.04	ga3(GL4ga3GL6)GA	4.07	1.08	1.08	ga3(GL4ga3GL6)GA
4.14	1.10	0.70	GL4ga3[ga3GL6)GA	4.12	1.08	0.68	GL4ga3[ga3GL6)GA												
		0.40	unk			0.40	unk	4.13	0.45	0.45	unk	4.11	0.38	0.38	unk	4.11	0.38	0.38	unk
4.21	0.87	0.87	unk	4.20	0.81	0.81	unk	4.20	0.81	0.81	unk	4.18	0.60	0.60	unk	4.18	0.66	0.66	unk
4.28	0.84	0.31	ga4GL2[ga4GL6)m	4.28	0.23			4.27	0.46			4.24	0.42			4.24	0.35		
		0.53	ga3(GL4ga4GL6)GA			0.23	ga3(GL4ga4GL6)GA			0.46	ga3(GL4ga4GL6)GA			0.42	ga3(GL4ga4GL6)GA			0.39	ga3(GL4ga4GL6)GA
4.38	0.21	0.21	ga3(ga3(fa4GL6))GA	4.37	0.19	0.19	ga3(ga3(fa4GL6))GA	4.36	0.12	0.12	ga3(ga3(fa4GL6))GA								
4.44	0.25	0.25	ga3[ga4(fa3GL6)]GA																
4.51	1.14	1.14	ul3ga4GL2m	4.49	1.08	1.08	ul3ga4GL2m	4.49	1.28	1.28	ul3ga4GL2m	4.46	1.40	1.40	ul3ga4GL2m	4.47	1.61	1.61	ul3ga4GL2m
4.56	0.40	0.30	ul3ga3GL2m	4.55	0.34	0.34	ul3ga3GL2m	4.56	0.34	0.34	ul3ga3GL2m	4.54	0.38	0.38	ul3ga3Gl2m	4.54	0.37	0.37	ul3ga3Gl2m
		0.10	ga3[(ga4)GL3)g3GA																
				4.76	0.14	0.14	GA3ga3[GA3ga4GL6)GA	4.80	0.24	0.24	GA3ga3[GA3ga4GL6)GA	4.76	0.22	0.22	GA3ga3[(GA3ga4GL6)GA	4.75	0.27	0.27	GA3ga3[GA3ga4GL6)GA
4.91	0.38	0.21	ga3[ga3GL4ga3GL6)GA	4.90	0.43	0.21	ga3[ga3GL4ga3GL6)GA	4.87	0.22			4.84	0.19			4.83	0.14		
		0.17	M3B			0.22	M3B			0.22	M3B			0.14	M3B			0.14	M3B
4.93	0.48	0.26	ga3[ga4GL4ga3GL6)GA	4.92	0.22			4.92	0.26			4.89	0.25			4.90	0.33		
		0.22	unk			0.22	unk			0.26	unk			0.25	unk			0.33	unk
		0.82	unk			0.82	unk			0.82	unk			0.82	unk			0.82	unk
4.99	1.20	0.38	A1	4.98	1.12	0.30	A1	4.98	1.20	0.38	A1	4.95	1.19	0.37	A1	4.95	1.31	0.49	A1
5.06	0.41	0.41	unk	5.05	0.38	0.38	unk	5.04	0.46	0.46	unk	5.01	0.43	0.43	unk	5.01	0.48	0.48	unk
5.15	1.16	1.16	unk	5.14	0.85	0.85	unk	5.14	0.99	0.99	unk	5.11	0.99	0.99	unk	5.11	1.05	1.05	unk
																5.49	0.29	0.29	A2
5.58	0.29	0.12	ga3GA3ga3(GA3ga4GL6)GA	5.59	0.12	0.12	ga3GA3ga3(GA3ga4GL6)GA												
		0.17	ga4GA3ga3(GA3ga4GL6)GA																
5.66	0.84	0.84	unk	5.65	0.62	0.62	unk	5.64	0.79	0.79	unk	5.61	0.87	0.87	unk	5.61	0.88	0.88	unk
																5.81	0.43	0.43	A2
6.17	1.67	1.67	M5	6.16	1.51	1.51	M5	6.15	1.66	1.66	M5	6.12	1.71	1.71	M5	6.11	1.99	1.99	M5
6.26	0.55	0.55	unk	6.24	0.44	0.44	unk	6.23	0.49	0.49	unk	6.20	0.78	0.78	unk	6.20	0.49	0.49	unk
6.36	0.47	0.47	unk	6.35	0.25	0.25	unk	6.33	0.45	0.45	unk	6.31	0.38	0.38	unk	6.31	0.48	0.48	unk
7.03	0.67	0.67	M6	7.00	0.34	0.34	M6	6.98	0.54	0.54	M6	6.96	0.61	0.61	M6	6.96	0.65	0.65	M6
7.89	0.30	0.30	M7	7.87	0.13	0.13	M7	7.82	0.46	0.46	M7	7.78	0.48	0.48	M7	7.77	0.43	0.43	M7
8.69	0.27	0.27	M8	8.66	0.20	0.20	M8	8.62	0.22	0.22	M8	8.58	0.26	0.26	M8	8.60	0.29	0.29	M8
9.35	0.50	0.50	M9	9.31	0.57	0.57	M9	9.26	0.43	0.43	M9	9.24	0.47	0.47	M9	9.25	0.51	0.51	M9
	100.03	100.03			99.99	99.99			100.02	100.02			100.00	100.00			99.92	99.92	

Appendix I (cont.). *Exoglycosidase digestion panel for healthy striatum pool.*

ABS+BTG+AMF+BKF+GUH				ABS+BTG+AMF+BKF+GUH+JBH				ABS+BTG+AMF+BKF+GUH+JBH+CBG				ABS+BTG+AMF+BKF+GUH+JBH+CBG+JBM			
GU	% Area		Structure	GU	% Area		Structure	GU	% Area		Structure	GU	% Area		Structure
	6.26	6.26	GA (galNAc)		5.10	5.10	GA (galNAc)		4.75	4.75	GA (galNAc)		3.63	3.63	GA (galNAc)
		23.09	gl (glucose)			23.39	gl (glucose)			22.11	gl (glucose)			22.07	gl (glucose)
1.02	38.59	2.00	junk	1.02	38.89	1.50	junk	1.01	37.36	1.25	junk	1.02	36.77	0.70	junk
		13.50	m (mannose)			14.00	m (mannose)			14.00	m (mannose)			14.00	m (mannose)
1.03	4.70	4.70	ga (galactose)	1.03	4.17	4.17	ga (galactose)	1.03	4.85	4.85	ga (galactose)	1.04	5.38	5.38	ga (galactose)
1.09	0.90	0.90	junk	1.09	0.90	0.90	junk	1.08	0.56	0.56	junk	1.09	0.37	0.37	junk
1.12	1.60	1.60	junk	1.12	1.29	1.29	junk	1.11	1.65	1.65	junk	1.15	4.06	4.06	junk
1.39	0.49	0.49	junk	1.39	0.54	0.54	junk	1.39	0.57	0.57	junk	1.41	0.72	0.72	junk
1.68	1.57	1.57	sga3GA***	1.68	1.80	1.80	sga3GA***	1.67	1.59	1.59	sga3GA***	1.70	1.54	1.54	sga3GA***
1.77	0.81	0.81	junk	1.77	0.81	0.81	junk	1.76	0.92	0.92	junk	1.81	1.45	1.45	junk
1.79	7.72	7.72	ga3GA (incomplete dig)	1.79	8.11	8.11	ga3GA (incomplete dig)	1.78	8.65	8.65	ga3GA (incomplete dig)			8.94	ga3GA (incomplete dig)
1.81	2.47	0.21	gaa3GA	1.80	2.52	0.26	gaa3GA	1.79	2.26			1.85	11.20		
		2.26	junk			2.26	junk			2.26	junk			2.26	junk
1.85	0.47	0.47	junk	1.85	0.52	0.52	junk	1.84	0.55	0.55	junk	1.88	0.89	0.59	junk
1.87	0.67	0.67	junk	1.86	0.27	0.27	junk	1.85	0.24	0.24	junk			0.30	junk
1.91	0.24	0.24	junk	1.91	0.24	0.24	junk	1.90	0.25	0.25	junk	1.95	1.13	1.13	junk
1.93	0.52	0.52	junk	1.93	0.59	0.59	junk	1.92	0.66	0.66	junk				
1.96	0.26	0.26	junk	1.96	0.60	0.60	junk	1.95	0.71	0.71	junk			1.64	junk
1.97	1.31	1.31	junk	1.97	1.29	1.29	junk	1.98	1.68	1.68	junk	1.97	1.64		
2.00	0.69	0.69	junk	1.99	0.70	0.70	junk								
2.05	1.21	1.21	junk	2.05	1.29	1.29	junk	2.04	1.22	1.22	junk	2.01	1.98	1.98	junk
2.34	0.56	0.56	junk	2.34	0.66	0.66	junk	2.32	0.76	0.76	junk	2.34	0.83	0.83	junk
2.47	0.22	0.22	junk	2.47	0.23	0.23	junk	2.47	0.20	0.20	junk				
2.51	0.19	0.19	junk	2.52	0.27	0.27	junk	2.50	0.44	0.44	junk	2.51	0.40	0.40	junk
2.52	0.20	0.20	junk	2.53	0.20	0.20	junk								
2.63	8.29	8.29	gl3ga4gl (glc core junk)	2.63	8.39	8.39	gl3ga4gl (glc core junk)	2.62	8.50	8.50	gl3ga4gl (glc core junk)	2.62	8.10	8.10	gl3ga4gl (glc core junk)
												2.67	3.48	3.48	M1
2.88	0.65	0.65	GL3ga4gl (glc core junk) (incomplete dig)	2.88	0.71	0.71	GL3ga4gl (glc core junk) (incomplete dig)	2.87	0.72	0.72	GL3ga4gl (glc core junk) (incomplete dig)	2.88	0.70	0.70	GL3ga4gl (glc core junk) (incomplete dig)
3.24	0.59	0.59	junk	3.24	0.70	0.70	junk	3.22	0.79	0.79	junk	3.24	1.34	1.34	junk
3.29	0.61	0.61	junk	3.29	0.61	0.61	junk	3.27	0.60	0.60	junk	3.27	0.62	0.62	junk
3.42	4.51	1.51	ga3[sGL6]GA	3.42	4.48	1.48	ga3[sGL6]GA	3.40	4.57	1.57	ga3[sGL6]GA	3.41	4.59	1.59	ga3[sGL6]GA
		3.00	junk			3.00	junk			3.00	junk			3.00	junk
3.47	0.45	0.45	junk	3.47	0.48	0.48	junk	3.45	0.49	0.49	junk	3.46	0.35	0.35	junk
3.57	0.77	0.77	junk	3.56	0.76	0.76	junk	3.54	0.89	0.89	junk	3.57	0.78	0.78	junk
3.65	0.25	0.25	ga3[(fa3GL6)]GA (incomp	3.64	0.23	0.23	ga3[(fa3GL6)]GA (incomp	3.63	0.28	0.28	ga3[(fa3GL6)]GA (incomplete d	3.64	0.30	0.30	ga3[(fa3GL6)]GA (incomplete dige
				3.86	0.18	0.18	junk					3.87	0.11	0.11	junk
								4.03	0.29	0.29	junk				
								4.09	0.52	0.52	junk	4.11	0.60	0.60	junk
4.12	0.36	0.36	junk	4.11	0.38	0.38	junk	4.16	0.67	0.67	junk	4.19	0.57	0.57	junk
4.19	0.58	0.58	junk	4.18	0.73	0.73	junk	4.24	0.28	0.28	junk	4.27	0.13	0.13	junk
												4.29	0.12	0.12	junk
4.40	0.59	0.59	M3	4.39	0.67	0.67	M3	4.37	0.49	0.49	M3				
4.48	1.67	1.67	sU3ga4GL2m	4.47	1.72	1.72	sU3ga4GL2m	4.44	1.80	1.80	sU3ga4GL2m	4.49	1.71	1.71	sU3ga4GL2m
		0.39	sU3ga3GL2m			0.41	sU3ga3GL2m			0.41	sU3ga3GL2m			0.36	sU3ga3GL2m
4.54	0.39			4.53	0.41			4.50	0.41			4.57	0.36		
4.74	0.20	0.20	GA3ga3[GA3ga4GL6]GA												
4.85	0.36	0.36	M3B	4.84	0.53	0.53	M3B	4.82	0.72	0.72	M3B				
4.90	0.28			4.89	0.34			4.87	0.28			4.92	0.26		
		0.28	junk			0.34	junk			0.28	junk			0.26	junk
4.96	0.82	0.82	junk	4.95	0.66	0.66	junk	4.92	0.77	0.77	junk	4.97	0.69	0.69	junk
5.02	0.32	0.32	junk	5.01	0.37	0.37	junk	4.98	0.34	0.34	junk	5.02	0.37	0.37	junk
5.12	1.00	1.00	junk	5.11	1.01	1.01	junk	5.08	0.99	0.99	junk	5.15	0.98	0.98	junk
5.63	0.80	0.80	junk	5.61	0.72	0.72	junk	5.58	0.82	0.82	junk	5.67	0.75	0.75	junk
5.82	0.48	0.48	A2B (incomplete digestion	5.81	0.36	0.36	A2B (incomplete digestion	5.78	0.23	0.23	A2B (incomplete digestion	5.87	0.27	0.27	A2B (incomplete digestion
6.13	1.75	1.75	M5	6.11	1.70	1.70	M5	6.08	1.73	1.73	M5				
6.21	0.38	0.38	junk	6.19	0.58	0.58	junk	6.16	0.61	0.61	junk	6.27	0.49	0.49	junk
6.32	0.41	0.41	junk	6.30	0.37	0.37	junk	6.27	0.38	0.38	junk	6.39	0.34	0.34	junk
6.97	0.60	0.60	M6	6.95	0.70	0.70	M6	6.92	0.56	0.56	M6				
7.84	0.34	0.34	M7	7.80	0.32	0.32	M7	7.77	0.43	0.43	M7				
8.62	0.34	0.34	M8	8.58	0.38	0.38	M8	8.56	0.39	0.39	M8				
9.27	0.54	0.54	M9	9.23	0.52	0.52	M9	9.20	0.58	0.58	M9				
	99.98	99.98			100.00	100.00			100.00	100.00			100.00	100.00	

Appendix I (cont.). *Exoglycosidase digestion panel for healthy striatum pool.*

											% Area
										% Area	JBM
									% Area	CBG	CBG
								% Area	JBH	JBH	JBH
							% Area	GUH	GUH	GUH	GUH
						% Area	BKF	BKF	BKF	BKF	BKF
				% Area	% Area	AMF	AMF	AMF	AMF	AMF	AMF
	% Area	% Area	% Area	SPG	BTG	BTG	BTG	BTG	BTG	BTG	BTG
	UND	NAN1	ABS	ABS	ABS	ABS	ABS	ABS	ABS	ABS	ABS
GalNAc					4.62	4.13	2.57	6.26	5.10	4.75	3.63
Gal	0.28	4.38	5.37	5.30	6.40	5.59	4.64	4.70	4.17	4.85	5.38
Glc	1.10	1.05	1.00	1.00	24.45	23.63	21.34	23.09	23.39	22.11	22.07
Man								13.50	14.00	14.00	14.00
Semi-digestible Disaccharides (ga3GA)		4.43	10.67	9.91	5.93	5.51	6.37	7.72	8.11	8.65	8.94
Digestible Disaccharides (others)	1.28	0.77	0.84	6.04	7.33	13.03	13.14	0.21	0.26		
Undigestible Disaccharides (sulfated)			1.56	2.01	1.26	1.40	1.23	1.57	1.80	1.59	1.54
Core 1 extensions	1.28	1.21	1.39	1.60	1.54	1.90	2.01				
Core 2 and extensions	2.86	3.32	3.30	3.29	3.93	4.12	4.63	0.45	0.23	0.28	0.30
Core 2 extensions with sulfation	1.75	1.60	1.37	1.48	1.35	1.29	1.78	1.51	1.48	1.57	1.59
Core 3 extensions	0.27	0.27	0.37	0.40							
Mannose Core 1 (linear trisaccharide and larger)	11.22	11.02	10.76	6.53	5.70						
Mannose Core 2 (branched trisaccharide and larger)	0.30	0.30	0.31	0.65	0.72	0.70	0.77				
SSTn, STn, ST antigens, peeling, disialyl	17.51	8.62									
STn and ST antigens (sulfated)	0.78	0.72									
Larger Sulfated Glycans that don't digest	1.50	1.46	1.44	1.42	1.62	1.78	1.98	2.06	2.13	2.21	2.07
High Mannose (N)	1.70	3.35	3.41	2.75	3.31	3.53	3.87	4.16	4.29	4.18	3.48
Hybrid (N)	0.50	0.55	0.55	0.52	0.60	0.56	1.35	0.84	0.89	0.95	0.27
Digestible hex chains	26.66	27.16	27.89	27.75	0.65	0.92	1.26	0.65	0.71	0.72	0.70
Undigestible hex chains	7.69	7.69	7.69	7.69	7.82	7.51	8.36	8.29	8.39	8.50	8.10
Integrated Junk	23.12	21.82	22.11	21.65	22.79	24.40	24.62	24.97	25.05	25.64	27.93
SUM	99.80	99.72	100.03	99.99	100.02	100.00	99.92	99.98	100.00	100.00	100.00

	% Area	Comment
Total GalNAc core	20.90	(including the 1 glycan in the larger sulfated glycans that doesn't digest)
Total Mannose core	13.02	(including the 1 glycan in the larger sulfated glycans that doesn't digest)
Total Hexose chain glycans (Glucose core)	35.45	inlcudes glc, gal-glc, glc-gal-glc and gal-gal-glc
Total Galactose core (probably peeling)	3.70	probably core 1 and 2 peeling
Total NeuNac core (polysialic)	1.61	One polysialic structure also seen in Gothenburg data
Total N glycans	2.20	leftover from Ana's digest or extra N-glycan that was released in microwave
Total Junk	23.12	Suspected tissue junk and junk present in blank
Total (should equal 100.00)	100.00	total % area from UND
Total Glycan	76.88	
Total Junk	23.12	

Appendix II. *Summary of % areas – Exoglycosidase digestion panel for healthy striatum pool.*

Structure (long name)	Structure (image)	BLANK				RT	UND				NAN1			
		GU	% Area		Structure		GU	% Area		Structure	GU	% Area		Structure
galNAc						1.73		0.14	0.14	GA (galNAc)		0.19	0.19	GA (galNAc)
glucose						1.95	1.01	2.35	2.35	gl (glucose)	1.01	2.41	2.41	gl (glucose)
mannose														
galactose						2.05	1.03	0.85	0.85	ga (galactose)	1.03	7.52	7.52	ga (galactose)
		1.08	2.01	2.01	junk	2.17	1.06	0.98	0.98	junk	1.06	0.20	0.20	junk
		1.11	13.93	13.93	junk	2.30	1.09	2.34	2.34	junk	1.09	2.90	2.90	junk
Galβ1-3-SO3-GalNAc						4.73	1.70	0.57	0.47	sga3GA***	1.70	0.59	0.49	sga3GA***
SO3-(NeuAcα2-6)GalNAc									0.10	S6sGA			0.10	S6sGA
NeuNAc-α2-3 NeuNAc/NeuAcα2-3-SO3-GalNAc		1.74	5.65	5.65	junk	4.95	1.76	2.40	2.40	S3S***				
Galβ1-3GalNAc						4.99			0.24	ga3GA			6.80	ga3GA
GlcNAcβ1-2Man														
Galα1-3GalNAc							1.77	0.72	0.24	gaa3GA	1.79	7.28	0.24	gaa3GA
SO3-GlcNAcβ1-2Man														
GlcNAcβ1-3GalNAc/GlcNAcβ1-6GalNAc									0.24	GL3/6GA			0.24	GL3/6GA
						5.14	1.81	1.93	1.93	junk	1.82	1.93	1.93	junk
						5.29	1.84	0.78	0.78	junk	1.86	0.25	0.25	junk
		1.79	13.43	13.43	junk	5.41	1.87	0.95	0.95	junk			0.52	junk
											1.90	0.52		
						5.63	1.93	0.30	0.30	junk	1.92	0.16	0.16	junk
Galβ1-3Glc		1.94	23.41	23.41	junk	5.74	1.95	5.74	5.07	ga3gl (glc core junk)	1.96	7.10	6.43	ga3gl (glc core junk)
									0.67	junk			0.67	junk
						5.89	1.99	1.35	1.35	junk	1.98	2.08	2.08	junk
		2.05	0.91	0.91	junk	6.16	2.06	0.72	0.72	junk	2.06	0.51	0.51	junk
		2.28	1.34	1.34	junk	7.22	2.33	0.55	0.55	junk	2.32	0.35	0.35	junk
NeuAcα2-3Gal		2.34	1.10	1.10	junk	7.32	2.35	1.13	0.65	S3ga* (split peak)	2.34	0.48		
NeuAcα2-6GalNAc									0.48	S6GA (split peak)			0.48	S6GA (split peak)
NeuAcα2-3Gal						7.53	2.41	9.57	7.07	S3ga*	2.40	2.50		
NeuAcα2-6GalNAc									2.50	S6GA			2.50	S6GA
Galβ1-3GlcNAcβ1-2Man											2.55	1.38	1.38	ga3GL2m
Galβ1-4GlcNAcβ1-2Man						8.32	2.61	3.55	0.65	ga4GL2m	2.61	8.49	5.59	ga4GL2m
Glcβ1-4Galβ1-3Glc		2.63	13.45	13.45	junk				2.90	gl4ga3gl (glc core junk)			2.90	gl4ga3gl (glc core junk)
Galβ1-3(GlcNAcβ1-6)GalNAc						8.44								
GlcNAcβ1-2[GlcNAcβ1-6]Man														
						8.85			0.22	junk	2.74	0.35	0.35	junk
							2.75	0.44	0.22	junk	2.79	0.45	0.45	junk
		2.76	2.30	2.30	junk									
Galβ1-3GlcNAcβ1-3GalNAc						9.26	2.87	0.18	0.18	junk	2.86	0.61	0.61	junk
NeuAcα2-3Galβ1-3GalNAc						9.59	2.96	0.91	0.20	S3ga3GA (split peak)	2.95	0.71		
NeuAcα2-6Galβ1-3GalNAc									0.71	S6ga3GA (split peak)			0.71	S6ga3GA (split peak)
NeuAcα2-3Galβ1-3GalNAc						9.83	3.03	4.77	4.00	S3ga3GA	3.01	0.77		
NeuAcα2-6Galβ1-3GalNAc									0.77	S6ga3GA			0.77	S6ga3GA
Galβ1-4(Fucα1-3GlcNAcβ1-2)Man						10.65			6.84	ga4(fa3GL2)m			8.20	ga4(fa3GL2)m
							3.27	7.61	0.77	junk	3.26	9.77	0.77	junk
Galβ1-3(Galβ1-4GlcNAcβ1-6)GalNAc													0.80	ga3[ga4GL6]GA
GlcNAcβ1-4Galβ1-3(GlcNAcβ1-6)GalNAc														
Galβ1-3[NeuAcα2-6]GalNAc		3.40	6.45	6.45	junk	10.98	3.37	3.31	2.40	ga3[S6]GA	3.37	13.55	12.64	ga3[S6]GA
Galβ1-3[SO3-GlcNAcβ1-6]GalNAc									0.91	ga3[sGL6]GA			0.91	ga3[sGL6]GA
Galβ1-4(Fucα1-3-SO3-GlcNAcβ1-2)Man						11.20	3.44	0.67	0.17	ga4(fa3sGL2)m	3.44	0.88	0.38	ga4(fa3sGL2)m
									0.50	junk			0.50	junk
SO3-Galβ1-3[NeuAcα2-6]GalNAc		3.56	1.70	1.70	junk	11.36	3.50	0.64	0.29	sga3[S6]GA	3.49	0.73	0.48	sga3[S6]GA
									0.35	junk			0.25	junk
						11.54								
							3.56	0.64	0.64	junk	3.56	0.54		
		3.62	1.69	1.69	junk								0.54	junk
											3.63	0.20	0.20	junk
									0.35	junk	3.68	0.49	0.49	junk
Galβ1-3[(Fucα1-3GlcNAcβ1-6)]GalNAc						12.01	3.72	4.89						
NeuAcα2-3Galβ1-4GlcNAcβ1-2Man									4.54	S3ga4GL2m				
NeuAcα2-3Galβ1-3GlcNAcβ1-2Man						12.16	3.77	1.11	1.11	S3ga3GL2m				
SO3-Galβ1-3[NeuAcα2-6]GalNAc						12.63	3.94	0.33	0.33	ga3[sS6]GA	3.94	0.84	0.64	ga3[sS6]GA
Galβ1-3GlcNAcβ1-3[NeuAcα2-6]GalNAc													0.20	ga3GL3[S6]GA
Galβ1-3[GlcNAcβ1-3Galβ1-3GlcNAcβ1-6]GalNAc														
GlcNAcβ1-4Galβ1-3[Galβ1-3GlcNAcβ1-6]GalNAc						12.93	4.05	0.25	0.25	GL4ga3[ga3GL6]GA	4.05	0.28	0.28	GL4ga3[ga3GL6]GA
Galβ1-3[GlcNAcβ1-4Galβ1-4GlcNAcβ1-6]GalNAc						13.01	4.08	1.15	0.81	ga3[GL4ga4GL6]GA	4.08	1.22	0.88	ga3[GL4ga4GL6]GA
									0.34	junk			0.34	junk
GlcNAcβ1-4Galβ1-3[Galβ1-4GlcNAcβ1-6]GalNAc		4.17	2.49	2.49	junk	13.13	4.13	0.49	0.17	GL4ga3[ga4GL6]GA	4.14	0.60	0.28	GL4ga3[ga4GL6]GA
									0.32	junk			0.32	junk
						13.23	4.17	0.42	0.42	junk	4.16	0.34	0.34	junk
Galβ1-4GlcNAcβ1-2[Galβ1-4GlcNAcβ1-6]Man											4.22	0.91	0.91	ga4GL2[ga4GL6]m
Galβ1-3[NeuAcα2-3Galβ1-4GlcNAcβ1-6]GalNAc						13.81			0.47	ga3[S3ga4GL6]GA				
Galβ1-3[(Galβ1-4Fucα1-3GlcNAcβ1-6)]GalNAc							4.40	1.18	0.30	ga3[ga4(fa3GL6)]GA	4.41	0.71	0.35	ga3[ga4(fa3GL6)]GA
M3									0.41	M3			0.36	M3
SO3-GlcAβ1-3Galβ1-4GlcNAcβ1-2Man						14.01	4.48	1.96	1.96	sU3ga4GL2m	4.48	1.83	1.83	sU3ga4GL2m
SO3-GlcAβ1-3Galβ1-3GlcNAcβ1-2Man						14.17	4.55	0.58	0.58	sU3ga3GL2m	4.55	0.72	0.72	sU3ga3GL2m
NeuAcα2-3Galβ1-3[Galβ1-4GlcNAcβ1-6]GalNAc						14.24	4.58	0.33	0.33	S3ga3[ga4GL6]GA				
NeuAcα2-3Galβ1-3[NeuAcα2-6]GalNAc						14.51	4.71	13.15	13.04	S3ga3[S6]GA				
GalNAcβ1-3Galβ1-3[GalNAcβ1-3Galβ1-4GlcNAcβ1-6]GalNAc														
		4.90	0.96	0.96	junk				0.11	junk			0.11	junk
Galβ1-3GlcNAcβ1-4Galβ1-3[Galβ1-3GlcNAcβ1-6]GalNAc						14.86			0.15	ga3GL4ga3[ga3GL6]GA			0.15	ga3GL4ga3[ga3GL6]GA
Galβ1-3[Galβ1-3GlcNAcβ1-4Galβ1-4GlcNAcβ1-6]GalNAc							4.86	0.81	0.25	ga3[ga3GL4ga3GL6]GA	4.86	1.20	0.55	ga3[ga3GL4ga3GL6]GA
Galβ1-4GlcNAcβ1-4Galβ1-3[Galβ1-3GlcNAcβ1-6]GalNAc									0.15	ga4GL4ga3[ga3GL6]GA			0.15	ga4GL4ga3[ga3GL6]GA
									0.26	junk			0.24	junk
α1		4.95	0.60	0.60	junk	15.11	4.98	1.63	0.81	A1	4.98	1.85	1.03	A1
									0.82	junk			0.82	junk
NeuAcα2-3Galβ1-3GlcNAcβ1-3[NeuAcα2-6]GalNAc							5.06	0.20	0.20	sU3ga3GL3[S6]GA				
						15.48	5.15	2.65	2.17	junk	5.15	2.17	2.17	junk
NeuAcα2-3(SO3-)Galβ1-3[NeuAcα2-6]GalNAc									0.48	S3sga3[S6]GA				
NeuAcα2-3Galβ1-4GlcNAcβ1-2[Galβ1-4GlcNAcβ1-6]Man						15.69	5.26	0.80	0.50	S3-ga4GL2[ga4GL6]m	5.21	0.20		
									0.30	junk			0.20	junk
NeuAcα2-6Galβ1-4GlcNAcβ1-2[Galβ1-4GlcNAcβ1-6]Man											5.33	0.32	0.32	S6-ga4GL2[ga4GL6]m
α1B														
α2						16.31	5.58	0.34	0.34	A2	5.58	0.62	0.62	A2
		5.65	0.34	0.34	junk	16.43	5.64	0.65	0.35	junk	5.65	0.77	0.40	junk
Galβ1-4GalNAcβ1-3Galβ1-3[GalNAcβ1-3Galβ1-4GlcNAcβ1-6]GalNAc									0.30	ga4GA3ga3[GA3ga4GL6]GA			0.37	ga4GA3ga3[GA3ga4GL6]GA
NeuAcα2-3(SO3-)Galβ1-3[NeuAcα2-6]GalNAc						16.70	5.79	0.40	0.40	S3ga3[sS6]GA				
NeuAcα2-6GlcNAcβ1-4Galβ1-3[NeuAcα2-3Galβ1-4GlcNAcβ1-6]GalNAc						16.92	5.91	0.39	0.13	S3GL4ga3[S3ga4GL6]GA	5.90	0.29		
									0.26	junk			0.29	junk
NeuAcα2-3-Galβ1-3[Galβ1-3GlcNAcβ1-4Galβ1-4GlcNAcβ1-6]GalNAc						16.98	5.94	0.37	0.21	S3-ga3[ga3GL4ga3GL6]GA	5.97	0.16		
									0.16	junk			0.16	junk
						17.18	6.05	0.30	0.30	junk	6.07	0.46	0.46	junk
M5						17.44	6.21	2.94	2.94	M5	6.21	2.55	2.55	M5
						17.60	6.30	0.72	0.72	junk	6.30	0.75	0.75	junk
						17.76	6.40	0.34	0.34	junk	6.41	0.36	0.36	junk
NeuAcα2-3Galβ1-4GlcNAcβ1-2[NeuAcα2-3Galβ1-4GlcNAcβ1-6]Man						17.89	6.47	0.42	0.42	S3/S6-ga4GL2[S3ga4GL6]m				
A4														
α2G2											6.96	0.25	0.25	A2G2
M6						18.89	7.11	1.15	1.15	M6	7.11	1.06	1.06	M6
α1F1G1Gal2														
						19.73	7.68	0.76	0.76	junk	7.69	0.78	0.78	junk
						20.02	7.89	0.42	0.42	junk	7.90	0.39	0.39	junk
M7						20.15	7.99	0.61	0.61	M7	8.04	0.49	0.49	M7
M8						21.29	8.88	0.52	0.52	M8	8.90	0.52	0.52	M8
						21.57	9.11	0.60	0.60	junk	9.13	0.61	0.61	junk
M9						22.11	9.58	1.05	1.05	M9	9.61	0.87	0.87	M9
								100.00	100.00			100.01	100.01	

Appendix III. *Exoglycosidase digestion panel for PD (stages 1-4) striatum pool.*

ABS				ABS+SPG				ABS+BTG				ABS+BTG+AMF				ABS+BTG+AMF+BKF			
GU	% Area		Structure	GU	% Area		Structure	GU	% Area		Structure	GU	% Area		Structure	GU	% Area		Structure
	0.16	0.16	GA (galNAc)		0.10	0.10	GA (galNAc)		5.76	5.76	GA (galNAc)		5.14	5.14	GA (galNAc)		4.61	4.61	GA (galNAc)
1.01	2.26	2.26	gl (glucose)	1.01	2.52	2.52	gl (glucose)	1.01	7.81	7.81	gl (glucose)	1.01	8.02	8.02	gl (glucose)	1.01	8.05	8.05	gl (glucose)
1.03	9.20	9.20	ga (galactose)	1.03	9.65	9.65	ga (galactose)	1.03	10.10	10.10	ga (galactose)	1.03	10.43	10.43	ga (galactose)	1.03	10.54	10.54	ga (galactose)
1.07	0.76	0.76	junk	1.07	0.93	0.93	junk	1.06	0.93	0.93	junk	1.07	0.89	0.89	junk	1.08	0.80	0.80	junk
1.09	1.38	1.38	junk	1.09	1.85	1.85	junk	1.08	1.84	1.84	junk	1.09	1.66	1.66	junk	1.09	1.67	1.67	junk
1.69	1.76	1.76	ga3GA***	1.70	1.60	1.60	ga3GA***	1.68	1.58	1.58	ga3GA***	1.69	1.27	1.27	ga3GA***	1.69	1.23	1.23	ga3GA***
		18.08	ga3GA			19.63	ga3GA			11.50	ga3GA (incomplete dig)			12.41	ga3GA (incomplete dig)			13.09	ga3GA (incomplete dig)
						6.16	GL2m			8.09	GL2m			16.26	GL2m			16.26	GL2m
1.78	18.56	0.24	gaa3GA	1.79	26.27	0.24	gaa3GA	1.78	20.27	0.24	gaa3GA	1.78	29.52	0.24	gaa3GA	1.78	30.20	0.24	gaa3GA
														0.17	sGL2m			0.17	sGL2m
		0.24	GL3/6GA			0.24	GL3/6GA			0.44	GL3/6GA			0.44	GL3/6GA			0.44	GL3/6GA
1.81	2.43	2.43	junk	1.81	2.20	2.20	junk	1.80	2.26	2.26	junk	1.81	2.43	2.43	junk	1.80	2.50	2.50	junk
1.85	2.20	2.20	junk	1.85	1.20	1.20	junk	1.84	1.52	1.52	junk	1.86	1.38	1.38	junk	1.86	1.38	1.38	junk
				1.91	0.77	0.77	junk	1.89	0.81	0.81	junk								
1.92	0.37	0.37	junk	1.93	0.52	0.52	junk	1.92	0.64	0.64	junk	1.92	0.77	0.77	junk	1.92	0.81	0.81	junk
1.95	6.34	5.67	ga3g (glc core junk)	1.96	6.40	5.73	ga3g (glc core junk)												
		0.67	junk			0.67	junk	1.94	0.67	0.67	junk	1.96	0.98	0.98	junk	1.96	1.06	1.06	junk
1.99	1.61	1.61	junk	1.99	1.40	1.40	junk	1.98	1.96	1.96	junk	1.98	2.14	2.14	junk	1.98	2.08	2.08	junk
2.06	1.13	1.13	junk	2.07	0.63	0.63	junk	2.05	0.97	0.97	junk	2.06	1.08	1.08	junk	2.06	0.98	0.98	junk
2.32	0.54	0.54	junk	2.33	0.37	0.37	junk	2.31	0.60	0.60	junk	2.31	0.34	0.34	junk	2.31	0.25	0.25	junk
2.54	2.38	2.38	ga3GL2m	2.54	1.93	1.93	ga3GL2m												
2.60	9.06	6.16	ga4GL2m	2.60	2.90														
		2.90	gl4ga3gl (glc core junk)			2.90	gl4ga3gl (glc core junk)	2.59	3.65	3.65	gl4ga3gl (glc core junk)	2.60	2.87	2.87	gl4ga3gl (glc core junk)	2.59	2.82	2.82	gl4ga3gl (glc core junk)
				2.63	0.92	0.92	ga3[GL6]GA	2.62	0.85	0.85	ga3[GL6]GA	2.63	1.49	1.49	ga3[GL6]GA	2.62	1.44	1.44	ga3[GL6]GA
				2.66	0.52	0.52	GL2[GL6]m	2.64	0.65	0.65	GL2[GL6]m	2.65	0.96	0.96	GL2[GL6]m	2.65	0.91	0.91	GL2[GL6]m
2.74	0.62	0.62	junk	2.74	0.33	0.33	junk	2.73	0.56	0.56	junk								
2.79	0.81	0.61	junk	2.79	0.68	0.48	junk	2.84	0.82			2.85	0.94			2.84	0.87		
		0.20	ga3GL3GA			0.20	ga3GL3GA												
2.85	0.72	0.72	junk	2.86	0.74	0.74	junk			0.82	junk			0.94	junk			0.87	junk
		7.82	ga4(fa3GL2)m			8.30	ga4(fa3GL2)m			8.17	ga4(fa3GL2)m								
3.26	9.39	0.77	junk	3.26	9.80	1.20	junk	3.23	9.54	1.37	junk	3.24	1.40	1.40	junk	3.23	1.36	1.36	junk
		0.80	ga3[ga4GL6]GA																
						0.30	GL4ga3[GL6]GA	3.29	1.45	1.45	GL4ga3[GL6]GA	3.30	1.57	1.57	GL4ga3[GL6]GA	3.30	1.61	1.61	GL4ga3[GL6]GA
3.38	1.31	1.31	ga3[sGL6]GA	3.38	0.91	0.91	ga3[sGL6]GA	3.36	0.90	0.90	ga3[sGL6]GA	3.37	0.77	0.77	ga3[sGL6]GA	3.37	0.72	0.72	ga3[sGL6]GA
3.44	1.07	0.57	ga4(fa3sGL2)m	3.43	0.78	0.28	ga4(fa3sGL2)m	3.40	0.67	0.17	ga4(fa3sGL2)m	3.42	0.50			3.41	0.49		
		0.50	junk			0.50	junk			0.50	junk			0.50	junk			0.49	junk
3.46	0.58			3.50	0.35			3.45	0.21			3.47	0.36			3.46	0.37		
		0.58	junk			0.35	junk			0.21	junk			0.36	junk			0.37	junk
3.54	0.79			3.54	0.85			3.52	0.97			3.53	0.87			3.53	0.77		
		0.79	junk			0.85	junk			0.97	junk			0.87	junk			0.77	junk
3.61	0.36	0.36	junk	3.60	0.33	0.33	junk	3.58	0.55			3.60	0.57			3.59	0.62		
3.65	0.50	0.50	junk	3.65	0.43	0.43	junk			0.55	junk			0.57	junk			0.62	junk
				3.71	0.36	0.36	ga3[(fa3GL6)]GA	3.68	0.20	0.20	ga3[(fa3GL6)]GA								
								3.90	0.54	0.54	ga3[GL4ga3GL6]GA	3.93	0.60	0.60	ga3[GL4ga3GL6]GA	3.92	0.56	0.56	ga3[GL4ga3GL6]GA
4.03	0.27	0.27	GL4ga3[ga3GL6]GA	4.01	0.38	0.38	GL4ga3[ga3GL6]GA												
4.06	1.36	1.02	ga3[GL4ga4GL6]GA	4.06	1.35	1.01	ga3[GL4ga4GL6]GA	4.00	1.31	0.97	ga3[GL4ga4GL6]GA	4.02	1.21	0.87	ga3[GL4ga4GL6]GA	4.01	1.17	0.83	ga3[GL4ga4GL6]GA
		0.34	junk			0.34	junk			0.34	junk			0.34	junk			0.34	junk
4.12	0.60	0.28	GL4ga3[ga4GL6]GA	4.12	0.32														
		0.32	junk			0.32	junk	4.08	0.82			4.11	0.47			4.12	0.54		
4.14	0.36	0.36	junk	4.14	0.26	0.26	junk			0.82	junk			0.47	junk			0.54	junk
4.20	1.10	1.10	ga4GL2[ga4GL6]m																
4.39	0.84	0.35	ga3[ga4(fa3GL6)]GA	4.39	0.54			4.30	0.45			4.33	0.29			4.32	0.32		
		0.49	M3			0.54	M3			0.45	M3			0.29	M3			0.32	M3
4.46	1.76	1.76	sU3ga4GL2m	4.46	2.00	2.00	sU3ga4GL2m	4.41	1.76	1.76	sU3ga4GL2m	4.43	1.95	1.95	sU3ga4GL2m	4.42	1.86	1.86	sU3ga4GL2m
4.52	0.76	0.76	sU3ga3GL2m	4.53	0.68	0.68	sU3ga3GL2m	4.47	0.66	0.66	sU3ga3GL2m	4.49	0.54	0.54	sU3ga3GL2m	4.48	0.66	0.66	sU3ga3GL2m
				4.70	0.50	0.39	GA3ga3[GA3ga4GL6]GA	4.69	0.46	0.35	GA3ga3[GA3ga4GL6]GA	4.72	0.36	0.25	GA3ga3[GA3ga4GL6]GA	4.70	0.45	0.34	GA3ga3[GA3ga4GL6]GA
		0.11	junk			0.11	junk			0.11	junk			0.11	junk			0.11	junk
		0.15	ga3GL4ga3[ga3GL6]GA			0.16	ga3GL4ga3[ga3GL6]GA												
4.82	1.38	0.55	ga3[ga3GL4ga3GL6]GA	4.83	0.98	0.55	ga3[ga3GL4ga3GL6]GA												
		0.15	ga4GL4ga3[ga3GL6]GA					4.82	0.42			4.84	0.20			4.84	0.26		
		0.42	junk			0.27	junk			0.42	junk			0.20	junk			0.26	junk
4.94	1.85	1.03	A1	4.95	1.76	0.94	A1	4.89	1.58	0.76	A1	4.91	1.68	0.86	A1	4.90	1.74	0.92	A1
		0.82	junk			0.82	junk			0.82	junk			0.82	junk			0.82	junk
5.11	2.02	2.02	junk	5.12	2.20	2.20	junk	5.05	1.94	1.94	junk	5.08	2.13	2.13	junk	5.07	2.08	2.08	junk
5.20	0.26	0.26	junk	5.20	0.36	0.36	junk	5.12	0.31	0.31	junk	5.16	0.28	0.28	junk	5.15	0.40	0.40	junk
								5.30	0.53	0.53	A1B	5.33	0.71	0.71	A1B	5.32	0.68	0.68	A1B
5.53	0.64	0.64	A2	5.54	0.78	0.78	A2	5.43	0.75	0.75	A2	5.45	0.63	0.63	A2	5.44	0.90	0.90	A2
5.60	0.44	0.44	junk	5.60	0.59	0.59	junk	5.59	0.49	0.49	junk	5.58	0.41	0.41	junk	5.55	0.50	0.50	junk
5.66	0.36	0.36	ga4GA3ga3[GA3ga4GL6]GA																
5.85	0.26			5.86	0.43			5.77	0.32			5.79	0.31			5.78	1.00		
		0.26	junk			0.43	junk			0.32	junk			0.31	junk			1.00	junk
5.91	0.13							5.82	0.32			5.85	0.53			5.86	0.48		
		0.13	junk	5.92	0.27	0.27	junk			0.32	junk			0.53	junk			0.48	junk
6.00	0.34	0.34	junk					5.91	0.23	0.23	junk	5.95	0.24	0.24	junk	5.93	0.28	0.28	junk
6.15	2.55	2.55	M5	6.15	2.58	2.58	M5	6.07	2.44	2.44	M5	6.09	2.52	2.52	M5	6.09	2.81	2.81	M5
6.23	0.76	0.76	junk	6.24	0.83	0.83	junk	6.15	0.89	0.89	junk	6.18	1.08	1.08	junk	6.28	0.36		
6.34	0.42	0.42	junk	6.35	0.28	0.28	junk	6.26	0.29	0.29	junk	6.28	0.46	0.46	junk			0.36	junk
																6.49	0.53	0.53	A4
6.89	0.30	0.30	A2G2					6.81	0.43	0.43	F(6)A4	6.84	0.57	0.57	F(6)A4				
7.03	1.43	1.43	M6	7.02	0.88	0.88	M6	6.94	0.97	0.97	M6	6.95	1.01	1.01	M6	6.95	1.12	1.12	M6
7.60	0.76	0.76	junk	7.60	0.81	0.81	junk	7.50	0.74	0.74	junk	7.51	0.60	0.60	junk	7.52	0.46	0.46	junk
7.81	0.43	0.43	junk	7.80	0.44	0.44	junk	7.70	0.55	0.55	junk	7.72	0.45	0.45	junk	7.72	0.48	0.48	junk
7.94	0.51	0.51	M7	7.91	0.65	0.65	M7	7.82	0.64	0.64	M7	7.83	0.86	0.86	M7	7.83	0.90	0.90	M7
8.79	0.48	0.48	M8	8.77	0.53	0.53	M8	8.66	0.55	0.55	M8	8.67	0.51	0.51	M8	8.67	0.56	0.56	M8
9.02	0.59	0.59	junk	9.02	0.61	0.61	junk	8.89	0.53	0.53	junk	8.91	0.52	0.52	junk	8.93	0.35	0.35	junk
9.47	0.75	0.75	M9	9.44	0.75	0.75	M9	9.33	0.97	0.97	M9	9.33	0.90	0.90	M9	9.34	0.77	0.77	M9
	100.00	100.00			100.00	100.00			100.63	100.63			100.37	100.37			100.36	100.36	

Appendix III (cont.). *Exoglycosidase digestion panel for PD (stages 1-4) striatum pool.*

ABS+BTG+AMF+BKF+GUH				ABS+BTG+AMF+BKF+GUH+JBH				ABS+BTG+AMF+BKF+GUH+JBH+CBG				ABS+BTG+AMF+BKF+GUH+JBH+CBG+JBM			
GU	% Area		Structure	GU	% Area		Structure	GU	% Area		Structure	GU	% Area		Structure
	7.84	7.84	GA (galNAc)		6.61	6.61	GA (galNAc)		6.03	6.03	GA (galNAc)		4.82	4.82	GA (galNAc)
1.02	25.04	8.04	gl (glucose)	1.02	24.97	7.97	gl (glucose)	1.02	25.19	8.19	gl (glucose)	1.02	25.00	8.00	gl (glucose)
		17.00	m (mannose)			17.00	m (mannose)			17.00	m (mannose)			17.00	m (mannose)
1.03	10.51	10.51	ga (galactose)	1.03	10.50	10.50	ga (galactose)	1.03	10.50	10.50	ga (galactose)	1.03	10.50	10.50	ga (galactose)
1.08	1.05	1.05	junk	1.08	2.55	0.86	junk	1.07	0.86	0.86	junk	1.06	0.12	0.12	junk
1.08	1.62	1.62	junk			1.69	junk	1.09	1.86	1.86	junk	1.09	1.37	1.37	junk
1.69	1.38	1.38	sga3GA***	1.69	1.43	1.43	sga3GA***	1.69	1.59	1.59	sga3GA***	1.69	1.86	1.86	sga3GA***
		13.67	ga3GA (incomplete dig)			14.41	ga3GA (incomplete dig)			15.91	ga3GA (incomplete dig)			16.45	ga3GA (incomplete dig)
1.78	14.08	0.24	gaa3GA	1.79	14.82	0.24	gaa3GA	1.78	16.08			1.79	16.62		
		0.17	sGL2m			0.17	sGL2m			0.17	sGL2m			0.17	sGL2m
1.81	2.10	2.10	junk	1.81	2.12	2.12	junk	1.80	2.25	2.25	junk	1.81	2.28	2.28	junk
1.86	0.73	0.73	junk	1.86	0.79	0.79	junk	1.86	0.56	0.56	junk	1.87	0.95	0.95	junk
1.92	0.95	0.95	junk	1.93	0.95	0.95	junk	1.92	0.79	0.79	junk	1.95	1.07	1.07	junk
1.96	1.22	1.22	junk	1.96	1.55	1.55	junk	1.96	1.64	1.64	junk	1.97	1.53	1.53	junk
1.98	2.14	2.14	junk	1.98	2.06	2.06	junk	1.98	1.85	1.85	junk	1.99	2.25	2.25	junk
2.06	1.01	1.01	junk	2.06	1.13	1.13	junk	2.06	0.86	0.86	junk	2.07	1.17	1.17	junk
2.31	0.25	0.25	junk	2.31	0.38	0.38	junk	2.30	0.23	0.23	junk	2.31	0.53	0.53	junk
2.59	2.74	2.74	gl4ga3gl (gtc core junk)	2.59	2.65	2.65	gl4ga3gl (gtc core junk)	2.59	2.48	2.48	gl4ga3gl (gtc core junk)	2.60	2.73	2.73	gl4ga3gl (gtc core junk)
												2.64	5.47	5.47	M1
2.73	0.50	0.50	junk	2.73	0.47	0.47	junk	2.73	0.44	0.44	junk	2.74	0.55	0.55	junk
2.84	0.87	0.87	junk	2.85	0.94	0.94	junk	2.85	0.80	0.80	junk	2.85	1.04	1.04	junk
3.01	0.32	0.32	junk	3.01	0.39	0.39	junk	3.00	0.32	0.32	junk				
3.23	1.53	1.53	junk	3.23	1.53	1.53	junk	3.23	1.54	1.54	junk	3.23	1.57	1.57	junk
3.37	0.77	0.77	ga3[sGL6]GA	3.37	0.73	0.73	ga3[sGL6]GA	3.37	0.92	0.92	ga3[sGL6]GA	3.38	0.95	0.95	ga3[sGL6]GA
3.41	0.60	0.60	junk	3.42	0.59	0.59	junk	3.41	0.46	0.46	junk	3.42	0.45	0.45	junk
3.45	0.26	0.26	junk	3.46	0.33	0.33	junk	3.46	0.46	0.46	junk	3.47	0.51	0.51	junk
3.53	0.90	0.90	junk	3.52	0.56	0.56	junk	3.53	0.74	0.74	junk	3.54	0.88	0.88	junk
				3.56	0.34	0.34	junk								
3.59	0.54	0.54	junk	3.59	0.62	0.62	junk	3.59	0.64	0.64	junk	3.60	0.66	0.66	junk
3.81	0.27	0.27	ga3[GL4ga3GL6]GA (incom	3.81	0.29	0.29	ga3[GL4ga3GL6]GA (incom	3.82	0.18	0.18	ga3[GL4ga3GL6]GA (incom	3.83	0.30	0.30	ga3[GL4ga3GL6]GA (incom
4.00	0.34	0.34	junk	4.02	0.47	0.47	junk	4.01	0.41	0.41	junk	4.02	0.52	0.52	junk
4.12	0.25	0.25	junk	4.12	0.21	0.21	junk	4.09	0.19	0.19	junk	4.11	0.27	0.27	junk
4.13	0.17	0.17	junk	4.13	0.26	0.26	junk	4.13	0.31	0.31	junk	4.15	0.44	0.44	junk
4.33	1.29	1.29	M3	4.34	1.35	1.35	M3	4.33	1.30	1.30	M3				
4.43	2.92	2.92	sU3ga4GL2m	4.43	3.13	3.13	sU3ga4GL2m	4.43	3.01	3.01	sU3ga4GL2m	4.46	3.04	3.04	sU3ga4GL2m
4.49	0.71	0.46	sU3ga3GL2m	4.49	0.73	0.73	sU3ga3GL2m	4.48	0.67	0.67	sU3ga3GL2m	4.52	0.81	0.81	sU3ga3GL2m
		0.25	GA3ga3[GA3ga4GL6]GA												
4.68	0.11	0.11	junk	4.68	0.31	0.31	junk	4.69	0.33	0.33	junk	4.72	0.30	0.30	junk
4.84	0.23	0.23	junk	4.85	0.41	0.41	junk	4.85	0.43	0.43	junk	4.88	0.39	0.39	junk
4.90	0.82	0.82	junk	4.91	0.66	0.66	junk	4.90	0.79	0.79	junk	4.95	0.91	0.91	junk
5.07	2.21	2.21	junk	5.08	2.10	2.10	junk	5.07	1.99	1.99	junk	5.11	2.08	2.08	junk
5.14	0.22	0.22	junk	5.15	0.23	0.23	junk	5.12	0.37	0.37	junk	5.19	0.23	0.23	junk
5.33	0.22	0.22	A1B (incomplete dig)	5.34	0.23	0.23	A1B (incomplete dig)	5.33	0.23	0.23	A1B (incomplete dig)	5.36	0.15	0.15	A1B (incomplete dig)
5.39	0.29	0.29	A2 (incomplete dig)	5.41	0.25	0.25	A2 (incomplete dig)	5.39	0.32	0.32	A2 (incomplete dig)	5.38	0.34	0.34	A2 (incomplete dig)
5.55	0.47	0.47	junk	5.57	0.51	0.51	junk	5.55	0.48	0.48	junk	5.60	0.52	0.52	junk
5.79	0.99	0.99	junk	5.80	0.96	0.96	junk	5.79	0.82	0.82	junk	5.84	0.81	0.81	junk
5.87	0.16	0.16	junk	5.90	0.19	0.19	junk	5.88	0.21	0.21	junk	5.93	0.18	0.18	junk
5.95	0.22	0.22	junk	6.04	0.39							6.01	0.60		
6.02	0.24	0.24	junk			0.39	junk	6.02	0.47	0.47	junk			0.60	junk
6.10	2.58	2.58	M5	6.12	2.51	2.51	M5	6.09	2.47	2.47	M5				
6.20	0.23	0.23	junk	6.22	0.26	0.26	junk	6.18	0.32	0.32	junk	6.24	0.52	0.52	junk
6.28	0.26	0.26	junk	6.32	0.24	0.24	junk	6.27	0.30	0.30	junk	6.36	0.26	0.26	junk
6.45	0.29	0.29	A4 (incomplete dig)	6.48	0.26	0.26	A4 (incomplete dig)	6.44	0.28	0.28	A4 (incomplete dig)	6.53	0.20	0.20	A4 (incomplete dig)
6.97	1.21	1.21	M6	7.00	1.07	1.07	M6	6.96	1.14	1.14	M6				
7.27	0.59	0.59	A1F1G1Gal2 (incomplete d	7.30	0.71	0.71	A1F1G1Gal2 (incomplete dig	7.26	0.63	0.63	A1F1G1Gal2 (incomplete dig	7.36	0.58	0.58	A1F1G1Gal2 (incomplete d
7.53	0.28	0.28	junk	7.56	0.23	0.23	junk	7.51	0.37	0.37	junk	7.62	0.41	0.41	junk
7.74	0.54	0.54	junk	7.77	0.54	0.54	junk	7.73	0.63	0.63	junk	7.83	0.62	0.62	junk
7.84	0.64	0.64	M7	7.86	0.57	0.57	M7	7.84	0.54	0.54	M7				
8.68	0.65	0.65	M8	8.74	0.56	0.56	M8	8.69	0.55	0.55	M8				
8.93	0.49	0.49	junk	8.97	0.38	0.38	junk	8.93	0.43	0.43	junk	9.05	0.62	0.62	junk
9.36	1.14	1.14	M9	9.41	0.97	0.97	M9	9.36	0.73	0.73	M9				
	99.98	99.98			99.99	99.99			99.99	99.99			99.98	99.98	

Appendix III (cont.). *Exoglycosidase digestion panel for PD (stages 1-4) striatum pool.*

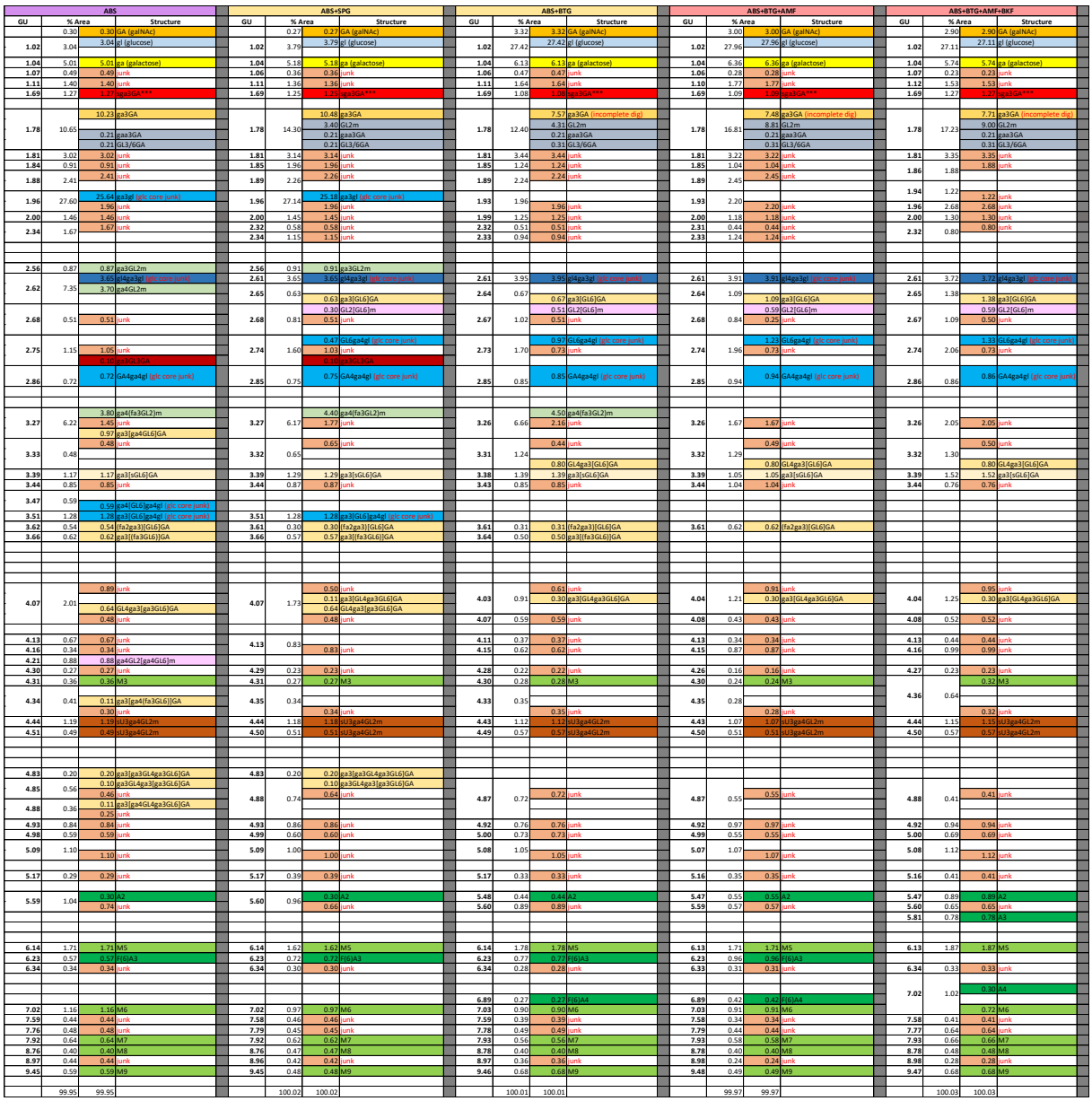
											% Area
										% Area	JBH
									% Area	CBG	CBG
								% Area	JBH	JBH	JBH
							% Area	GUH	GUH	GUH	GUH
						% Area	BKF	BKF	BKF	BKF	BKF
				% Area	% Area	AMF	AMF	AMF	AMF	AMF	AMF
	% Area	% Area	% Area	SPG	BTG	BTG	BTG	BTG	BTG	BTG	BTG
	UND	NAN1	ABS	ABS	ABS	ABS	ABS	ABS	ABS	ABS	ABS
GalNAc	0.14	0.19	0.16	0.10	5.76	5.14	4.61	7.84	6.61	6.03	4.82
Gal	0.85	7.52	9.20	9.65	10.10	10.43	10.54	10.51	10.50	10.50	10.50
Glc	2.35	2.41	2.26	2.52	7.81	8.02	8.05	8.04	7.97	8.19	8.00
Man								17.00	17.00	17.00	17.00
Semi-digestible Disaccharides (ga3GA)	0.24	6.80	18.08	19.63	11.50	12.41	13.09	13.67	14.41	15.91	16.45
Digestible Disaccharides (others)	0.48	0.48	0.48	6.64	8.77	16.94	16.94	0.24	0.24		
Undigestible Disaccharides (sulfated)	0.47	0.49	1.76	1.60	1.58	1.44	1.40	1.55	1.60	1.76	2.03
Core 1 extensions											
Core 2 and extensions	3.52	3.81	3.93	4.07	4.36	4.78	4.78	0.52	0.29	0.18	0.30
Core 2 extensions with sulfation	0.91	0.91	1.31	0.91	0.90	0.77	0.72	0.77	0.73	0.92	0.95
Core 3 extensions	0.20	0.20	0.20	0.20							
Mannose Core 1 (linear trisaccharide and larger)	13.31	15.55	16.93	10.51	8.34						
Mannose Core 2 (branched trisaccharide and larger)	0.92	1.23	1.10	0.52	0.65	0.96	0.91				
SSTn, STn, ST antigens, peeling, disialyl	34.22	17.10									
STn and ST antigens (sulfated)	1.60	1.22									
Larger Sulfated Glycans that don't digest	2.54	2.55	2.52	2.68	2.42	2.49	2.52	3.38	3.86	3.68	3.85
High Mannose (N)	6.68	5.85	6.21	5.93	6.02	6.09	6.48	7.51	7.03	6.73	5.47
Hybrid (N)	1.15	1.90	1.97	1.72	2.47	2.77	3.03	1.39	1.45	1.46	1.27
Digestible hex chains	5.07	6.43	5.67	5.73							
Undigestible hex chains	2.90	2.90	2.90	2.90	3.65	2.87	2.82	2.74	2.65	2.48	2.73
Integrated Junk	22.45	22.47	25.32	24.69	26.30	25.26	24.47	24.82	25.65	25.15	26.61
SUM	100.00	100.01	100.00	100.00	100.63	100.37	100.36	99.98	99.99	99.99	99.98

	% Area	Comment
Total GalNAc core	31.66	(including the 1 glycan in the larger sulfated glycans that doesn't digest)
Total Mannose core	16.77	(including the 1 glycan in the larger sulfated glycans that doesn't digest)
Total Hexose chain glycans (Glucose core)	10.32	inlcudes glc, gal-glc, glc-gal-glc and gal-gal-glc
Total Galactose core (probably peeling)	8.57	probably core 1 and 2 peeling
Total NeuNac core (polysialic)	2.40	One polysialic structure also seen in Gothenburg data
Total N glycans	7.83	leftover from Ana's digest or extra N-glycan that was released in microwave
Total Junk	22.45	Suspected tissue junk and junk present in blank
Total (should equal 100.00)	100.00	total % area from UND
Total Glycan	77.55	
Total Junk	22.45	

Appendix IV. *Summary of % areas – Exoglycosidase digestion panel for PD (stages 1-4) striatum pool.*

Structure (long name)	Structure (image)	BLANK				RT	UND			NAN1			
		GU	% Area		Structure		GU	% Area	Structure	GU	% Area		Structure
galNAc						1.76		0.62	0.62 GA (galNAc)		0.16	0.16	GA (galNAc)
glucose						1.99			3.80 gl (glucose)			2.98	gl (glucose)
mannose		1.02	8.89	8.89	junk		1.02	3.80		1.02	2.98		
galactose						2.07	1.04	0.73	0.73 ga (galactose)			3.14	ga (galactose)
						2.19	1.07	2.00	2.00 junk	1.04	3.53	0.39	junk
										1.11	1.86	1.86	junk
Galβ1-3-SO3-GalNAc													
NeuNAc-α2-3 NeuNAc						4.98	1.75	2.00	2.00 S3S				
Galβ1-3GalNAc						5.03			0.20 ga3GA			4.41	ga3GA
GlcNAcβ1-2Man													
Galα1-3GalNAc		1.73	1.69	1.69	junk		1.76	0.62	0.21 ga1GA	1.78	4.83	0.21	ga1GA
GlcNAcβ1-3GalNAc/GlcNAcβ1-6GalNAc									0.21 GL3/6GA			0.21	GL3/6GA
						5.19	1.80	3.07	3.07 junk	1.81	3.02	3.02	junk
		1.78	3.92	3.92	junk	5.32	1.84	0.68	0.68 junk			0.98	junk
						5.45	1.87	0.86	0.86 junk	1.88	0.98		
						5.52	1.88	0.55	0.55 junk				
Galβ1-3Glc						5.83	1.96	29.81	27.85 ga3gl (glc core junk)	1.97	28.28	26.32	ga3gl (glc core junk)
		1.93	65.83	65.83	junk				1.96 junk			1.96	junk
						5.98	2.00	0.92	0.92 junk	1.99	3.24	3.24	junk
		2.32	2.73	2.73	junk	7.38	2.34	1.64	1.64 junk	2.34	1.92	1.92	junk
NeuAcα2-3Gal		2.45	1.45	1.45	junk	7.63	2.41	5.29	2.41 S3ga*				
NeuAcα2-6GalNAc									2.88 S6GA	2.40	2.88	2.88	S6GA
Galβ1-3GlcNAcβ1-2Man										2.56	0.58	0.58	ga3GL2m
Glcβ1-4Galβ1-3Glc						8.45			3.68 gl4ga3gl (glc core junk)			3.68	gl4ga3gl (glc core junk)
Galβ1-4GlcNAcβ1-2Man		2.62	3.7	3.70	junk		2.62	4.55	0.90 ga4GL2m	2.62	7.33	3.68	ga4GL2m
Galβ1-3[GlcNAcβ1-6]GalNAc													
GlcNAcβ1-2[GlcNAcβ1-6]Man													
						8.66	2.67	0.53	0.53 junk	2.68	0.65	0.65	junk
M1													
GlcNAcβ1-4Galβ1-3Glc													
		2.75	2.11	2.11	junk	8.97	2.76	0.44	0.44 junk	2.75	1.08	1.08	junk
Galβ1-3GlcNAcβ1-3GalNAc													
GalNAcβ1-4Galβ1-3Glc										2.85	0.65	0.65	GAga4gl (glc core junk)
NeuAcα2-6Galβ1-3GalNAc						9.65	2.94	0.51	0.51 S6ga3GA	2.94	0.64	0.64	S6ga3GA
NeuAcα2-3Galβ1-3GalNAc						9.93	3.02	2.83	2.83 S3ga3GA				
Galβ1-4(Fuca1-3GlcNAcβ1-2)Man		3.31	0.41	0.41	junk	10.77			4.08 ga4(fa3GL2)m			4.08	ga4(fa3GL2)m
		3.33	0.4	0.40	junk		3.27	4.98	0.90 junk	3.27	6.53	1.43	junk
Galβ1-3[Galβ1-4GlcNAcβ1-6]GalNAc												1.02	ga3[ga4GL6]GA
		3.39	0.62	0.62	junk	11.06			0.48 junk			0.48	junk
Galβ1-3[NeuAcα2-6]GalNAc							3.36	1.93	1.45 ga3[S6]GA	3.36	7.13	6.65	ga3[S6]GA
GlcNAcβ1-4Galβ1-3[GlcNAcβ1-6]GalNAc													
Galβ1-3[SO3-GlcNAcβ1-6]GalNAc		3.42	0.43	0.43	junk	11.16	3.39	0.91	0.91 ga3[sGL6]GA	3.38	0.93	0.93	ga3[sGL6]GA
		3.49	1.39	1.39	junk	11.29	3.43	0.81	0.81 junk	3.43	0.96	0.96	junk
SO3-Galβ1-3[NeuAcα2-6]GalNAc					junk	11.43						0.15	sga3[S6]GA
Galβ1-4[GlcNAcβ1-6]Galβ1-4Glc		3.54	0.93	0.93	junk		3.48	0.36	0.36 ga4[GL6]ga4gl (glc core junk)	3.47	0.65	0.50	ga4[GL6]ga4gl (glc core junk)
Galβ1-3[GlcNAcβ1-6]Galβ1-4Glc		3.6	1.29	1.29	junk	11.55	3.52	1.04	1.04 ga3[GL6]ga4gl (glc core junk)	3.51	0.98	0.98	ga3[GL6]ga4gl (glc core junk)
(Fucα1-2Galβ1-3)[GlcNAcβ1-6]GalNAc		3.64	0.59	0.59	junk								
Galβ1-3[(Fucα1-3)GlcNAcβ1-6]GalNAc						11.96	3.66	0.18	0.18 ga3'[fa3GL6]GA	3.66	0.51	0.51	ga3'[fa3GL6]GA
NeuAcα2-3Galβ1-4GlcNAcβ1-2Man						12.08	3.70	2.71	2.71 S3ga4GL2m				
NeuAcα2-3Galβ1-3GlcNAcβ1-2Man		3.85	0.34	0.34	junk	12.21	3.75	0.75	0.75 S3ga3GL2m				
SO3-Galβ1-3[NeuAcα2-6]GalNAc						12.74						0.17	ga3[sS6]GA
Galβ1-3GlcNAcβ1-3[NeuAcα2-6]GalNAc		4.01	0.31	0.31	junk					3.93	0.27	0.10	ga3GL3[S6]GA
						13.02			0.49 junk			0.82	junk
Galβ1-3[GlcNAcβ1-4Galβ1-3GlcNAcβ1-6]GalNAc		4.05	0.77	0.77	junk		4.05	1.18		4.07	2.04		
GlcNAcβ1-4Galβ1-3[Galβ1-3GlcNAcβ1-6]GalNAc									0.29 GL4ga3[ga3GL6]GA			0.64	GL4ga3[ga3GL6]GA
						13.11			0.40 junk			0.58	junk
NeuAcα2-3GalNAcβ1-4Galβ1-3Glc		4.16	0.46	0.46	junk		4.08	0.72	0.72 S3GA4ga4gl (glc core junk)				
						13.29	4.15	0.57		4.14	0.90		
									0.57 junk			0.90	junk
Galβ1-4GlcNAcβ1-2[Galβ1-4GlcNAcβ1-6]Man		4.25	0.18	0.18	junk					4.21	0.67	0.67	ga4GL2[ga4GL6]m
		4.27	0.39	0.39	junk	13.63	4.29	0.20	0.20 junk	4.28	0.28	0.28	junk
M3						13.69	4.31	0.21	0.21 M3	4.31	0.36	0.36	M3
Galβ1-3[NeuAcα2-3Galβ1-4GlcNAcβ1-6]GalNAc						13.82			0.39 ga3[S3ga4GL6]GA				
Galβ1-3[(Galβ1-4Fucα1-3GlcNAcβ1-6)]GalNAc							4.37	0.79	0.10 ga3[ga4(fa3GL6)]GA	4.37	0.46	0.16	ga3[ga4(fa3GL6)]GA
		4.48	0.33	0.33	junk				0.30 junk			0.30	junk
SO3-GlcAβ1-3Galβ1-4GlcNAcβ1-2Man						13.99	4.44	0.96	0.96 sU3ga4GL2m	4.44	1.06	1.06	sU3ga4GL2m
SO3-GlcAβ1-3Galβ1-3GlcNAcβ1-2Man						14.12	4.50	0.26	0.26 sU3ga4GL2m	4.52	0.43	0.43	sU3ga4GL2m
NeuAcα2-3Galβ1-3[Galβ1-4GlcNAcβ1-6]GalNAc						14.22	4.54	0.54	0.54 S3ga3[ga4GL6]GA				
NeuAcα2-3Galβ1-3[NeuAcα2-6]GalNAc						14.49	4.66	6.64	6.64 S3ga3[S6]GA				
NeuAcα2-6-(Fucα1-2Galβ1-3)[GlcNAcβ1-6]GalNAc						14.70	4.75	0.05	0.05 S6-(fa2ga3)[GL6]GA	4.76	0.27	0.27	S6-(fa2ga3)[GL6]GA
Galβ1-3[Galβ1-3GlcNAcβ1-4Galβ1-4GlcNAcβ1-6]GalNAc						14.83				4.83	0.20	0.20	ga3[ga3GL4ga3GL6]GA
Galβ1-3GlcNAcβ1-4Galβ1-3[Galβ1-3GlcNAcβ1-6]GalNAc						14.89	4.84	0.45	0.10 ga3GL4ga3[ga3GL6]GA			0.10	ga3GL4ga3[ga3GL6]GA
									0.35 junk	4.88	0.95	0.74	junk
Galβ1-3[Galβ1-4GlcNAcβ1-4Galβ1-4GlcNAcβ1-6]GalNAc						14.96	4.87	0.32	0.11 ga3[ga4GL4ga3GL6]GA			0.11	ga3[ga4GL4ga3GL6]GA
		4.91	0.14	0.14	junk				0.21 junk				
						15.05	4.92	0.82	0.82 junk	4.93	0.89	0.89	junk
						15.19	4.99	0.39	0.39 junk	4.99	0.59	0.59	junk
NeuAcα2-3Galβ1-3GlcNAcβ1-3[NeuAcα2-6]GalNAc		4.95	0.54	0.54	junk				0.10 S3ga3GL3[S6]GA				
						15.38	5.08	1.13	0.90 junk	5.08	1.15	1.00	junk
NeuAcα2-3(SO3-)Galβ1-3[NeuAcα2-6]GalNAc									0.13 S3ga3[S6]GA				
						15.62	5.20	0.51	0.15 junk			0.15	junk
NeuAcα2-3Galβ1-4GlcNAcβ1-2[Galβ1-4GlcNAcβ1-6]Man									0.36 S3-ga4GL2[ga4GL6]m				
α2							5.59	0.89	0.30 A2	5.59	0.93	0.30	A2
						16.36			0.59 junk			0.63	junk
NeuAcα2-3(SO3-)Galβ1-3[NeuAcα2-6]GalNAc						16.78			0.25 S3ga3[sS6]GA				
NeuAcα2-6GlcNAcβ1-4Galβ1-3[NeuAcα2-3Galβ1-4GlcNAcβ1-6]GalNAc							5.83	0.80	0.35 S3Gl4ga3[S3ga3GL6]GA				
NeuAcα2-3-Galβ1-3GlcNAcβ1-4Galβ1-3[Galβ1-4GlcNAcβ1-6]GalNAc									0.20 S3-ga3GL4ga3[ga3GL6]GA				
M5						17.29	6.13	1.77	1.77 M5	6.14	1.58	1.58	M5
F(6)A3						17.43	6.21	0.63	0.63 F(6)A3	6.23	0.63	0.63	F(6)A3
						17.62	6.33	0.21	0.21 junk	6.34	0.26	0.26	junk
NeuAcα2-3Galβ1-4GlcNAcβ1-2[NeuAcα2-3Galβ1-4GlcNAcβ1-6]Man						17.67	6.36	0.25	0.25 S3ga4GL2[S3ga4GL6]m				
A4													
F(6)A4													
M6						18.65	7.01	1.17	1.17 M6	7.03	0.93	0.93	M6
						19.47	7.59	0.42	0.42 junk	7.60	0.38	0.38	junk
						19.69	7.76	0.45	0.45 junk	7.77	0.47	0.47	junk
M7						20.01	8.00	0.80	0.80 M7	7.93	0.55	0.55	M7
M8						20.92	8.75	0.37	0.37 M8	8.78	0.34	0.34	M8
						21.17	8.96	0.70	0.70 junk	8.98	0.47	0.47	junk
M9						21.70	9.44	0.71	0.71 M9	9.46	0.63	0.63	M9
								100.03	100.02		99.99	99.99	

Appendix V. *Exoglycosidase digestion panel for healthy substantia nigra pool.*



ABS				ABS+5PG				ABS+BTG				ABS+BTG+AMF				ABS+BTG+AMF+BKF			
GU	% Area		Structure	GU	% Area		Structure	GU	% Area		Structure	GU	% Area		Structure	GU	% Area		Structure
	0.30	0.30	GA (galNAc)		0.27	0.27	GA (galNAc)		3.32	3.32	GA (galNAc)		3.00	3.00	GA (galNAc)		2.90	2.90	GA (galNAc)
1.02	3.04	3.04	gl (glucose)	1.02	3.79	3.79	gl (glucose)	1.02	27.42	27.42	gl (glucose)	1.02	27.96	27.96	gl (glucose)	1.02	27.11	27.11	gl (glucose)
1.04	5.01	5.01	ga (galactose)	1.04	5.18	5.18	ga (galactose)	1.04	6.13	6.13	ga (galactose)	1.04	6.36	6.36	ga (galactose)	1.04	5.74	5.74	ga (galactose)
1.07	0.49	0.49	junk	1.06	0.36	0.36	junk	1.06	0.47	0.47	junk	1.06	0.28	0.28	junk	1.07	0.23	0.23	junk
1.11	1.40	1.40	junk	1.11	1.36	1.36	junk	1.11	1.64	1.64	junk	1.10	1.77	1.77	junk	1.12	1.53	1.53	junk
1.69	1.27	1.27	ga3GA***	1.69	1.25	1.25	ga3GA***	1.69	1.08	1.08	ga3GA***	1.69	1.09	1.09	ga3GA***	1.69	1.27	1.27	ga3GA***
		10.23	ga3GA			10.48	ga3GA			7.57	ga3GA (incomplete dig)			7.48	ga3GA (incomplete dig)			7.71	ga3GA (incomplete dig)
1.78	10.65			1.78	14.30	3.40	GL2m	1.78	12.40	4.31	GL2m	1.78	16.81	8.81	GL2m	1.78	17.23	9.00	GL2m
		0.21	gaa3GA			0.21	gaa3GA			0.21	gaa3GA			0.21	gaa3GA			0.21	gaa3GA
		0.21	GL3/6GA			0.21	GL3/6GA			0.31	GL3/6GA			0.31	GL3/6GA			0.31	GL3/6GA
1.81	3.02	3.02	junk	1.81	3.14	3.14	junk	1.81	3.44	3.44	junk	1.81	3.22	3.22	junk	1.81	3.35	3.35	junk
1.84	0.91	0.91	junk	1.85	1.96	1.96	junk	1.85	1.24	1.24	junk	1.85	1.04	1.04	junk	1.86	1.88	1.88	junk
1.88	2.41	2.41	junk	1.89	2.26	2.26	junk	1.89	2.24	2.24	junk	1.89	2.45	2.45	junk				
		25.64	ga3gl (glc core junk)			25.53	ga3gl (glc core junk)									1.94	1.22	1.22	junk
1.96	27.60	1.96	junk	1.96	27.14			1.93	1.96	1.96	junk	1.93	2.20	2.20	junk	1.96	2.68	2.68	junk
2.00	1.46	1.46	junk	2.00	1.45	1.45	junk	1.99	1.25	1.25	junk	2.00	1.18	1.18	junk	2.00	1.30	1.30	junk
2.34	1.67	1.67	junk	2.32	0.58	0.58	junk	2.32	0.51	0.51	junk	2.31	0.44	0.44	junk	2.32	0.80	0.80	junk
				2.34	1.15	1.15	junk	2.33	0.94	0.94	junk	2.33	1.24	1.24	junk				
2.56	0.87	0.87	ga3Gl2m	2.56	0.91	0.91	ga3GL2m												
		3.65	gl4ga3gl (glc core junk)	2.61	3.65	3.65	gl4ga3gl (glc core junk)	2.61	3.95	3.95	gl4ga3gl (glc core junk)	2.61	3.91	3.91	gl4ga3gl (glc core junk)	2.61	3.72	3.72	gl4ga3gl (glc core junk)
2.62	7.35	3.70	ga4GL2m																
				2.65	0.63			2.64	0.67			2.64	1.09			2.65	1.38		
						0.63	ga3[GL6]GA			0.67	ga3[GL6]GA			1.09	ga3[GL6]GA			1.38	ga3[GL6]GA
						0.30	GL2[GL6]m			0.51	GL2[GL6]m			0.59	GL2[GL6]m			0.59	GL2[GL6]m
2.68	0.51	0.51	junk	2.68	0.81	0.51	junk	2.67	1.02	0.51	junk	2.68	0.84	0.25	junk	2.67	1.09	0.50	junk
						0.47	GL6ga4gl (glc core junk)			0.97	GL6ga4gl (glc core junk)			1.23	GL6ga4gl (glc core junk)			1.33	GL6ga4gl (glc core junk)
2.75	1.15	1.05	junk	2.74	1.60	1.03	junk	2.73	1.70	0.73	junk	2.74	1.96	0.73	junk	2.74	2.06	0.73	junk
		0.10	a3GL3GA			0.10	a3GL3GA												
2.86	0.72	0.72	GA4ga4gl (glc core junk)	2.85	0.75	0.75	GA4ga4gl (glc core junk)	2.85	0.85	0.85	GA4ga4gl (glc core junk)	2.85	0.94	0.94	GA4ga4gl (glc core junk)	2.86	0.86	0.86	GA4ga4gl (glc core junk)
		3.80	ga4(fa3GL2)m			4.40	ga4(fa3GL2)m			4.50	ga4(fa3GL2)m								
3.27	6.22	1.45	junk	3.27	6.17	1.77	junk	3.26	6.66	2.16	junk	3.26	1.67	1.67	junk	3.26	2.05	2.05	junk
		0.97	ga3[ga4GL6]GA																
3.33	0.48	0.48	junk	3.32	0.65	0.65	junk	3.31	1.24	0.44	junk	3.32	1.29	0.49	junk	3.32	1.30	0.50	junk
										0.80	GL4ga3[GL6]GA			0.80	GL4ga3[GL6]GA			0.80	GL4ga3[GL6]GA
3.39	1.17	1.17	ga3[sGL6]GA	3.39	1.29	1.29	ga3[sGL6]GA	3.38	1.39	1.39	ga3[sGL6]GA	3.39	1.05	1.05	ga3[sGL6]GA	3.39	1.52	1.52	ga3[sGL6]GA
3.44	0.85	0.85	junk	3.44	0.87	0.87	junk	3.43	0.85	0.85	junk	3.44	1.04	1.04	junk	3.44	0.76	0.76	junk
3.47	0.59	0.59	ga4[GL6]ga4gl (glc core junk)																
3.51	1.28	1.28	ga3[GL6]ga4gl (glc core junk)	3.51	1.28	1.28	ga3[GL6]ga4gl (glc core junk)												
3.62	0.54	0.54	fa2ga3)[GL6]GA	3.61	0.30	0.30	(fa2ga3)[GL6]GA	3.61	0.31	0.31	(fa2ga3)[GL6]GA	3.61	0.62	0.62	fa2ga3)[GL6]GA				
3.66	0.62	0.62	ga3[(fa3GL6)]GA	3.66	0.57	0.57	ga3[(fa3GL6)]GA	3.64	0.50	0.50	ga3[(fa3GL6)]GA								
		0.89	junk			0.50	junk			0.61	junk			0.91	junk			0.95	junk
4.07	2.01			4.07	1.73	0.11	ga3[GL4ga3GL6]GA	4.03	0.91	0.30	ga3[GL4ga3GL6]GA	4.04	1.21	0.30	ga3[GL4ga3GL6]GA	4.04	1.25	0.30	ga3[GL4ga3GL6]GA
		0.64	GL4ga3[ga3GL6]GA			0.64	GL4ga3[ga3GL6]GA												
		0.48	junk			0.48	junk	4.07	0.59	0.59	junk	4.08	0.43	0.43	junk	4.08	0.52	0.52	junk
4.13	0.67	0.67	junk	4.13	0.83			4.11	0.37	0.37	junk	4.13	0.34	0.34	junk	4.13	0.44	0.44	junk
4.16	0.34	0.34	junk			0.83	junk	4.15	0.62	0.62	junk	4.15	0.87	0.87	junk	4.16	0.99	0.99	junk
4.21	0.88	0.88	ga4GL2[ga4GL6]m																
4.30	0.27	0.27	junk	4.29	0.23	0.23	junk	4.28	0.22	0.22	junk	4.26	0.16	0.16	junk	4.27	0.23	0.23	junk
4.31	0.36	0.36	M3	4.31	0.27	0.27	M3	4.30	0.28	0.28	M3	4.30	0.24	0.24	M3			0.32	M3
4.34	0.41	0.11	ga3[ga4(fa3GL6)]GA	4.35	0.34			4.33	0.35			4.35	0.28			4.36	0.64		
		0.30	junk			0.34	junk			0.35	junk			0.28	junk			0.32	junk
4.44	1.19	1.19	sU3ga4GL2m	4.44	1.18	1.18	sU3ga4GL2m	4.43	1.12	1.12	sU3ga4GL2m	4.43	1.07	1.07	sU3ga4GL2m	4.44	1.15	1.15	sU3ga4GL2m
4.51	0.49	0.49	sU3ga4GL2m	4.50	0.51	0.51	sU3ga4GL2m	4.49	0.57	0.57	sU3ga4GL2m	4.50	0.51	0.51	sU3ga4GL2m	4.50	0.57	0.57	sU3ga4GL2m
4.83	0.20	0.20	ga3[ga3GL4ga3GL6]GA	4.83	0.20	0.20	ga3[ga3GL4ga3GL6]GA												
4.85	0.56	0.10	ga3GL4ga3[ga3GL6]GA			0.10	ga3GL4ga3[ga3GL6]GA												
		0.46	junk	4.88	0.74	0.64	junk	4.87	0.72	0.72	junk	4.87	0.55	0.55	junk	4.88	0.41	0.41	junk
4.88	0.36	0.11	ga3[ga4GL4ga3GL6]GA																
		0.25	junk																
4.93	0.84	0.84	junk	4.93	0.86	0.86	junk	4.92	0.76	0.76	junk	4.92	0.97	0.97	junk	4.92	0.94	0.94	junk
4.98	0.59	0.59	junk	4.99	0.60	0.60	junk	5.00	0.73	0.73	junk	4.99	0.55	0.55	junk	5.00	0.69	0.69	junk
5.09	1.10	1.10	junk	5.09	1.00	1.00	junk	5.08	1.05	1.05	junk	5.07	1.07	1.07	junk	5.08	1.12	1.12	junk
5.17	0.29	0.29	junk	5.17	0.39	0.39	junk	5.17	0.33	0.33	junk	5.16	0.35	0.35	junk	5.16	0.41	0.41	junk
5.59	1.04	0.30	A2	5.60	0.96	0.30	A2	5.48	0.44	0.44	A2	5.47	0.55	0.55	A2	5.47	0.89	0.89	A2
		0.74	junk			0.66	junk	5.60	0.89	0.89	junk	5.59	0.57	0.57	junk	5.60	0.65	0.65	junk
																5.81	0.78	0.78	A3
6.14	1.71	1.71	M5	6.14	1.62	1.62	M5	6.14	1.78	1.78	M5	6.13	1.71	1.71	M5	6.13	1.87	1.87	M5
6.23	0.57	0.57	F(6)A3	6.23	0.72	0.72	F(6)A3	6.23	0.77	0.77	F(6)A3	6.23	0.96	0.96	F(6)A3				
6.34	0.34	0.34	junk	6.34	0.30	0.30	junk	6.34	0.28	0.28	junk	6.33	0.31	0.31	junk	6.34	0.33	0.33	junk
																7.02	1.02	0.30	A3
								6.89	0.27	0.27	F(6)A4	6.89	0.42	0.42	F(6)A4				
7.02	1.16	1.16	M6	7.02	0.97	0.97	M6	7.03	0.90	0.90	M6	7.03	0.91	0.91	M6			0.72	M6
7.59	0.44	0.44	junk	7.58	0.46	0.46	junk	7.59	0.39	0.39	junk	7.58	0.34	0.34	junk	7.58	0.41	0.41	junk
7.76	0.48	0.48	junk	7.79	0.45	0.45	junk	7.78	0.49	0.49	junk	7.79	0.44	0.44	junk	7.77	0.64	0.64	junk
7.92	0.64	0.64	M7	7.92	0.62	0.62	M7	7.93	0.56	0.56	M7	7.93	0.58	0.58	M7	7.93	0.66	0.66	M7
8.76	0.40	0.40	M8	8.76	0.47	0.47	M8	8.78	0.40	0.40	M8	8.78	0.40	0.40	M8	8.78	0.48	0.48	M8
8.97	0.44	0.44	junk	8.96	0.42	0.42	junk	8.97	0.36	0.36	junk	8.98	0.24	0.24	junk	8.98	0.28	0.28	junk
9.45	0.59	0.59	M9	9.45	0.48	0.48	M9	9.46	0.68	0.68	M9	9.48	0.49	0.49	M9	9.47	0.68	0.68	M9
	99.95	99.95			100.02	100.02			100.01	100.01			99.97	99.97			100.03	100.03	

Appendix V (cont.). *Exoglycosidase digestion panel for healthy substantia nigra pool.*

ABS+BTG+AMF+BKF+GUH				ABS+BTG+AMF+BKF+GUH+JBH				ABS+BTG+AMF+BKF+GUH+JBH+CBG				ABS+BTG+AMF+BKF+GUH+JBH+CBG+JBM			
GU	% Area		Structure	GU	% Area		Structure	GU	% Area		Structure	GU	% Area		Structure
	4.87	4.87	GA (galNAc)		3.85	3.85	GA (galNAc)		3.73	3.73	GA (galNAc)		3.09	3.09	GA (galNAc)
1.02	37.50	27.50	gl (glucose)	1.02	37.03	27.03	gl (glucose)	1.02	38.09	28.09	gl (glucose)	1.02	38.58	28.33	gl (glucose)
		10.00	m (mannose)			10.00	m (mannose)			10.00	m (mannose)			10.25	m (mannose)
1.04	5.95	5.95	ga (galactose)	1.04	5.29	5.29	ga (galactose)	1.04	5.89	5.89	ga (galactose)	1.04	6.27	6.27	ga (galactose)
1.07	0.30	0.30	junk	1.07	0.36	0.36	junk	1.07	0.14	0.14	junk	1.09	0.21	0.21	junk
1.10	1.39	1.39	junk	1.10	1.52	1.52	junk	1.10	1.57	1.57	junk	1.11	1.15	1.15	junk
1.69	1.38	1.38	sga3GA***	1.68	1.68	1.68	sga3GA***	1.68	1.63	1.63	sga3GA***	1.70	1.48	1.48	sga3GA***
		8.06	ga3GA (incomplete dig)			8.99	ga3GA (incomplete dig)			9.52	ga3GA (incomplete dig)			10.05	ga3GA (incomplete dig)
1.78	8.27			1.77	9.20			1.77	9.52			1.81	10.05		
		0.21	gaa3GA			0.21	gaa3GA								
1.81	3.27	3.27	junk	1.80	3.07	3.07	junk	1.80	3.14	3.14	junk	1.83	3.27	3.27	junk
1.86	1.84	1.84	junk	1.87	1.34	1.34	junk	1.86	1.15	1.15	junk	1.88	1.03	1.03	junk
1.93	1.48			1.93	1.79			1.92	1.97			1.96	2.56		
1.97	2.44	1.48	junk	1.98	3.56	1.79	junk	1.98	3.30	1.97	junk	2.00	2.34	2.56	junk
1.99	1.22	2.44	junk	2.00	0.71	3.56	junk	1.99	0.77	3.30	junk	2.01	0.75	2.34	junk
2.32	0.97	1.22	junk	2.32	0.99	0.71	junk	2.31	0.94	0.77	junk	2.35	0.80	0.75	junk
		0.97	junk			0.99	junk			0.94	junk			0.80	junk
2.62	3.83	3.83	gl4ga3gl (glc core junk)	2.61	3.98	3.98	gl4ga3gl (glc core junk)	2.60	4.19	4.19	gl4ga3gl (glc core junk)	2.63	3.78	3.78	gl4ga3gl (glc core junk)
2.68	0.65	0.65	junk	2.67	0.61	0.61	junk	2.67	0.53	0.53	junk	2.67	6.68	0.50	junk
														6.18	M1
2.76	0.73	0.73	junk	2.75	0.73	0.73	junk	2.74	0.66	0.66	junk	2.77	1.14	1.14	junk
2.86	0.78	0.78	GA4ga4gl (glc core junk)	2.85	0.92	0.92	GA4ga4gl (glc core junk) (incomplete dig)	2.85	0.82	0.82	GA4ga4gl (glc core junk) (incomplete dig)	2.87	0.81	0.81	GA4ga4gl (glc core junk) (incomplete dig)
3.26	1.91	1.91	junk	3.25	1.97	1.97	junk	3.25	1.90	1.90	junk	3.28	2.63	2.63	junk
		0.78	junk			0.83	junk			0.71	junk			0.53	junk
3.32	0.78			3.32	0.83			3.32	0.71			3.35	0.53		
3.39	1.34	1.34	ga3[sGL6]GA	3.38	1.25	1.25	ga3[sGL6]GA	3.38	1.26	1.26	ga3[sGL6]GA	3.41	1.28	1.28	ga3[sGL6]GA
3.44	0.84	0.84	junk	3.43	1.08	1.08	junk	3.43	1.13	1.13	junk	3.46	1.11	1.11	junk
		0.97	junk			1.07	junk								
4.02	0.97			4.01	1.07										
4.09	0.48	0.48	junk	4.07	0.52	0.52	junk	4.08	0.51	0.51	junk	4.11	0.49	0.49	junk
4.12	0.37	0.37	junk	4.12	0.41	0.41	junk	4.12	0.34	0.34	junk	4.15	0.09	0.09	junk
4.16	0.97	0.97	junk	4.14	1.05	1.05	junk	4.15	1.18	1.18	junk	4.17	1.11	1.11	junk
4.28	0.26	0.26	junk	4.27	0.28	0.28	junk	4.28	0.24	0.24	junk	4.30	0.23	0.23	junk
		2.26	M3			2.28	M3			2.09	M3				
4.36	2.90			4.35	2.92			4.36	2.73						
		0.64	junk			0.64	junk			0.64	junk	4.47	1.41	0.30	junk
4.44	1.30	1.30	sU3ga4GL2m	4.43	1.68	1.68	sU3ga4GL2m	4.44	1.59	1.59	sU3ga4GL2m			1.11	sU3ga4GL2m
4.50	0.66	0.66	sU3ga4GL2m	4.49	0.63	0.63	sU3ga4GL2m	4.51	0.63	0.63	sU3ga4GL2m	4.54	0.59	0.59	sU3ga4GL2m
4.87	0.55	0.55	junk	4.87	0.52	0.52	junk	4.88	0.48	0.48	junk	4.90	0.56	0.56	junk
4.93	0.95	0.95	junk	4.92	0.47	0.47	junk	4.92	0.54	0.54	junk	4.96	0.65	0.65	junk
4.99	0.59	0.59	junk	4.98	0.61	0.61	junk	4.99	0.58	0.58	junk	5.02	0.80	0.80	junk
5.08	1.27	1.27	junk	5.07	1.21	1.21	junk	5.08	1.23	1.23	junk	5.12	1.26	1.26	junk
5.16	0.52	0.52	junk	5.15	0.57	0.57	junk	5.17	0.45	0.45	junk	5.21	0.24	0.24	junk
5.60	0.75	0.75	junk	5.59	0.76	0.76	junk	5.60	0.89	0.89	junk	5.61	1.04	1.04	junk
6.14	1.73	1.73	M5	6.13	1.82	1.82	M5	6.14	1.71	1.71	M5				
6.34	0.34	0.34	junk	6.33	0.33	0.33	junk	6.35	0.37	0.37	junk	6.38	0.28	0.28	junk
7.04	0.85	0.85	M6	7.03	0.86	0.86	M6	7.04	0.89	0.89	M6				
7.58	0.54	0.54	junk	7.56	0.42	0.42	junk	7.58	0.36	0.36	junk	7.59	0.28	0.28	junk
7.79	0.44	0.44	junk	7.79	0.44	0.44	junk	7.79	0.41	0.41	junk	7.83	0.68	0.68	junk
7.94	0.42	0.42	M7	7.93	0.44	0.44	M7	7.94	0.47	0.47	M7				
8.79	0.48	0.48	M8	8.79	0.44	0.44	M8	8.79	0.45	0.45	M8				
8.99	0.36	0.36	junk	8.98	0.30	0.30	junk	8.99	0.39	0.39	junk	9.03	0.76	0.76	junk
9.48	0.58	0.58	M9	9.47	0.57	0.57	M9	9.48	0.55	0.55	M9				
	100.02	100.02			100.08	100.08			100.03	100.03			100.01	100.01	

Appendix V (cont.). *Exoglycosidase digestion panel for healthy substantia nigra pool.*

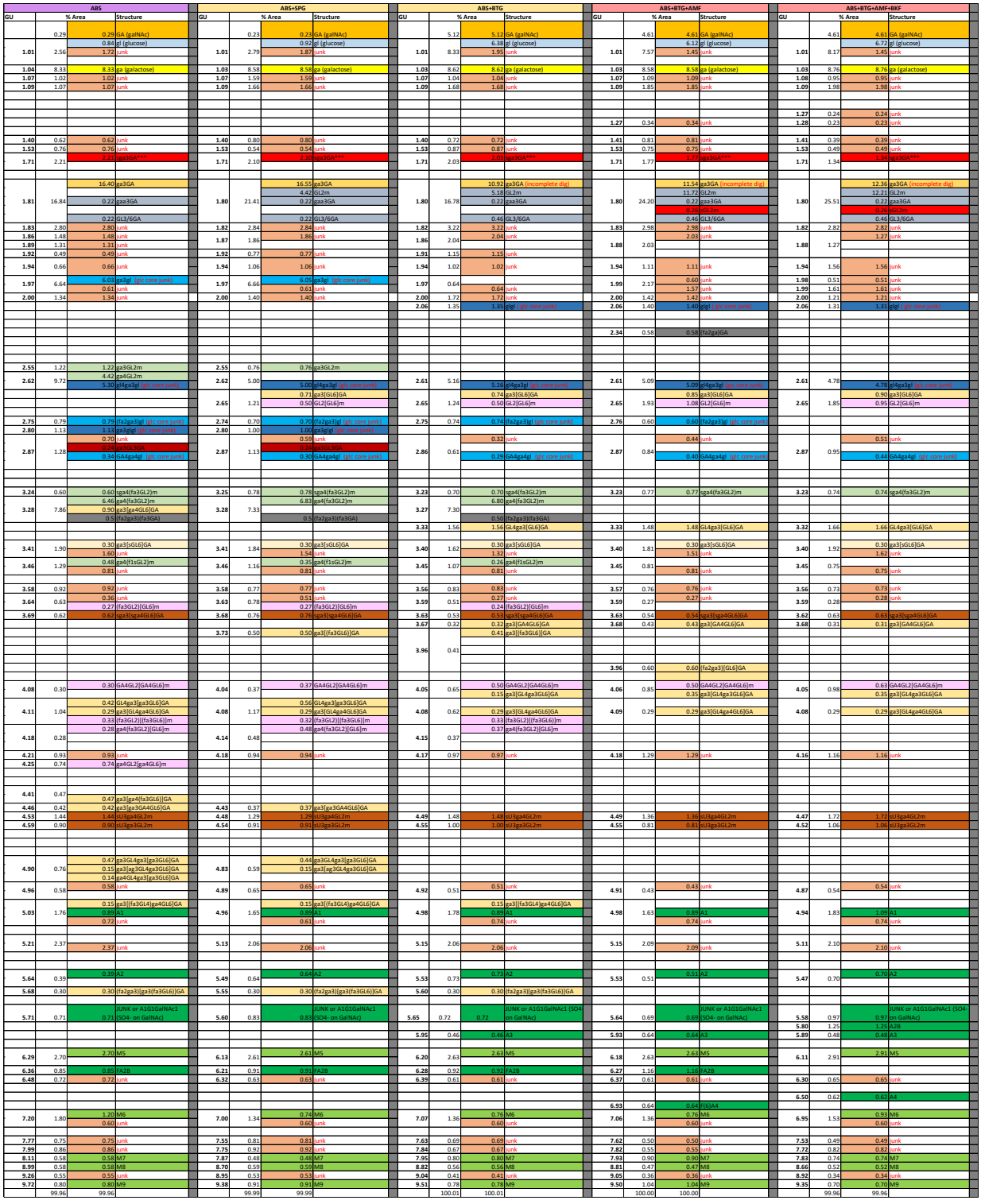
											% Area
										% Area	**JBM**
									% Area	**CBG**	**CBG**
								% Area	**JBH**	**JBH**	**JBH**
							% Area	**GUH**	**GUH**	**GUH**	**GUH**
						% Area	**BKF**	**BKF**	**BKF**	**BKF**	**BKF**
				% Area	% Area	**AMF**	**AMF**	**AMF**	**AMF**	**AMF**	**AMF**
	% Area	% Area	% Area	**SPG**	**BTG**	**BTG**	**BTG**	**BTG**	**BTG**	**BTG**	**BTG**
	UND	**NAN1**	**ABS**	**ABS**	**ABS**	**ABS**	**ABS**	**ABS**	**ABS**	**ABS**	**ABS**
GalNAc	0.62	0.16	0.30	0.27	3.32	3.00	2.90	4.87	3.85	3.73	3.09
Gal	0.73	3.14	5.01	5.18	6.13	6.36	5.74	5.95	5.29	5.89	6.27
Glc	3.80	2.98	3.04	3.79	27.42	27.96	27.11	27.50	27.03	28.09	28.33
Man								10.00	10.00	10.00	10.25
Semi-digestible Disaccharides (ga3GA)	0.20	4.41	10.23	10.48	7.57	7.48	7.71	8.06	8.99	9.52	10.05
Digestible Disaccharides (others)	0.41	0.42	0.42	3.82	4.83	9.33	9.52	0.21	0.21		
Undigestible Disaccharides (sulfated)			1.27	1.25	1.08	1.09	1.27	1.38	1.68	1.63	1.48
Core 1 extensions											
Core 2 and extensions	2.31	3.01	3.29	2.55	2.58	2.81	2.48				
Core 2 extensions with sulfation	0.91	0.93	1.17	1.29	1.39	1.05	1.52	1.34	1.25	1.26	1.28
Core 3 extensions	0.10	0.10	0.10	0.10							
Mannose Core 1 (linear trisaccharide and larger)	8.44	8.34	8.37	5.31	4.50						
Mannose Core 2 (branched trisaccharide and larger)	0.61	0.67	0.88	0.30	0.51	0.59	0.59				
SSTn, STn, ST antigens, peeling, disialyl	18.72	10.17									
STn and ST antigens (sulfated)	0.38	0.32									
Larger Sulfated Glycans that don't digest	1.22	1.49	1.68	1.69	1.69	1.58	1.72	1.96	2.31	2.22	1.70
High Mannose (N)	5.03	4.39	4.86	4.43	4.60	4.33	4.73	6.32	6.41	6.16	6.18
Hybrid (N)	0.93	0.93	0.87	1.02	1.48	1.93	1.97				
Digestible hex chains	29.97	28.45	28.23	27.68	1.82	2.17	2.19	0.78	0.92	0.82	0.81
Undigestible hex chains	3.65	3.65	3.65	3.65	3.95	3.91	3.72	3.83	3.98	4.19	3.78
Integrated Junk	21.99	26.43	26.58	27.21	27.14	26.38	26.86	27.82	28.16	26.52	26.79
SUM	100.02	99.99	99.95	100.02	100.01	99.97	100.03	100.02	100.08	100.03	100.01

	% Area	Comment
Total GalNAc core	19.24	(including the 1 glycan in the larger sulfated glycans that doesn't digest)
Total Mannose core	10.27	(including the 1 glycan in the larger sulfated glycans that doesn't digest)
Total Hexose chain glycans (Glucose core)	37.42	inlcudes glc, gal-glc, glc-gal-glc and gal-gal-glc
Total Galactose core (probably peeling)	3.14	probably core 1 and 2 peeling
Total NeuNac core (polysialic)	2.00	One polysialic structure also seen in Gothenburg data
Total N glycans	5.75	leftover from Ana's digest or extra N-glycan that was released in microwave
Total Junk	22.20	Suspected tissue junk and junk present in blank
Total (should equal 100.02)	100.02	total % area from UND
Total Glycan	77.82	
Total Junk	22.20	

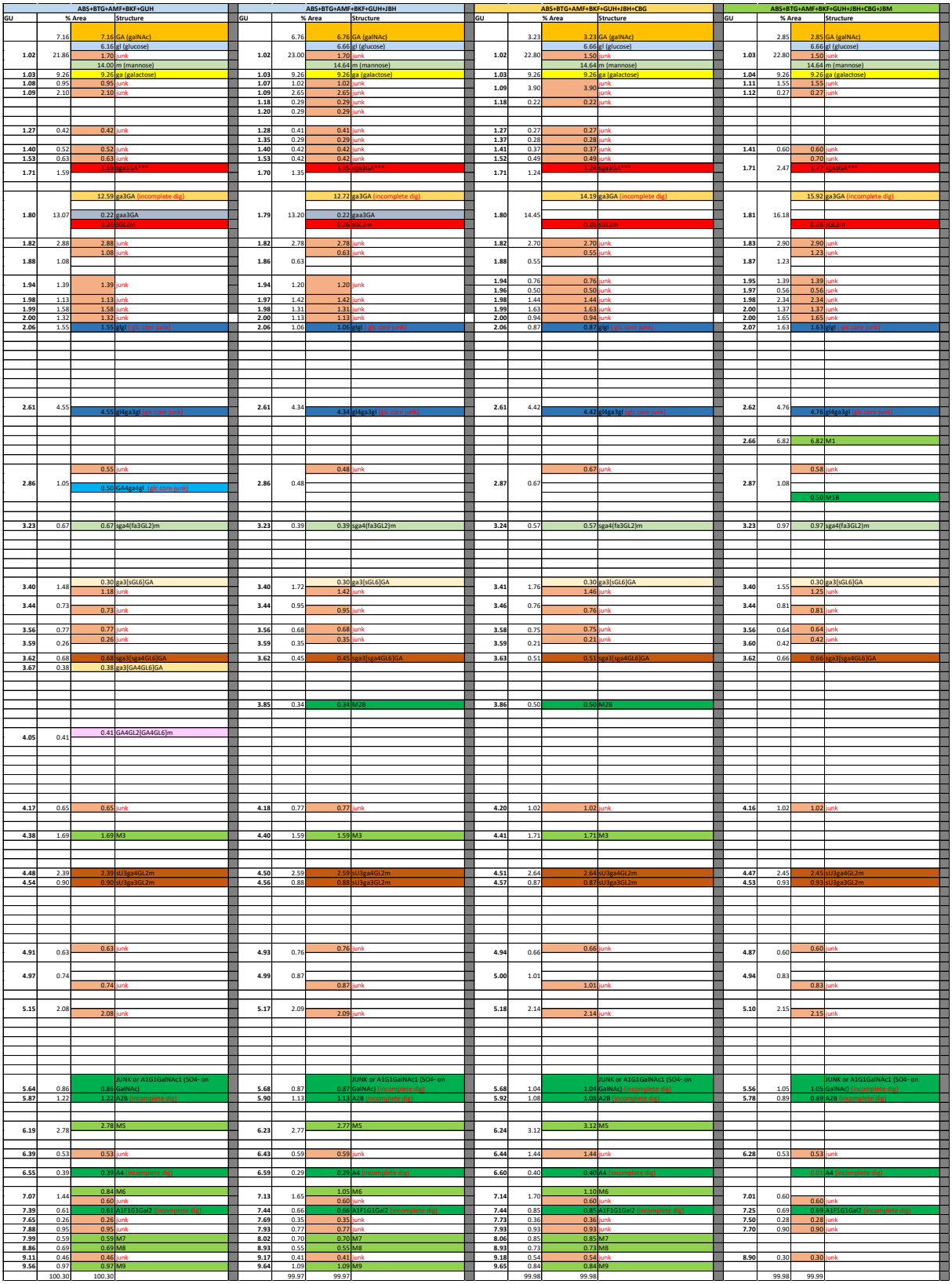
Appendix VI. *Summary of % areas – Exoglycosidase digestion panel for healthy substantia nigra pool.*

Structure (long name)	Structure (image)	RT	BLANK				UND				NAN1			
			GU	% Area		Structure	GU	% Area			GU	% Area		
galNAc		1.75						0.19	0.19	GA (galNAc)		0.37	0.37	GA (galNAc)
glucose									0.97	gl (glucose)			1.11	gl (glucose)
		1.93	**1.01**	8.25	8.25	junk	**1.01**	2.95	1.98	junk	**1.01**	3.37	2.26	junk
mannose														
galactose		2.03					**1.03**	0.90	0.90	ga (galactose)	**1.03**	7.62	7.62	ga (galactose)
		2.15	**1.08**	2.01	2.01	junk	**1.06**	1.16	1.16	junk	**1.08**	0.88	0.88	junk
		2.27					**1.09**	2.06	2.06	junk	**1.09**	3.04	3.04	junk
			1.11	13.85	13.85	junk								
		3.23	**1.20**	0.08	0.08	junk								
							1.33	0.32	0.32	junk	**1.35**	0.71	0.71	junk
		4.02					**1.53**	0.79	0.79	junk	**1.53**	0.50	0.50	junk
Galβ1-3-SO3-GalNAc		4.67	**1.74**	5.65	5.65	junk			0.81	ga3GA***			0.81	ga3GA***
SO3-(NeuAcα2-6)GalNAc							**1.70**	0.91	0.10	S6sGA	**1.72**	0.71	0.10	S6sGA
NeuNAc-α2-3 NeuNAc							**1.77**	3.44	3.44	S3S**				
Galβ1-3GalNAc									0.22	ga3GA			6.84	ga3GA
GlcNAcβ1-2Man														
Galα1-3GalNAc		4.98					**1.78**	0.66	0.22	gaa3GA	**1.79**	7.28	0.22	gaa3GA
SO3-GlcNAcβ1-2Man														
GlcNAcβ1-3GalNAc/GlcNAcβ1-6GalNAc									0.22	GL3/6GA			0.22	GL3/6GA
		5.12					**1.81**	2.57	2.57	junk	**1.82**	2.49	2.49	junk
			1.79	13.43	13.43	junk	**1.85**	0.97	0.97	junk	**1.85**	0.90	0.90	junk
		5.39					**1.88**	1.12	1.12	junk				
													0.93	junk
											1.97	7.22		
Galβ1-3Glc		5.70	**1.94**	23.41	23.41	junk			6.10	ga3gl (gtc core junk)			5.68	ga3gl (gtc core junk)
							1.96	6.71	0.61	junk			0.61	junk
		5.86					**2.00**	1.27	1.27	junk	**2.00**	2.60	2.60	junk
			2.05	0.91	0.91	junk								
			2.28	1.34	1.34	junk								
(Fucα1-2Galβ1-3)GalNAc			**2.34**	1.10	1.10	junk								
NeuAcα2-3Gal		7.28					**2.36**	0.86	0.86	S3ga* (split peak)				
NeuAcα2-3Gal		7.48					**2.41**	8.90	6.24	S3ga*	**2.41**	2.66		
NeuAcα2-6GalNAc									2.66	S6GA			2.66	S6GA
Galβ1-3GlcNAcβ1-2Man											**2.55**	0.88	0.88	ga3GL2m
Galβ1-4GlcNAcβ1-2Man		8.26	**2.63**	13.45			**2.62**	5.22	0.65	ga4GL2m	**2.61**	9.19	4.19	ga4GL2m
Glcβ1-4Galβ1-3Glc					13.45	junk			4.57	g4ga3gl (gtc core junk)			5.00	g4ga3gl (gtc core junk)
Galβ1-3GlcNAcβ1-6)GalNAc														
GlcNAcβ1-2[GlcNAcβ1-6]Man														
M1														
(Fucα1-2Galβ1-3)Glc		8.75					**2.75**	0.51	0.51	fa2ga3gl (gtc core junk)	**2.75**	0.69	0.69	fa2ga3gl (gtc core junk)
Galβ1-4Galβ1-3Glc		8.95				junk	**2.80**	0.82	0.82	ga3glg! (gtc core junk)	**2.79**	1.03	1.03	ga3glg! (gtc core junk)
		9.19	**2.76**	2.30	2.30	junk			0.46	junk			0.46	junk
Galβ1-3GlcNAcβ1-3GalNAc							**2.87**	0.46			**2.86**	0.73		
GalNAcβ1-4Galβ1-3Glc						junk							0.27	GA4ga4gl (gtc core junk)
M1B														
NeuAcα2-3Galβ1-3GalNAc		9.51					**2.96**	0.65	0.65	S3ga3GA (split peak)				
NeuAcα2-3Galβ1-3GalNAc		9.75					**3.03**	4.63	4.63	S3ga3GA				
SO3-Galβ1-4(Fucα1-3GlcNAcβ1-2)Man		10.43					**3.23**	0.20	0.20	sga4(fa3GL2)m	**3.23**	0.60	0.60	sga4(fa3GL2)m
Galβ1-4(Fucα1-3GlcNAcβ1-2)Man									5.91	ga4(fa3GL2)m			6.74	ga4(fa3GL2)m
Galβ1-3[Galβ1-4Galβ1-3GlcNAcβ1-6]GalNAc		10.56					**3.27**	6.41			**3.27**	7.94	0.70	ga3[ga4GL6]GA
(Fucα1-2Galβ1-3)(Fucα1-3GalNAc)									0.50	(fa2ga3)(fa3GA)			0.5	(fa2ga3)(fa3GA)
GlcNAcβ1-4Galβ1-3[GlcNAcβ1-6]GalNAc														
Galβ1-3(NeuAcα2-6)GalNAc		10.90					**3.38**	2.44	2.44	ga3[S6]GA			10.52	ga3[S6]GA
Galβ1-3[SO3-GlcNAcβ1-6]GalNAc		10.96	**3.36**	4.82			**3.40**	1.60	0.30	ga3[sGL6]GA	**3.37**	12.12	0.30	ga3[sGL6]GA
					4.82	junk			1.30	junk			1.30	junk
Galβ1-4(Fucα1-3-SO3-GlcNAcβ1-2)Man		11.11	**3.44**	1.63			**3.45**	0.90	0.09	ga4(f1sGL2)m	**3.44**	0.87	0.06	ga4(f1sGL2)m
					1.63	junk			0.81	junk			0.81	junk
SO3-Galβ1-3(NeuAcα2-6)GalNAc		11.28					**3.50**	0.88	0.88	sga3[S6]GA	**3.50**	1.23	1.23	sga3[S6]GA
		11.46	**3.56**	1.70	1.70	junk	**3.56**	0.60	0.60	junk	**3.56**	0.77	0.77	junk
		11.65	**3.62**	1.69	1.69	junk			0.20	junk			0.25	junk
(Fucα1-3GlcNAcβ1-2)[GlcNAcβ1-6]Man							**3.63**	0.47	0.27	(fa3GL2)[GL6]m	**3.61**	0.52	0.27	(fa3GL2)[GL6]m
SO3-Galβ1-3[SO3-Galβ1-4GlcNAcβ1-6]GalNAc		11.79					**3.68**	0.27	0.27	sga3[sga4GL6]GA	**3.68**	0.58	0.58	sga3[sga4GL6]GA
Galβ1-3[GalNAcβ1-4GlcNAcβ1-6]GalNAc														
Galβ1-3[(Fucα1-3GlcNAcβ1-6)]GalNAc														
NeuAcα2-3Galβ1-4GlcNAcβ1-2Man		11.93					**3.72**	3.49	3.49	S3ga4GL2m				
NeuAcα2-3Galβ1-3GlcNAcβ1-2Man		12.10					**3.78**	0.78	0.78	S3ga3GL2m				
M2B														
SO3-Galβ1-3[NeuAcα2-6]GalNAc													0.32	ga3[sS6]GA
Galβ1-3GlcNAcβ1-3[NeuAcα2-6]GalNAc											**3.95**	0.56	0.24	ga3GL3[S6]GA
GalNAcβ1-4GlcNAcβ1-2[GalNAcβ1-4GlcNAcβ1-6]Man													0.29	GA4GL2[GA4GL6]m
Galβ1-3[GlcNAcβ1-4Galβ1-3GlcNAcβ1-6]GalNAc											**4.05**	0.29		
GlcNAcβ1-4Galβ1-3[Galβ1-3GlcNAcβ1-6]GalNAc		12.93							0.07	GL4ga3[ga3GL6]GA			0.14	GL4ga3[ga3GL6]GA
Galβ1-3[GlcNAcβ1-4Galβ1-4GlcNAcβ1-6]GalNAc						junk	**4.09**	0.69	0.29	ga3[GL4ga4GL6]GA	**4.09**	0.76	0.29	ga3[GL4ga4GL6]GA
(Fucα1-3GlcNAcβ1-2)[(Fucα1-3GlcNAcβ1-6)]Man									0.33	(fa3GL2)[(fa3GL6)]m			0.33	(fa3GL2)[(fa3GL6)]m
Galβ1-4-Fucα1-3-GlcNAcβ1-2[GlcNAcβ1-6]Man		13.03							0.28	ga4(fa3GL2)[GL6]m			0.28	ga4(fa3GL2)[GL6]m
NeuAcα2-6GalNAcβ1-4Galβ1-3Glc			**4.17**	2.49	2.49	junk	**4.13**	0.59	0.07	S6GA4ga4gl (gtc core junk)	**4.16**	0.35	0.07	S6GA4ga4gl (gtc core junk)
NeuAcα2-3GalNAcβ1-4Galβ1-3Glc						junk			0.24	S3GA4ga4gl (gtc core junk)				
		13.15				junk	**4.18**	0.66	0.66	junk	**4.18**	0.91	0.91	junk
Galβ1-4GlcNAcβ1-2[Galβ1-4GlcNAcβ1-6]Man						junk					**4.24**	0.57	0.57	ga4GL2[ga4GL6]m
Galβ1-3[NeuAcα2-6Galβ1-4GlcNAcβ1-6]GalNAc											**4.37**	0.25	0.25	ga3[S6ga4GL6]GA
NeuAcα2-3Galβ1-4(Fucα1-3GlcNAcβ1-2)Man									0.26	S3ga3(fa3GL2)m				
Galβ1-3[NeuAcα2-3Galβ1-4GlcNAcβ1-6]GalNAc		13.72					**4.41**	1.06	0.30	ga3[S3ga4GL6]GA	**4.40**	0.40		
Galβ1-3[(Galβ1-4Fucα1-3GlcNAcβ1-6)]GalNAc									0.50	ga3[ga4(fa3GL6)]GA			0.40	ga3[ga4(fa3GL6)]GA
Galβ1-3[Galβ1-3GalNAcβ1-4GlcNAcβ1-6]GalNAc		13.83					**4.46**	0.32	0.32	ga3[ga3GA4GL6]GA	**4.46**	0.21	0.21	ga3[ga3GA4GL6]GA
SO3-GlcAβ1-3Galβ1-4GlcNAcβ1-2Man		13.93					**4.50**	0.97	0.97	sU3ga4GL2m	**4.50**	1.23	1.23	sU3ga4GL2m
SO3-GlcAβ1-3Galβ1-3GlcNAcβ1-2Man		14.08					**4.56**	0.74	0.74	sU3ga3GL2m	**4.57**	0.76	0.76	sU3ga3GL2m
NeuAcα2-3Galβ1-3[Galβ1-4GlcNAcβ1-6]GalNAc		14.19							0.40	S3ga3[ga4GL6]GA				
NeuAcα2-6Galβ1-4(Fucα1-3-SO3-GlcNAcβ1-2)Man							**4.61**	0.56	0.16	S6ga4(f1sGL2)m	**4.62**	0.16	0.16	S6ga4(f1sGL2)m
NeuAcα2-3Galβ1-3[NeuAcα2-6]GalNAc		14.43					**4.72**	9.72	9.72	S3ga3[S6]GA				
Galβ1-3GlcNAcβ1-4Galβ1-3[Galβ1-3GlcNAcβ1-6]GalNAc		14.75							0.46	ga3GL4ga3[ga3GL6]GA			0.55	ga3GL4ga3[ga3GL6]GA
Galβ1-3[Galβ1-3GlcNAcβ1-4Galβ1-4GlcNAcβ1-6]GalNAc							**4.86**	0.74	0.15	ga3[ag3GL4ga3GL6]GA	**4.86**	0.87	0.15	ga3[ag3GL4ga3GL6]GA
Galβ1-4GlcNAcβ1-4Galβ1-3[Galβ1-3GlcNAcβ1-6]GalNAc									0.13	ga4GL4ga3[ga3GL6]GA			0.17	ga4GL4ga3[ga3GL6]GA
		14.88	**4.90**	0.96	0.96	junk	**4.92**	0.53	0.53	junk	**4.91**	0.94	0.53	junk
NeuAcα2-6GlcNAcβ1-4Galβ1-3[Galβ1-4GlcNAcβ1-6]GalNAc													0.41	S6GL4ga3[ga3GL6]GA
Galβ1-3[(Fucα1-3GlcNAcβ1-4)Galβ1-4GlcNAcβ1-6)]GalNAc		15.02							0.14	ga3[(fa3GL4)ga4GL6]GA			0.12	ga3[(fa3GL4)ga4GL6]GA
α1			**4.95**	0.60			**4.99**	1.32	0.74	A1	**4.98**	1.31	0.74	A1
					0.60	junk			0.44	junk			0.44	junk
NeuAcα2-3Galβ1-3GlcNAcβ1-3[NeuAcα2-6]GalNAc							**5.05**	0.24	0.24	S3ga3GL3[S6]GA				
NeuAcα2-3[SO3-]Galβ1-3[NeuAcα2-6]GalNAc		15.38					**5.16**	2.39	0.32	S3sga3[S6]GA	**5.16**	2.07		
									2.07	junk			2.07	junk
NeuAcα2-3Galβ1-4GlcNAcβ1-2[Galβ1-4GlcNAcβ1-6]Man		15.54					**5.24**	0.49	0.29	S3-ga4GL2[ga4GL6]m				
NeuAcα2-3Galβ1-3[NeuAcα2-6Galβ1-4GlcNAcβ1-6]GalNAc									0.20	S3ga3[S6ga4GL6]GA				
α2		16.34							0.49	A2			0.57	A2
NeuAcα2-6-Galβ1-3[Galβ1-4(Fucα1-3GlcNAcβ1-6)]GalNAc			**5.65**	0.34	0.34	junk	**5.65**	0.59	0.10	S6-ga3[ga4(fa3GL6)]GA	**5.65**	0.67	0.10	S6-ga3[ga4(fa3GL6)]GA
Fucα1-2Galβ1-3[(Galβ1-4Fucα1-3GlcNAcβ1-6)]GalNAc							**5.68**	0.30	0.30	(fa2ga3)[ga3(fa3GL6)]GA	**5.68**	0.30	0.30	(fa2ga3)[ga3(fa3GL6)]GA
NeuAcα2-3Galβ1-3[NeuAcα2-3Galβ1-4GlcNAcβ1-6]GalNAc							**5.78**	0.38	0.38	S3ga3[sS6]GA				
JUNK or α1G1GalNAc1 (SO4- on GalNAc)		16.57												
α2B														
NeuAcα2-6GlcNAcβ1-4Galβ1-3[NeuAcα2-3Galβ1-4GlcNAcβ1-6]GalNAc		16.87							0.41	S6GL4ga3[S3ga3GL6]GA				
NeuAcα2-3-SO3-Galβ1-3[NeuAcα2-3-SO3-Galβ1-4GlcNAcβ1-6]GalNAc							**5.94**	0.74	0.33	S3sga3[S3sga4GL6]GA				
M5		17.32							2.44	M5			2.64	M5
NeuAcα2-3GalNAcβ1-4GlcNAcβ1-2[NeuAcα2-3GalNAcβ1-4GlcNAcβ1-6]Man							**6.20**	2.73	0.29	S3GA4GL2[S3GA4GL6]m	**6.20**	2.64		
Fa2B		17.45					**6.28**	0.88	0.88	FA2B	**6.29**	0.90	0.90	FA2B
		17.62					**6.38**	0.56	0.56	junk	**6.39**	0.61	0.61	junk
NeuAcα2-3Galβ1-4GlcNAcβ1-2[NeuAcα2-3Galβ1-4GlcNAcβ1-6]Man		17.75					**6.45**	0.24	0.24	S3ga4GL2[S3ga4GL6]m				
M6		18.75					**7.08**	1.58	0.98	M6	**7.07**	1.51	0.91	M6
									0.60	junk			0.60	junk
α1F1G1Gal2														
		19.57					**7.64**	0.67	0.67	junk	**7.66**	0.71	0.71	junk
		19.87					**7.86**	0.69	0.69	junk	**7.86**	0.74	0.74	junk
M7		20.01					**7.96**	0.69	0.69	M7	**7.95**	0.57	0.57	M7
M8		21.12					**8.83**	0.66	0.66	M8	**8.84**	0.63	0.63	M8
		21.38					**9.04**	0.88	0.88	junk	**9.08**	0.58	0.58	junk
M9		21.92					**9.52**	1.32	1.32	M9	**9.54**	1.03	1.03	M9
								100.04	100.04			100.03	100.03	

Appendix VII. *Exoglycosidase digestion panel for PD (stages 1-4) substantia nigra pool.*



Appendix VII (cont.). *Exoglycosidase digestion panel for PD (stages 1-4) substantia nigra pool.*



Appendix VII (cont.). *Exoglycosidase digestion panel for PD (stages 1-4) substantia nigra pool.*

											% Area
										% Area	JBH
									% Area	CBG	CBG
								% Area	JBH	JBH	JBH
							% Area	GUH	GUH	GUH	GUH
						% Area	BKF	BKF	BKF	BKF	BKF
				% Area	% Area	AMF	AMF	AMF	AMF	AMF	AMF
	% Area	% Area	% Area	SPG	BTG	BTG	BTG	BTG	BTG	BTG	BTG
	UND	NAN1	ABS	ABS	ABS	ABS	ABS	ABS	ABS	ABS	ABS
GalNAc	0.19	0.37	0.29	0.23	5.12	4.61	4.61	7.16	6.76	3.23	2.85
Gal	0.90	7.62	8.33	8.58	8.62	8.58	8.76	9.26	9.26	9.26	9.26
Glc	0.97	1.11	0.84	0.92	6.38	6.12	6.72	6.16	6.66	6.66	6.66
Man								14.00	14.64	14.64	14.64
Semi-digestible Disaccharides (ga3GA)	0.22	6.84	16.40	16.55	10.92	11.54	12.36	12.59	12.72	14.19	15.92
Digestible Disaccharides (others)	0.44	0.44	0.44	4.86	5.86	12.40	12.89	0.22	0.22		
Undigestible Disaccharides (sulfated)	0.81	0.61	2.21	2.10	2.03	2.03	1.60	1.85	1.61	1.50	2.03
Core 1 extensions	0.50	0.50	0.50	0.50	0.50	0.58					
Core 2 and extensions	3.77	3.79	3.71	3.47	3.92	4.00	3.51	0.38			
Core 2 extensions with sulfation	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30	0.30
Core 3 extensions	0.24	0.24	0.24	0.24							
Mannose Core 1 (linear trisaccharide and larger)	11.34	12.03	12.58	7.94	7.06						
Mannose Core 2 (branched trisaccharide and larger)	1.70	1.74	1.92	1.94	1.94	1.58	1.58	0.41			
SSTn, STn, ST antigens, peeling, disialyl	30.64	13.18									
STn and ST antigens (sulfated)	1.68	1.65									
Larger Sulfated Glycans that don't digest	2.51	3.17	3.56	3.74	3.71	3.48	4.15	4.64	4.31	4.59	5.01
High Mannose (N)	6.09	5.78	5.86	5.33	5.53	5.80	5.80	7.56	7.75	8.35	6.82
Hybrid (N)	2.11	2.22	2.84	3.27	3.72	4.53	5.11	3.08	3.29	3.87	3.13
Digestible hex chains	6.92	6.71	7.16	7.05	1.03	1.00	0.44	0.50			
Undigestible hex chains	5.39	6.03	6.43	6.00	6.51	6.49	6.09	6.10	5.40	5.29	6.39
Integrated Junk	23.32	25.70	26.35	26.97	26.86	26.96	26.04	26.09	27.05	28.10	26.97
SUM	100.04	100.03	99.96	99.99	100.01	100.00	99.96	100.30	99.97	99.98	99.98

	% Area	Comment
Total GalNAc core	28.85	(including the 1 glycan in the larger sulfated glycans that doesn't digest)
Total Mannose core	14.95	(including the 1 glycan in the larger sulfated glycans that doesn't digest)
Total Hexose chain glycans (Glucose core)	13.28	inlcudes glc, gal-glc, glc-gal-glc and gal-gal-glc
Total Galactose core (probably peeling)	8.00	probably core 1 and 2 peeling
Total NeuNac core (polysialic)	3.44	One polysialic structure also seen in Gothenburg data
Total N glycans	8.20	leftover from Ana's digest or extra N-glycan that was released in microwave
Total Junk	23.32	Suspected tissue junk and junk present in blank
Total (should equal 100.04)	100.04	total % area from UND
Total Glycan	76.72	
Total Junk	23.32	

Appendix VIII. *Summary of % areas – Exoglycosidase digestion panel for PD (stages 1-4) substantia nigra pool.*

Structures and Naming				MS info		GOTHENBURG MS ION TRAP								
						Healthy striatum		PD striatum		Healthy substantia nigra		PD substantia nigra		
Tentative structure	Short name	SNFG	[M-xH]^{x} (Reduced Alditol) Theoretical	Charge	[M-xH]^{x} (Reduced Alditol) Experimental	Int(%)	[M-xH]^{x} (Reduced Alditol) Experimental	Int(%)	[M-xH]^{x} (Reduced Alditol) Experimental	Int(%)	[M-xH]^{x} (Reduced Alditol) Experimental	Int(%)		
Glc	gl		181.0717	nd	oor	oor	oor	oor	oor	oor	oor	oor		
Gal	ga		181.0717	nd	oor	oor	oor	oor	oor	oor	oor	oor		
GalNAc	GA		181.0717	nd	oor	oor	oor	oor	oor	oor	oor	oor		
Galβ1-3Glc	^{x}ga3gl		343.1246	nd	oor	oor	oor	oor	oor	oor	oor	oor		
Galβ1-3GalNAc	ga3GA		384.1511	1	384.3333	1.50	384.2500	1.49	384.2500	2.33	384.3333	2.66		
Galα1-3GalNAc	gaa3GA		384.1511	1	384.3333	0.75	384.2500	0.62	384.2500	0.71	384.3333	0.95		
GlcNAcβ1-3GalNAc	GL3GA		425.1777	1	425.3333	1.04	425.3333	1.00	425.3333	0.87	425.3333	0.90		
SO3-3(6)Galβ1-3GalNAc	s3ga3GA		464.1079	1	464.2500	4.07	464.2500	1.78	464.2500	1.93	464.2500	3.32		
SO3-6(3)Galβ1-3GalNAc	s6ga3GA		464.1079	1	464.3333	0.80	464.2500	0.23	464.2500	0.42	nd	nd		
NeuAcα2-3Gal	S3ga		^{x}472.1672	1	472.4167	0.16	472.4167	0.35	473.3333	0.13	472.3333	0.21		
(Fucα1-2Galβ1-3)Glc	(fa2ga3)gl		^{x}489.1825	nd	nd	nd	nd	nd	nd	nd	nd	nd		
Glcβ1-4Galβ1-3Glc	^{x}gl4ga3gl		^{x}505.1774	1	505.3333	0.15	505.2500	0.03	nd	nd	nd	nd		
Glcβ1-3Galβ1-3Glc	^{x}g3ga4gl		^{x}505.1774	1	505.3333	0.05			nd	nd	nd	nd		
NeuAcα2-6GalNAc	S6GA		513.1937	1	513.4167	1.25	513.2500	0.36	nd	nd	513.4167	0.55		
Galβ1-4GlcNAcβ1-2Man	ga4Gl2m		546.2039	1	546.4167	2.56	546.4167	0.57	nd	nd	nd	nd		
Galβ1-3GlcNAcβ1-2Man	ga3Gl2m		546.2039	1										
Galβ1-3GlcNAcβ1-3GalNAc	ga3GL3GA		^{x}587.2305	1							nd	nd		
Galβ1-3(Glcβ1-6)GalNAc	ga3(Gl6)GA		^{x}587.2305	1	587.4200	0.15	587.3333	0.07	586.3333	0.18	nd	nd		
GlcNAcβ1-4Galβ1-3GalNAc	GL4ga3GA		^{x}587.2305	1							nd	nd		
GlcNAcβ1-3Galβ1-3GalNAc	GL3ga3GA		^{x}587.2305	1							nd	nd		
SO3-(NeuAcα2-6)GalNAc	s-[S6]GA		^{x}593.1505	nd	nd	nd	nd	nd	nd	nd	nd	nd		
NeuAcα2-3NeuAc	S3S		^{x}601.2097	1	601.4167	0.10	601.4167	0.09	601.4167	0.30	601.4167	0.09		
SO3-Galβ1-3(Glcβ1-6)GalNAc	ga3(sGl6)GA		667.1873	nd	nd	nd	nd	nd	nd	nd	nd	nd		
Galβ1-3(NeuAcα2-6)GalNAc	ga3[S6]GA		675.2465	1	675.5000	23.20	675.4167	18.15	675.4167	18.73	675.4167	22.27		
NeuAcα2-3Galβ1-3GalNAc	S3ga3GA		675.2465	1	675.4167	24.25	675.5000	16.92	675.4167	16.97	675.4167	21.28		
NeuAcα2-6Galβ1-3GalNAc	S6ga3GA		675.2465	1										
(Fucα1-2Galβ1-3)(Fucα1-3GalNAc)	(fa3ga3)(fa6GA)		676.2669	nd	nd	nd	nd	nd	nd	nd	nd	nd		
Galβ1-4(Glcβ1-6)Galβ1-4Glc	ga4(Gl6)ga4gl		^{x}708.2568	1	692.4167	0.45	692.4167	0.29	692.4167	0.36	692.4167	0.45		
Galβ1-4(Fucα1-3GlcNAcβ1-2)Man	ga4(fa3Gl2)m		692.2618	1										
Galβ1-3(Glcβ1-6)Galβ1-4Glc	^{x}ga3(Gl6)ga4gl		^{x}708.2568	1	707.3333	0.14	707.3333	0.08	707.2500	0.10	707.3333	0.11		
Galβ1-3(Fucα1-3GlcNAcβ1-6)GalNAc	ga3(fa3Gl6)GA		733.2884	nd	nd	nd	nd	nd	nd	nd	nd	nd		
Fucα1-3-GlcNAcβ1-2[Glcβ1-6]Man	(fa3Gl2)[Gl6]m		733.2884	nd	nd	nd	nd	nd	nd	nd	nd	nd		
Galβ1-3[Galβ1-4GlcNAcβ1-6]GalNAc	ga3[ga4Gl6]GA		^{x}749.2833	1	749.4167	0.06	749.4167	0.04	nd	nd	749.1667	0.06		
Galβ1-3GlcNAcβ1-4Galβ1-3GalNAc	ga3Gl4ga3GA		^{x}749.2833	1										
Galβ1-4GlcNAcβ1-4Galβ1-3GalNAc	ga4Gl3ga3GA		^{x}749.2833	1										
[NeuAc] + SO3-Galβ1-3GalNAc	S-ga3GA		755.2033	1	755.3333	3.33	755.2500	2.28	755.3333	2.15	755.3333	3.42		
[NeuAc] + SO3-Galβ1-3GalNAc	s5-ga3GA		755.2033	1	755.3333	0.47	755.3333	0.61	755.3333	0.56	nd	nd		
Galβ1-4(SO3-Fucα1-3GlcNAcβ1-2)Man	ga4(fa3sGl2)m		772.2187	1	772.3333	0.07	772.4167	0.03	772.3333	0.04	772.3333	0.07		
SO3-Galβ1-4(Fucα1-3GlcNAcβ1-2)Man	sga4(fa3Gl2)m		772.2187	1										
Galβ1-4(SO3-Fucα1-3GlcNAcβ1-2)Man(lactone form)	ga4(fa3sGl2)m-one		790.2340	1	790.3333	0.08	790.3333	0.02	nd	nd	790.4167	0.05		
SO3-GlcAβ1-3Galβ1-4Glcβ1-2Man	sU3ga4Gl2m		802.1928	1	802.3333	0.08	802.3333	0.13	802.3333	0.13	802.3333	0.18		
SO3-GlcAβ1-3Galβ1-3Glcβ1-2Man	sU3ga3Gl2m		802.1928	1										
NeuAcα2-3Galβ1-4Glcβ1-2Man	S3ga4Gl2m		837.2994	1										
NeuAcα2-3Galβ1-3Glcβ1-2Man	S3ga3Gl2m		837.2994	1	837.4167	1.56	837.5000	2.65	837.5000	2.70	837.5000	3.52		
NeuAcα2-6Galβ1-4Glcβ1-2Man	S6ga4Gl2m		837.2994	1										
NeuAcα2-6Galβ1-3Glcβ1-2Man	S6ga3Gl2m		837.2994	1										
NeuAcα2-3GalNAcβ1-4Galβ1-3Glc	^{x}S3GA4ga4gl		837.2994	nd	nd	nd	nd	nd	nd	nd	nd	nd		
NeuAcα2-6GalNAcβ1-4Galβ1-3Glc	^{x}S6GA4ga4gl		837.2994	nd	nd	nd	nd	nd	nd	nd	nd	nd		

Appendix IX. *Ion-trap MS data outlining m/z values as well as relative intensities and charge state observed. "x" – Gal core contamination, "y" – precursor ions discovered with no MS^2 fragmentation spectra. "oor" – out of range. "nd" – not detected.*

Structures and Naming			GOTHENBURG MS ION TRAP									
			MS info	Healthy striatum			PD striatum		Healthy substantia nigra		PD substantia nigra	
Tentative structure	Short name	SNFG	[M-xH]$^{x+}$ (Reduced Alditol) Theoretical	Charge	[M-xH]$^{x+}$ (Reduced Alditol) Experimental	Int(%)	[M-xH]$^{x+}$ (Reduced Alditol) Experimental	Int(%)	[M-xH]$^{x+}$ (Reduced Alditol) Experimental	Int(%)	[M-xH]$^{x+}$ (Reduced Alditol) Experimental	Int(%)
(Fucα1-3GlcNAcβ1-2)[(Fucα1-3GlcNAcβ1-6)]Man	(fa3GL2)[(fa3GL6)]m		879.3463	nd	nd	nd	nd	nd	nd	nd	nd	nd
Galβ1-3(Galβ1-4(Fucα1-3GlcNAcβ1-6)GalNAc	ga3]ga4[fa3GL6]GA		895.3412	1	895.5000	0.43	895.4167	0.29	895.3333	0.41	895.3333	0.60
Galβ1-3(Galβ1-3(Fucα1-4GlcNAcβ1-6)GalNAc	ga3]ga4[fa4GL6]GA		895.3412	nd	nd	nd	nd	nd	nd	nd	nd	nd
Galβ1-4-Fucα1-3-GlcNAcβ1-2-[GlcNAcβ1-6]Man	ga4[fa3GL2][GL6]m		^{y}895.3412	1	nd	nd	nd	nd	895.2500	0.03	nd	nd
SO3-Galβ1-3)SO3-Galβ1-4GlcNAcβ1-6)GalNAc	sga3]sga4GL6]GA		909.1969	nd	nd	nd	nd	nd	nd	nd	nd	nd
GlcNAcβ1-4Galβ1-3(Galβ1-3GlcNAcβ1-6)GalNAc	GL4ga3]ga3GL6]GA		^{y}952.3627	1	nd	nd	nd	nd			nd	nd
GlcNAcβ1-4Galβ1-3(Galβ1-4GlcNAcβ1-6)GalNAc	GL4ga3]ga4GL6]GA		^{y}952.3627	1	nd	nd	nd	nd	952.4167	0.05	nd	nd
Galβ1-3(GlcNAcβ1-4Galβ1-3GlcNAcβ1-6)GalNAc	ga3[GL4ga3]GL6]GA		^{y}952.3627	1	nd	nd	nd	nd			nd	nd
Galβ1-3(GlcNAcβ1-4Galβ1-4GlcNAcβ1-6)GalNAc	ga3[GL4ga4GL6]GA		^{y}952.3627	1	nd	nd	nd	nd			nd	nd
Galβ1-3[Galβ1-3GalNAcβ1-4GlcNAcβ1-6)GalNAc	ga3[ga3GA4GL6]GA		^{y}952.3627	1	nd	nd	nd	nd			nd	nd
NeuAcα2-3Galβ1-3(NeuAcα2-6)GalNAc	S3ga3[S6]GA		966.3419	1	966.5000	30.07	966.4167	49.13	966.5000	47.91	966.4167	35.37
NeuAcα2-3Galβ1-4(Fucα1-3GlcNAcβ1-2)Man	S3ga4[fa3GL2]m		983.3573	nd	nd	nd	nd	nd	nd	nd	nd	nd
GalNAcβ1-4GlcNAcβ1-2[GalNAcβ1-4GlcNAcβ1-6)Man	GA4GL2[GA4GL6]m		993.3892	nd	nd	nd	nd	nd	nd	nd	nd	nd
NeuAcα2-6-(Fucα1-2Galβ1-3)[GlcNAcβ1-6)GalNAc	S6-(fa2ga3)[GL6]GA		1024.3838	nd	nd	nd	nd	nd	nd	nd	nd	nd
NeuAcα2-3Galβ1-3(Galβ1-4GlcNAcβ1-6)GalNAc	S3ga3[ga4GL6]GA		1040.3787	1	1040.5834	0.42	1040.5000	0.38	1040.5000	0.84	1040.5000	0.91
Galβ1-3(NeuAcα2-3Galβ1-4GlcNAcβ1-6)GalNAc	ga3[S3ga4GL6]GA		1040.3787	1	1040.5834	0.36	1040.5000	0.52	1040.5000	0.85	1040.5000	0.98
Fucα1-2Galβ1-3)[Galβ1-4Fucα1-3GlcNAcβ1-6)GalNAc	(fa2ga3)[ga3(fa3GL6)]GA		1041.3991	nd	nd	nd	nd	nd	nd	nd	nd	nd
NeuAcα2-3)SO3-Galβ1-3(NeuAcα2-6)GalNAc	S3ga3[S6]GA		1046.2988	1	1046.3334	0.94	1046.1667	0.11	nd	nd	1046.3334	0.33
NeuAcα2-3(SO3-)Galβ1-3(NeuAcα2-6)GalNAc	S3ga3[S6]GA		1046.2988	1	1046.4167	0.80	1046.3334	0.72	1046.4167	0.38	1046.3334	0.50
[SO3] + NeuAcα2-3Galβ1-3(NeuAcα2-6)GalNAc	s-S3ga3[S6]GA		1046.2988	1	1046.4167	0.23	nd	nd	nd	nd	1046.4167	0.30
NeuAcα2-6Galβ1-4(Fucα1-3-SO3-GlcNAcβ1-2)Man	S6ga4(fa3sGL2)m		^{y}1063.3141	1	nd	nd	1063.4167	0.03	nd	nd	1063.1667	0.05
Galβ1-3)[(Fucα1-3GlcNAcβ1-4)Galβ1-4GlcNAcβ1-6)GalNAc	ga3[(fa3GL4)ga4GL6]GA		1098.4206	nd	nd	nd	nd	nd	nd	nd	nd	nd
Galβ1-4GlcNAcβ1-4Galβ1-3(Galβ1-4GlcNAcβ1-6)GalNAc	ga4GL4ga3]ga3GL6]GA		1114.4155	nd	nd	nd	nd	nd	nd	nd	nd	nd
Galβ1-3GlcNAcβ1-4Galβ1-3(Galβ1-4GlcNAcβ1-6)GalNAc	ga3GL4ga3]ga3GL6]GA		1114.4155	nd	nd	nd	nd	nd	nd	nd	nd	nd
Galβ1-3(Galβ1-3GlcNAcβ1-4Galβ1-4GlcNAcβ1-6)GalNAc	ga3[ga3GL4ga4GL6]GA		1114.4155	nd	nd	nd	nd	nd	nd	nd	nd	nd
Galβ1-3(Galβ1-3GlcNAcβ1-4Galβ1-4GlcNAcβ1-6)GalNAc	ga3[ga4GL4ga4GL6]GA		1114.4155	nd	nd	nd	nd	nd	nd	nd	nd	nd
NeuAcα2-3Galβ1-3GlcNAcβ1-3(NeuAcα2-6)GalNAc	S3ga3GL3[S6]GA		^{y}1169.4213 and 584.2070	1-2	583.5000	0.11	1169.2500 and 583.5	0.13	583.4167	0.09	583.5833	0.08
NeuAc-Galβ1-3(Galβ1-4(Fucα1-3GlcNAcβ1-6)GalNAc	S-ga3]ga4(fa3GL6)]GA		^{y}1186.4366	1	1186.5834	0.04	1186.5000	0.05	1186.5000	0.05	1186.6667	0.05
NeuAcα2-3Galβ1-4GlcNAcβ1-2[Galβ1-4GlcNAcβ1-6)Man	S3ga4GL2]ga4GL6]m		^{y}1202.4315	1	nd	nd	1202.5000	0.07	1202.5834	0.07	1202.4167	0.06
Galβ1-3GalNAcβ1-3Galβ1-3(GalNAcβ1-4GlcNAcβ1-6)GalNAc	ga3GA3ga3]GA3ga4GL6]GA		1317.4949	nd	nd	nd	nd	nd	nd	nd	nd	nd
Galβ1-4GalNAcβ1-3Galβ1-3(GalNAcβ1-3Galβ1-4GlcNAcβ1-6)GalNAc	ga4GA3ga3]GA3ga4GL6]GA		1317.4949	nd	nd	nd	nd	nd	nd	nd	nd	nd
NeuAcα2-3Galβ1-3(NeuAcα2-3Galβ1-4GlcNAcβ1-6)GalNAc	S3ga3[S3ga4GL6]GA		1331.4741	1	1331.4167	0.16	1331.5834	0.97	1331.4167	0.34	1331.5834	0.72
NeuAcα2-3Galβ1-3(NeuAcα2-6Galβ1-4GlcNAcβ1-6)GalNAc	S3ga3[S6ga4GL6]GA		1331.4741	1								
NeuAcα2-3-Galβ1-3(Galβ1-3GlcNAcβ1-4Galβ1-4GlcNAcβ1-6)GalNAc	S3-ga3[ga3GL4ga4GL6]GA		1405.5109	nd	nd	nd	nd	nd	nd	nd	nd	nd
NeuAcα2-3-Galβ1-3GlcNAcβ1-4Galβ1-3(Galβ1-4GlcNAcβ1-6)GalNAc	S3-ga3GL4ga3]ga3GL6]GA		1405.5109	nd	nd	nd	nd	nd	nd	nd	nd	nd
NeuAcα2-3-SO3-Galβ1-3(NeuAcα2-3-SO3-Galβ1-4GlcNAcβ1-6)GalNAc	S3sga3[S3sga4GL6]GA		1491.3878	nd	nd	nd	nd	nd	nd	nd	nd	nd
NeuAcα2-3Galβ1-4GlcNAcβ1-2[NeuAcα2-3Galβ1-4GlcNAcβ1-6)Man	S3ga4GL2[S3ga4GL6]m		1493.5270 and 746.2599	1-2	nd	nd	746.5000 and 1493.6667	0.16	746.9167 and 1493.6667	0.12	nd	nd
NeuAcα2-3Galβ1-4GlcNAcβ1-2[NeuAcα2-6Galβ1-4GlcNAcβ1-6)Man	S3ga4GL2[S6ga4GL6]m		1493.5270 and 746.2599	1-2	nd	nd					nd	nd
NeuAcα2-6GlcNAcβ1-4Galβ1-3(NeuAcα2-3Galβ1-4GlcNAcβ1-6)Man	S6GL4ga3]S3ga3GL6]GA		^{y}1534.5535	2	nd	nd	766.3333	0.02	nd	nd	nd	nd
NeuAcα2-3GlcNAcβ1-4Galβ1-3(NeuAcα2-3Galβ1-4GlcNAcβ1-6)GalNAc	S3GL4ga3]S3ga3GL6]GA		^{y}1534.5535	2								
NeuAcα2-3GalNAcβ1-4GlcNAcβ1-2[NeuAcα2-3GalNAcβ1-4GlcNAcβ1-6)Man	S3GA4GL2[S3GA4GL6]m		1575.5801	nd	nd	nd	nd	nd	nd	nd	nd	nd

Appendix IX (cont.). *Ion-trap MS data outlining m/z values as well as relative intensities and charge state observed. "y" – precursor ions discovered with no MS^2 fragmentation spectra. "oor" – out of range. "nd" – not detected.*

Structures and Naming				NIBRT MS XEVO QTOF Entire Pool (all 4 groups)			
Tentative structure	Short name	SNFG	$[M-H]^{-x}$ (2AB Labelled) Theoretical	Charge	$[M-xH]^{-x}$ (2AB Labelled) Experimental	Error (ppm)	RT TIC
Glc	**g**l		299.1248	nd	nd	nd	nd
Gal	**g**a		299.1248	nd	nd	nd	nd
GalNAc	**G**A		340.1514	nd	nd	nd	nd
Galβ1-3Glc	x**g**a3**g**l		461.1777	nd	nd	nd	nd
Galβ1-3GalNAc	g**a**3**GA**		502.2042	nd	nd	nd	nd
Galα1-3GalNAc	g**a**a3**GA**		502.2042	nd	nd	nd	nd
GlcNAcβ1-3GalNAc	**GL**3**GA**		1543.2308	1	543.2400	16.94	4.53
SO3-3(6)Galβ1-3GalNAc	s3ga3**GA**		582.1610	1	582.1530	13.74	4.63
SO3-6(3)Galβ1-3GalNAc	s6ga3**GA**		582.1610	1			
NeuAcα2-3Gal	S3**g**a		590.2203	1	590.2156	7.96	7.75
(Fucα1-2Galβ1-3)Glc	(fa2ga3)**g**l		607.2356	nd	nd	nd	nd
Glcβ1-4Galβ1-3Glc	x**g**l4ga3**g**l		623.2305	1	623.2308	0.48	8.60
Glcβ1-3Galβ1-3Glc	x**g**a3ga4**g**l		623.2305	1			
NeuAcα2-6GalNAc	S6**GA**		631.2468	1	631.2471	0.48	4.78
Galβ1-4GlcNAcβ1-2Man	ga4GL2**m**		664.2570	1	664.2530	6.02	8.67
Galβ1-3GlcNAcβ1-2Man	ga3GL2**m**		664.2570	1			
Galβ1-3GlcNAcβ1-3GalNAc	ga3GL3**GA**		705.2836	nd	nd	nd	nd
Galβ1-3[GlcNAcβ1-6]GalNAc	ga3[GL6]**GA**		705.2836	nd	nd	nd	nd
GlcNAcβ1-4Galβ1-3GalNAc	GL4ga3**GA**		705.2836	nd	nd	nd	nd
GlcNAcβ1-3Galβ1-3GalNAc	GL3ga3**GA**		705.2836	nd	nd	nd	nd
SO3-(NeuAcα2-6)GalNAc	s-[S6]**GA**		711.2036	1	711.1962	10.40	4.55
NeuAcα2-3NeuAc	S3**S**		719.2628	1	719.2731	14.32	4.51
SO3-Galβ1-3[GlcNAcβ1-6]GalNAc	ga3[sGL6]**GA**		785.2404	1	785.2429	3.18	11.40
Galβ1-3(NeuAcα2-6)GalNAc	ga3[S6]**GA**		793.2996	1	793.3065	8.70	10.25
NeuAcα2-3Galβ1-3GalNAc	S3ga3**GA**		793.2996	1	793.3065	8.70	11.40
NeuAcα2-6Galβ1-3GalNAc	S6ga3**GA**		793.2996	1			
(Fucα1-2Galβ1-3)(Fucα1-3GalNAc)	(fa3ga3)(fa6**GA**)		794.3200	nd	nd	nd	nd
Galβ1-4(Fucα1-3GlcNAcβ1-2)Man	ga4(fa3GL2)**m**		810.3149	1	810.3229	9.87	11.18
Galβ1-4[GlcNAcβ1-6]Galβ1-4Glc	x**g**a4[GL6]g**a**4**g**l		826.3099	nd	nd	nd	nd
Galβ1-3[GlcNAcβ1-6]Galβ1-4Glc	x**g**a3[GL6]ga4**g**l		826.3099	nd	nd	nd	nd
Galβ1-3[(Fucα1-3GlcNAcβ1-6)]GalNAc	ga3[(fa3GL6)]**GA**		851.3415	nd	nd	nd	nd
Fucα1-3-GlcNAcβ1-2[GlcNAcβ1-6]Man	(fa3GL2)[GL6]**m**		851.3415	nd	nd	nd	nd
Galβ1-3[Galβ1-4GlcNAcβ1-6]GalNAc	ga3[ga4GL6]**GA**		867.3364	nd	nd	nd	nd
Galβ1-3GlcNAcβ1-4Galβ1-3GalNAc	ga3GL4ga3**GA**		867.3364	nd	nd	nd	nd
Galβ1-4GlcNAcβ1-4Galβ1-3GalNAc	ga4GL3ga3**GA**		867.3364	nd	nd	nd	nd
[NeuAc] + SO3-Galβ1-3GalNAc	S-sga3**GA**		873.2564	nd	nd	nd	nd
[NeuAc] + SO3-Galβ1-3GalNAc	sS-ga3**GA**		873.2564	nd	nd	nd	nd
Galβ1-4[SO3-Fucα1-3GlcNAcβ1-2]Man	ga4(fa3sGL2)**m**		890.2718	1	890.2792	8.31	11.61
SO3-Galβ1-4(Fucα1-3GlcNAcβ1-2)Man	sg4(fa3GL2)**m**		890.2718	1	890.273	1.35	10.75
Galβ1-4(SO3-Fucα1-3GlcNAcβ1-2)Man(lactone form)	ga4(fa3sGL2)**m-one**		908.2871	nd	nd	nd	nd
SO3-GlcAβ1-3Galβ1-4GlcNAcβ1-2Man	sU3g**a**4GL2**m**		920.2459	1	920.2567	11.74	14.64
SO3-GlcAβ1-3Galβ1-3GlcNAcβ1-2Man	sU3ga3GL2**m**		920.2459	1			
NeuAcα2-3Galβ1-4GlcNAcβ1-2Man	S3g**a**4GL2**m**		955.3525	1	955.3639	11.93	12.48
NeuAcα2-3Galβ1-3GlcNAcβ1-2Man	S3ga3GL2**m**		955.3525	1	955.3448	8.06	13.48
NeuAcα2-6Galβ1-4GlcNAcβ1-2Man	S6g**a**4GL2**m**		955.3525	1	May be a part of the above two glycans		
NeuAcα2-6Galβ1-3GlcNAcβ1-2Man	S6ga3GL2**m**		955.3525	1			
NeuAcα2-3GalNAcβ1-4Galβ1-3Glc	x**S**3**GA**4g**a**4**g**l		957.3525	nd	nd	nd	nd
NeuAcα2-6GalNAcβ1-4Galβ1-3Glc	x**S**6**GA**4ga4**g**l		958.3525	nd	nd	nd	nd

Appendix X. *QToF MS data outlining m/z values as well as relative charge state observed, error (ppm) and RT (TIC) of glycans. "x" – Gal core contamination. "nd" – not detected.*

Structures and Naming				NIBRT MS XEVO QTOF Entire Pool (all 4 groups)				
Tentative structure	Short name	SNFG	[M-H]$^{-x}$ (2AB Labelled) Theoretical	Charge	[M-xH]$^{-x}$ (2AB Labelled) Experimental	Error (ppm)	RT TIC
(Fuca1-3GlcNAcβ1-2)[(Fuca1-3GlcNAcβ1-6)]Man	(fa3GL2)[(fa3GL6)]m		997.3994	nd	nd	nd	nd
Galβ1-3[Galβ1-4[Fuca1-3GlcNAcβ1-6)]GlcNAc	ga3[ga4(fa3GL6)]GA		506.1935	2			
Galβ1-3[Galβ1-3[Fuca1-4GlcNAcβ1-6)]GlcNAc	g a3[ga4(fa4GL6)]GA		506.1935	2	506.1916	3.75	15.86
Galβ1-4-Fuca1-3-GlcNAcβ1-2[GlcNAcβ1-6)]Man	ga4(fa3GL2)[GL6]m		506.1935	2			
SO3-Galβ1-3[SO3-Galβ1-4GlcNAcβ1-6)]GalNAc	sga3[sga4GL6]GA		1027.2500	1	1027.2668	16.35	12.10
GlcNAcβ1-4Galβ1-3[Galβ1-3GlcNAcβ1-6)]GalNAc	GL4ga3[ga3GL6]GA		1070.4158	nd	nd	nd	nd
GlcNAcβ1-4Galβ1-3[Galβ1-4GlcNAcβ1-6)]GalNAc	GL4ga3[ga4GL6]GA		1070.4158	nd	nd	nd	nd
Galβ1-3[GlcNAcβ1-4Galβ1-3GlcNAcβ1-6)]GalNAc	ga3[GL4ga3GL6]GA		1070.4158	nd	nd	nd	nd
Galβ1-3[GlcNAcβ1-4Galβ1-4GlcNAcβ1-6)]GalNAc	ga3[GL4ga4GL6]GA		1070.4158	nd	nd	nd	nd
Galβ1-3[Galβ1-3GalNAcβ1-4GlcNAcβ1-6)]GalNAc	ga3[ga3GA4GL6]GA		1070.4158	nd	nd	nd	nd
NeuAcα2-3Galβ1-3(NeuAcα2-6)GalNAc	S3ga3[S6]GA		541.6939	2	541.6866	13.48	15.22
NeuAcα2-3Galβ1-4[Fuca1-3GlcNAcβ1-2)]Man	S3ga4(fa3GL2)m		1101.4104	1	1101.4104	0.00	14.38
GalNAcβ1-4GlcNAcβ1-2[GalNAcβ1-4GlcNAcβ1-6)]Man	GA4GL2[GA4GL6]m		555.2175	2	555.2166	1.62	15.82
NeuAcα2-6-[Fuca1-2Galβ1-3)]GlcNAcβ1-6)GalNAc	S6-[fa2ga3)][GL6]GA		1142.4369	nd	nd	nd	nd
NeuAcα2-3Galβ1-3(Galβ1-4GlcNAcβ1-6)GalNAc	S3ga3[ga4GL6]GA		1158.4318	nd	nd	nd	nd
Galβ1-3[NeuAcα2-3Galβ1-4GlcNAcβ1-6)]GalNAc	ga3[S3ga4GL6]GA		578.7123	2	578.7152	5.01	13.20
Fuca1-2Galβ1-3[(Galβ1-4Fuca1-3GlcNAcβ1-6)]GalNAc	(fa2ga3)[ga3(fa3GL6)]GA		1159.4522	1	1159.4525	0.26	16.45
NeuAcα2-3(SO3-)Galβ1-3(NeuAcα2-6)GalNAc	S3sga3[S6]GA		581.6723	2			
NeuAcα2-3(SO3-)Galβ1-3(NeuAcα2-6)GalNAc	S3sga3[S6]GA		581.6723	2	581.6775	8.94	15.86
[SO3] + NeuAcα2-3Galβ1-3(NeuAcα2-6)GalNAc	s-S3ga3[S6]GA		581.6723	2			
NeuAcα2-6Galβ1-4[Fuca1-3-SO3-GlcNAcβ1-2)]Man	S6ga4(fa3sGL2)m		1181.3672	nd	nd	nd	nd
Galβ1-3[(Fuca1-3GlcNAcβ1-4)Galβ1-4GlcNAcβ1-6)]GalNAc	ga3[(fa3GL4)ga4GL6]GA		1216.4737	nd	nd	nd	nd
Galβ1-4GlcNAcβ1-4Galβ1-3(Galβ1-4GlcNAcβ1-6)GalNAc	ga4GL4ga3[ga3GL6]GA		615.7307	2			
Galβ1-3GlcNAcβ1-4Galβ1-3(Galβ1-4GlcNAcβ1-6)GalNAc	ga3GL4ga3[ga3GL6]GA		615.7307	2			
Galβ1-3[Galβ1-3GlcNAcβ1-4Galβ1-4GlcNAcβ1-6)]GalNAc	ga3[ga3GL4ga4GL6]GA		615.7307	2	615.7292	2.44	15.82
Galβ1-3[Galβ1-3GlcNAcβ1-4Galβ1-4GlcNAcβ1-6)]GalNAc	ga3[ga4GL4ga4GL6]GA		615.7307	2			
NeuAcα2-3Galβ1-3GlcNAcβ1-3(NeuAcα2-6)GalNAc	S3ga3GL3[S6]GA		643.2336	2	643.2452	18.03	15.80
NeuAc-Galβ1-3[Galβ1-4[Fuca1-3GlcNAcβ1-6)]GalNAc	S-ga3[ga4(fa3GL6)]GA		651.7412	2	651.7399	1.99	16.17
NeuAcα2-3Galβ1-4GlcNAcβ1-2[Galβ1-4GlcNAcβ1-6)]Man	S3ga4GL2[ga4GL6]m		1320.4846	nd	nd	nd	nd
Galβ1-3GalNAcβ1-3Galβ1-3[GalNAcβ1-3Galβ1-4GlcNAcβ1-6)]GalNAc	ga3GA3ga3[GA3ga4GL6]GA		1435.5480	nd	nd	nd	nd
Galβ1-4GalNAcβ1-3Galβ1-3[GalNAcβ1-3Galβ1-4GlcNAcβ1-6)]GalNAc	ga4GA3ga3[GA3ga4GL6]GA		1435.5480	nd	nd	nd	nd
NeuAcα2-3Galβ1-3(NeuAcα2-3Galβ1-4GlcNAcβ1-6)GalNAc	S3ga3[S3ga4GL6]GA		1449.5272	nd	nd	nd	nd
NeuAcα2-3Galβ1-3(NeuAcα2-6Galβ1-4GlcNAcβ1-6)GalNAc	S3ga3[S6ga4GL6]GA		1449.5272	nd	nd	nd	nd
NeuAcα2-3-Galβ1-3(Galβ1-3GlcNAcβ1-4Galβ1-4GlcNAcβ1-6)GalNAc	S3-ga3[ga3GL4ga4GL6]GA		1523.5640	nd	nd	nd	nd
NeuAcα2-3-Galβ1-3GlcNAcβ1-4Galβ1-3(Galβ1-4GlcNAcβ1-6)GalNAc	S3-ga3GL4ga3[ga3GL6]GA		1523.5640	nd	nd	nd	nd
NeuAcα2-3-SO3-Galβ1-3(NeuAcα2-3-SO3-Galβ1-4GlcNAcβ1-6)GalNAc	S3sga3[S3sga4GL6]GA		1609.4409	nd	nd	nd	nd
NeuAcα2-3Galβ1-4GlcNAcβ1-2[NeuAcα2-3Galβ1-4GlcNAcβ1-6)]Man	S3ga4GL2[S3ga4GL6]m		1611.5801	nd	nd	nd	nd
NeuAcα2-3Galβ1-4GlcNAcβ1-2[NeuAcα2-6Galβ1-4GlcNAcβ1-6)]Man	S3ga4GL2[S6ga4GL6]m		1611.5801	nd	nd	nd	nd
NeuAcα2-6GlcNAcβ1-4Galβ1-3[NeuAcα2-3Galβ1-4GlcNAcβ1-6)]GalNAc	S6Gl4ga3[S3ga3GL6]GA		1652.6066	nd	nd	nd	nd
NeuAcα2-3GlcNAcβ1-4Galβ1-3[NeuAcα2-3Galβ1-4GlcNAcβ1-6)]GalNAc	S3Gl4ga3[S3ga3GL6]GA		1652.6066	nd	nd	nd	nd
NeuAcα2-3Galβ1-4GlcNAcβ1-2[NeuAcα2-3GalNAcβ1-4GlcNAcβ1-6)]Man	S3GA4GL2[S3GA4GL6]m		563.8729	3	563.8661	4.11	15.93

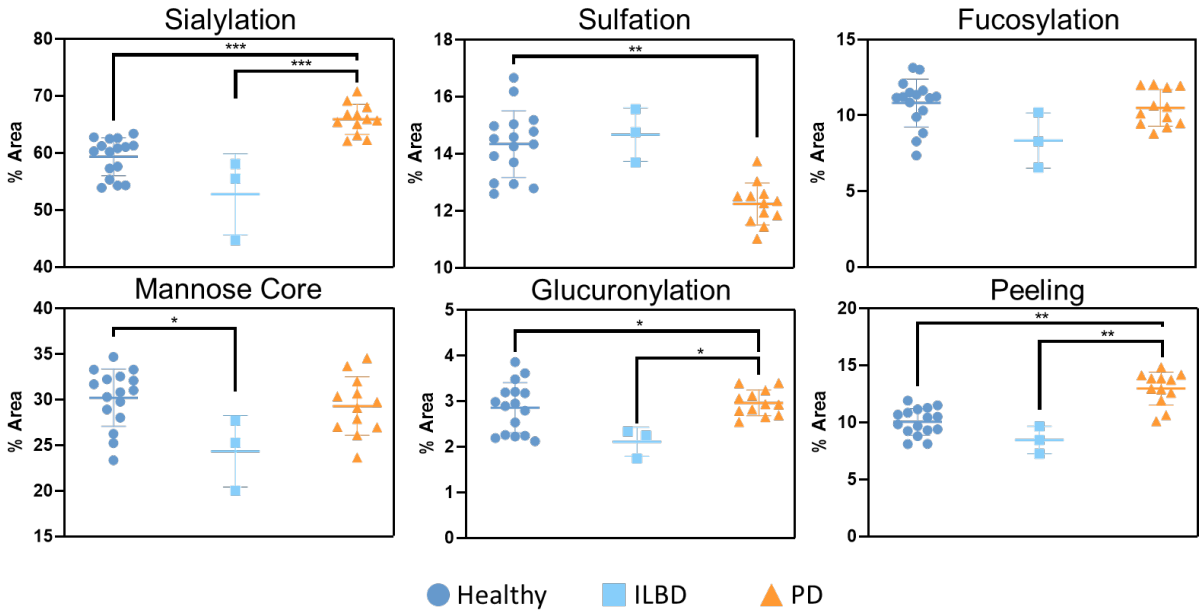
Appendix X (cont.). *QToF MS data outlining m/z values as well as relative charge state observed, error (ppm) and RT (TIC) of glycans. "x" – Gal core contamination. "nd" – not detected.*

Structures and Naming			NIBRT UPLC								
			Healthy striatum		PD striatum		Healthy substantia nigra		PD substantia nigra		
Tentative structure	Short name	SNFG	GU	% Area	GU	% Area	GU	% Area	GU	% Area	
Glc	**gl**		1.02	1.10	1.01	2.35	1.02	3.80	1.01	0.97	
Gal	**ga**		1.04	0.28	1.03	0.85	1.04	0.73	1.03	0.90	
GalNAc	**GA**		[z]0.98	digestion	[z]0.98	0.14	[z]0.98	0.62	[z]0.98	0.19	
Galβ1-3Glc	x**ga3gl**		1.97	25.69	1.95	5.07	1.96	27.85	1.96	6.10	
Galβ1-3GalNAc	**ga3GA**		1.80	digestion	1.77	0.24	1.76	0.20	1.78	0.22	
Galα1-3GalNAc	**gaa3GA**		1.81	0.68	1.77	0.24	1.76	0.21	1.78	0.22	
GlcNAcβ1-3GalNAc	**GL3GA**		1.85	0.60	1.77	0.24	1.76	0.21	1.78	0.22	
SO3-3(6)Galβ1-3GalNAc	**s3ga3GA**		1.69	digestion	1.70	0.47	1.69	digestion	1.70	0.81	
SO3-6(3)Galβ1-3GalNAc	**s6ga3GA**										
NeuAcα2-3Gal	**S3ga**		2.43	3.42	2.35 and 2.41	7.72	2.41	2.41	2.36 and 2.41	7.10	
(Fucα1-2Galβ1-3)Glc	(fa2ga3)**gl**		not present		not present		not present		2.75	0.51	
Glcβ1-4Galβ1-3Glc	x**gl4ga3gl**		2.64	7.69	2.61	2.90	2.62	3.65	2.62	4.57	
Glcβ1-3Galβ1-3Glc	x**ga3ga4gl**		2.79	0.47	not present		not present		2.80	0.82	
NeuAcα2-6GalNAc	**S6GA**		2.43	1.50	2.35 and 2.41	2.98	2.41	2.88	2.41	2.66	
Galβ1-4GlcNAcβ1-2Man	**ga4GL2m**		2.64	1.40	2.61	0.65	2.62	0.90	2.62	0.65	
Galβ1-3GlcNAcβ1-2Man	**ga3GL2m**		2.59	0.20	2.55	digestion	2.56	digestion	2.55	digestion	
Galβ1-3GlcNAcβ1-3GalNAc	**ga3GL3GA**		2.83	0.27	2.79	digestion	2.75	digestion	2.87	digestion	
Galβ1-3[GlcNAcβ1-6]GalNAc	**ga3[GL6]GA**		2.64	digestion	2.63	digestion	2.65	digestion	2.65	digestion	
GlcNAcβ1-4Galβ1-3GalNAc	**GL4ga3GA**		2.77	digestion	not present		not present		not present		
GlcNAcβ1-3Galβ1-3GalNAc	**GL3ga3GA**		2.71	digestion	not present		not present		not present		
SO3-(NeuAcα2-6)GalNAc	**s-[S6]GA**		not present		1.70	0.10	not present		1.70	0.10	
NeuAcα2-3NeuAc	**S3S**		1.77	1.61	1.76	2.40	1.75	2.00	1.77	3.44	
SO3-Galβ1-3[GlcNAcβ1-6]GalNAc	**ga3[sGL6]GA**		3.43	1.75	3.37	0.91	3.39	0.91	3.40	0.30	
Galβ1-3(NeuAcα2-6)GalNAc	**ga3[S6]GA**		3.41	1.23	3.37	2.40	3.36	1.45	3.38	2.44	
NeuAcα2-3Galβ1-3GalNAc	**S3ga3GA**		3.07	3.02	2.96 and 3.03	4.20	3.02	2.83	2.96 and 3.03	5.28	
NeuAcα2-6Galβ1-3GalNAc	**S6ga3GA**		2.98	0.65	2.96 and 3.03	1.48	2.94	0.51	not present		
(Fucα1-2Galβ1-3)(Fucα1-3GalNAc)	(fa3ga3)(fa6**GA**)		not present		not present		not present		3.27	0.50	
Galβ1-4(Fucα1-3GlcNAcβ1-2)Man	**ga4(**fa3GL2)**m**		3.32	4.92	3.27	6.84	3.27	4.08	3.27	5.91	
Galβ1-4[GlcNAcβ1-6]Galβ1-4Glc	x**ga4(**GL6)**ga4gl**		not present		not present		3.48	0.36	not present		
Galβ1-3(GlcNAcβ1-6)Galβ1-4Glc	x**ga3(**GL6)**ga4gl**		not present		not present		3.52	1.04	not present		
Galβ1-3[(Fucα1-3GlcNAcβ1-6)]GalNAc	**ga3[(**fa3GL6)**]GA**		3.66	digestion	3.71	digestion	3.66	0.18	3.73	digestion	
Fucα1-3-GlcNAcβ1-2[GlcNAcβ1-6]Man	(fa3GL2)[GL6]**m**		not present		not present		not present		3.63	0.27	
Galβ1-3[Galβ1-4GlcNAcβ1-6]GalNAc	ga3[ga4GL6]**GA**		3.33	digestion	3.26	digestion	3.27	digestion	3.27	digestion	
Galβ1-3GlcNAcβ1-4Galβ1-3GalNAc	**ga3GL4ga3GA**		3.56	0.85	not present		not present		not present		
Galβ1-4GlcNAcβ1-4Galβ1-3GalNAc	**ga4GL3ga3GA**		3.52	0.33	not present		not present		not present		
[NeuAc] + SO3-Galβ1-3GalNAc	**S-ga3GA**		3.52	digestion	3.50	0.29	3.47	digestion	3.50	0.88	
[NeuAc] + SO3-Galβ1-3GalNAc	**sS-ga3GA**		3.97	0.30	3.94	0.33	3.93	digestion	3.95	digestion	
Galβ1-4(SO3-Fucα1-3GlcNAcβ1-2)Man	ga4(fa3sGL2)**m**		not present		3.44	0.17	not present		3.45	0.09	
SO3-Galβ1-4(Fucα1-3GlcNAcβ1-2)Man	sga4(fa3GL2)**m**		not present		not present		not present		3.23	0.20	
Galβ1-4(SO3-Fucα1-3GlcNAcβ1-2)Man(lactone form)	ga4(fa3sGL2)**m-one**		not present		not present		not present		not present		
SO3-GlcAβ1-3Galβ1-4GlcNAcβ1-2Man	sU3ga4GL2**m**		4.48	1.18	4.48	1.96	4.44	0.96	4.50	0.97	
SO3-GlcAβ1-3Galβ1-3GlcNAcβ1-2Man	sU3ga3GL2**m**		4.56	0.32	4.55	0.58	4.50	0.26	4.56	0.74	
NeuAcα2-3Galβ1-4GlcNAcβ1-2Man	**S3ga4GL2m**		3.74	2.89	3.72	4.54	3.70	2.71	3.72	3.49	
NeuAcα2-3Galβ1-3GlcNAcβ1-2Man	**S3ga3GL2m**		3.80	0.79	3.77	1.11	3.75	0.75	3.78	0.78	
NeuAcα2-6Galβ1-4GlcNAcβ1-2Man	**S6ga4GL2m**		3.74	0.27	not present		not present		not present		
NeuAcα2-6Galβ1-3GlcNAcβ1-2Man	**S6ga3GL2m**		3.63	0.53	not present		not present		not present		
NeuAcα2-3GalNAcβ1-4Galβ1-3Glc	x**S3GA4ga4gl**		4.09	0.50	not present		not present		4.13	0.24	
NeuAcα2-6GalNAcβ1-4Galβ1-3Glc	x**S6GA4ga4gl**		not present		not present		not present		4.13	0.07	

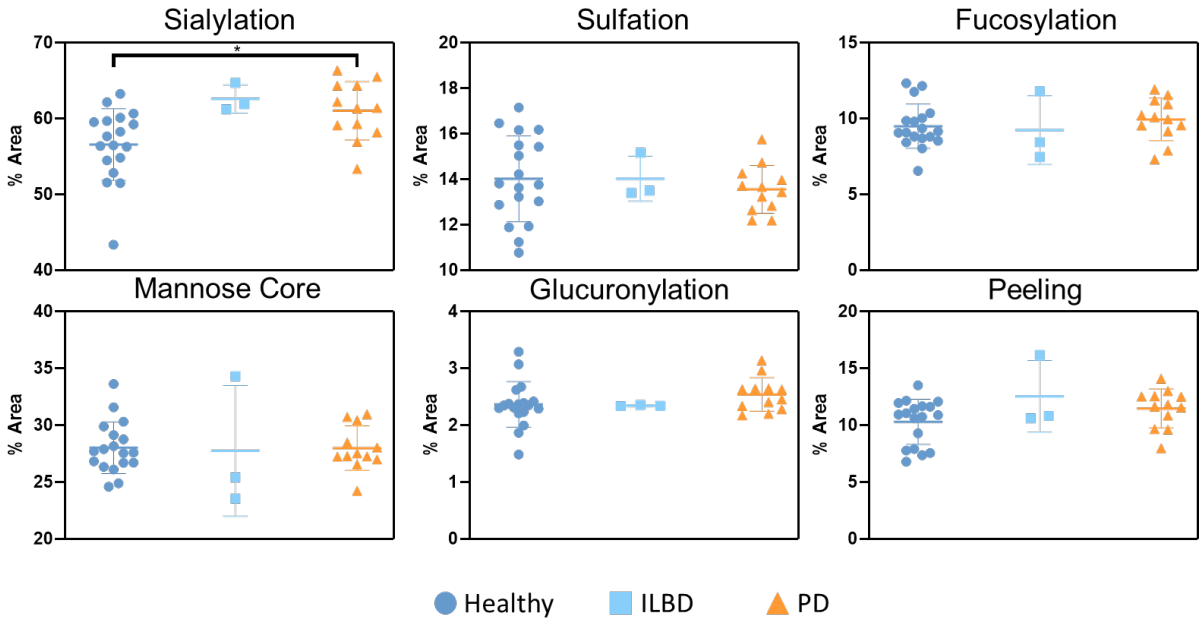
Appendix XI. *FLR-UPLC data of 2AB labelled glycans outlining GU values and relative area. "x" – Gal core contamination. "z" – GU value based on values in GlycoStore as dextran was unable to provide GU values lower than 1.*

Structures and Naming			NIBRT UPLC							
			Healthy striatum		PD striatum		Healthy substantia nigra		PD substantia nigra	
Tentative structure	Short name	SNFG	GU	% Area	GU	% Area	GU	% Area	GU	% Area
(Fucα1-3GlcNAcβ1-2)[(Fucα1-3GlcNAcβ1-6)]Man	(fa3GL2)[(fa3GL6)]m		not present		not present		not present		4.09	0.33
Galβ1-3[Galβ1-4(Fucα1-3GlcNAcβ1-6)]GalNAc	ga3[ga4(fa3GL6)]GA		4.42	0.32	4.40	0.30	4.37	0.10	4.41	0.50
Galβ1-3[Galβ1-3(Fucα1-4GlcNAcβ1-6)]GalNAc	ga3[ga4(fa4GL6)]GA		4.36	0.22	not present		not present		not present	
Galβ1-4-Fucα1-3-GlcNAcβ1-2[GlcNAcβ1-6)]Man	ga4(fa3GL2)[GL6]m		not present		not present		not present		4.13	0.28
SO3-Galβ1-3[SO3-Galβ1-4GlcNAcβ1-6)]GalNAc	sga3[sga4GL6]GA		not present		not present		not present		3.68	0.27
GlcNAcβ1-4Galβ1-3[Galβ1-3GlcNAcβ1-6)]GalNAc	GL4ga3[ga3GL6]GA		4.12	0.60	4.05	0.25	4.05	0.29	4.09	0.07
GlcNAcβ1-4Galβ1-3[Galβ1-4GlcNAcβ1-6)]GalNAc	GL4ga3[ga4GL6]GA		not present		4.13	0.17	not present		not present	
Galβ1-3(GlcNAcβ1-4Galβ1-3GlcNAcβ1-6)GalNAc	ga3[GL4ga3GL6]GA		4.08	digestion	3.90	digestion	4.07	digestion	4.05	digestion
Galβ1-3(GlcNAcβ1-4Galβ1-4GlcNAcβ1-6)GalNAc	ga3[GL4ga4GL6]GA		4.27	0.33	4.08	0.81	not present		4.09	0.29
Galβ1-3[Galβ1-3GalNAcβ1-4GlcNAcβ1-6)]GalNAc	ga3[ga3GA4GL6]GA		not present		not present		not present		4.46	0.32
NeuAcα2-3Galβ1-3(NeuAcα2-6)GalNAc	S3ga3[S6]GA		4.71	6.08	4.71	13.04	4.66	6.64	4.72	9.72
NeuAcα2-3Galβ1-4[Fucα1-3GlcNAcβ1-2)]Man	S3ga4[fa3GL2]m		4.42	0.22	not present		not present		4.41	0.26
GalNAcβ1-4GlcNAcβ1-2[GalNAcβ1-4GlcNAcβ1-6)]Man	GA4GL2[GA4GL6]m		not present		not present		not present		4.05	digestion
NeuAcα2-6-{Fucα1-2Galβ1-3)GlcNAcβ1-6)GalNAc	S6-{fa2ga3}[GL6]GA		not present		not present		4.75	0.05	not present	
NeuAcα2-3Galβ1-3(Galβ1-4GlcNAcβ1-6)GalNAc	S3ga3[ga4GL6]GA		4.56	0.25	4.58	0.33	4.54	0.54	4.61	0.40
Galβ1-3(NeuAcα2-3Galβ1-4GlcNAcβ1-6)GalNAc	ga3[S3ga4GL6]GA		4.42	0.26	4.40	0.47	4.37	0.39	4.41	0.30
Fucα1-2Galβ1-3(Galβ1-4Fucα1-3GlcNAcβ1-6)]GalNAc	(fa2ga3)[ga3(fa3GL6)]GA		not present		not present		not present		5.68	0.30
NeuAcα2-3(SO3-)Galβ1-3(NeuAcα2-6)GalNAc	S3sga3[S6]GA	5.11 and 5.19	0.29 and 0.06	5.15	0.48	5.08	0.13	5.16	0.32
NeuAcα2-3(SO3-)Galβ1-3(NeuAcα2-6)GalNAc	S3sga3[S6]GA									
[SO3] + NeuAcα2-3Galβ1-3(NeuAcα2-6)GalNAc	s-S3ga3[S6]GA		5.74	0.13	5.79	0.40	5.83	0.25	5.78	0.38
NeuAcα2-6Galβ1-4(Fucα1-3-SO3-GlcNAcβ1-2)Man	S6ga4(fa3sGL2)m		not present		not present		not present		4.61	0.16
Galβ1-3([Fucα1-3GlcNAcβ1-4)Galβ1-4GlcNAcβ1-6)GalNAc	ga3[(fa3GL4)ga4GL6]GA		not present		not present		not present		4.99	0.14
Galβ1-4GlcNAcβ1-4Galβ1-3(Galβ1-4GlcNAcβ1-6)GalNAc	ga4GL4ga3[ga3GL6]GA		not present		4.86	0.15	not present		4.86	0.13
Galβ1-3GlcNAcβ1-4Galβ1-3(Galβ1-4GlcNAcβ1-6)GalNAc	ga3GL4ga3[ga3GL6]GA		not present		4.86	0.15	4.84	0.10	4.86	0.46
Galβ1-3[Galβ1-3GlcNAcβ1-4Galβ1-4GlcNAcβ1-6)GalNAc	ga3[ga3GL4ga4GL6]GA		4.92	digestion	4.86	0.25	4.83	digestion	5.05	0.24
Galβ1-3[Galβ1-3GlcNAcβ1-4Galβ1-4GlcNAcβ1-6)GalNAc	ga3[ga4GL4ga4GL6]GA		4.90	0.28	not present		4.87	0.11	not present	
NeuAcα2-3Galβ1-3GlcNAcβ1-3(NeuAcα2-6)GalNAc	S3ga3GL3[S6]GA		5.05	0.20	5.06	0.20	5.08	0.10	4.86	0.15
NeuAc-Galβ1-3-[Galβ1-4(Fucα1-3GlcNAcβ1-6)]GalNAc	S-ga3[ga4(fa3GL6)]GA		not present		not present		not present		5.65	0.10
NeuAcα2-3Galβ1-4GlcNAcβ1-2-[Galβ1-4GlcNAcβ1-6)]Man	S3ga4GL2[ga4GL6]m		5.23	0.30	5.26	0.50	5.20	0.36	5.24	0.29
Galβ1-3GalNAcβ1-3Galβ1-3(GalNAcβ1-3GlcNAcβ1-6)GalNAc	ga3GA3ga3[GA3ga4GL6]GA		5.57	0.11	not present		not present		not present	
Galβ1-4GalNAcβ1-3Galβ1-3[GalNAcβ1-3Galβ1-4GlcNAcβ1-6)GalNAc	ga4GA3ga3[GA3ga4GL6]GA		5.57	0.15	5.64	0.30	not present		not present	
NeuAcα2-3Galβ1-3(NeuAcα2-3Galβ1-4GlcNAcβ1-6)GalNAc	S3ga3[S3ga4GL6]GA		5.30	0.07	not present		not present		not present	
NeuAcα2-3Galβ1-3(NeuAcα2-6Galβ1-4GlcNAcβ1-6)GalNAc	S3ga3[S6ga4GL6]GA		not present		not present		not present		5.24	0.20
NeuAcα2-3-Galβ1-3[Galβ1-3GlcNAcβ1-4Galβ1-4GlcNAcβ1-6)GalNAc	S3-ga3[ga3GL4ga4GL6]GA		5.84	0.15	5.94	0.21	not present		not present	
NeuAcα2-3-Galβ1-3GlcNAcβ1-4Galβ1-3(Galβ1-4GlcNAcβ1-6)GalNAc	S3-ga3GL4ga3[ga3GL6]GA		not present		not present		5.83	0.20	not present	
NeuAcα2-3-SO3-Galβ1-3(NeuAcα2-3-SO3-Galβ1-4GlcNAcβ1-6)GalNAc	S3sga3[S3sga4GL6]GA		not present		not present		not present		5.94	0.33
NeuAcα2-3Galβ1-4GlcNAcβ1-2[NeuAcα2-3Galβ1-4GlcNAcβ1-6)Man	S3ga4GL2[S3ga4GL6]m		not present		not present		6.36	0.25	6.45	0.24
NeuAcα2-3Galβ1-4GlcNAcβ1-2[NeuAcα2-6Galβ1-4GlcNAcβ1-6)Man	S3ga4GL2[S6ga4GL6]m		not present		6.47	0.42	not present		not present	
NeuAcα2-6GlcNAcβ1-4Galβ1-3[NeuAcα2-3Galβ1-4GlcNAcβ1-6)GalNAc	S6GL4ga3[S3ga3GL6]GA		5.89	0.12	not present		not present		5.94	0.41
NeuAcα2-3GlcNAcβ1-4Galβ1-3(NeuAcα2-3Galβ1-4GlcNAcβ1-6)GalNAc	S3GL4ga3[S3ga3GL6]GA		not present		5.91	0.13	5.83	0.35	not present	
NeuAcα2-3GalNAcβ1-4GlcNAcβ1-2[NeuAcα2-3GalNAcβ1-4GlcNAcβ1-6)Man	S3GA4GL2[S3GA4GL6]m		not present		not present		not present		6.20	0.29

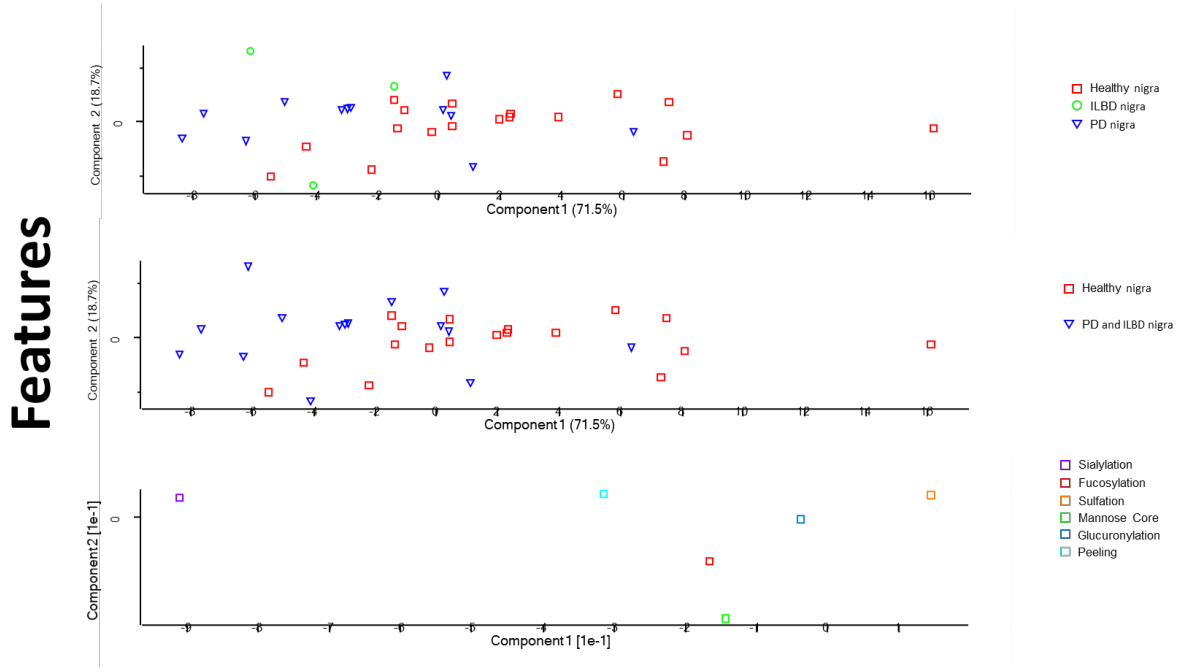
Appendix XI (cont.). *FLR-UPLC data of 2AB labelled glycans outlining GU values and relative area. "x" – Gal core contamination. "z" – GU value based on values in GlycoStore as dextran was unable to provide GU values lower than 1.*



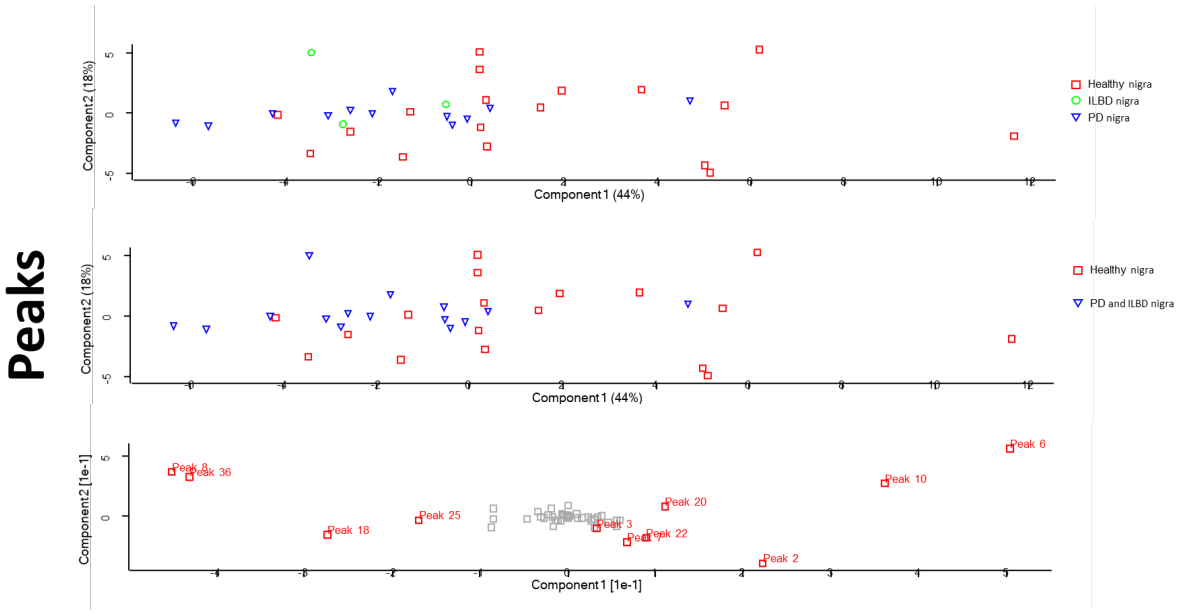
Appendix XII. *Nested tables showing % area for individual samples in each group compared against disease states in striatum. Significance across groups is written as follows: * ($p < 0.05$), ** ($p < 0.01$), *** ($p < 0.001$).*



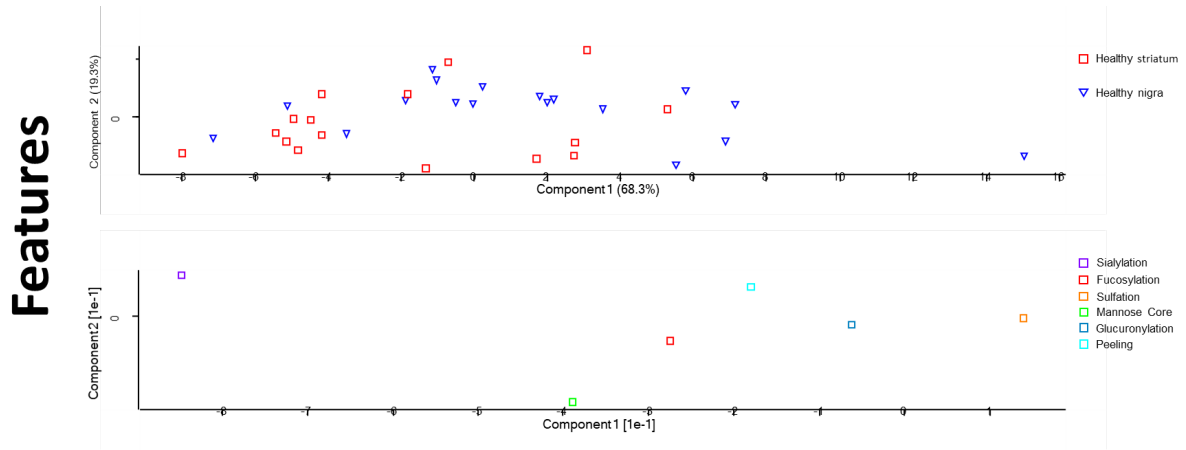
Appendix XIII. *Nested tables showing % area for individual samples in each group compared against disease states in substantia nigra. Significance across groups is written as follows: * ($p < 0.05$).*



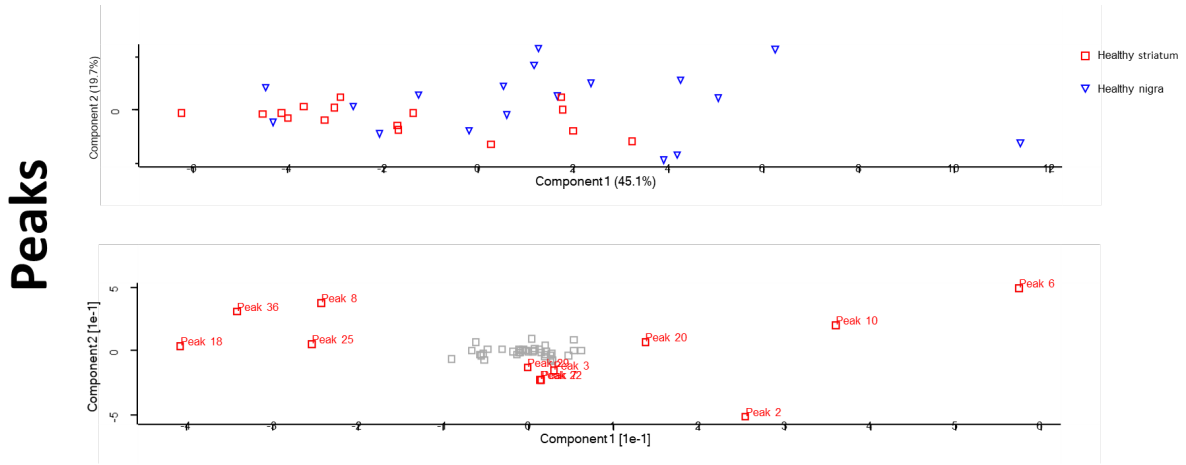
Appendix XIV. *PCA comparing glycan features in healthy control PD, ILDB substantia nigra.*



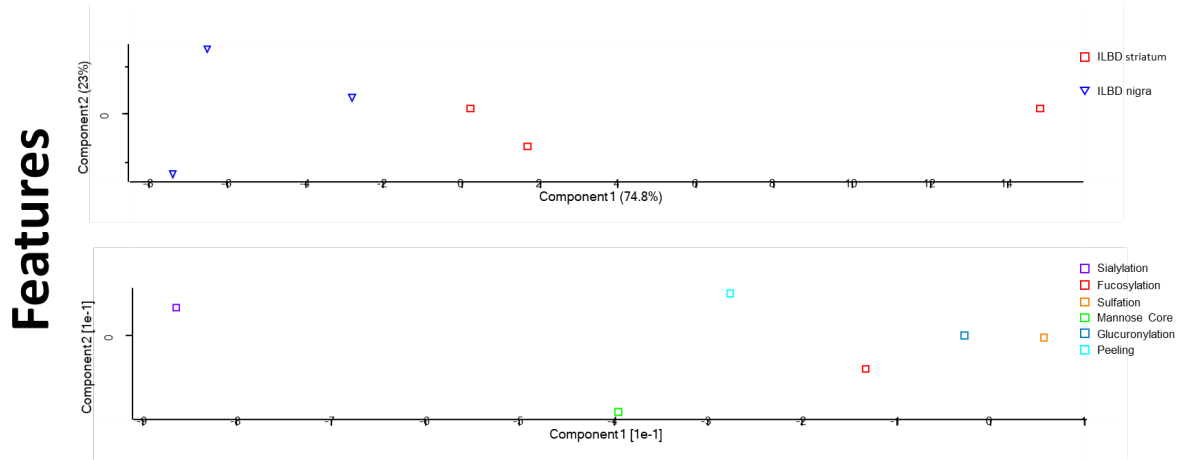
Appendix XV. *PCA comparing glycan peaks in healthy control PD, ILDB substantia nigra.*



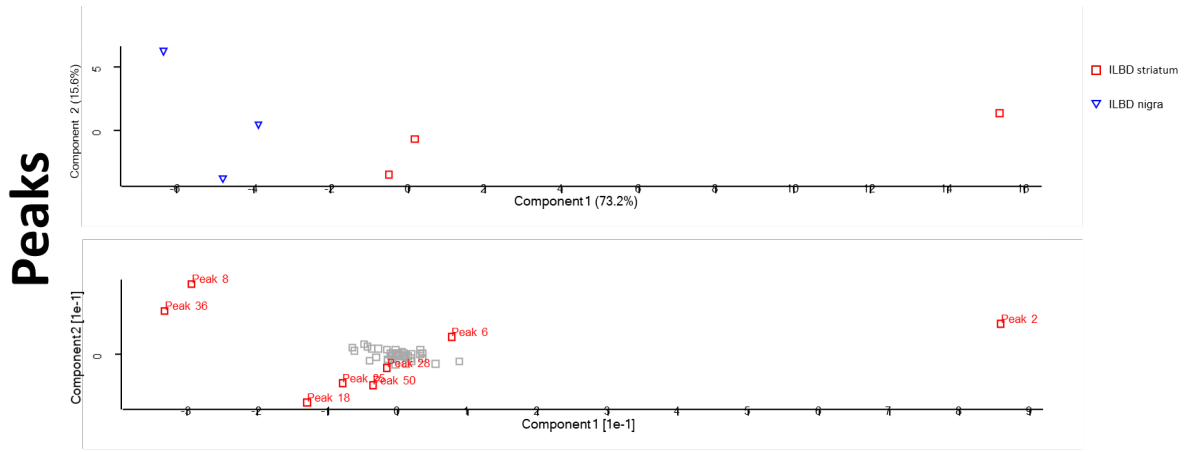
Appendix XVI. *PCA comparing glycan features between healthy striatum and substantia nigra.*



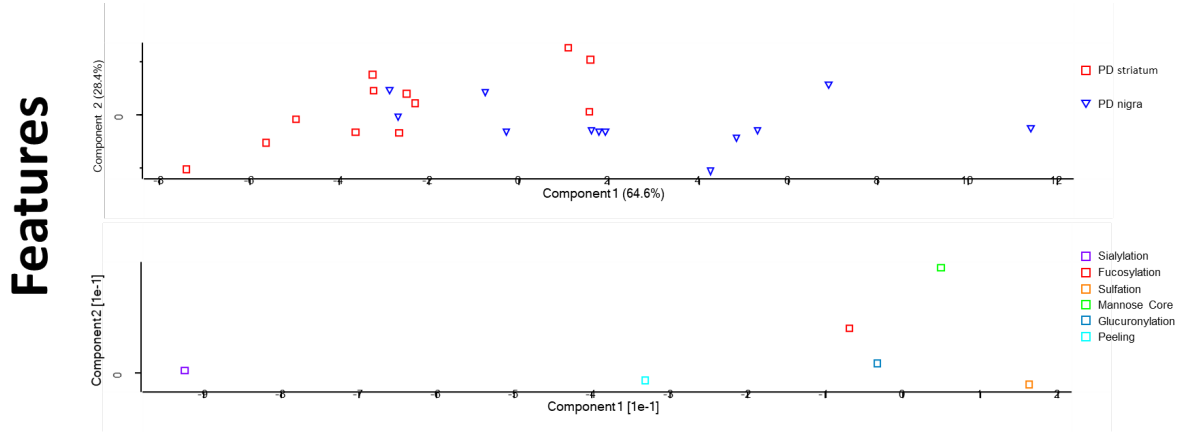
Appendix XVII. *PCA comparing glycan peaks between healthy striatum and substantia nigra.*



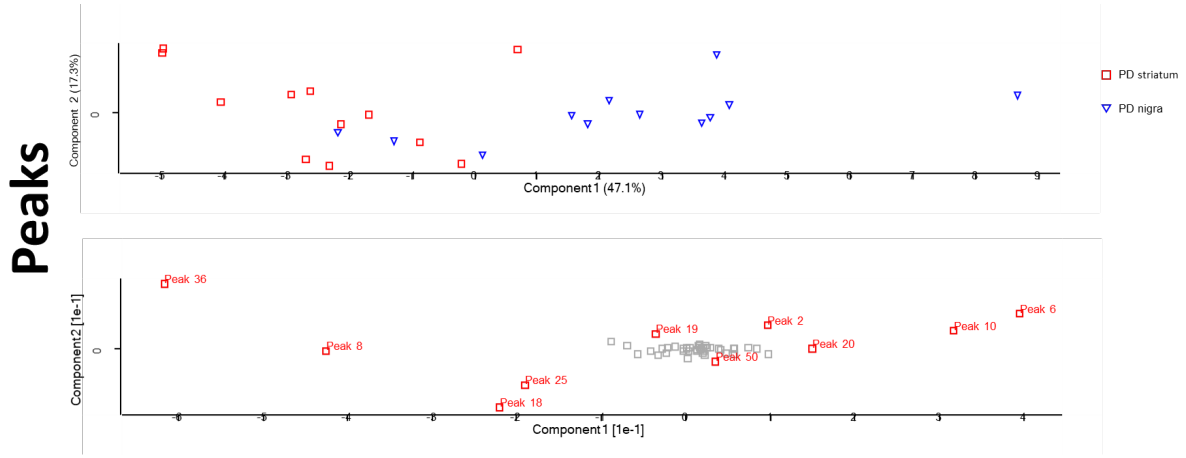
Appendix XVIII. *PCA comparing glycan features between ILBD striatum and substantia nigra.*



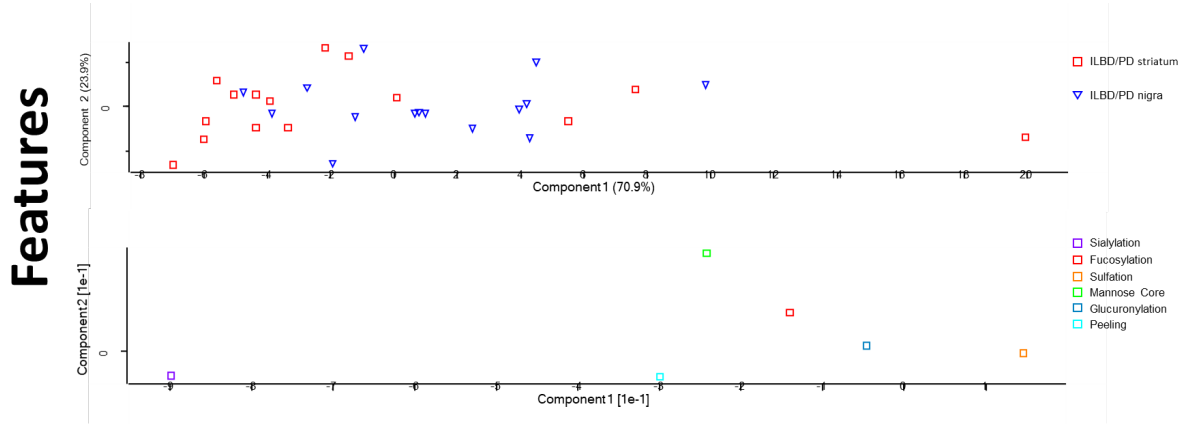
Appendix XIX. *PCA comparing glycan peaks between ILBD striatum and substantia nigra.*



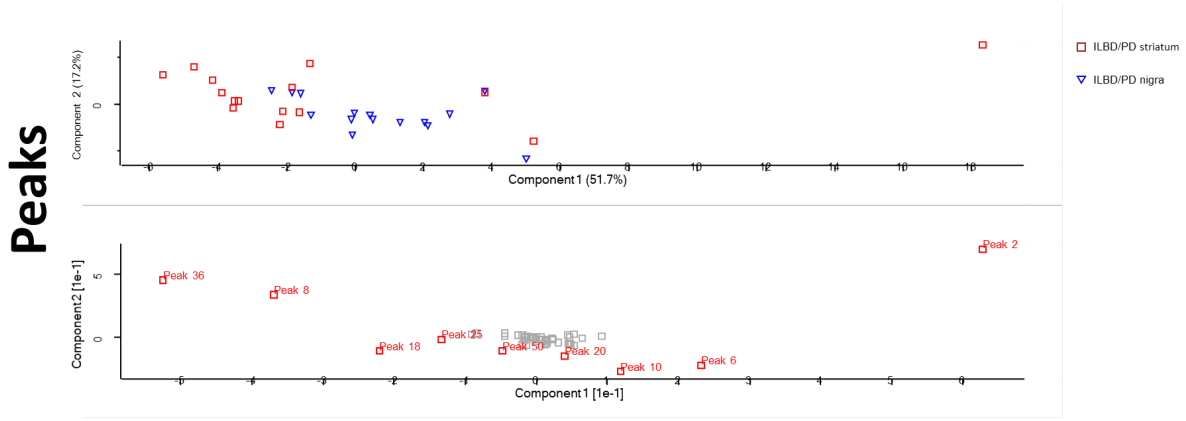
Appendix XX. *PCA comparing glycan features between PD striatum and substantia nigra.*



Appendix XXI. *PCA comparing glycan peaks between PD striatum and substantia nigra.*



Appendix XXII. *PCA comparing glycan features between ILBD/PD striatum and substantia nigra.*



Appendix XXIII. *PCA comparing glycan peaks between ILBD/PD striatum and substantia nigra.*

compared to positive mode ionization strategies. Analysis in negative mode was not accompanied by rearrangements such as fucose migration or in-source decay leading to loss of N¬-acetyl glucosamine residues leading to more confident identifications. In addition, MS2 fragmentation in negative mode generated more structurally informative spectra due to the increased propensity for cross-ring fragments to be formed. Released N-glycans from several glycoprotein sources have been labelled using 3-ASA and analysed in negative ionization mass spectrometry. More structurally relevant diagnostic ions were observed in glycans labelled with 3-ASA compared to other commercially available labels which led to enhanced identification and characterization glycan structures.
Enhanced structural characterization of released N-glycans using a novel fluorescent label in a manner accessible to the biopharma industry.

391. SLC35A2 has the potential to form complexes with functionally distinct glycosylation enzymes

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Glycosylation is crucial for nearly all aspects of life, including the development of multicellular organisms. Substrates which are used in the glycosylation process are nucleotide sugars. They are synthesized in the cytosol. To be delivered into the ER and Golgi lumen, where glycosylation occurs, they need to be translocated across membranes of these organelles. Nucleotide sugar transporters (NSTs) play an essential role in this process. The mutations in the gene encoding SLC35A2 which has a UDP-galactose transporting activity, causes a variant of congenital disorder of glycosylation.
Here we employed a novel protein fragment complementation assay called NanoBiT to explore the potential of SLC35A2 to form complexes with selected glycosylation enzymes. We previously showed that SLC35A2 forms heteromers with N-acetylglucosaminyltransferases (MGATs 1-5). Here, for the first time, we demonstrate an interaction between SLC35A2 and B4GalT1, which is the second discovered interaction of this NST with a galactosylotransferase. Moreover, we show a previously unacknowledged interaction between SLC35A2 and a sialyltransferase (ST6Gal1). Furthermore, we show that Mgat2 interacts with B4GalT1. Given the fact that B4GalT1 interacts with ST6Gal1 and SLC35A2 interacts with Mgat2, B4GalT1 and ST6Gal1, we postulate that all listed proteins, which are crucial for process of N-glycosylation, could be in close proximity. We also demonstrate a close proximity between SLC35A2 and galactosyltransferases involved in O-glycosylation (C1GalT1) and keratan sulfate chain elongation (B4GalT4).
Our study suggests that the biological consequences of the SLC35A2 depletion in CDG patients might not be limited to a defect in UDP-galactose translocation across the Golgi membrane. Additionally, we suggest the existence of multiprotein glycosylation-related complexes, whose spatial organization might be orchestrated by NSTs.

392. Method development of o-glycan analyses for use in regenerative medicine

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Glycosylation is the most common post-translational modification of proteins. Glycosylation patterns may differ between control and disease tissue and have potential to be used as biomarkers for these diseases. Regenerative implants and therapeutics developed from this knowledge may be utilised to re-establish a healthy glyco-environment. As glycans help determine protein stability and function, an understanding into the role of glycoproteins in Parkinson's Disease (PD) is of great interest.
Two main classes of glycans present on these proteins are N-glycans and O-glycans. The enzyme PNGase F cleaves the common core of N-glycans from the glycoprotein for subsequent analysis, however, no single enzymatic approach exists to release O-glycans due to a lack of common O-glycan core, so chemical methods are used. The downside to this is a "peeling" reaction which degrades the glycan making analysis difficult and non-reproducible.
This work focuses on optimising a non-reductive ®-elimination for O-glycan release that utilises a base, 40% dimethylamine containing ammonium carbonate, to release and convert the glycan to a glycosylamine to prevent peeling. The addition of microwave radiation has been utilised to speed up a release reaction that typically requires 16-40 hours of incubation to 8-12 minutes. The optimisation of the method includes optimal release of O-glycans of fetuin from fetal bovine serum suspended in a bis-acrylamide gel by observing a range of sample preparation conditions such as reducing and alkylating the protein, limiting glycan heterogeneity by releasing N-glycans first, finding ideal reaction reagents, identifying ideal sample concentration and finding the most suitable microwave reaction conditions, as well as adjusting analytical techniques using hydrophilic interaction liquid chromatography-ultra performance liquid chromatography (HILIC-UPLC) and mass spectrometry (MS) for subsequent analysis.
After the method is developed, O-glycans will be released from nigrostriatal tissue regions of the human brain (healthy and PD) as well as from rat models with LPS induced neuroinflammation, forming the basis of a comparative glycan analysis for potential biomarker identification.
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393. Glycan structures and their recognition roles in the human blood group abh/ii, lea, b, x, y and sialyl lea,x active cyst glycoproteins

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Human ovarian cyst glycoproteins (HOC , cyst gps) isolated from pseudomucinous type of human ovarian cyst fluids is one of the most rich and pioneer sources for studying brosynthesis, structures and function roles of blood group **ABH, Lea,b,x,y, sLea** and **sLex** active glycoproteins. After 70+ years exploration, four top highlights are shared. (i) an updated concept of glycotopes and their internal structures in cyst gps was composited; (ii) the unknown codes of new genes in secreted cyst gps were unlocked as **Lex** and **Ley** ; (iii) recognition profiles of cyst glycans and a sialic acid rich (18%) glycan with lectins and antibodies were shown. (iv) co-expression of Blood Group **A/ A-Leb/y** and **B/B-Leb/y** active Glycotopes in glycans were isolated and illustrated. These are the most advanced achievements since 1980.

394. Both catalytic and non-catalytic domains of the cmp-sia synthetase are essential for embryogenesis

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Sialic acid (Sia) residues are expressed on glycan chains of vertebrate glycoproteins and glycolipids in a developmental stage-specific and organ-specific manners. It has been reported that disturbance of Sia biosynthesis leads to many human diseases, For example, Biallelic mutations

Appendix XXIV. *Abstract from Conference Proceeding. The 25th International Symposium for Glycoconjugates – 24th-31st August, 2019 – Milan, Italy*

EPE ACTIVITY	TITLE / DETAILS	DATE	VENUE (if applicable)	AUDIENCE	LINK (if one exists)
Contribution to a website, online portal or creation of a blog linked to your research and/or creation and upkeep of social media profile (on appropriate channels available to a lay audience)	Twitter (A.S.A.P Speak)	Current	N/A	Lay	https://twitter.com/asapspeak
Development of online/social media portals, games, apps, websites and/or cross-platform projects	Website to accompany my poster presentation for Glyco25, Milan, Italy	25th August 2019	N/A	Researchers	https://glycobiologyoglycan.wordpress.com/
Participation in training programs to build the capacity of researchers to communicate to and engage with the public	"Writing for Public Engagement" workshop, CURAM Annual Retreat, 25th June	25th June 2019	Sheraton Hotel, Athlone	Lay Audience and Researchers	
Development of and/or delivery of small scale school interactions and open lab days, including prop development	Transition Year Program (NIBRT)	31st January 2019	National Institute for Bioprocessing, Research and Training, Dublin	Transition Year students	
Creation of teaching materials and/or development of teaching methods to support teachers and students at primary and post primary level	Science Communication Videos for transition year students	30th August 2020	N/A	Secondary School Students	
Collaboration and/or participation in festivals and events run within your research institution	Cheltenham Science Festival - Famelab International	3rd June 2019 - 7th June 2019	Cheltenham Town Hall, UK	Lay	https://www.britishcouncil.ie/events/famelab-international-finals-2015
Collaboration and/or participation in public engagement activities with museums, galleries or public access venues	FameLab International Echoes	15th November 2019	Prague Library, Prague	Lay/Researchers/students	
Collaboration and/or participation in public engagement activities with museums, galleries or public access venues	Famelab Dublin Final	11th April 2019	Science Gallery, Dublin	Lay	https://www.ucd.ie/newsandopinion/news/2019/april/12/ucdstudentcompetesfortitleofworldsbestsciencecommunicatoratfamelab/
Contribution to broadcast or media productions including TV, film, animation or radio helping embed research and discovery in the everyday lexicon of Irish culture	Newstalk (Futureproof) Radio	25th May 2019	Newstalk Headquarters, Dublin	Lay	https://www.newstalk.com/futureproof/famelab-science-audience-futureproof-864398 https://www.newstalk.com/podcasts/futureproof-with-jonathan-mccrea/famelab-finalist-hayden-wilkinson
STEM public engagement workshops, events, debates and discussions	Postgrad/Postdoc Poetry Slam	25th June 2019	Dark Horse Pub, Athlone	Lay	
Creation of situations facilitating dialogue with policy makers and wider stakeholders to ensure research informs or influences changes in policy and legislation	SCI:COM Science Communication conference - Interaction with stakeholders across media, schools, universities about best practice to encourage science communication and engagement across schools and universities	3rd December 2019	Aviva Stadium, Dublin	Lay/stakeholders	https://www.scicom.ie/
Provision of opportunities to engage the public and wider stakeholders to ensure research informs or influences changes in behavior or practice	"Why Research Matters" Video Competition	30th June 2020	N/A	Lay/stakeholders	https://twitter.com/TheUSI/status/1275821022231609346/photo/1
Consultation with stakeholders/communities affected by the research to inform the research process	PwC/U21 Innovation Challenge	27th March 2020	N/A	Lay/Researchers /Stakeholder	https://universitas21.com/news-and-events/news/winners-u21-pwc-innovation-challenge-2019-20-announced

Appendix XXV. *Education and Public Engagement activities undertaken during the project.*

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